

# CHAPTER 6

## USE OF VIBRATION SIGNATURE ANALYSIS TO DIAGNOSE MACHINE PROBLEMS

### 6.0 USE OF VIBRATION SIGNATURE ANALYSIS

Probably the greatest shortcoming in today's predictive maintenance programs is the ability to diagnose the mechanical and electrical problems within the machine that are evidenced in the vibration signatures if the vibration exceeds preset overall and spectral alarm levels. For example, an ever increasing number of plants have PMP data collectors and software and successfully build large databases and capture vibration measurements at great numbers of points. However, surveys prove that less than 15% of such plants know how to properly set overall and spectral alarm levels. Then, even a smaller minority know how to diagnose the array of potential problems from the vibration spectra and related variables (i.e. spike energy) which are faithfully printed out by the PMP software for those points that exceed alarm. Therefore, the overriding purpose of this section is to begin to acquaint the reader on how to diagnose such problems from this data. Much valuable information is contained within these vibration spectra, but is only of use if the analyst can unlock its "secrets".

Therefore, Table 6.0 has been developed to put many of these "secrets" right at the fingertips of the analyst. Several hundred hours of research have gone into the development of this four page diagnostic chart. Please note that this chart not only provides text elaborating on vibration symptoms for various machine problems, but it also includes illustrations of "typical vibration spectra" for each problem covered. In addition, drawings are included to illustrate how phase reacts when such problems are predominant. Table 6.0 represents the best understanding to date of the author on how these problems are best diagnosed, based on approximately 16 years field experience in vibration signature analysis and research on a wide range of articles which have been written on the subject.

There are several key items included in Table 6.0. First, the plots under "TYPICAL SPECTRUM" column reveal invaluable information as to the source of the problem. When looking at such spectra, the analyst should ask questions similar to the following:

1. Which frequencies are present in spectrum and how do they relate to machine operating speed (that is, are the peaks present equal to 1X, 2X, 3X, 5.78X RPM or what)?
2. What are the amplitudes of each peak?
3. How do the frequency peaks relate to one another? (i.e., "2X RPM is much higher than 1X RPM"; "there is a large peak at 7.43X RPM"; "there are large number of operating speed harmonics present"; "there are high amplitude sidebands around gear mesh frequency"; "there are 7200 CPM sidebands around a large peak at 46X RPM"; etc.).
4. Finally, if there are significant amplitude peaks, what exactly is their source ("is 7.43X RPM a bearing defect frequency"; "is the 46X RPM peak equal to the number of rotor bars RPM"?).

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As its column name implies, "TYPICAL SPECTRUM" is meant to be a representative signature for each problem listed in Table 6.0. These spectra are not intended to be all inclusive. For example, referring to "REMARKS" for the Angular Misalignment problem, note that while the typical spectrum for this problem shows high amplitude 1X RPM and 2X RPM peaks in the axial direction, the discussion shows that it is not unusual for either 1X, 2X or 3X RPM to dominate the whole spectrum when angular misalignment is the problem. In addition, it is not unusual for a machine to have two or more problems present at any one time. For example, if a machine simultaneously had both mechanical looseness and rotor unbalance, they each would contribute frequencies to its spectra which might show high 1X RPM in addition to multiple running speed harmonics.

The next column in Table 6.0 is entitled "PHASE RELATIONSHIP". Information on phase is provided for several of the problem sources listed. Amplitude reveals how much something is vibrating. Frequency relates how many cycles occur per unit of time. Phase completes the picture by showing just how the machine is vibrating. Of great importance, phase is a powerful tool in helping to differentiate which of several problem sources are dominant. For example, there are a large number of problems that generate vibration at 1X and 2X RPM. Using phase, one learns how the machine is vibrating, and in the process, helps zero in on just which problem is present. For example, Table 6.0 shows how phase reacts during the following scenarios:

1. **Force (or "static") unbalance** is evidenced by nearly identical phase in the radial direction on each bearing of a machine rotor.
2. **Couple unbalance** shows approximately a 180° out-of-phase relationship when comparing the outboard and inboard horizontal, or the outboard and inboard vertical direction phase on the same machine.
3. **Dynamic unbalance** is indicated when the phase difference is well removed from either 0° or 180°, but importantly is nearly the same in the horizontal and vertical directions. That is, the horizontal phase difference could be almost anything between the outboard and inboard bearings; but, the key point is that the vertical phase difference should then be almost identical to the horizontal phase difference ( $\pm 30^\circ$ ). For example, if the horizontal phase difference between the outboard and inboard bearings is 60°, and the dominant problem is dynamic unbalance, the vertical phase difference between these two bearings should be about 60° ( $\pm 30^\circ$ ). If the horizontal phase difference varies greatly from the vertical phase difference, this strongly suggests the dominant problem is not unbalance.
4. **Angular misalignment** is indicated by approximately a 180° phase difference across the coupling, with measurements in the axial direction.
5. **Parallel misalignment** causes radial direction phase across the coupling to be approximately 180° out of phase with respect to one another.
6. **Bent shaft** causes axial phase on the same shaft of a machine to approach a 180° difference when comparing measurements on the outboard and inboard bearings of the same machine rotor.
7. **Resonance** is shown by a 90° phase change at the point when the forcing frequency coincides with a natural frequency, and approaches a full 180° phase change when the machine passes through the natural frequency (depending on the amount of damping present).
8. **Rotor rub** causes significant, instantaneous changes in phase.

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9. **Mechanical looseness/weakness due to base/frame problems or loose hold-down bolts** is indicated by nearly a 180° phase change when one moves his transducer from the machine foot down to its baseplate and then down to its support base.
  10. **Mechanical looseness due to a cracked frame, loose bearing or loose rotor** causes phase to be unsteady with probable widely differing phase measurements from one measurement to the next. The phase measurement may noticeably differ every time you start up the machine, particularly if the rotor itself is loose and rotates on the shaft a few degrees with each startup.

Often, even though phase measurement capability is now offered by most data collectors, users do not use this powerful tool. If not used, this will severely limit the diagnostic capabilities of any program. However, currently it would be impractical to make phase measurements on all machinery during regular PMP surveys. Its greatest use comes into play when performing diagnostics on machines which have developed high vibration at 1X, 2X or 3X RPM, requiring investigation to detect the predominate cause(s) prior to taking corrective actions.

Note that "PHASE RELATIONSHIP" is illustrated in each of the first 8 problems of Table 6.0 since it is primarily with these problems that phase can be used to differentiate which problem(s) dominate. Phase is then discussed in many of the remarks for the remaining problems in Table 6.0, although it is not illustrated.

Finally, a remarks column is included in Table 6.0 to provide further explanatory information on machine problem symptoms and diagnostics. For example, there is a warning under the remarks column for the "bent shaft" problem source to be sure and account for transducer orientation when taking axial phase measurements.

It is hoped that this illustrated chart will help users in diagnosing a wide variety of machine problems. Further information is now being researched and field tested which may soon be added to the diagnostic chart as we constantly learn more and more about how machines react when subjected to a whole series of problems and how we can "read" these reactive responses via diagnostic techniques.

Following on the next pages will be separate discussions on each of the problems outlined in Table 6.0. Later, real-world case histories will be presented giving real-world examples of each of these problems.

Table 6.0

# ILLUSTRATED VIBRATION DIAGNOSTIC CHART

| PROBLEM SOURCE                                 | TYPICAL SPECTRUM             | PHASE RELATIONSHIP | REMARKS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------------|------------------------------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MASS UNBALANCE</b><br>A. FORCE UNBALANCE    | 1X RADIAL                    |                    | Force Unbalance will be in-phase and steady. Amplitude due to unbalance will increase by the square of speed below first rotor critical (a 3X speed increase = 9X higher vibration). 1X RPM always present and normally dominates spectrum. Can be corrected by placement of only one balance correction weight in one plane at Rotor center of gravity (CG). Approx. 0° phase difference should exist between OB & IB horizontals, as well as between OB & IB verticals. Also, approx. 90° phase difference between horizontal & vertical readings usually occurs on each bearing of unbalanced rotor ( $\pm 30^\circ$ ).                                                                                                                                                                       |
| B. COUPLE UNBALANCE                            | 1X RADIAL                    |                    | Couple Unbalance results in 180° out-of-phase motion on same shaft. 1X RPM always present and normally dominates spectrum. Amplitude varies with square of increasing speed below first rotor critical speed. May cause high axial vibration as well as radial. Correction requires placement of balance weights in at least 2 planes. Note that approx. 180° phase difference should exist between OB & IB horizontals, as well as between OB & IB verticals. Also, approx. a 90° difference between the horizontal & vertical phase readings on each bearing usually occurs ( $\pm 30^\circ$ ).                                                                                                                                                                                                |
| C. DYNAMIC UNBALANCE                           | 1X RADIAL                    |                    | Dynamic Unbalance is the dominant type of unbalance found and is a combination of both force and couple unbalance. 1X RPM dominates the spectrum, and truly requires 2 plane correction. Here, the radial phase difference between outboard and inboard bearings can range anywhere from 0° to 180°. However, the horizontal phase difference should closely match the vertical phase difference, when comparing outboard and inboard bearing measurements ( $\pm 30^\circ$ ). Secondly, if unbalance predominates, roughly a 90° phase difference usually results between the horizontal and vertical readings on each bearing ( $\pm 30^\circ$ ).                                                                                                                                              |
| D. OVERHUNG ROTOR UNBALANCE                    | 1X AXIAL & RADIAL            |                    | Overhung Rotor Unbalance causes high 1X RPM in both Axial and Radial directions. Axial readings tend to be in-phase whereas radial phase readings might be unsteady. However, the horizontal phase differences will usually match the vertical phase differences on the unbalanced rotor ( $\pm 30^\circ$ ). Overhung rotors have both force and couple unbalance, each of which will likely require correction. Thus, correction weights will most always have to be placed in 2 planes to counteract both force and couple unbalance.                                                                                                                                                                                                                                                          |
| <b>ECCENTRIC ROTOR</b>                         | 1X FAN<br>1X MOTOR<br>RADIAL |                    | Eccentricity occurs when center of rotation is offset from geometric centerline of a pulley, gear, bearing, motor armature, etc. Largest vibration occurs at 1X RPM of eccentric component in a direction thru centerlines of the two rotors. Comparative horizontal and vertical phase readings usually differ either by 0° or by 180° (each of which indicate straight-line motion). Attempts to balance eccentric rotors often result in reducing vibration in one radial direction, but increasing it in the other radial direction (depending on amount of eccentricity).                                                                                                                                                                                                                   |
| <b>BENT SHAFT</b>                              | 1X AXIAL<br>2X               |                    | Bent shaft problems cause high axial vibration with axial phase differences tending towards 180° on the same machine component. Dominant vibration normally occurs at 1X if bent near shaft center, but at 2X if bent near the coupling. (Be careful to account for transducer orientation for each axial measurement if you reverse probe direction.) Use dial indicators to confirm bent shaft.                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>MISALIGNMENT</b><br>A. ANGULAR MISALIGNMENT | 1X AXIAL<br>2X<br>3X         |                    | Angular Misalignment is characterized by high axial vibration, 180° out-of-phase across the coupling. Typically will have high axial vibration with both 1X and 2X RPM. However, not unusual for either 1X, 2X or 3X to dominate. These symptoms may also indicate coupling problems as well. Severe angular misalignment may excite many 1X RPM harmonics. Unlike Mechanical Looseness Type 3, these multiple harmonics do not typically have a raised noise floor on the spectra.                                                                                                                                                                                                                                                                                                              |
| B. PARALLEL MISALIGNMENT                       | 1X RADIAL<br>2X<br>3X        |                    | Offset Misalignment has similar vibration symptoms to Angular, but shows high radial vibration which approaches 180° out-of-phase across coupling. 2X often larger than 1X, but its height relative to 1X is often dictated by coupling type and construction. When either Angular or Radial Misalignment becomes severe, they can generate either high amplitude peaks at much higher harmonics (4X-8X), or even a whole series of high frequency harmonics similar in appearance to mechanical looseness. Coupling type and material will often greatly influence the entire spectrum when misalignment is severe. Does not typically have raised noise floor.                                                                                                                                 |
| C. MISALIGNED BEARING COCKED ON SHAFT          | 1X AXIAL<br>2X<br>3X         |                    | Cocked Bearing will generate considerable axial vibration. Will cause Twisting Motion with approximately 180° phase shift top to bottom and/or side to side as measured in axial direction on same bearing housing. Attempts to align coupling or balance the rotor will not alleviate problem. Bearing usually must be removed and correctly installed.                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>RESONANCE</b>                               |                              |                    | Resonance occurs when a Forcing Frequency coincides with a System Natural Frequency, and can cause dramatic amplitude amplification, which might result in premature, or even catastrophic failure. This may be a natural frequency of the rotor, but can often originate from support frame, foundation, gearbox or even drive belts. If a rotor is at or near resonance, it can be almost impossible to balance due to the great phase shift it experiences (90° at resonance; nearly 180° when passes thru). Often requires changing natural frequency to a higher or lower frequency. Natural Frequencies do not generally change with a change in speed which helps facilitate their identification (unless on a large plain bearing machine or on a rotor which has significant overhang). |

Table 6.0

# ILLUSTRATED VIBRATION DIAGNOSTIC CHART

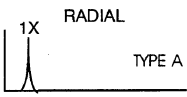
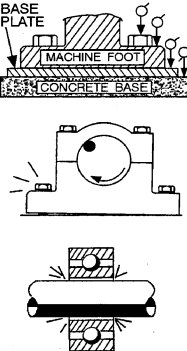
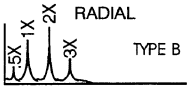
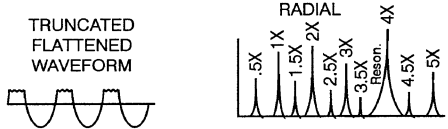
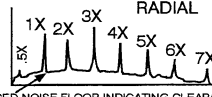
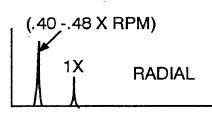
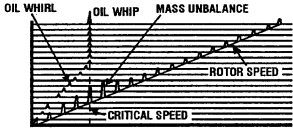
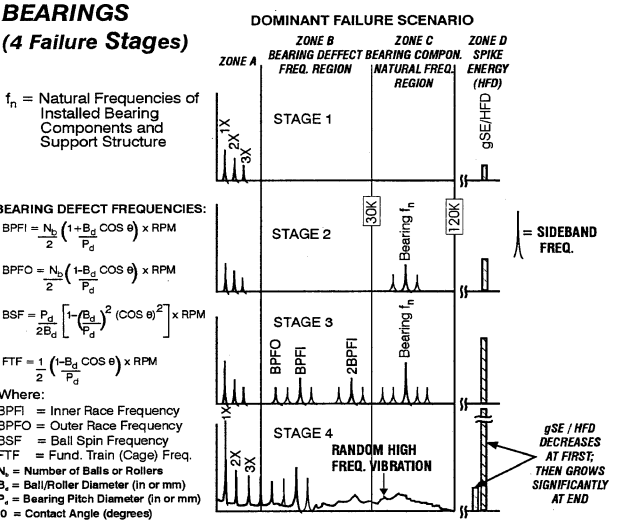
| PROBLEM SOURCE                    | TYPICAL SPECTRUM                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | PHASE RELATIONSHIP                                                                  | REMARKS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MECHANICAL LOOSENESS</b>       |  <p>1X RADIAL<br/>TYPE A</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    | <p>Mechanical Looseness is indicated by either Type A, B or C vibration spectra. <b>Type A</b> is caused by Structural looseness/weakness of machine feet, baseplate or foundation; also by deteriorated grouting, loose hold-down bolts at the base; and distortion of the frame or base (i.e., soft foot). Phase analysis may reveal approx. 90° to 180° phase difference between vertical measurements on bolt, machine foot, baseplate or base itself.</p> <p><b>Type B</b> is generally caused by loose pillowblock bolts, cracks in frame structure or in bearing pedestal.</p> <p><b>Type C</b> is normally generated by improper fit between component parts which will cause many harmonics due to nonlinear response of loose parts to dynamic forces from rotor. Causes a truncation of time waveform and a raised noise floor in the spectrum. Type C is often caused by a bearing liner loose in its cap, a bearing loose and turning on its shaft, excessive clearance in either a sleeve or rolling element bearing, a loose impeller on a shaft, etc. Type C Phase is often unstable and may vary widely from one measurement to next, particularly if rotor shifts position on shaft from one startup to next. Mechanical Looseness is often highly directional and may cause very different readings when comparing levels at 30° increments in radial direction all the way around one bearing housing. Also, note that looseness will often cause subharmonic multiples at exactly 1/2 or 1/3X RPM (.5X, 1.5X, 2.5X, etc.).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                   |  <p>1X 2X 3X 4X 5X 6X 7X 8X 9X 10X RADIAL<br/>TYPE B</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                   |  <p>NOTE RAISED NOISE FLOOR INDICATING LOOSENESS</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>ROTOR RUB</b>                  |  <p>TRUNCATED FLATTENED WAVEFORM</p> <p>5X 1X 1.5X 2X 2.5X 3X 3.5X 4X 4.5X 5X RADIAL</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                     | <p>Rotor Rub produces similar spectra to Mechanical Looseness when rotating parts contact stationary components. Rub may be either partial or throughout the entire shaft revolution. Usually generates a series of frequencies, often exciting one or more resonances. Often excites integer fraction subharmonics of running speed (1/2, 1/3, 1/4, 1/5...1/n), depending on location of rotor natural frequencies. Rotor rub can excite many high frequencies (similar to wide-band noise when chalk is chalk along a blackboard). It can be very serious and of short duration if caused by shaft contacting bearing babbit. A full annular rub throughout an entire shaft revolution can induce "reverse precession" with the rotor whirling at critical speed in a direction opposite shaft rotation (inherently unstable which can lead to catastrophic failure).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>JOURNAL BEARINGS</b>           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>A. WEAR/CLEARANCE PROBLEMS</b> |  <p>1X 2X 3X 4X 5X 6X 7X RADIAL</p> <p>NOTE RAISED NOISE FLOOR INDICATING CLEARANCE/LOOSENESS.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                     | <p>Latter stages of journal bearing wear are normally evidenced by presence of whole series of running speed harmonics (up to 10 or 20). Wiped journal bearings often will allow high vertical amplitudes compared to horizontal, but, may show only one pronounced peak at 1X RPM. Journal bearings with excessive clearance may allow a minor unbalance and/or misalignment to cause high vibration which would be much lower if bearing clearances were set to spec.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>B. OIL WHIRL INSTABILITY</b>   |  <p>(40 - 48 X RPM)<br/>1X RADIAL</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  | <p>Oil Whirl instability occurs at .40 - .48X RPM and is often quite severe. Considered excessive when amplitude exceeds 40% of bearing clearances. Oil Whirl is an oil film excited vibration where deviations in normal operating conditions (attitude angle and eccentricity ratio) cause oil wedge to "push" shaft around within bearing. Destabilizing force in direction of rotation results in a whirl (or forwards precession). Oil Whirl is unstable since it increases centrifugal forces which increase whirl forces. Can cause oil to no longer support shaft and can become unstable when whirl frequency coincides with a rotor natural frequency. Changes in oil viscosity, lube pressure and external preloads can affect oil whirl.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>C. OIL WHIP INSTABILITY</b>    |  <p>OIL WHIRL OIL WHIP MASS UNBALANCE</p> <p>A spectral map showing Oil Whirl becoming Oil Whip instability as shaft speed passes thru 2X critical.</p> <p>CRITICAL SPEED FREQUENCY</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                     | <p>Oil Whip may occur if machine operated at or above 2X rotor critical frequency. When rotor brought up to twice critical speed, whirl will be very close to rotor critical and may cause excessive vibration that oil film may no longer be capable of supporting. Whirl speed will actually "lock onto" rotor critical and this peak will not pass through it even if machine is brought to higher and higher speeds. Produces a lateral forward precessional subharmonic vibration at rotor critical frequency. Inherently unstable which can lead to catastrophic failure.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>ROLLING ELEMENT BEARINGS</b>   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>(4 Failure Stages)</b>         |  <p><b>DOMINANT FAILURE SCENARIO</b></p> <p>ZONE A BEARING DEFECT FREQ. REGION</p> <p>ZONE B BEARING DEFECT FREQ. REGION</p> <p>ZONE C NATURAL FREQ. REGION</p> <p>ZONE D SPIKE ENERGY (HFD)</p> <p>STAGE 1</p> <p>STAGE 2</p> <p>STAGE 3</p> <p>STAGE 4</p> <p>RANDOM HIGH FREQ. VIBRATION</p> <p>gSE / HFD DECREASES AT FIRST; THEN GROWS SIGNIFICANTLY AT END</p> <p><math>f_n</math> = Natural Frequencies of Installed Bearing Components and Support Structure</p> <p>BEARING DEFECT FREQUENCIES:</p> <p><math>BPFI = \frac{N_b}{2} (1 + B_d \cos \theta) \times \text{RPM}</math></p> <p><math>BPFO = \frac{N_b}{2} (1 - B_d \cos \theta) \times \text{RPM}</math></p> <p><math>BSF = \frac{P_d}{2B_d} [1 - (\frac{B_d}{P_d})^2 (\cos \theta)^2] \times \text{RPM}</math></p> <p><math>FTF = \frac{1}{2} (1 - B_d \cos \theta) \times \text{RPM}</math></p> <p>Where:</p> <p>BPFI = Inner Race Frequency</p> <p>BPFO = Outer Race Frequency</p> <p>BSF = Ball Spin Frequency</p> <p>FTF = Fund. Train (Cage) Freq.</p> <p><math>N_b</math> = Number of Balls or Rollers</p> <p><math>B_d</math> = Ball/Roller Diameter (in or mm)</p> <p><math>P_d</math> = Bearing Pitch Diameter (in or mm)</p> <p><math>\theta</math> = Contact Angle (degrees)</p> |                                                                                     | <p><b>4 ROLLING ELEMENT BEARING FAILURE STAGES</b></p> <p><b>STAGE 1:</b> Earliest indications of bearing problems appear in ultrasonic frequencies ranging from about 250,000 - 350,000 Hz; later, as wear increases, usually drops to approximately 20,000 - 60,000 Hz (1,200,000 - 3,600,000 CPM). These are frequencies evaluated by Spike Energy (gSE), HFD(g) and Shock Pulse (dB). For example, spike energy may first appear at about 25 gSE in Stage 1 (actual value depending on measurement location and machine speed). Acquiring high frequency enveloped spectra confirms whether or not bearing is in Failure Stage 1.</p> <p><b>STAGE 2:</b> Slight bearing defects begin to "ring" bearing component natural frequencies (<math>f_n</math>) which predominantly occur in 30K - 120K CPM range. Such natural frequencies may also be resonances of bearing support structures. Sideband frequencies appear above and below natural frequency peak at end of Stage 2. Overall spike energy grows (for example, from 25 to .50 gSE).</p> <p><b>STAGE 3:</b> Bearing defect frequencies and harmonics appear. When wear progresses, more defect frequency harmonics appear and number of sidebands grow, both around these and bearing component natural frequencies. Overall spike energy continues to increase (for example, from .5 to over 1 gSE). Wear is now usually visible and may extend throughout periphery of bearing, particularly when many well formed sidebands accompany bearing defect frequency harmonics. High frequency demodulated and enveloped spectra help confirm Stage III. <b>Replace bearings now!</b> (independent of bearing defect frequency amplitudes in vibration spectra).</p> <p><b>STAGE 4:</b> Towards the end, amplitude of 1X RPM is even effected. It grows, and normally causes growth of many running speed harmonics. Discrete bearing defect and component natural frequencies actually begin to "disappear" and are replaced by random, broadband high frequency "noise floor". In addition, amplitudes of both high frequency noise floor and spike energy may in fact decrease; but just prior to failure, spike energy and HFD will usually grow to excessive amplitudes.</p> |

Table 6.0

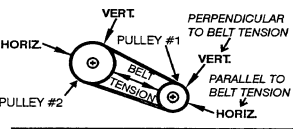
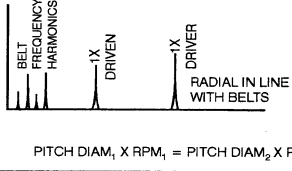
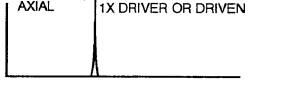
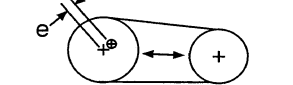
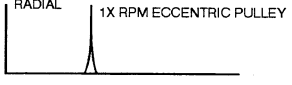
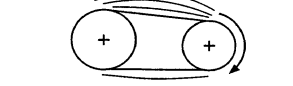
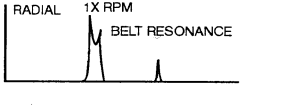
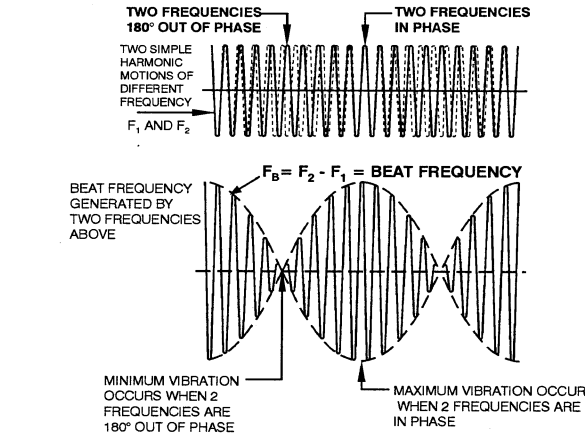
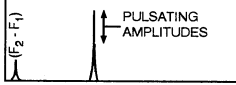
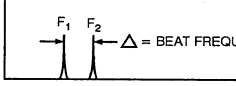
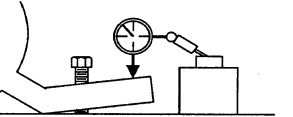
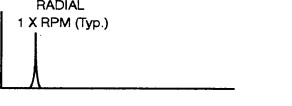
# ILLUSTRATED VIBRATION DIAGNOSTIC CHART

| PROBLEM SOURCE                           | TYPICAL SPECTRUM                                                                                                                                                                                                                                                                                                                                               | REMARKS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>HYDRAULIC AND AERODYNAMIC FORCES</b>  |                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>A. BLADE PASS &amp; VANE PASS</b>     | <p>BPF = # BLADES X RPM</p> <p>1X 2X BPF 2BPF 1X RPM SIDEBANDS</p>                                                                                                                                                                                                                                                                                             | <p>Blade Pass Frequency (BPF) = No. of Blades (or Vanes) X RPM. This frequency is inherent in pumps, fans and compressors, and normally does not present a problem. However, large amplitude BPF (and harmonics) can be generated in pump if gap between rotating vanes and stationary diffusers is not equal all the way around. Also, BPF (or harmonic) sometimes can coincide with a system natural frequency causing high vibration. High BPF can be generated if impeller wear ring seizes on shaft, or if welds fastening diffuser vanes fail. Also, high BPF can be caused by abrupt bends in pipe (or duct), obstructions which disturb flow, damper settings or if pump or fan rotor is positioned eccentrically within housing.</p>                                                                                                                                                             |
| <b>B. FLOW TURBULENCE</b>                | <p>RANDOM VIBRATION</p> <p>1X BPF</p> <p>BPF = BLADE OR VANE PASS FREQUENCY</p>                                                                                                                                                                                                                                                                                | <p>Flow Turbulence often occurs in blowers due to variations in pressure or velocity of the air passing thru the fan or connected ductwork. This flow disruption causes turbulence which will generate random, low frequency vibration, typically in the range of 50 to 2000 CPM. If surging occurs within a compressor, random broadband high frequency vibration can occur. Excessive turbulence can also excite broadband high frequency.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>C. CAVITATION</b>                     | <p>RANDOM HIGH FREQ. VIBRATION</p> <p>1X BPF 120K CPM</p>                                                                                                                                                                                                                                                                                                      | <p>Cavitation normally generates random, higher frequency broadband energy which is sometimes superimposed with blade pass frequency harmonics. Normally indicates insufficient suction pressure (starvation). Cavitation can be quite destructive to pump internals if left uncorrected. It can particularly erode impeller vanes. When present, it often sounds as if "gravel" is passing thru pump. Cavitation is usually caused by insufficient inlet flow. Can occur during one survey, and be absent the next survey (if changes in suction valve settings are made).</p>                                                                                                                                                                                                                                                                                                                           |
| <b>GEARS</b>                             |                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>A. NORMAL SPECTRUM</b>                | <p>GMF = <math>T_d \times \text{RPM}_d = T_p \times \text{RPM}_p</math></p> <p>1X GEAR 1X PINION GMF 2GMF 3GMF 3.25X GMF</p>                                                                                                                                                                                                                                   | <p>Normal Spectrum shows Gear &amp; Pinion Speeds, along with Gear Mesh Frequency (GMF) and very small GMF harmonics. GMF harmonics commonly will have running speed sidebands around them. All peaks are of low amplitude, and no natural frequencies of gears are excited. <math>F_{max}</math> recommended at 3.25X GMF (minimum) when # teeth are known. If tooth count is not known, set <math>F_{max}</math> at 200X RPM on each shaft.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>B. TOOTH WEAR</b>                     | <p>51T 26T</p> <p>1X GEAR 1X PINION GEAR NATURAL FREQUENCY GMF 2GMF 3GMF 3.25X GMF</p>                                                                                                                                                                                                                                                                         | <p>Key indicator of Tooth Wear is excitation of Gear Natural Frequency (<math>f_n</math>), along with sidebands around it spaced at the running speed of the bad gear. Gear Mesh Frequency (GMF) may or may not change in amplitude, although high amplitude sidebands and number of sidebands surrounding GMF usually occur when wear is noticeable. Sidebands may be better wear indicator than GMF frequencies themselves. Also, high amplitudes commonly occur at either 2XGMF or 3XGMF (esp. 3XGMF), even when GMF amplitude is acceptable.</p>                                                                                                                                                                                                                                                                                                                                                      |
| <b>C. TOOTH LOAD</b>                     | <p>1X GEAR 1X PINION GMF 2GMF 3GMF 3.25X GMF</p>                                                                                                                                                                                                                                                                                                               | <p>Gear Mesh Frequencies are often very sensitive to load. High GMF amplitudes do not necessarily indicate a problem, particularly if sideband frequencies remain low level, and no gear natural frequencies are excited. Each Analysis should be performed with system at maximum operating load for meaningful spectral comparisons.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>D. GEAR ECCENTRICITY AND BACKLASH</b> | <p>1X GEAR 1X PINION GMF 2GMF 3GMF 3.25X GMF</p>                                                                                                                                                                                                                                                                                                               | <p>Fairly high amplitude sidebands around GMF harmonics often suggest gear eccentricity, backlash, or non-parallel shafts which allow the rotation of one gear to "modulate" either the GMF amplitude or the running speed of the other gear. The gear with the problem is indicated by the spacing of the sideband frequencies. Also, 1X RPM level of eccentric gear will normally be high if eccentricity is the dominant problem. Improper backlash normally excites GMF harmonics and Gear Natural Frequency, both of which will be sidebanded at 1X RPM. GMF amplitudes will often decrease with increasing load if backlash is the problem.</p>                                                                                                                                                                                                                                                     |
| <b>E. GEAR MISALIGNMENT</b>              | <p>1X GEAR 1X PINION GMF 2GMF 3GMF 3.25X GMF</p>                                                                                                                                                                                                                                                                                                               | <p>Gear Misalignment almost always excites second order or higher GMF harmonics which are sidebanded at running speed. Often will show only small amplitude 1X GMF, but much higher levels at 2X or 3X GMF. Important to set <math>F_{max}</math> high enough to capture at least 3 GMF harmonics. Also, sidebands around 2XGMF will often be spaced at 2X RPM. Note that sideband amplitudes often are not equal on left and right side of GMF and GMF harmonics due to the tooth misalignment. Causes uneven wear pattern.</p>                                                                                                                                                                                                                                                                                                                                                                          |
| <b>F. CRACKED/BROKEN TOOTH</b>           | <p>1X GEAR RPM GEAR <math>f_n</math> TIME WAVEFORM</p> <p><math>\Delta = \text{RPM OF GEAR WITH BROKEN OR CRACKED TOOTH}</math></p>                                                                                                                                                                                                                            | <p>A Cracked or Broken Tooth will generate a high amplitude at 1X RPM of this gear only in the time waveform, plus it will excite gear natural frequency (<math>f_n</math>) sidebanded at its running speed. It is best detected in Time Waveform which will show a pronounced spike every time the problem tooth tries to mesh with teeth on the mating gear. Time between impacts (<math>\Delta</math>) will correspond to 1/RPM of gear with the problem. Amplitudes of Impact Spikes in Time Waveform often will be 10X to 20X higher than that at 1X RPM in the FFT!</p>                                                                                                                                                                                                                                                                                                                             |
| <b>G. GEAR ASSEMBLY PHASE PROBLEMS</b>   | <p>GAPF = <math>\frac{GMF}{N_A}</math></p> <p>600 RPM 15T = <math>T_d</math> 25T = <math>T_p</math> 15T = <math>T_d</math> 25T = <math>T_p</math> 1000 RPM</p> <p>GAPF = <math>\frac{GMF}{N_A} = \frac{15,000}{5}</math></p> <p>GAPF = 3000 CPM = 0.20X GMF (FRACTIONAL GMF)</p> <p>1X GEAR 1X PINION GAPF 2GAPF 3GAPF 4GAPF GMF 15,000 CPM</p>                | <p>Gear Assembly Phase Freq. (GAPF) can result in Fractional Gear Mesh Frequencies (if <math>N_A &gt; 1</math>). It literally means <math>(T_d/N_A)</math> gear teeth will contact <math>(T_p/N_A)</math> pinion teeth and will generate <math>N_A</math> wear patterns, where <math>N_A</math> in a given tooth combination equals the product of prime factors common to the number of teeth on the gear and pinion (<math>N_A</math> = Assembly Phase Factor). GAPF (or harmonics) can show up right from the beginning if there were manufacturing problems. Also, its sudden appearance in a periodic survey spectrum can indicate damage if contaminate particles pass through the mesh, resulting in damage to the teeth in mesh at the time of ingestion just as they enter and leave meshing or that gears have been reoriented.</p>                                                             |
| <b>H. HUNTING TOOTH PROBLEMS</b>         | <p>DRIVER 1000 RPM DRIVEN 657 RPM</p> <p>6 = 1X2X3 7 = 1X7 <math>N_A = 1</math></p> <p><math>f_{HT} = \frac{(6 \times 1000)(1)}{(6)(7)} = 143 \text{ CPM}</math> (One Pulse Per 7 Pinion Revolutions)</p> <p><math>f_{HT} = \frac{(GMF)(N_A)}{(T_{GEAR})(T_{PINION})}</math></p> <p><math>N_A = 1</math> is the ideal assembly phase factor in gear design</p> | <p>Hunting Tooth Frequency (<math>f_{HT}</math>) occurs when faults are present on both the gear and pinion which might have occurred during the manufacturing process, due to mishandling, or in the field. It can cause quite high vibration, but since it occurs at low frequencies predominately less than 600 CPM, it is often missed. A gear set with this tooth repeat problem normally emits a "growing" sound from the drive. The maximum effect occurs when the faulty pinion and gear teeth both enter mesh at the same time (on some drives, this may occur only 1 of every 10 to 20 revolutions, depending on the <math>f_{HT}</math> formula). Note that <math>T_{GEAR}</math> and <math>T_{PINION}</math> refer to number of teeth on the gear and pinion, respectively. <math>N_A</math> is the Assembly Phase Factor defined above. Will often modulate both GMF and Gear RPM peaks.</p> |

**Table 6.0**  
**ILLUSTRATED VIBRATION DIAGNOSTIC CHART**

| PROBLEM SOURCE                                                                                                                                                                                                                    | TYPICAL SPECTRUM | REMARKS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GEARS (CONTINUED)</b>                                                                                                                                                                                                          |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>I. LOOSE BEARING FIT</b>                                                                                                                                                                                                       |                  | Excessive Clearance of bearings supporting the gears can not only excite many running speed harmonics, but will often cause high amplitude response at GMF, 2GMF and/or 3GMF. These high GMF amplitudes are actually a response to, and not the cause of, looseness within the bearings supporting the gearing. Such excessive clearance can be caused either by extensive bearing wear or by improper bearing fit onto the journal during installation. Left uncorrected, it can cause excessive gear wear and damage to other components.                                                                                                                                                                                                                                                        |
| <b>AC INDUCTION MOTORS</b>                                                                                                                                                                                                        |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>A. STATOR ECCENTRICITY, SHORTED LAMINATIONS OR LOOSE IRON</b>                                                                                                                                                                  |                  | Stator problems generate high vibration at 2X line frequency (2FL). Stator eccentricity produces uneven stationary air gap between rotor and stator which produces very directional vibration. Differential Air Gap should not exceed 5% for induction motors and 10% for synchronous motors. Soft foot and warped bases can produce an eccentric stator. Loose iron is due to stator support weakness or looseness. Shorted stator laminations can cause uneven, localized heating which can distort the stator itself. This produces thermally-induced vibration which can significantly grow with operating time causing stator distortion and static air gap problems.                                                                                                                         |
| <b>B. ECCENTRIC ROTOR (Variable Air Gap)</b><br>$F_L$ = Electrical Line Freq.<br>$N_s$ = Synch. Speed = $\frac{120F_L}{P}$<br>$F_s$ = Slip Freq. = $N_s - \text{RPM}$<br>$F_p$ = Pole Pass Freq. = $F_s \times P$<br>$P$ = #Poles |                  | Eccentric Rotors produce a rotating variable air gap between the rotor and stator which induces pulsating vibration (normally between 2FL and closest running speed harmonic). Often requires "zoom" spectrum to separate 2FL and running speed harmonic. Eccentric rotors generate 2FL surrounded by Pole Pass frequency sidebands (Fp), as well as Fp sidebands around running speed. Fp appears itself at low frequency (Pole Pass Frequency = Slip Frequency X #Poles). Common values of Fp range from about 20 to 120 CPM (0.3 - 2.0 Hz). Soft foot or misalignment often induces a variable air gap due to distortion (actually a mechanical problem; not electrical).                                                                                                                       |
| <b>C. ROTOR PROBLEMS</b>                                                                                                                                                                                                          |                  | Broken or Cracked rotor bars or shorting rings; bad joints between rotor bars and shorting rings; or shorted rotor laminations will produce high 1X running speed vibration with pole pass frequency sidebands (Fp). In addition, these problems will often generate Fp sidebands around the second, third, fourth and fifth running speed harmonics. Loose or open rotor bars are indicated by 2X line freq. (2FL) sidebands surrounding Rotor Bar Pass Frequency (RBPF) and/or its harmonics (RBPF = Number of Bars X RPM). Often will cause high levels at 2X RBPF, with only a small amplitude at 1X RBPF. Electrically induced arcing between loose rotor bars and end rings will often show high levels at 2X RBPF (with 2FL sidebands); but little or no increase in amplitudes at 1X RBPF. |
| <b>D. PHASING PROBLEM (Loose Connector)</b>                                                                                                                                                                                       |                  | Phasing problems due to loose or broken connectors can cause excessive vibration at 2X Line Freq. (2FL) which will have sidebands around it spaced at 1/3 Line Freq. (1/3 Fp). Levels at 2FL can exceed 1.0 in/sec if left uncorrected. This is particularly a problem if the defective connector is only sporadically making contact. Loose or broken connectors must be repaired to prevent catastrophic failure.                                                                                                                                                                                                                                                                                                                                                                                |
| <b>AC SYNCHRONOUS MOTORS</b>                                                                                                                                                                                                      |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>(Loose Stator Coils)</b>                                                                                                                                                                                                       |                  | Loose stator coils in synchronous motors will generate fairly high vibration at Coil Pass Freq. (CPF) which equals the number of stator coils X RPM (#Stator Coils = #Poles X #Coils/Pole). The Coil Pass Frequency will be surrounded by 1X RPM sidebands. Synchronous motor problems may also be indicated by high amplitude peaks at approx. 60,000 to 90,000 CPM, accompanied by 2FL sidebands. Take at least one spectrum up to 90,000 CPM on each motor bearing housing.                                                                                                                                                                                                                                                                                                                     |
| <b>DC MOTORS AND CONTROLS</b>                                                                                                                                                                                                     |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>A. NORMAL SPECTRUM</b>                                                                                                                                                                                                         |                  | Many DC Motor and Control Problems can be detected by vibration analysis. Full-wave rectified, motors (6 SCR's) generate a signal at 6X Line Frequency (6FL = 360 Hz = 21,600 CPM); while half-wave rectified DC motors (3 SCR's) generate 3X Line Freq. (3FL = 180 Hz = 10,800 CPM). The SCR firing Frequency is normally present in a DC Motor Spectrum, but at low amplitude. Note the absence of other peaks at multiples of FL.                                                                                                                                                                                                                                                                                                                                                               |
| <b>B. BROKEN ARMATURE WINDINGS, GROUNDING PROBLEMS OR FAULTY SYSTEM TUNING</b>                                                                                                                                                    |                  | When DC Motor spectra are dominated by high levels at SCR or 2X SCR, this normally indicates either Broken Motor Windings or Faulty Tuning of the Electrical Control System. Proper tuning alone can lower vibration at SCR and 2X SCR significantly if control problems predominate. High amplitudes at these frequencies would normally be above approximately .10 in/sec, peak at 1X SCR and about .04 in/sec at 2X SCR Firing Freq.                                                                                                                                                                                                                                                                                                                                                            |
| <b>C. FAULTY FIRING CARD OR BLOWN FUSE</b>                                                                                                                                                                                        |                  | When one firing card fails to fire, then 1/3 of power is lost, and can cause repeated momentary speed changes in the motor. This can lead to high amplitudes at 1/3X and 2/3X SCR Frequency (1/3X SCR Freq. = 1X FL for half-wave rectified, but 2X FL for a full-wave rectified SCR).<br>Caution: Card/SCR configuration should be known before troubleshooting motor (#SCR's, #Firing Cards, etc.).                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>D. FAULTY SCR, SHORTED CONTROL CARD, LOOSE CONNECTIONS AND/OR BLOWN FUSE</b>                                                                                                                                                   |                  | Faulty SCR's, Shorted Control Cards and/or Loose Connections can generate noticeable amplitude peaks at many combinations of line frequency (FL) and SCR firing frequency. Normally, 1 bad SCR can cause high levels at FL and/or 5FL in 6 SCR motors. The point to be made is that neither FL, 2FL, 4FL nor 5FL should be present in DC Motors spectra.                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>E. FAULTY COMPARATOR CARD</b>                                                                                                                                                                                                  |                  | Faulty Comparator Cards cause problems with RPM fluctuation or "hunting". This causes a constant collapsing and regenerating of the magnetic field. These sidebands often approximate the RPM fluctuation and require a high resolution FFT to even detect them. Such sidebands could also be due to generation and regeneration of the magnetic field.                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>F. ELECTRICAL CURRENT PASSAGE THRU DC MOTOR BEARINGS</b>                                                                                                                                                                       |                  | Electrically-induced Fluting is normally detected by a series of difference frequencies with the spacing most often at the outer race defect frequency (BPFO), even if such fluting is present on both the outer and inner races. They most often show up in a range centered at about 100,000 to 150,000 CPM. A 180K CPM spectrum with 1600 lines is recommended for detection with measurements on both the OB and IB DC motor bearings.                                                                                                                                                                                                                                                                                                                                                         |

**Table 6.0**  
**ILLUSTRATED VIBRATION DIAGNOSTIC CHART**

| PROBLEM SOURCE                                                                                                                                  | TYPICAL SPECTRUM                                                                    | REMARKS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>BELT DRIVE PROBLEMS</b>                                                                                                                      |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>A. WORN, LOOSE OR MISMATCHED BELTS</b><br>                  |    | $\text{BELT FREQ.} = \frac{3.142 \times \text{PULLEY RPM} \times \text{PITCH DIAM.}}{\text{BELT LENGTH}}$<br>$\text{TIMING BELT FREQ.} = \text{BELT FREQ.} \times \# \text{BELT TEETH}$<br>$= \text{PULLEY RPM} \times \# \text{PULLEY TEETH}$<br><p>Belt frequencies are below the RPM of either the motor or the driven machine. When they are worn, loose or mismatched, they normally cause 3 to 4 multiples of belt frequency. Often 2X belt freq. is the dominant peak. Amplitudes are normally unsteady, sometimes pulsing with either driver or driven RPM. On timing belt drives, wear or pulley misalignment is indicated by high amplitudes at the Timing Belt Frequency. Chain drives will indicate problems at Chain Pass Frequency which equals #Sprocket Teeth X RPM.</p>                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>B. BELT/PULLEY MISALIGNMENT</b><br>                         |    | <p>Misalignment of pulley produces high vibration at 1X RPM predominantly in the axial direction. The ratio of amplitudes of driver to driven RPM depends on where the data is taken, as well as on relative mass and frame stiffness. Often with pulley misalignment, the highest axial vibration on the motor will be at fan RPM, or vice versa. Can be confirmed by phase measurements by setting Phase Filter at RPM of pulley with highest axial amplitude; then compare phase at this particular frequency on each rotor in the axial direction.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>C. ECCENTRIC PULLEYS</b><br>                                |    | <p>Eccentric pulleys cause high vibration at 1X RPM of the eccentric pulley. The amplitude is normally highest in line with the belts, and should show up on both driver and driven bearings. It is sometimes possible to balance eccentric pulleys by attaching washers to taper-lock bolts. However, even if balanced, the eccentricity will still induce vibration and reversible fatigue stresses in the belt. Pulley eccentricity can be confirmed by phase analysis showing horizontal &amp; vertical phase differences of nearly 0° or 180°.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>D. BELT RESONANCE</b><br>                                  |   | <p>Belt Resonance can cause high amplitudes if the belt natural frequency should happen to approach, or coincide with, either the motor or driven RPM. Belt natural frequency can be altered by changing either the belt tension, belt length or cross section. Can be detected by tensioning and then releasing belt while measuring the response on pulleys or bearings. However, when operating, belt natural frequencies will tend to be slightly higher on the tight side and lower on the slack side.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>BEAT VIBRATION</b>                                                                                                                           |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                              |                                                                                     | <p>A Beat Frequency is the result of two closely spaced frequencies going into and out of synchronization with one another. The wideband spectrum normally will show one peak pulsating up and down. When you zoom into this peak (lower spectrum below), it actually shows two closely spaced peaks. The difference in these two peaks (<math>F_2 - F_1</math>) is the beat frequency which appears itself in the wideband spectrum. The beat frequency is not commonly seen in normal frequency range measurements since it is inherently low frequency, usually ranging from only approximately 5 to 100 CPM.</p> <p>Maximum vibration will result when the time waveform of one frequency (<math>F_1</math>) comes into phase with the waveform of the other frequency (<math>F_2</math>). Minimum vibration occurs when waveforms of these two frequencies line up 180° out of phase.</p> <div style="display: flex; justify-content: space-around;"> <div>  <p>WIDEBAND SPECTRUM</p> </div> <div>  <p>ZOOM SPECTRUM</p> </div> </div> |
| <b>SOFT FOOT, SPRUNG FOOT AND FOOT-RELATED RESONANCE</b><br> |  | <p><b>"Soft Foot"</b> occurs when a machine's foot or frame deflects greatly when a hold-down bolt is loosened to hand tightness, causing the foot to rise more than approximately .002 - .003 inch. This does not always cause a great vibration increase. However, it can do so if the soft foot affects alignment or motor air gap concentricity.</p> <p><b>"Sprung Foot"</b> can cause great frame distortion, resulting in increased vibration, force and stress in the frame, bearing housing, etc. This can occur when a hold-down bolt is forcefully torqued down on the sprung foot in an attempt to level the foot.</p> <p><b>"Foot-Related Resonance"</b> can cause dramatic amplitude increases from 5X to 15X or more, as compared with that when the bolt (or combination of bolts) is loosened to hand tightness. When tight, this bolt can notably change the natural frequency of the foot or machine frame itself.</p> <p>Soft Foot, Sprung Foot or Foot-Related Resonance most often affects vibration at 1X RPM, but can also do so at 2X RPM, 3X RPM, 2X line frequency, blade pass frequency, etc. (particularly Foot-Related Resonance).</p>                                                 |

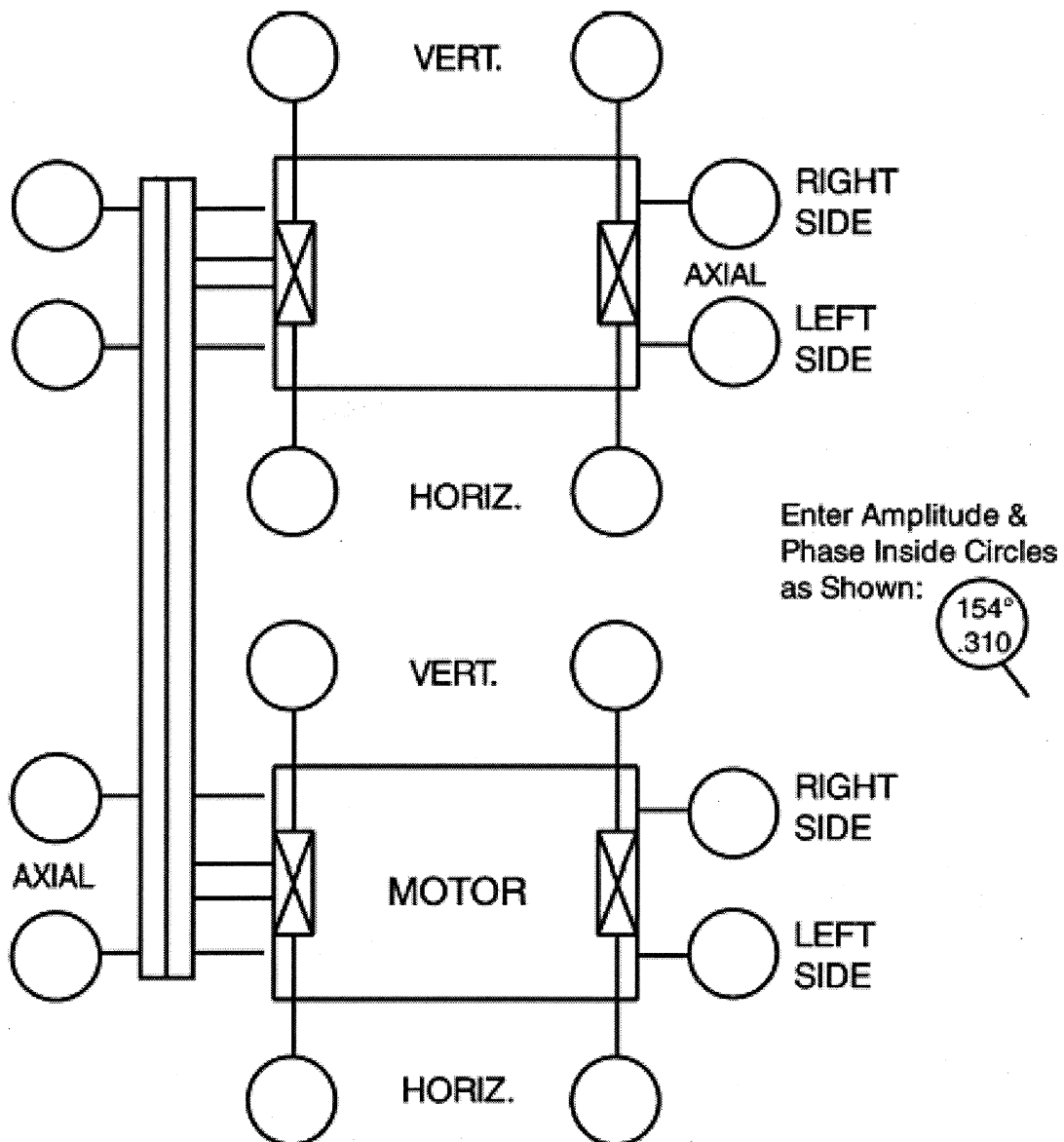
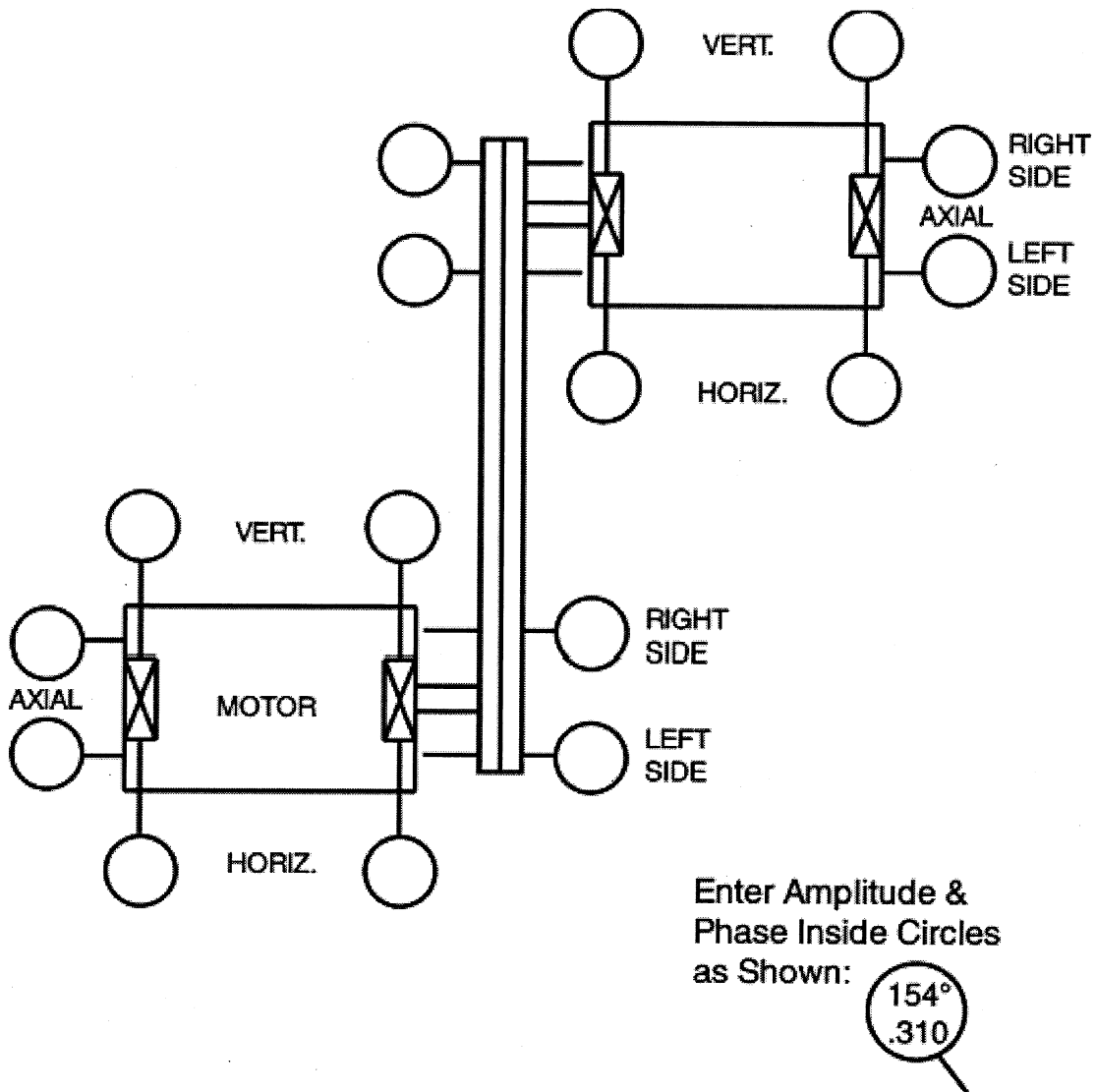
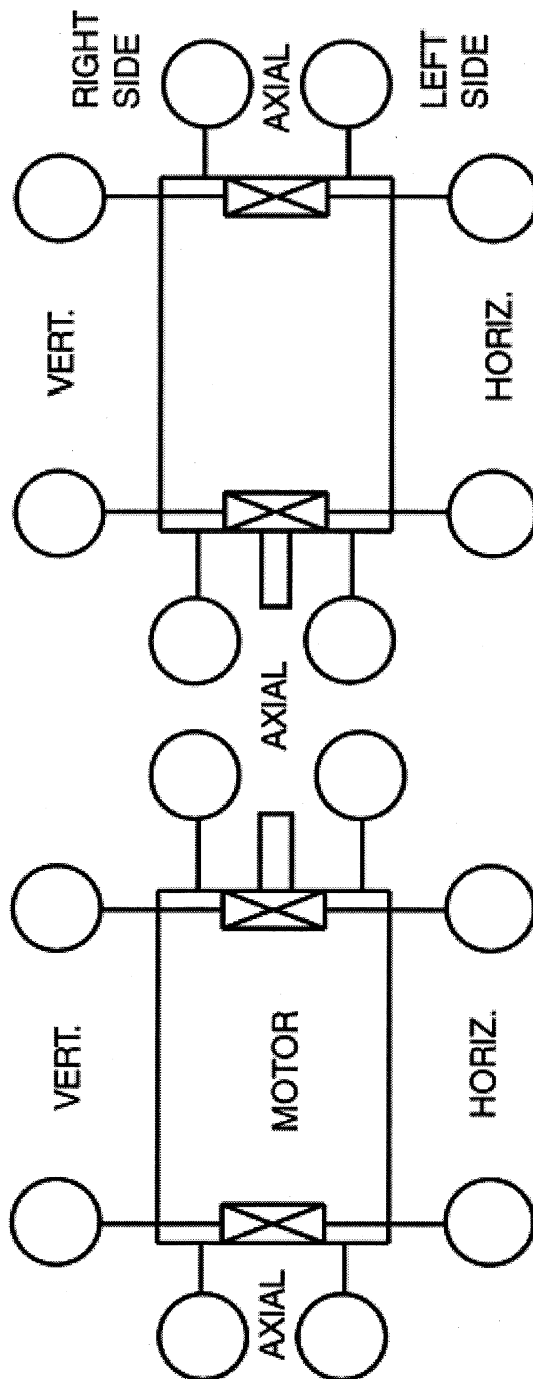


FIGURE 6.0A  
PHASE ANALYSIS DIAGRAM BDB-1



**FIGURE 6.0B**  
**PHASE ANALYSIS DIAGRAM BDB-2**



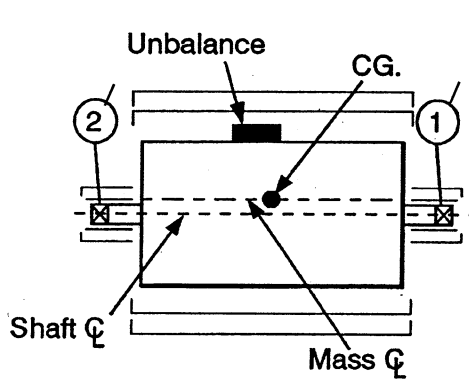
Enter Amplitude &  
Phase Inside Circles  
as Shown:

154°  
.310

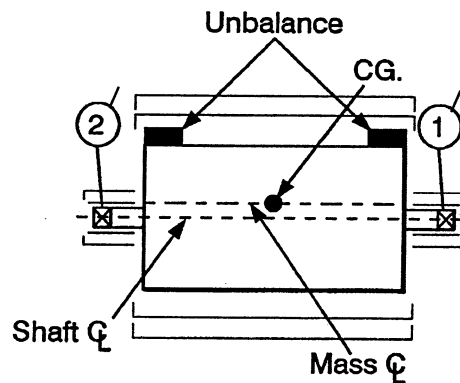
FIGURE 6.0C  
PHASE ANALYSIS DIAGRAM DC-1

## 6.01 MASS UNBALANCE

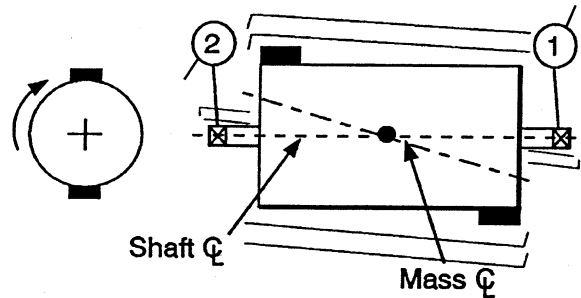
Unbalance occurs when the mass centerline does not coincide with the shaft centerline as shown in Figures 6.01A thru 6.01D. Some degree of unbalance exists in all rotors whether they are a cooling tower fan or a precision grinding wheel. The key is to know how much unbalance is acceptable for the particular type of machine at its specific operating speed which will be discussed in Section 6.015 "Allowable Residual Unbalance".



**FIGURE 6.01A  
FORCE UNBALANCE**

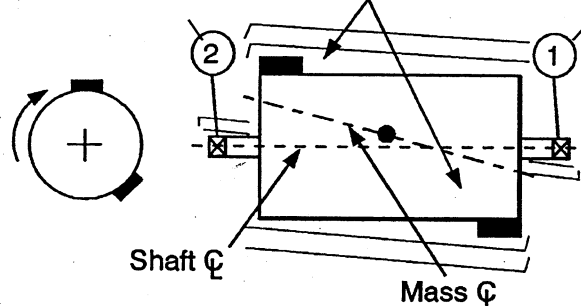


**FIGURE 6.01B  
FORCE UNBALANCE ALSO**



**FIGURE 6.01C  
COUPLE UNBALANCE**

Unbalance Masses Not  
Diametrically Opposed



**FIGURE 6.01D  
DYNAMIC UNBALANCE**

---

Unbalanced rotors exhibit each of the following characteristics:

1. Unbalance is always indicated by high vibration at 1X RPM of the unbalanced part (but, vibration at 1X RPM is not always unbalance). Normally, this 1X RPM peak will dominate the spectrum.
2. The amplitude at 1X RPM will normally be greater than or equal to 80% of the overall amplitude when the problem is limited to unbalance (may be only 50% to 80% if other problems exist in addition to unbalance).
3. The amplitude of vibration is proportional to how far the mass center is displaced from the shaft center. For example, when operating below the first rotor critical speed, amplitude will vary with the square of RPM (that is, tripling the speed will result in an increase in unbalance vibration by a factor of 9 times).
4. Mass unbalance generates a uniform rotating force which is continually changing direction, but is evenly applied in all radial directions. As a result, the shaft and supporting bearings tend to move in somewhat a circular orbit. However, due to the fact that vertical bearing stiffness is normally higher than that in the horizontal direction, the normal response is a slightly elliptical orbit. Subsequently, horizontal vibration is normally somewhat higher than that in the vertical commonly ranging between 2 and 3 times higher. When the ratio of horizontal to vertical is higher than about 6 to 1, it normally indicates other problems, particularly resonance.
5. When unbalance dominates over other problems, there will normally be about a 90° phase difference between horizontal and vertical directions on a bearing ( $\pm 30^\circ$ ). Therefore, if there is a high vibration at 1X RPM, but this phase difference is either 0° or close to 180°, it normally points to another problem source such as eccentricity.
6. Probably an even greater indicator of unbalance than the approximately 90° phase shift between horizontal and vertical is the fact that when significant unbalance exists, the horizontal phase difference between outboard and inboard bearings should be close to the difference in phase in the vertical direction. That is, instead of comparing horizontal and vertical phase on the same bearing, compare outboard and inboard horizontal phase difference with outboard and inboard vertical phase difference. For example, please refer to Table A of Figure 6.01E which shows a machine having dominant force unbalance. Note that the horizontal phase difference between the #1 and #2 bearings is about 5° (30° minus 25°) compared to a vertical phase difference of about 10° (120° minus 110°). Similarly, over on the pump, the horizontal phase difference (position 3) is about 10° and the vertical phase difference is about 15°. This is the expected phase response with dominant force unbalance.
7. When unbalance is dominant, radial vibration (horizontal and vertical) will normally be quite much higher than that in the axial direction (except for overhung rotors which will be discussed in Section 6.014).
8. Unbalanced rotors normally exhibit steady and repeatable phase in radial directions. When the rotor is trim balanced, the phase can begin to "dwell" back and forth under a strobe light as you achieve a better and better balance, particularly if other problems are present. However, if there is high unbalance, and other problems are not significant, the phase should be steady and repeatable.
9. The effects of unbalance may sometimes be amplified by resonance. This will be discussed in Section 6.05.

10. Unbalance can be a great contributor to high looseness vibration. In fact, on a rotor with unbalance and looseness, if it is possible to balance the rotor, this may substantially reduce the looseness vibration although it will often return when even the least little unbalance component returns. Often, it is not even possible to balance rotors having noticeable looseness.

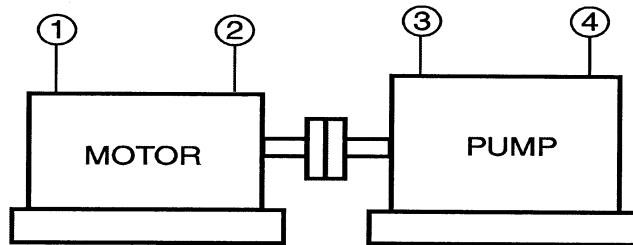


TABLE A.  
PHASE AT 1X RPM FOR A MACHINE  
WITH DOMINANT FORCE UNBALANCE

| DIR. | 1    | 2     | 3    | 4     |
|------|------|-------|------|-------|
| A    | 60°  | * 70° | 60°  | * 80° |
| H    | 30°  | 25°   | 30°  | 40°   |
| V    | 120° | 110°  | 120° | 135°  |

\* Corrected Phase Values Accounting For 180°  
Orientation of Accelerometer.

TABLE B.  
PHASE AT 1X RPM FOR A MACHINE  
WITH DOMINANT COUPLE UNBALANCE

| DIR. | 1    | 2     | 3    | 4     |
|------|------|-------|------|-------|
| A    | 60°  | * 70° | 60°  | * 80° |
| H    | 30°  | 210°  | 200° | 180°  |
| V    | 120° | 295°  | 280° | 300°  |

\* Corrected Phase Values Accounting For 180°  
Orientation of Accelerometer.

TABLE C.  
PHASE AT 1X RPM FOR A MACHINE  
WITH SIGNIFICANT DYNAMIC UNBALANCE

| DIR. | 1    | 2     | 3    | 4     |
|------|------|-------|------|-------|
| A    | 60°  | * 70° | 60°  | * 80° |
| H    | 30°  | 90°   | 80°  | 70°   |
| V    | 120° | 180°  | 170° | 165°  |

\* Corrected Phase Values Accounting For 180°  
Orientation of Accelerometer.

FIGURE 6.01E  
TYPICAL PHASE MEASUREMENTS WHICH WOULD INDICATE EITHER  
FORCE, COUPLE OR DYNAMIC UNBALANCE

---

There are 3 major types of unbalance including force, couple and dynamic unbalance which will be discussed in Sections 6.011, 6.012 and 6.013, respectively, which follow:

#### **6.011 Force Unbalance:**

Force unbalance is sometimes known as "static unbalance". Force unbalance is a condition where the mass centerline is displaced from and parallel to the shaft centerline as shown in Figure 6.01A. This is the type of unbalance that has been classically corrected for many years by placing a fan rotor on knife edges or within its bearings and allowing it to "roll to the bottom". That is, when the fan wheel is released, if the heavy spot is angularly displaced from the bottom (6:00 position), it would tend to roll to the bottom hopefully ending up in the 6:00 position, assuming the rotor was sufficiently free within its bearings to rotate. So-called correction of this force unbalance was then accomplished by placing a weight opposite this location (or at about 12:00).

Actually, there are two types of force unbalance as shown in Figure 6.01A and Figure 6.01B. In the case of Figure 6.01A, only one heavy spot exists and is located close to the rotor center of gravity (CG). This is corrected by simply placing an equal weight 180° opposite the angular position of the heavy spot. Figure 6.01B likewise illustrates force unbalance even though it shows heavy spots acting on both the outboard and inboard planes (angularly parallel to one another). In this case, it can either be corrected by placing correction weights either at the CG, or by placing equal and opposite weights at each of the two planes (if corrected at the CG, it would of course require double the correction weight in this case).

Characteristics common to force unbalance can be summarized as follows:

1. Approximately the same unbalance forces at 1X RPM are normally present both on the outboard and inboard rotor bearing housings (however, horizontal and vertical responses may differ somewhat depending on the support stiffness in each direction).
2. With pure force unbalance, the outboard horizontal phase will equal the inboard horizontal phase on the same shaft (that is, if the horizontal phase on the outboard bearing were at 6:00, the inboard reading should likewise be about 6:00 since the two shaft ends are moving together).
3. Likewise, the outboard vertical phase should approximately equal the inboard vertical phase on the same shaft.
4. Force unbalance only requires a single plane correction with the counterweight acting through the rotor CG.
5. The difference in horizontal outboard and inboard phase should approximately equal the phase difference in outboard and inboard vertical phase and the phase change across the coupling should be small (less than 60° to 90°) if force unbalance were dominant.

#### **6.012 Couple Unbalance:**

Couple unbalance is a condition where the mass centerline axis intersects the shaft centerline axis at the rotor's center of gravity as shown in Figure 6.01F. Here, a couple is created by equal heavy spots at each end of the rotor, but 180° opposite each other. Significant couple unbalance can introduce severe instability to the rotor causing it to wobble back and forth (like a "seesaw" with the fulcrum at the rotor CG).

---

Couple unbalance exhibits each of the following characteristics:

1. In pure couple unbalance, the rotor is statically balanced and will not roll to the bottom when the rotor is placed on knife edges. That is, referring to Figure 6.01C, since the heavy spot at position 1 is equal to that at position 2, this meets the requirement for force or static balance. Still, this rotor will also generate considerable vibration at 1X RPM.
2. Couple unbalance generates high amplitude vibration at 1X RPM on both the outboard and inboard bearing housings, but it may be somewhat higher on one bearing than on the other.
3. Substantial couple unbalance can sometimes generate high axial vibration.
4. The horizontal phase difference between the outboard and inboard bearings will approximate  $180^\circ$  (that is, if the outboard horizontal phase were at 6:00, then the inboard horizontal phase will probably be about 12:00 since the two ends are moving opposite each other in a rocking motion).
5. Similarly, the vertical phase difference between outboard and inboard bearings will approximate  $180^\circ$ .
6. Refer to Table B of Figure 6.01E illustrating how phase should react to couple unbalance. Note the  $180^\circ$  phase difference between position 1 and 2 horizontal ( $210^\circ - 30^\circ$ ), and the  $175^\circ$  phase difference between position 1 and 2 vertical ( $295^\circ - 120^\circ$ ). This shows that if the problem is couple unbalance (and not misalignment), both the horizontal and vertical phase differences should roughly be equal to one another - both approximately  $180^\circ$  difference between the outboard and inboard bearings.

#### **6.013 Dynamic Unbalance:**

Dynamic unbalance is by far the most common type of unbalance as compared to either purely force or couple unbalance and is defined as "that condition in which the mass centerline is neither parallel to nor intersects the shaft centerline axis". In essence, dynamic unbalance is a combination of both force and couple unbalance. It requires correction in at least 2 planes perpendicular to the shaft centerline axis.

Dynamic unbalance exhibits each of the following characteristics:

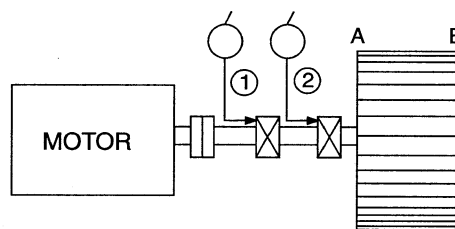
1. Dynamic unbalance generates high vibration at 1X RPM, but the amplitude on the outboard bearing may be somewhat different than that on the inboard bearing housing. Still, they should be within the same order of magnitude, or below about 3 to 1 assuming there are no other significant problems present.
2. Like force and couple unbalance, phase is still steady and repeatable when dynamic unbalance dominates.
3. Although the horizontal phase difference between outboard and inboard bearings could be anything from  $0^\circ$  to  $180^\circ$ , this difference should still approximately equal the vertical phase difference. For example, if the horizontal phase difference was about  $60^\circ$ , the vertical phase difference should likewise be about  $60^\circ$  ( $\pm 30^\circ$ ) as illustrated in Table C of Figure 6.01E. Here, in this example, notice that the phase difference in both horizontal and vertical directions at positions 1 and 2 is about  $60^\circ$  and that the phase difference across the coupling does not approach  $180^\circ$ . Dynamic unbalance requires correction in at least 2 planes.

- Whether or not force or couple balance dominates, the horizontal phase difference at bearings 1 and 2 should approximately equal the vertical phase difference at these two bearings (if the horizontal phase difference is about  $150^\circ$  showing high couple unbalance, the vertical phase difference will approximate  $150^\circ$  as well) .

#### 6.014 Overhung Rotor Unbalance:

Figure 6.01F shows an overhung rotor. In this case, the driven rotor is placed outboard of bearings 1 & 2 (rotors which are placed between bearings are known as simply supported rotors). Overhung rotors can cause some interesting vibration symptoms and often can present real problems to the analyst when he attempts to balance one. Overhung rotors display the following characteristics:

- Overhung rotors can generate large axial forces at 1X RPM which can cause axial vibration to be equal to or greater than radial vibration amplitudes.
- Overhung rotors often generate a high degree of couple unbalance in addition to force unbalance, both of which must be corrected.
- Referring to Figure 6.01F, for pure unbalance of an overhung rotor, the axial phase at bearing 1 will approximately equal that at bearing 2 ( $\pm 30^\circ$ ). Here again, this phase difference depends on how dominant the unbalance problem is as compared to others such as misalignment, resonance, etc.
- Normally, overhung rotor unbalance can be corrected by first taking care of the force unbalance component which would leave the remainder as couple unbalance with phase differences approaching  $180^\circ$ . The couple component would then require placement of correction weights in 2 planes  $180^\circ$  opposite one another.



**FIGURE 6.01F**  
**BALANCING OF AN OVERHUNG ROTOR**

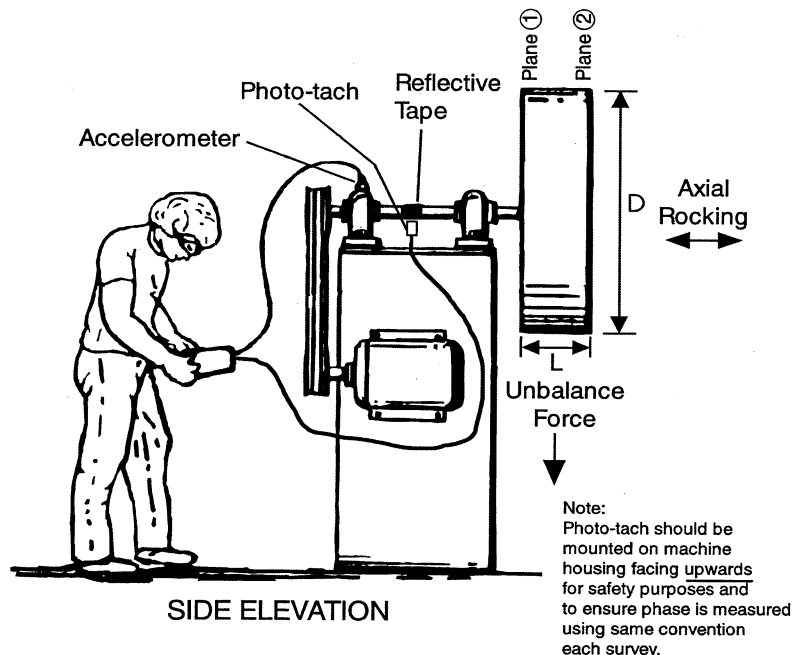
#### 6.0141 Summary of Procedures for Balancing Overhung Rotors

Overhung rotors are machine configurations like that shown in Figure 6.01G where the fan wheel to be balanced is outboard of its two supporting bearings. This configuration is very often found with machines such as blowers, pumps, etc. Because the planes where balance correction weights must be attached are outside the supporting bearings, these rotors will often not respond to standard single and two-plane balancing techniques. In addition, because the unbalance planes are outside the support bearings, even a static unbalance alone will create a couple unbalance proportional to the distance of the unbalance plane from the rotor CG. Therefore, when attempting to balance overhung rotors, the analyst needs to take into account both static and couple unbalance forces, and treat them accordingly.

When balancing an overhung rotor, one of the two following procedures should be taken:

**1. Balancing Overhung Rotors by Classic Single-Plane Static-Couple Method:**

Figure 6.01G helps explain methods of balancing overhung rotors. Classically, Bearing A is most sensitive to static unbalance whereas the bearing farthest from the fan wheel to be balanced (Bearing B) is most sensitive to couple unbalance. Since Plane 1 is closest to the rotor center of gravity (CG), static corrections should be made in this plane while measuring the response on Bearing A. On the other hand, measurements should be made on Bearing B when making couple corrections in Plane 2. However, placing a trial weight in Plane 2 will destroy the static balance achieved at Bearing A. Therefore, in order to maintain the static balance at Bearing A, a trial weight placement which will generate a couple must be used. Thus, a trial weight of identical size should be placed in Plane 1 at an angle  $180^\circ$  opposite the trial weight location in Plane 2.



**FIGURE 6.01G**  
**FIELD INSTRUMENT SETUP FOR BALANCING OVERHUNG ROTORS**

Therefore, either the data collector can be used using single-plane balance software or the single-plane graphic technique previously explained can be successfully employed on many overhung rotors, particularly if the ratio of the rotor length-to-diameter ( $L/D$ ) is less than approximately .50 (where  $L$  is length of the rotating component on which correction weights will be placed and  $D$  is the diameter of this component - see Figure 6.01G).

Following below will be a description of this classic single-plane balancing technique for overhung rotors:

- a. **Set Up Data Collector and/or Spectrum Analyzer Instruments** - The data collector, phototach, accelerometer and so forth should be set up as previously described under Section D and Figure 6.01G showing the two-plane balancing procedure. Alternatively, the analyst may wish to employ either a swept-filter analyzer which drives a strobe light (like an IRD 350 or IRD 880), or a spectrum analyzer which will fire a phototach for phase measurement.

- 
- b. Take Initial Measurements** - Take initial measurements of 1X RPM amplitude, frequency and phase before adding any trial weights. Measurements should be taken on both the outboard and inboard bearings in both vertical and horizontal directions. The radial direction measurement having the highest amplitude will normally be employed for initial balancing (however, after correcting unbalance in the radial direction, measurements will have to be taken in the other radial direction to ensure amplitudes in it are likewise acceptable).
- c. Determine if the Dominant Problem is Either Static or Couple Unbalance** - Looking at the amplitude and phase measurements taken on both bearings in the radial and horizontal directions, determine if the problem is dominated by either static or couple unbalance. If phase differences between the outboard and inboard bearing are approximately  $140^\circ$  or more in both the vertical and horizontal directions, the dominant problem will be couple unbalance. On the other hand, if these differences are both anywhere from  $0^\circ$  to approximately  $40^\circ$ , static unbalance is dominant. Of course, phase differences ranging from approximately  $40^\circ$  to  $140^\circ$  are truly dynamic balance once again with a combination of static and couple. If the problem appears to be mostly couple unbalance, use couple unbalance procedures outlined below. However, if the problem appears to be predominantly static or dynamic unbalance, employ static balance procedures. For now, we will assume that the problem is mostly static.
- d. Make a Single-Plane Static Balance** - Referring to Figure 6.01G, use single-plane techniques taking measurements on Bearing A and placing trial and correction weights in Plane 1.
- e. Determine if Resultant Vibration Amplitudes Meet Required Criteria** - After completing the single-plane static balance using Plane 1, repeat vibration measurements on both the outboard and inboard bearings in each direction (including axial) and ensure that amplitudes now meet allowable criteria.
- f. If Considerable Couple Unbalance Now Remains, Continue With Single-Plane Balance From Bearing B** - Overhung rotors often have large cross-effects which means that single-plane balancing from Plane 1 will often cause high vibration over at Bearing B. Therefore, the analyst will perform another single-plane balance, this time making his measurements from Bearing B farthest from the component to be balanced. When he arrives at the single-plane correction weight solution, he should place this weight in Plane 2; and then place an identical size correction weight over in Plane 1 some  $180^\circ$  away from the weight location in Plane 2.
- g. Determine if Amplitudes Now Meet all Criteria** - After completing the single-plane couple correction, the analyst must again make measurements in horizontal, vertical and axial directions on each bearing and determine that all amplitudes now meet allowable criteria. Often, further balancing must be done at this point beginning with another single-plane balance using Bearing A and Plane 1 which might possibly be followed by another couple balance correction.
- h. If Allowable Criteria Cannot be Met in all Three Directions of Each Bearing, Proceed to Two-Plane Balance Procedure Outlined Below** - Sometimes, this single plane approach will not successfully reduce amplitudes below allowable criteria in all three directions on each bearing, particularly if the L/D ratio is greater than .50 or if the component to be balanced is located far away from the closest bearing. If this happens, two-plane techniques outlined below will have to be taken.

---

## 2. Balancing Overhung Rotors by Classic Two-Plane Static-Couple Method :

Due to the significant cross-effects which are often present in overhung rotors, two-plane balance correction techniques often are more successful than those employing single-plane methods. However, one of the problems with two-plane methods is that it can sometimes be a little confusing on deciding which bearing is the left and which is the right bearing; similarly, which plane is the left and which is the right plane? (Some data collectors refer to these as the near and far planes as opposed to left and right; terminology does not matter - only that the analyst remain consistent in his convention.) Referring to Figure 6.01G, when using two-plane techniques, Bearing A will be considered the bearing closest to the overhung rotor while Bearing B will be closest to the pulley. Similarly, Plane 1 will be on the inboard side of the wheel closest to the bearings whereas Plane 2 will be outboard.

Here again, a static/couple solution will be employed when the two-plane correction weight calculations are completed. Since most overhung rotors are so sensitive to static unbalance, only the static correction weight will be placed when this static/couple solution is obtained. Then, after trim balancing, if considerable couple unbalance remains, the analyst will proceed to correct this as well. He should follow the procedure outlined below:

**a. Set Up Instruments as Outlined in Two-Plane Balance Method in Figure 6.01G -**

Here again, this same procedure can be used with either data collectors, swept-filter analyzers or real-time analyzers. However, if using either a swept-filter or real-time analyzer, the analyst should have a two-plane calculator program that is capable of providing static/couple solutions.

**b. Take Initial Measurements on Both Bearings -** Here again, 1X RPM amplitude, frequency and phase should be measured in horizontal, vertical and axial directions on both the outboard and inboard bearings.

**c. Complete a Two-Plane Balancing Procedure, But Do Not Yet Place Balance Correction Weights -** A two-plane balance procedure like that outlined in Section D should be employed, but final correction weights not put in place. Instead, when the trial weights sizes and locations are calculated for each plane, the analyst should ask for a static/ couple solution and should initially only make the static correction. For example, if the static solution called for 1 oz. in Plane 1 whereas the couple solution called for a 2 oz. correction in Planes 1 & 2 180° opposite one another, make only the static correction at this point.

**d. Determine if Amplitudes Now Meet Allowable Criteria -** After making the static correction in Plane 1, see if amplitudes in all three directions on each bearing are now within compliance with allowable criteria. If not, trim as required. Again, when the two-plane corrections are determined, ask for the static/couple solution and once again, make only the static correction. Most of the time, the problems are resolved at this point. However, if considerable couple unbalance still remains, complete another two-plane procedure again asking for the static/couple solution - this time making the couple correction called for, and not the static correction.

**e. Determine if Amplitudes Now Meet Allowable Criteria -** After each of the two trials making these static corrections and the single trial making the couple correction, compare amplitudes in horizontal, vertical and axial directions on both the outboard and inboard bearings with allowable criteria. A small percentage of the time, the couple correction will throw the static balance back off. If this is the case, it may require one more static correction before the rotor is successfully balanced.

---

### **6.015 Allowable Residual Unbalance and ISO Balance Quality Grade:**

When balancing a rotor, one needs to know to what precision he is required to balance the unit. In reality, it is not enough to simply say that it will be acceptable to balance the machine to a level of .10 in/sec or 1.0 mil at a certain speed. While one rotor may satisfactorily be balanced at such levels, another will not. This was recognized back in the 1950's by experts involved in the balance field. They recognized that the residual unbalance is truly proportional to the amount and radius of the remaining rotor eccentricity as well as the weight of the rotor itself and its operating speed. Therefore, they developed a series of balance tolerances known as ISO Standard No. 1940 on "Balance Quality of Rotating Rigid Bodies". Table 6.01A provides the balance quality grades as per these standards for a whole group of rotor types. Then, Table 6.01B provides the numerical standards for each of the ISO balance quality grades (ISO G-1, ISO G-2.5, ISO G-6.3, etc.). Note that the lower the G tolerance, the more precision the balance quality grade. Also note that it is based on the rotor RPM (horizontal axis) as well as the residual unbalance per pound of rotor weight (vertical axis). Table 6.01C is provided showing these same balance quality grades delineated by bands separating one balance quality grade from another. Table 6.01C also shows common nominal RPM's in the United States (1200, 1800 and 3600 RPM).

When balancing a machine, you first refer to the tabulated information in Table 6.01A to determine to what tolerance you should balance it. For example, if balancing an automobile crankshaft, this falls under ISO G-16 quality grade. On the other hand, fans fall under G-6.3, and grinding-machine drives fall under G-1. Note that the quality grade number itself represents the maximum permissible circular velocity of the rotor center of gravity expressed in millimeters per second (mm/sec). For example, a quality grade G-6.3 corresponds to rotor velocity of 6.3 mm/sec RMS which corresponds to an equivalent .248 in/sec RMS (.351 in/sec peak).

It is the experience of the author that this ISO Standard, which was established in 1966, is a little too conservative, possibly because it might be based on the technology available during that day and time. It is recommended that when using the ISO tolerance, that you should use one quality grade better than that specified for the specific machine you wish to balance (that is, if the standard calls for G-6.3, we would recommend using ISO G-2.5). For example, if balancing a fan wheel, note that Table 6.01A calls for an ISO quality grade G-6.3 for fans. In this case, we would recommend ISO grade G-2.5. Following below will be a procedure on how to determine the allowable residual unbalance, the ISO balance quality grade that you have achieved, and the rotor balance sensitivity:

#### **How to Determine Residual Unbalance Remaining in a Rotor After Balancing**

When field balancing, one must know when to determine that the job is complete. He will know this not only when he has achieved low vibration levels, but also when he knows he has balanced the rotor within allowable specifications. To know this, he must determine the residual unbalance remaining in the rotor. This can be accomplished by following the procedure below (refer to Figure 6.01H):

- a. Make original measurements of amplitude and phase and graph this to scale on polar coordinate paper. Call this vector the "O" vector.
- b. Attach a trial weight and document the trial weight size (oz) and radius (in) to which it is attached. ( $mr = \text{trial weight size} \times \text{trial weight radius}$ )
- c. After attaching the trial weight, spin the rotor and measure amplitude and phase. Graph this on the polar coordinate paper as the "O + T" vector.

- d. Draw a vector called "T" from the end of vector "O" to the end of vector "O + T". Vector "T" represents the effect of the trial weight alone. Measure the length of vector "T" to the same scale as that used for vectors "O" and "O + T". Using this scale, determine the equivalent vibration level (mils).
- e. Calculate Rotor Sensitivity as per the following equation:

$$\text{Rotor Sensitivity} = \frac{(\text{Trial Wt. Size})(\text{Trial Wt. Radius})}{\text{Trial Weight Effect}}$$

(oz-in/mil)                      mils

(Eqn. 6.01A)

- f. Calculate Residual Unbalance using Equation 6.01B. If Residual Unbalance not brought within tolerances, trim balance using current correction weight as the trial weight for the trim run. Continue trim balancing until Residual Unbalance is reduced within required balance tolerances:

$$\text{Residual Unbalance} = \text{Rotor Sens.} \times \text{Vib. Ampl. After Bal.}$$

(oz-in)                      (oz-in/mil)                      (mils)

(Eqn. 6.01B)

**Example (see Figure 6.01H):**

Given:

Required ISO Balance Quality = G 2.5  
Rotor Weight = 100 lb  
Rotor Speed = 800 RPM  
Amplitude After Balancing = 2.0 mils

Therefore, Required  $U_{\text{per}} = 1.76 \text{ oz-in total (single-plane balance)}$

- a. Original reading = 10 mils @  $240^\circ$  = "O" vector.
- b. Trial weight of 3 oz is attached in the balance plane at a 6 inch radius  
( $mr = 3 \text{ oz} \times 6 \text{ in} = 18 \text{ oz-in}$ )
- c. Trial run reading = 8 mils @  $120^\circ$  = "O + T" vector
- d. Effect of trial weight alone = T = 15.5 mils (from Figure 6.01H)
- e. Rotor Sensitivity =  $\frac{18 \text{ oz-in}}{15.5 \text{ mils}} = 1.16 \text{ oz-in/mil}$
- f. Residual Unbalance =  $\frac{(1.16 \text{ oz-in/mil})(8 \text{ mils})}{1} = 9.28 \text{ oz-in}$   
(not within specs)

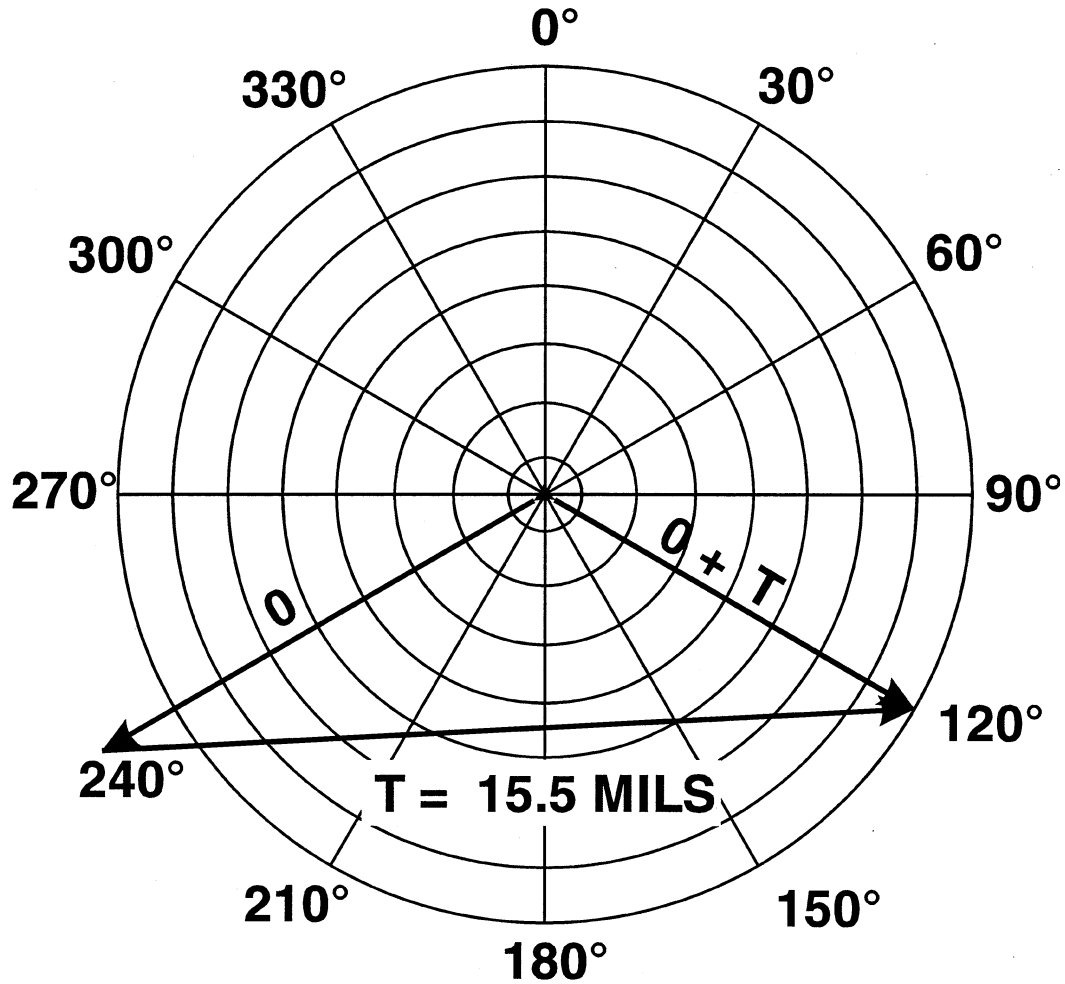
Continued balancing and reduced vibration to 1.0 mil

$$\text{Residual Unbalance} = \frac{(1.16 \text{ oz-in/mil})(1.0 \text{ mil})}{1} = 1.16 \text{ oz-in}$$

(in compliance)

**$0 = 10 \text{ MILS @ } 240^\circ$**

**$0 + T = 8 \text{ MILS @ } 120^\circ$**



**FIGURE 6.01H**  
**STANDARD SINGLE-PLANE VECTOR SOLUTION**

**TABLE 6.01A\***  
**BALANCE QUALITY GRADES FOR VARIOUS GROUPS OF REPRESENTATIVE**  
**RIGID ROTORS IN ACCORDANCE WITH ISO 1940 AND ANSI S2.19-1975**

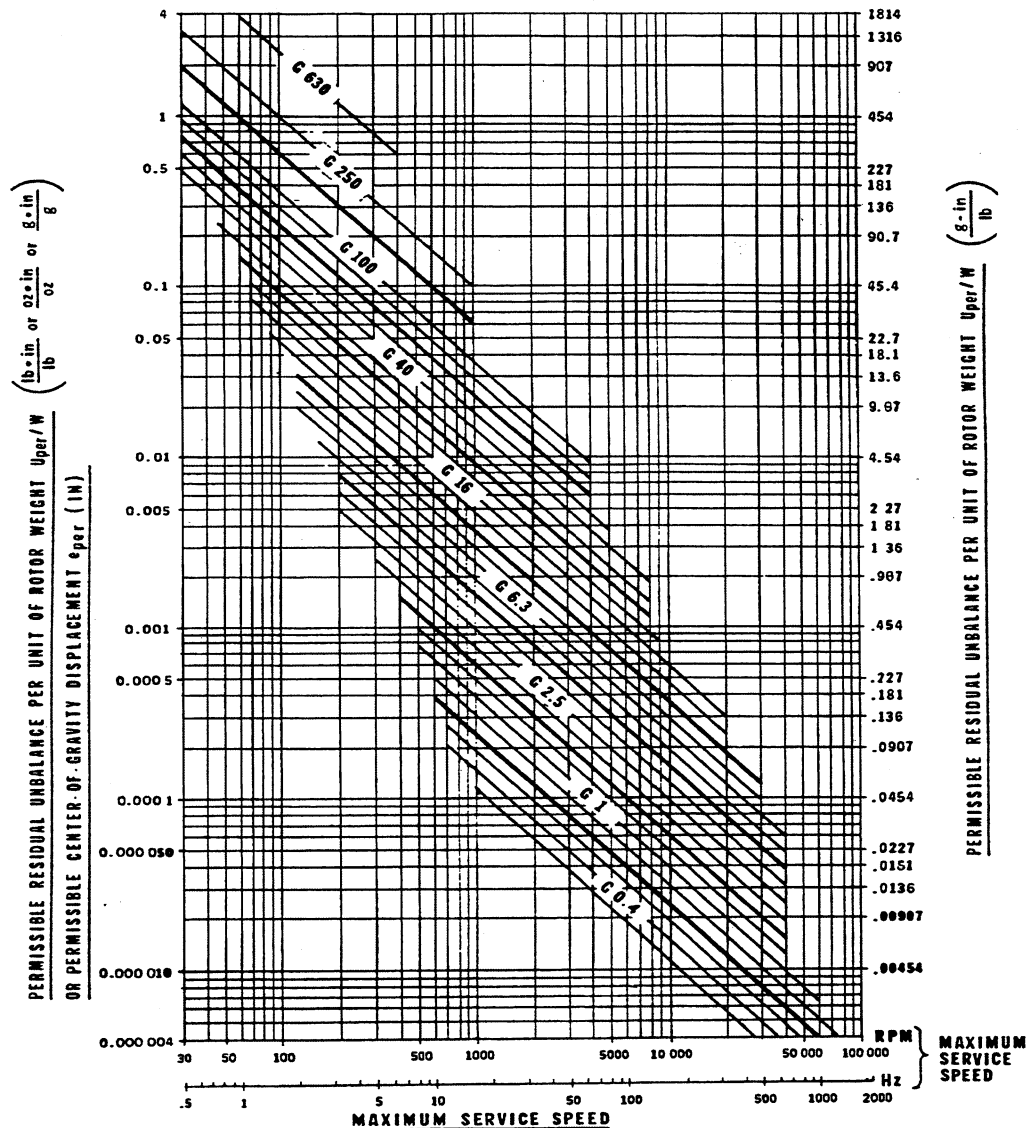
\*(Reference 20)

| Balance Quality Grades G | EW 1, 2 (mm/sec) | Rotor Types - General Examples                                                                                                                                                                                                                                                                                                                       |
|--------------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| G 4000                   | 4000             | Crankshaft drives <sup>3</sup> of rigidly mounted slow marine diesel engines with uneven number of cylinders. <sup>4</sup>                                                                                                                                                                                                                           |
| G 1600                   | 1600             | Crankshaft drives of rigidly mounted large two-cycle engines.                                                                                                                                                                                                                                                                                        |
| G 630                    | 630              | Crankshaft drives of rigidly mounted large four-cycle engines.<br>Crankshaft drives of elastically mounted marine diesel engines.                                                                                                                                                                                                                    |
| G 250                    | 250              | Crankshaft drives of rigidly mounted fast four-cylinder diesel engines. <sup>4</sup>                                                                                                                                                                                                                                                                 |
| G 100                    | 100              | Crankshaft drives of fast diesel engines with six or more cylinders. <sup>4</sup> Complete engines (gasoline or diesel) for cars, trucks and locomotives. <sup>5</sup>                                                                                                                                                                               |
| G 40                     | 40               | Car wheels, wheel rims, wheel sets, drive shafts. Crankshaft drives or elastically mounted fast four-cycle engines (gasoline or diesel) with six or more cylinders. <sup>4</sup> Crankshaft drives for engines of cars, trucks or locomotives.                                                                                                       |
| G 16                     | 16               | Drive shafts (propeller shafts, cardan shafts) with special requirements. Parts of crushing machinery. Parts of agricultural machinery. Individual components of engines (gasoline or diesel) for cars, trucks, and locomotives. Crankshaft drives of engines with six or more cylinders under special requirements. Slurry or dredge pump impeller. |
| G 6.3                    | 6.3              | Parts or process plant machines. Marine main turbine gears (merchant service). Centrifuge drums. Fans. Assembly aircraft gas turbine rotors. Fly wheels. Pump impellers. Machine-tool and general machinery parts. Normal electrical armatures. Individual components of engines under special requirements.                                         |
| G 2.5                    | 2.5              | Gas and steam turbines, including marine main turbines (merchant service). Rigid turbo-generator rotors. Rotors. Turbo-compressors. Machine-tool drives. Medium and large electrical armatures with special requirements. Small electrical armatures. Turbine-driven pumps.                                                                          |
| G 1                      | 1                | Tape recorder and phonograph (gramophone) drives. Grinding machine drives. Small electrical armatures with special requirements.                                                                                                                                                                                                                     |
| G 0.4                    | 0.4              | Spindles, disks, and armatures of precision grinders. Gyroscopes.                                                                                                                                                                                                                                                                                    |

1. The quality grade number represents the maximum permissible circular velocity of the center of gravity (CG) in mm/sec.
2.  $\omega = 2\pi n/60 \sim n/10$ , if  $n$  is measured in revolutions per minute and  $\omega$  in radians per second.
3. In general, for rigid rotors with two correction planes, one-half of the recommended residual unbalance is to be taken for each plane; these values apply usually for any two arbitrarily chosen planes, but the state of unbalance may be improved upon at the bearings. For disk-shaped rotors the full recommended value holds for one plane.
4. A crankshaft-drive is an assembly which includes the crankshaft, a flywheel, clutch, pulley, vibration damper, rotating portion of connecting rod, etc.
5. For purposes of this Standard, slow diesel engines are those with a piston velocity of less than 9 mm/sec; fast diesel engines are those with a piston velocity of greater than 9 m/sec.
6. In complete engines, the rotor mass comprises the sum of all masses belonging to the crankshaft drive described in Note 3 above.

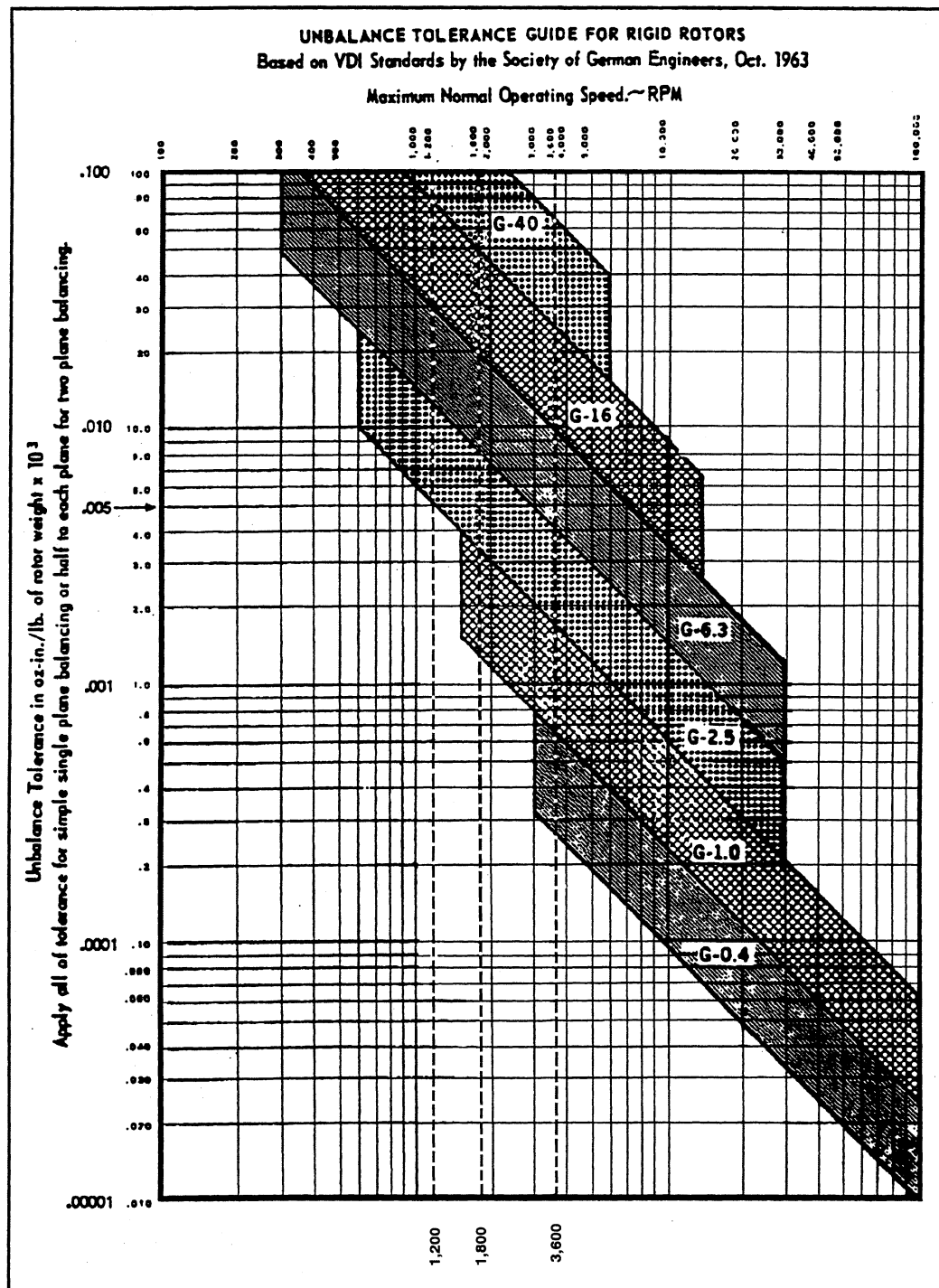
**TABLE 6.01B\***  
**MAXIMUM PERMISSIBLE RESIDUAL SPECIFIC UNBALANCE**  
**CORRESPONDING TO VARIOUS BALANCE QUALITY GRADES G,**  
**IN ACCORDANCE WITH ISO 1940**  
 \*(Reference 20)



**NOTES:**

- (1) In general, for rigid rotors with two correction planes, one half of the recommended residual unbalance is allowed for each plane; these values apply usually for any two arbitrarily chosen planes. (See paragraph on Applying Tolerances to Two-plane Rotors). For disc-shaped rotors the full recommended value applies to one plane.
- (2) To obtain acceptable residual unbalance in oz-in per lb of rotor weight multiply lb-in value by 16.
- (3) To avoid need for multiplication use appropriate Balance Tolerance Nomogram on following pages.

**TABLE 6.01C\***  
**EQUIVALENT ISO QUALITY GRADES SHOWN IN BANDED REGIONS**  
 \*(Reference 21)



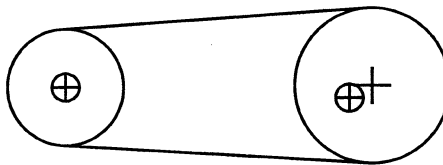
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## 6.02 ECCENTRIC ROTORS

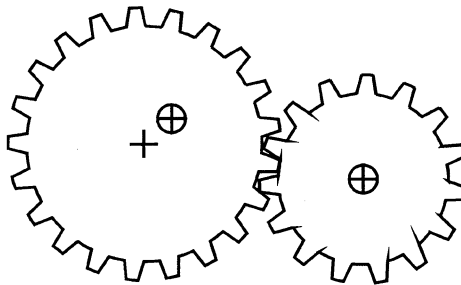
McGraw Hill's "Dictionary of Mechanical and Design Engineering" defines Eccentricity as:

*"the distance of the geometric center of a revolving body from the axis of rotation".*

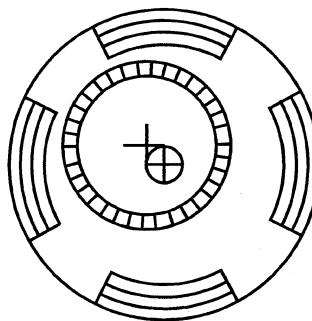
In other words, referring to Figures 6.02A through 6.02C, an eccentric rotor is one in which the shaft centerline does not line up with the rotor centerline. This results in more weight being on one side of the rotating centerline than the other and causes the shaft to wobble in an irregular orbit. This is inherently unstable and can be the source of troublesome vibration. Sometimes, it is possible to "balance out" part of the effect of eccentricity, but much of the displaced motion still remains. In other cases, it is not even possible to perform a good balance on rotors having more eccentricity. Today with the emphasis on higher and higher rotating speeds, it is very important that eccentricity be minimized.



**FIGURE 6.02A  
ECCENTRIC SHEAVE**

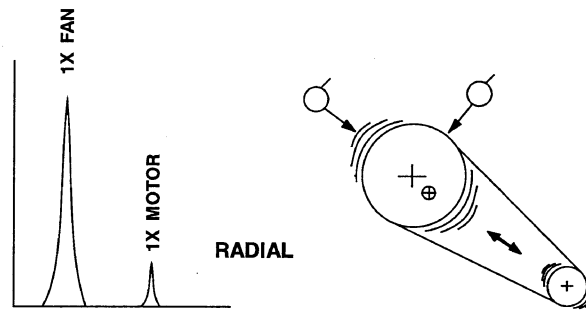


**FIGURE 6.02B  
ECCENTRIC GEAR**



**FIGURE 6.02C  
ECCENTRIC MOTOR ARMATURE**

The classic spectrum of an eccentric rotor is shown on Figure 6.02D. Note that, like unbalance, the spectrum is dominated by the frequency at 1X RPM of the eccentric component, particularly when the measurement is taken in the direction through the centers of the two rotors. Looking at Figure 6.02D which shows a motor driving a fan with an eccentric pulley, note that the motor operating speed peak will be much lower than that at fan speed, particularly when taken in line with the belt direction. This eccentricity causes a very highly directional loading so that vibration at 1X RPM can be very much higher in one radial direction than in the other (depending on the amount of eccentricity).



**FIGURE 6.02D**  
**TYPICAL SPECTRUM OF AN ECCENTRIC ROTOR**

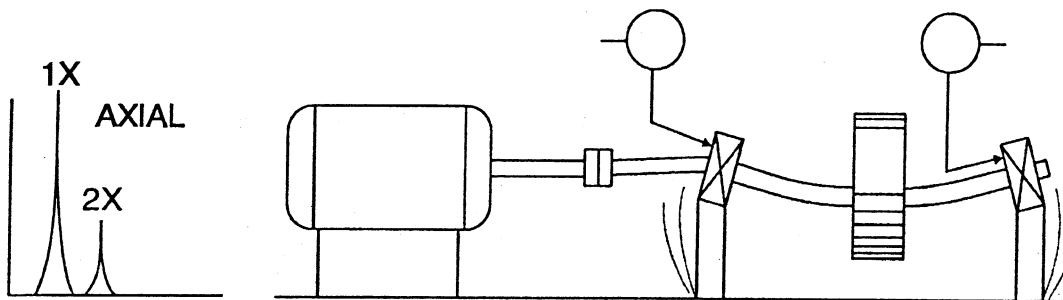
An eccentric rotor exhibits each of the following characteristics:

1. Some of the more common types of eccentric rotors include eccentric pulleys, gears, motor rotors and pump impellers:
  - a. Figure 6.02A shows an eccentric pulley. In these units, the largest vibration most often occurs in the direction of belt tension and at the frequency of 1X RPM of the eccentric pulley. Eccentric pulleys represent one of the most troublesome sources of undesirable vibration in belt drives today. Unfortunately, the industry to date has not sufficiently policed itself to minimize eccentricity in common pulleys. Often, attempts are made to overcome pulley eccentricity after the fact by balancing. Even when this is done, balancing alone will not significantly lower the back-and-forth belt motion which results in continuous belt tension variation, depending on the position of the eccentric pulley at any instant. Plants need to protect themselves by writing eccentricity specifications into their belt drive orders if they want to maximize the life of their machinery and lower their vibration.
  - b. Figure 6.02B shows an eccentric gear in which the largest vibration will occur in a direction in line with the centers of the two gears, and at a frequency of 1X RPM of the eccentric gear. The vibration signature will appear like unbalance of this gear, but it is not. If the eccentricity is significant, it can induce very high dynamic loads on gear teeth as they are forced into and out of a bind with the mating gear. Phase analysis can be used on gears having high 1X RPM vibration to differentiate whether unbalance or eccentricity is the source (See characteristic #3 below). Not only do eccentric gears result in higher 1X RPM vibration, but they also can generate high amplitude gear mesh frequencies and harmonics which will be accompanied by higher than normal amplitude sideband frequencies spaced around the gear mesh frequency at the eccentric gear RPM. Sometimes, these sidebands will be at 2X RPM of the eccentric gear. These sidebands will modulate the amplitude of gear mesh frequencies themselves.

- 
- c. Figure 6.02C shows an eccentric motor rotor. Eccentric rotors produce a rotating variable air gap between the rotor and stator which induces pulsating vibration between 2X line frequency (7200 CPM) and its closest running speed harmonic as well as generating pole pass frequency ( $F_p$ ) sidebands around 2X line frequency (see "Electrical Problem Vibration Symptoms" in Section 6.12). That is, for a 3580 RPM motor, this would be between 2X running speed and 2X line frequency, whereas for a 1780 RPM unit, it would be between 4X RPM and 7200 CPM. Section 6.12 will show that an eccentric motor rotor will also generate pole pass frequency sidebands around 2X line frequency (where pole pass frequency,  $F_p$ , equals #Poles times slip frequency). Finally, the eccentric rotor motion itself will cause a variation in the magnetic field between the stator poles and rotor, thereby inducing 1X RPM vibration between the rotor and stator.
- d. Eccentric pump impellers can result in unequal hydraulic forces distributed between the rotating impeller and stationary diffuser vanes. This can result not only in high vibration at pump RPM, but also at vane pass frequency and multiples (# vanes times RPM and multiples) due to a "hydraulic unbalance" induced by the eccentric impeller.
2. Attempts to balance eccentric rotors will often result in reducing vibration in one direction, but increasing it in the other radial direction.
3. Eccentric rotors may cause significantly higher vibration in one radial direction than in the other (as does resonance, wiped bearings and sometimes looseness as well). Phase analysis can be employed as an effective tool to detect whether or not the source of high vibration at 1X RPM is from eccentricity or from another 1X RPM source such as unbalance. Comparative horizontal and vertical phases usually differ by approximately 0° or 180° since the force induced by eccentricity is highly directional (rather than a 90° phase difference in horizontal and vertical as in the case of dominant unbalance problems).

### 6.03 BENT SHAFT

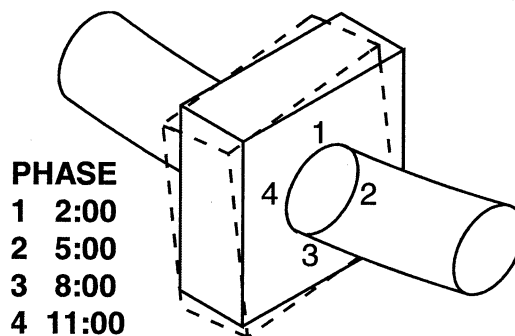
A bowed or bent shaft can generate excessive vibration in a machine, depending on the amount and location of the bend. Like eccentric shafts, the effects can sometimes be decreased by balancing. However, more often than not, it is not possible to achieve a satisfactory balance in a shaft which has any noticeable bend. Analysts are sometimes successful in removing the bend by various techniques sometimes involving thermal treatments. In these cases, however, one must be careful not to introduce residual stresses which might later lead to shaft fatigue.



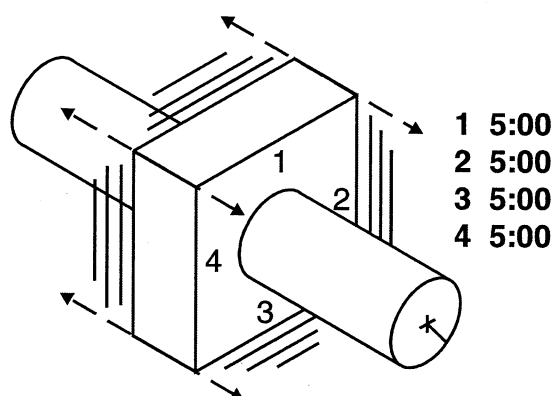
**FIGURE 6.03A**  
**BENT SHAFT SPECTRAL AND PHASE RESPONSE**

Bent shafts exhibit the following characteristics:

1. Figure 1 shows that high axial vibration is generated by the rocking motion induced by the bent shaft. Dominant vibration normally is at 1X RPM if bent near the shaft center, but a higher than normal 2X RPM component can also be produced, particularly if bent near the coupling.
2. Axial phase change between two bearings on the same component (motor, fan, pump, etc.) approaches  $180^\circ$ , dependent on the amount of the bend (as shown in Figure 1). In addition, if one makes several measurements on the same bearing at various points in the axial direction, he will normally find that phase differences approaching  $180^\circ$  occur between that measured on the left and right hand side of the bearing, and also between the upper and lower sides of the same bearing.
3. Amplitudes of 1X RPM and 2X RPM will normally be steady, assuming that 2X RPM is not located close to twice line frequency (7200 CPM) which might induce a beat of the 2X RPM component with 2X line frequency if there is high electromagnetic vibration present.
4. Please note the axial phase measurements on 4 points of a bearing housing pictured in Figure 6.03B. If the shaft is bowed through or very near a bearing, you get a twisting motion by the bearing housing itself which will result in significantly different phase readings on this bearing housing in the axial direction as pictured in Drawing A of Figure 6.03B. Drawing B of this figure shows the axial phase which results from a true, straight shaft.
5. When much runout is present at the rotating mass, it appears as unbalance. When run-out at the coupling occurs, it appears as misalignment.



**A: AXIAL PHASE MEASUREMENTS INDICATING TWISTING MOTION DUE TO A BENT SHAFT**



**B: AXIAL PHASE MEASUREMENTS INDICATING A TRUE SHAFT IN PROPER MOTION**

**FIGURE 6.03B**

6. In bent shafts, amplitude can vary with the square of speed and preload. If unbalance is more of the problem than bow, vibration will decrease abruptly if operating below the first critical speed. However, if the rotor is brought above its first critical speed, unbalance amplitude will change only a small amount, whereas if the dominant problem is a bent shaft, the amplitude will again drop significantly as the speed is dropped towards the first critical speed.
7. If a rotor is located between bearings and should operate at or close to its fundamental natural frequency, it will appear to be a "bent" shaft and will display these symptoms (see Figure 6.05E in Section 6.05 on "Resonant Vibration"). However, this is only temporary. When the machine is stopped or at another non-resonant speed, it will then "straighten out".
8. When electric motors have problems such as shorted laminations, they will thermally induce a bend as the machine heats up, with the resultant vibration getting higher and higher as the rotor heats. This again will introduce bent shaft symptoms (see Figure 6.12F in Section 6.12 on "Electrical Vibration"). In this case, the shaft again will straighten when allowed to come back to room temperature if the plastic limit of the shaft material has not been exceeded. This will be covered later in the electrical problems Section 6.12.

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## 6.04 MISALIGNMENT

Likely the most widespread mechanical problem in industry today is misalignment. Many plants have begun to resolve a number of their unbalance problems as more and more data collectors having this capability, as well as other analyzers are acquired by plants, and balance actions are taken. However, new instruments are also now becoming available to resolve alignment problems. These include optical as well as the newer laser devices. With these newer instruments, we can now realize that machines have been operating for quite some time with much higher levels of misalignment than had previously been thought. In fact, we are learning that it is not uncommon at all to have 30% to 50% or more of machines in any plant that have high degrees of misalignment.

The trouble with such high levels of misalignment is that it induces high vibration levels leading to premature failure of expensive machine components and increased energy demands as well. Misalignment is now probably one of the leading causes of bearing failures as well.

Although vibration responds to the degree of misalignment, there is not a direct 1-for-1 relationship between the amount of misalignment (angularity and offset) and the amount of vibration. As John Mitchell states on page 182 of Reference 2:

*"the vibration characteristics associated with misaligned flexible couplings are not a direct measure of the amount of misalignment but of the coupled system's ability to accommodate misalignment. Thus, the external symptoms of misalignment, in addition to being a function of the offset between shafts, are also affected by speed, torque, or any other condition such as corrosion or sludging which may alter the coupling's stiffness and hence its ability to accommodate a given offset."*

The first page of the Vibration Diagnostic Chart (Table 6.0) shows there are 3 types of alignment concerns including angular misalignment, parallel misalignment and a misaligned bearing cocked on a shaft. Each of these will be covered separately along with a section on coupling problems later. First, since misalignment problems are so prevalent today, a number of key facts should be considered about it including what effect it has on component lives; where it directs its potentially harmful forces; what are its spectral characteristics (harmonic content); what are its directional characteristics; what are its phase characteristics; and finally, what should be done to monitor alignment:

1. Component Failures Due To Misalignment - Misalignment can of course cause the coupling to fail, but other machine components as well. For example, if the coupling is stronger than the adjacent bearing, it can subject the bearing to excessive forces with little or no damage to the coupling. Similarly, such misalignment can detrimentally affect other components including gears, belts, sheaves, blading, etc.
2. Reaction On Free (or Outboard) End - It is possible for the highest reaction to misalignment to occur not on the bearing closest to the coupling, but on the free or outboard machine end. In these cases, incoming forces from the coupling may be strong enough to stabilize this system adjacent to the coupling and suppress the symptoms on this end.
3. Axial Vibration - Misalignment normally causes both high axial and radial vibration (as opposed to unbalance which acts mostly in the radial direction with the exception of overhung rotors).

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4. Other Sources Of High Axial Vibration - Again, while misalignment is probably the most common source of high axial vibration, there are several other sources which can generate it including:
    - a. Bent Shafts;
    - b. Shafts in Resonant Whirl;
    - c. Misaligned Bearings Cocked on the Shaft (See Section 6.043);
    - d. Resonance of Some Component in the Axial Direction;
    - e. Worn Thrust Bearings;
    - f. Worn Helical or Bevel Gears;
    - g. A Sleeve Bearing Motor Hunting for its Magnetic Center;
    - h. Couple Component of a Dynamic Unbalance.

Therefore, when high axial vibration occurs, do not quickly jump to the conclusion that the problem is misalignment. Instead, refer especially to phase; and then to the vibration spectrum.

5. Low Axial Vibration During Misalignment - Although misalignment is classically categorized as having high axial vibration, it does not always occur. For example, the writer has experienced some cases where misalignment was the problem even though axial levels were only about 1/4 of those in the radial direction. This is quite possible for machines with predominately parallel offset versus angular misalignment.
6. Comparable Horizontal And Vertical Amplitudes - Since it has been pointed out that it is possible for a machine to have good horizontal alignment but poor vertical, it is quite possible for misaligned machines to have much higher vibration in one radial direction versus another.
7. Radial Vibration Response To Misalignment - One would think that if driver and driven shafts were horizontally offset, it would cause high horizontal forces. Although this is sometimes the case, Reference 4 states that in most cases, high horizontal amplitudes are primarily the result of vertical misalignment and vice versa.
8. 2X RPM Vibration - Often, misalignment generates a higher than normal 2X RPM vibration which can act not only in the axial direction, but also in the radial. This second operating speed harmonic is caused by asymmetric stiffness in the machine and its supports, or in the coupling. That is, there is often quite a difference in stiffness around the supporting housing, frame, foundation and coupling itself which can allow a "back-and-forth" motion with each revolution, thereby resulting in 2X RPM vibration.
9. Higher Harmonics - Misalignment can also cause large numbers of harmonics which will make the spectrum appear like looseness/excessive clearance problems. The key distinguishing feature still appears to be the high level at 2X RPM in the axial direction. Several tests have been conducted purposely misaligning units and measuring their response (Reference 3). During these referenced tests, multiple harmonics often begin to appear when the misalignment became more and more severe.
10. Phase Is Best Indicator - When high vibration occurs on a machine predominately at 1X RPM and 2X RPM, the best overall indicator of misalignment problems is phase (that is, how the machine is shaking). Phase will differentiate between a number of other potential 1X RPM and 2X RPM vibration sources. Phase behavior in response to misalignment can be summarized as follows:
  - a. Probably the best indicator of misalignment problems is evaluation of phase across the coupling. Here, one is checking how the driver shaft and its coupling half is

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reacting relative to the driven shaft with its coupling half. When this phase difference across the coupling approaches  $180^\circ$  ( $\pm 40^\circ$  to  $50^\circ$ ), misalignment is often indicated, particularly when other misalignment symptoms are present. The higher the severity of misalignment, the closer this difference will approach  $180^\circ$ . Similarly, the less significant are other problems such as unbalance, eccentricity, resonance, etc., the more this difference will approach  $180^\circ$ . Here, it is important that not only vibration spectra, but also phase measurements be captured in horizontal, vertical and axial directions on each of the bearing housings which are accessible.

- b. Since it is possible for shafts to have good horizontal alignment, but poor vertical alignment (or vice versa), it is common in these cases for the horizontal phase difference to be quite different from the vertical phase difference. In fact, this is the case most of the time. In the special case where shafts have good horizontal, but poor vertical alignment, the shaft orbit itself would tend to be greatly elliptical which may make the alignment problem to appear to be eccentricity, resonance or a similar problem. That is, the amplitudes may be greatly different in one radial direction versus another. However, examination of the phase differences throughout the machine will indicate the misalignment problem.
- c. When examining the phase difference on one of the rotors (just the motor, pump, fan, etc.), the radial phase differences for significant misalignment will be either  $0^\circ$  or  $180^\circ$  ( $\pm 30^\circ$ ). This is unlike unbalance in which such phase differences could be most anything (i.e., both the horizontal and vertical phase differences might be  $70^\circ$ ). The key here is that misalignment phase differences will approach either  $0^\circ$  or  $180^\circ$ .
- d. When comparing horizontal phase differences with vertical phase differences on the same rotor, about 90% of misaligned machines will show a difference approaching  $180^\circ$  between the vertical and horizontal. For example, if the horizontal phase differed about  $30^\circ$  between the outboard and inboard bearings, the vertical phase difference would be about  $210^\circ$  for most misaligned rotors. An unbalanced rotor will not show this phase behavior since whatever phase difference occurs on the horizontal direction will be very close to that in the vertical direction.

In summary, phase data should always be taken if possible on machines having high vibration at 1X and 2X RPM since phase will be the key indicator in differentiating whether the dominant problem source is misalignment as opposed to other problems of similar symptoms. While other symptoms such as high axial vibration and harmonic vibration are also good symptoms, these should not receive as much weight as phase (for example, if phase does indicate misalignment, but axial vibration does not, one should give more weight to the data provided him by phase) if the vibration is high.

- 11. Effect Of Other Problem Sources - When other problems such as unbalance, bent shaft, resonance, etc. are present, along with misalignment, this can affect not only the vibration spectrum, but also phase behavior. For example, if both unbalance and misalignment are present, it might show high levels at both 1X RPM and 2X RPM, plus radial phase differences which may or may not approach  $150^\circ$  to  $180^\circ$ , depending on the severity of each problem (in this case, axial phase differences across the coupling will still likely approach  $180^\circ$ ). Reference 4 suggests that when several problem sources are all present, each of them will contribute vectorially. That is, if one had polar coordinate graph paper, you might show the contribution of unbalance as a 3 mil level at  $30^\circ$ ; the misalignment a 2 mil level at  $60^\circ$ ; and a simultaneous eccentricity problem contributing 1 mil at  $0^\circ$ . The resultant vector would not show a phase at any of these 3 individual angles, but instead would produce a vector somewhere on the order of 4 mils at about  $40^\circ$ . This would still not be radically different in any of the original phase angles.

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However, if each of these 3 were at widely differing phase angles, the resultant phase angle and magnitude could be quite different. In this case, one would first have to solve one problem at a time (in this case, he should likely solve first the eccentricity problem; then the misalignment, and finally balance the rotor). As each problem is solved, the vibration spectra and phase will react accordingly.

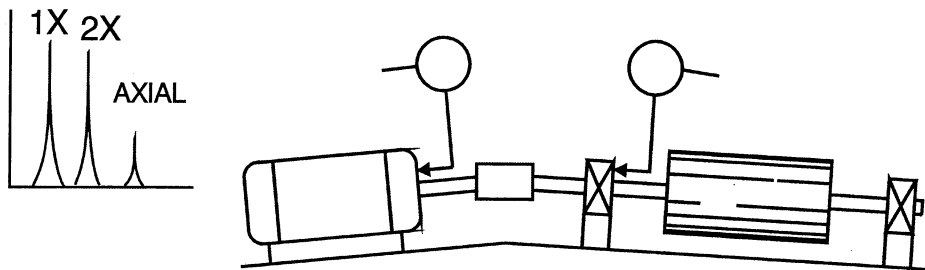
12. **Monitoring Alignment Change** - When aligning especially critical machines, it is often helpful to monitor the alignment and see how it might change. When doing so, it is especially important to monitor phase in all 3 directions at each bearing on each machine involved in the alignment. In most machines that have thermal offsets, if the machine is brought up to speed from room temperature, it should display misalignment symptoms in the beginning which should disappear as the machine comes up to full operating temperature. For example, the phase difference across the coupling should initially be on the order of 150° to 180°, but should drop to close to 0° to 30° in the end. In addition to phase, one should monitor how the vibration spectrum changes as well as other non-vibration related variables such as bearing temperature, temperature of the support legs and oil film pressure. When monitoring alignment change, each of the following should be considered:
- a. **1X RPM** - It might be better to monitor vibration at higher harmonics of 2X up to 4X RPM rather than 1X RPM since the first harmonic will be effected by so many other things (unbalance, resonance, eccentricity, bent shaft, etc.).
  - b. **2X RPM** - The 2X component should be a much better indicator of alignment than 1X assuming that this is not a 3600 RPM nominal motor (if so, 2X RPM will likely be very close to 2X line frequency which will "contaminate" the apparent 2X amplitude unless one is able to separate 2X RPM from 2X line frequency).
  - c. **3X RPM** - This 3X component may be the best indicator of alignment change if this particular machine does in fact cause an increase at 3X RPM with an alignment change. This is often the case. In such cases, it is not necessary for the 3X component to be larger than either 1X or 2X RPM, just that it be sensitive itself to alignment change.
  - d. **4X RPM** - The same analogy applies to 4X RPM as does 3X RPM with the exception being that this is not an 1800 RPM nominal speed machine (in which case 4X RPM would closely approach 2X line frequency at 7200 CPM).
  - e. **Number of Coupling Grids (or Segments) X RPM** - Some coupling types include a number of grids or segments which often cause vibration at the number of grids (or segments) on one coupling half times the RPM, particularly when misalignment becomes severe. In these cases, where the coupling components themselves are effected and do respond to misalignment, this frequency will be an excellent choice to monitor since it will be well removed from any effects of unbalance, bent shaft, eccentricity or any other such source other than alignment.

#### **6.041 Angular Misalignment:**

Angular misalignment is pictured in Figure 6.04A. Each of the following characteristics are demonstrated by angular misalignment:

- 1. Angular misalignment primarily generates high axial vibration, particularly at 1X and 2X RPM. However, it is not unusual for either one of these peaks (1X, 2X or 3X RPM) to dominate alone.

2. However typically, when the amplitude of either 2X RPM or 3X RPM exceeds approximately 30% to 50% of that at 1X RPM in the axial direction, angular misalignment is indicated. This assumes that there is high vibration (that is, misalignment may be of concern if  $1X = .30$  ips and  $2X = .20$  ips; but not if  $1X = .03$  and  $2X = .02$  ips).
3. Angular misalignment is best detected by  $180^\circ$  phase change across the coupling in the axial direction as pictured in Figure 6.04A. If each of the bearings on one of the side are moving one way, while those on the other side are moving in the opposite direction, angular alignment is highly suspect.

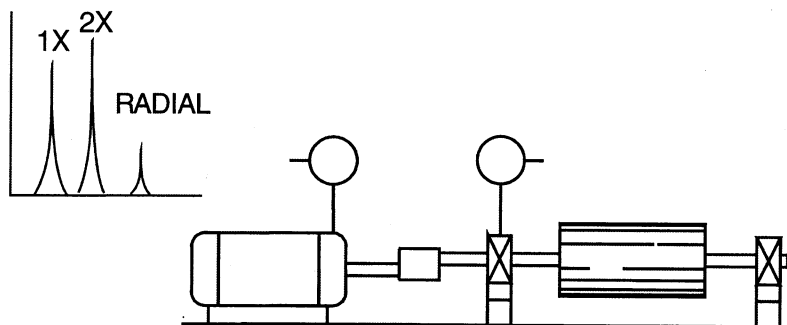


**FIGURE 6.04A**  
**ANGULAR MISALIGNMENT SPECTRAL AND PHASE RESPONSE**

#### **6.042 Parallel Misalignment (Also known as Radial Offset Misalignment):**

Radial misalignment is pictured in Figure 6.04B. It displays each of the following characteristics:

1. Parallel misalignment primarily affects radial vibration as opposed to angular which affects axial.
2. Like angular alignment problems, parallel misalignment causes phase to approach  $180^\circ$  difference across the coupling, but in the radial direction (horizontal or vertical).
3. Radial misalignment is often indicated in a spectrum when 2X RPM exceeds approximately 50% of the amplitude at 1X RPM, but its height relative to 1X RPM is often dictated by the coupling type and construction. It is not uncommon for 2X RPM to exceed that at 1X RPM, particularly when the parallel misalignment becomes severe.
4. When either angular or parallel misalignment becomes severe, each can generate an array of harmonics ranging up to and including the 4th through the 8th harmonic. In this case, the severe misalignment spectrum can appear to be mechanical looseness (see page 1 of Table 6.0 Diagnostics Chart).

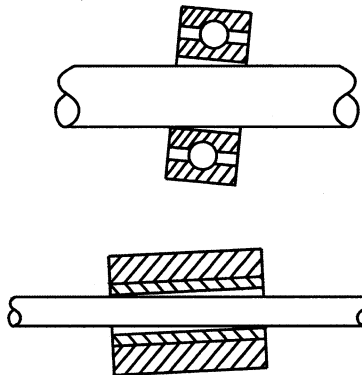


**FIGURE 6.04B**  
**PARALLEL MISALIGNMENT SPECTRAL AND PHASE RESPONSE**

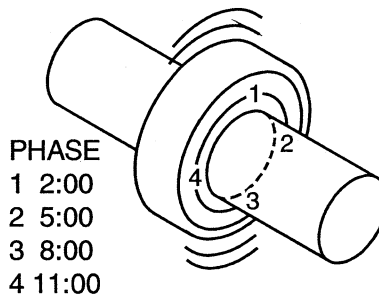
### 6.043 Misaligned Bearing Cocked on the Shaft:

When either a sleeve or rolling element bearing is misaligned or cocked on the shaft, it can cause high vibration and unusual loading. If it is detected, it should quickly be resolved before causing premature component failures. This problem is pictured in Figure 6.04C. Each of the following characteristics are indicative of a misaligned bearing on a shaft:

1. A cocked bearing will normally generate considerable axial vibration which can affect not only that at 1X RPM, but also 2X RPM as well.
2. If phase is measured in the axial direction at each of 4 points 90° apart from each other as shown in Figure 6.04D, a cocked bearing will be indicated by a 180° phase shift from top to bottom or from side to side.
3. Attempts to align the coupling or balance the rotor will not alleviate the problem. The effected bearing must be removed and correctly installed.



**FIGURE 6.04C**  
**MISALIGNED BEARINGS COCKED ON SHAFT**



**FIGURE 6.04D**  
**AXIAL PHASE MEASUREMENTS INDICATING**  
**A COCKED BEARING ON A SHAFT**

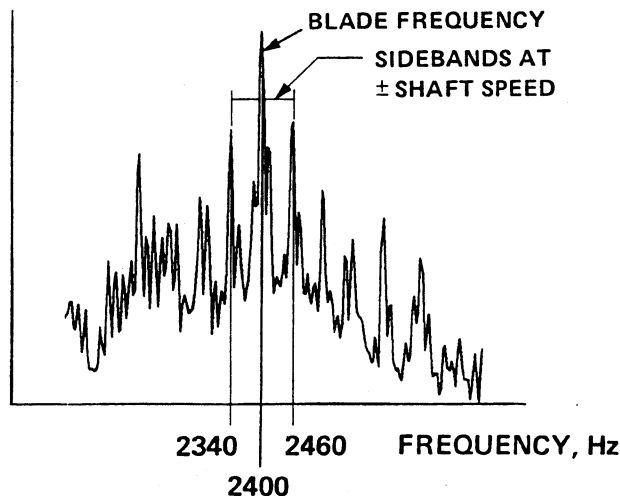
### 6.044 Coupling Problems:

It is often difficult to tell from vibration signatures or phase analysis whether the problem is misalignment or a coupling problem. Each of the many types of couplings has a different effect on the response of the machinery to which it is coupled. Other factors affecting its response include spacing between shafts, shaft diameter and bearing type. However, problem couplings do display the following characteristics:

1. 3X RPM will often respond to a coupling problem, particularly for a coupling having a spacer that is too long or too short. In these cases, the radial spectrum will normally indicate a fairly noticeable 3X running speed vibration, and that in the axial direction will show a much higher 3X RPM component. These problems are resolved by either correctly sizing the spacer or repositioning either the driver or driven equipment.
2. Gear Type Couplings can experience coupling lockup where the frictional force developed at gear teeth is greater than the applied force causing the coupling to become a rigid member. Friction welding of teeth can occur at this point, particularly if there is a lack of lubrication. A locked coupling can cause severe problems and may lead to thrust bearing failure if it results in the thrust load of 2 machines being applied to only one thrust bearing. Also, if the teeth do weld together and then break loose, it leaves pit marks on the coupling teeth. Coupling lockup can be broken temporarily either due to a change in load or by striking the coupling with a mallet or a piece of wood. However, this coupling should be closely inspected as soon as possible looking for tooth damage, lubrication problems and alignment problems, replacing if necessary.

Coupling lockup will normally cause an increase in both axial and radial vibration with axial vibration normally being higher. Most of the time, the 1X RPM is most effected. However, certain types of couplings will generate a frequency distribution resembling a "Christmas Tree" effect. In these cases, many harmonics can appear with the vibration dropping approximately 25% from one harmonic to the next (Reference 5). What gives the spectrum a "Christmas Tree" effect is that there is a fairly uniform drop of about 25% all the way from the 2nd through the 5th or 6th harmonic.

3. A loose coupling is likely to cause sidebands around blade pass frequencies ( $\# \text{blades} \times \text{RPM}$ ) and mesh frequencies ( $\# \text{teeth} \times \text{RPM}$ ) as shown in Figure 6.04E (however, sidebanding of blade pass and mesh frequencies does not always indicate a loose coupling). This is caused by the fact that a loose coupling does not drive the rotating equipment at a uniform speed, but rather, pulses at multiples of the shaft speed causing its running speed to modulate these other frequencies. Therefore, a signal similar to that shown in Figure 6.04E with equally spaced sidebands at coupling RPM can mean the coupling is loose (either from a poor fit on the shaft or from worn coupling components).



**FIGURE 6.04E**  
**LOOSENESS OF COUPLING INDICATED BY COUPLING RUNNING SPEED**  
**SIDE BANDS ABOUT BLADE PASS FREQUENCY**

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## 6.05 MACHINERY FAILURES DUE TO RESONANT VIBRATION

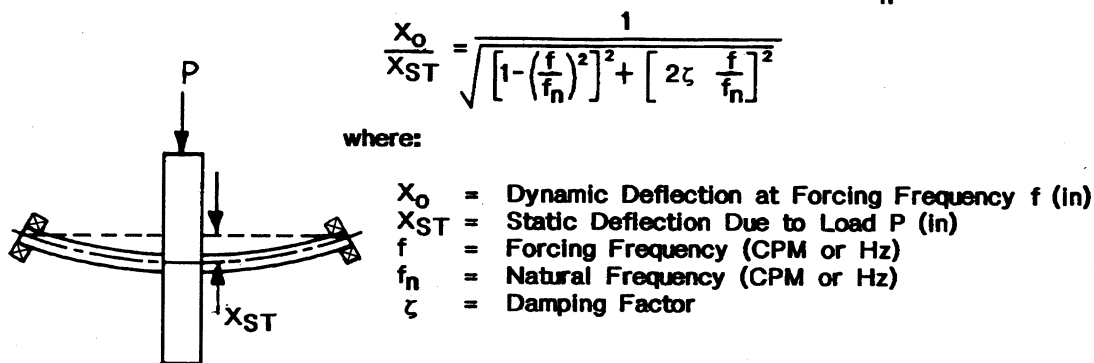
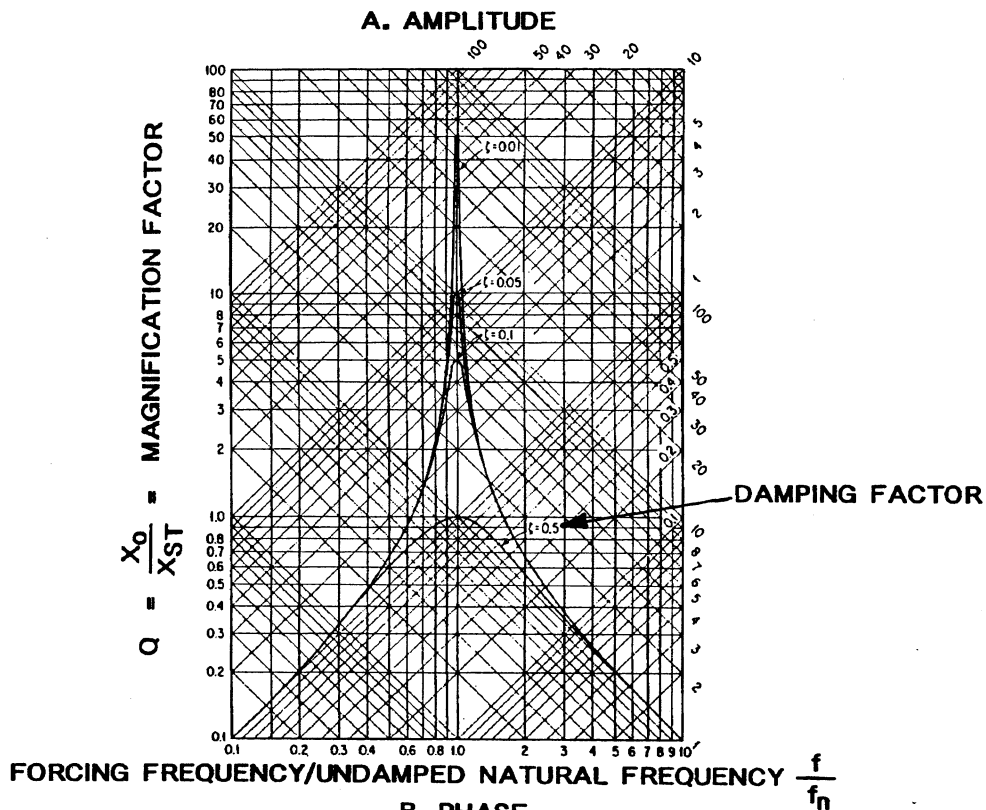
Resonant vibration problems are much more commonplace on today's machinery than it generally has been assumed. As a result, machinery components are often subjected to excessive vibration which often leads to catastrophic failure or, as a minimum, premature failure long before the parts should have failed. Therefore, it is important that industry learn both how to detect resonance and how to solve resonant vibration problems.

Resonance occurs when a forcing frequency coincides with a natural frequency. This may be a natural frequency of the rotor, but often can be a natural frequency of the support frame, foundation, or even drive belts. Forcing frequencies include those from sources such as unbalance, misalignment, looseness, bearing defects, gear defects, belt wear, etc.

Figure 6.05A helps illustrate resonance. It shows a graph of magnification factor ( $Q$ ) on the vertical axis versus frequency ratio ( $f/f_n$ ) on the horizontal axis. The magnification factor,  $Q$ , is actually an amplitude ratio which relates how much vibration is amplified when a machine passes through resonance. As the drawing on Figure 6.05A shows, the magnification factor is a ratio of dynamic deflection at a forcing frequency ( $X_d$ ) to static deflection due to load ( $X_{st}$ ). In other words, it compares shaft deflection at rest versus that with it rotating. The frequency ratio is the ratio of forcing frequency ( $f$ ) to natural frequency ( $f_n$ ). The upper graph in Figure 6.05A shows that when the frequency ratio approaches 1.0 (or when the forcing frequency "tunes in" to a natural frequency), the vibration amplitude can become extremely high, depending only on the amount of damping in the system. In fact, the equation below the graph shows that the only thing which limits an infinite vibration amplitude when at resonance is the system damping (where damping is  $\zeta$ , also known as the damping factor). In common machine structures made of steel, aluminum, cast iron and so forth, the damping factor is normally 0.05 or less, which shows that resonant amplification will be 10X for this damping factor, and can even approach amplifications of 50X for systems with lower damping (see upper graph in Figure 6.05A). Therefore, such vibration can easily lead to premature, or even catastrophic machine failure.

Please note the lower graph in Figure 6.05A plotting Phase Lag versus frequency ratio for various levels of damping. Note that when a machine passes through resonance, it will see a phase change of  $90^\circ$  when right on the natural frequency, and phase will continue to rapidly change when still in the vicinity of the natural frequency. Finally, phase will change almost  $180^\circ$  when completely through the resonance. The total phase change and the rate at which phase will change for given distances away from the natural frequency will be governed by the amount of damping which can be seen in Figure 6.05A. The lower the damping, the greater will be the rate of phase change.

Figure 6.05B shows a diagram of something known as a Bode Plot. Here, both vibration amplitude and phase are plotted on the vertical axis versus shaft speed (RPM) on the horizontal, as a machine is either brought from rest to full operating speed or when it is shut down and allowed to coast to a stop. Looking at the amplitude versus RPM curve, note that this particular machine went through two resonant events as it coasted down from approximately 6500 RPM. First, the amplitude increased from only .125 mil to about .872 mil when it reached about 4850 RPM; then it quickly dropped back down to about .130 mil again. Then, it excited another natural frequency at about 2450 RPM when levels increased up to about .600 mil. Finally, it dropped again and continued to decrease in amplitude until it came to rest. This is a typical Bode Plot for common rotating machinery. In this example, this machine would run well if it were operated from approximately 0 to 2200 RPM; or from 2700 to 4300 RPM; or from 5300 to 6500 RPM. However, it would have serious resonant amplification if it were operated either from about 2300 to 2600 or from 4600 to 5100 RPM.



**FIGURE 6.05A**  
**RESONANT AMPLIFICATION CURVE AND PHASE CHANGE AT RESONANCE**

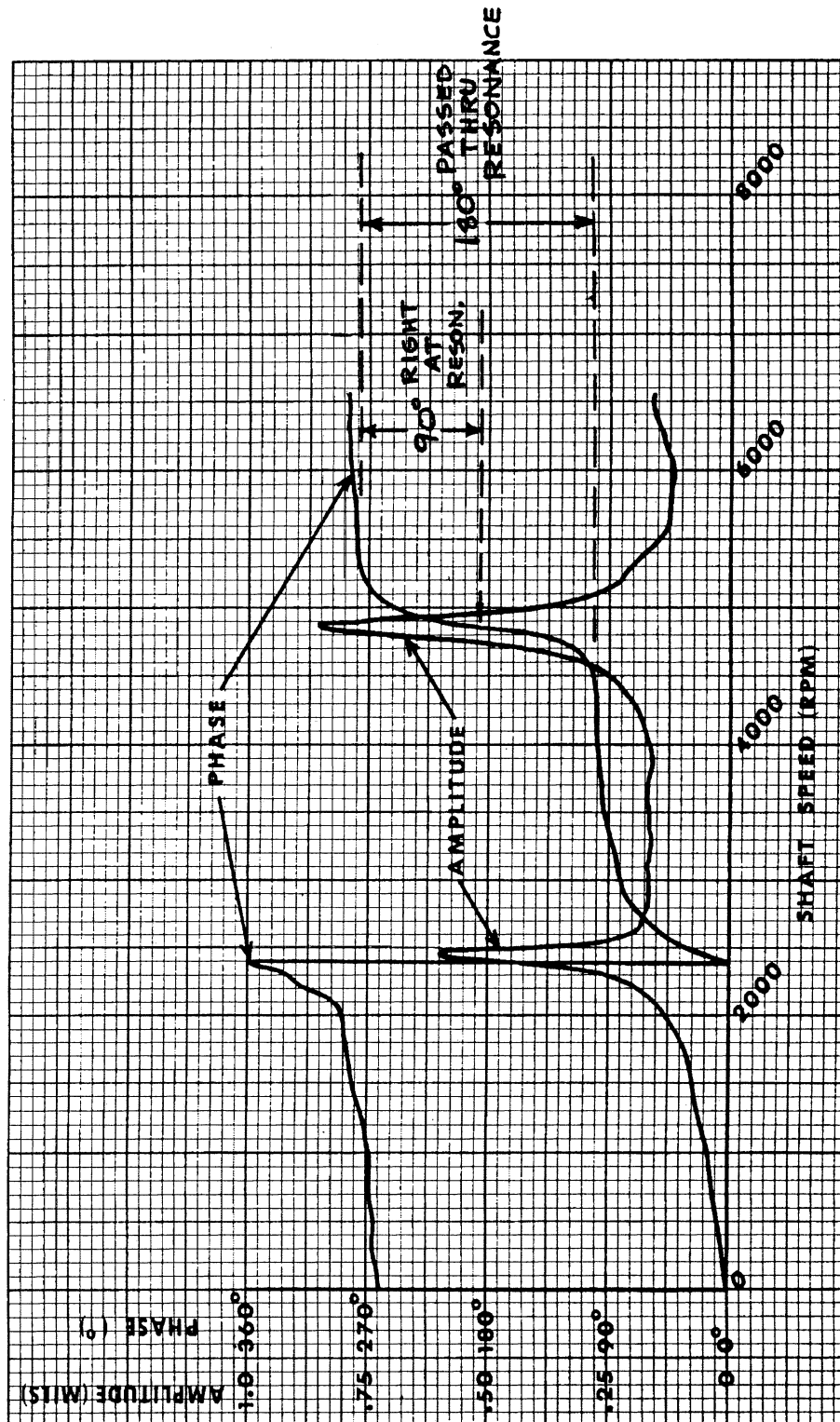


FIGURE 6.05B  
BODE' PLOT - VIBRATION AMPLITUDE AND PHASE VERSUS RPM

$$\frac{X_0}{X_{ST}} = \frac{1}{\sqrt{\left[1 - \left(\frac{f_i}{f_n}\right)^2\right]^2 + \left[2d \frac{f_i}{f_n}\right]^2}}$$

When Forcing Freq. = Natl. Freq. ( $f_i = f_n$ ):

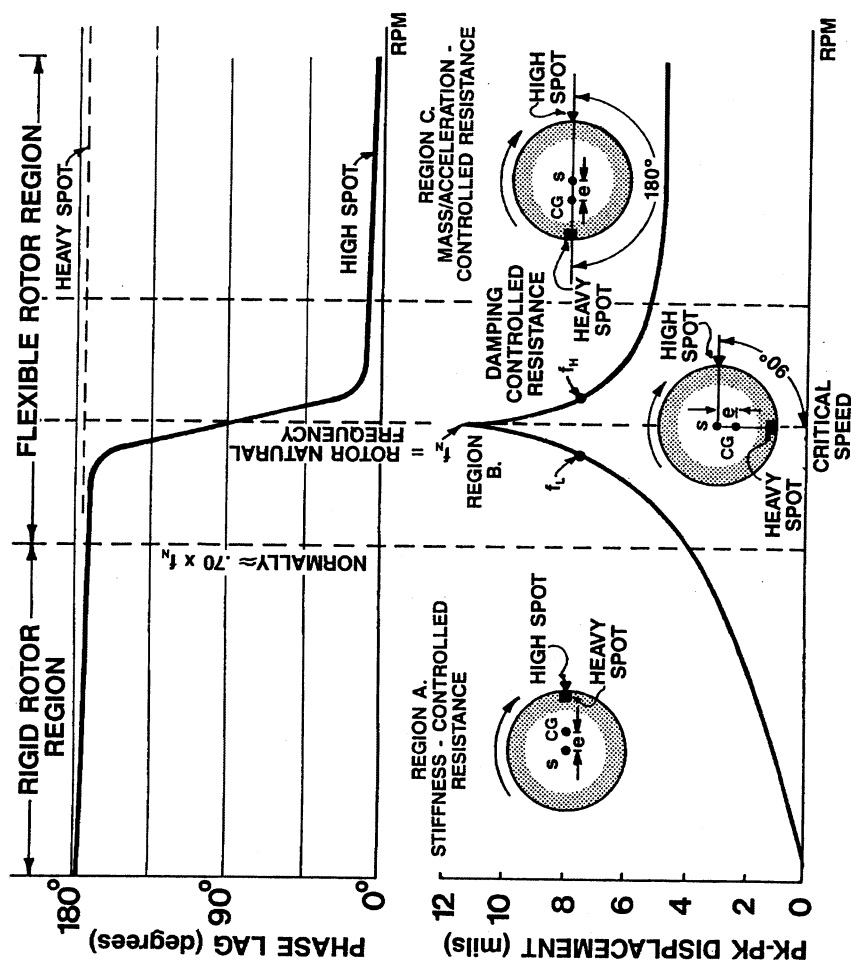
$$\frac{X_0}{X_{ST}} = \frac{1}{2d} = Q = \frac{f_n}{(f_n - f_i)}$$

WHERE:

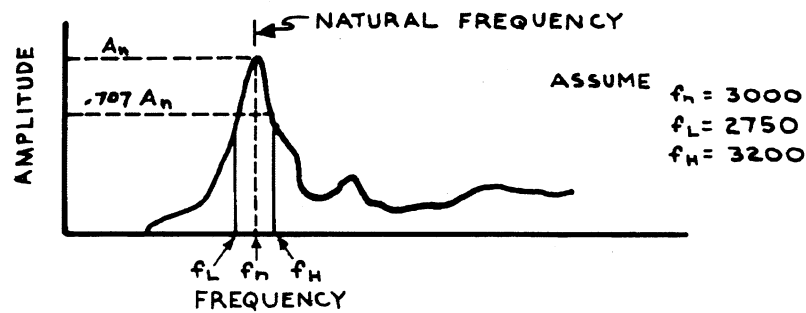
$X_0$  = Dynamic Deflection at Frequency  $f_i$  (mils)  
 $X_{ST}$  = Static Deflection (mils)  
 $f_i$  = Forcing Frequency (CPM)  
 $f_n$  = Natural Frequency (CPM)  
 $d$  = Damping Factor (lb-sec/in)  
 $Q$  = Resonant Amplification Factor  
 $f_H$  = .707  $A_n$  = Upper Half-Power Freq. (CPM)  
 $f_L$  = .707  $A_n$  = Lower Half-Power Freq. (CPM)  
 $A_n$  = Amplitude of Natural Frequency (mils)

#### LEGEND

CG = Rotor Center of Gravity  
 S = Shaft Geometric Centerline  
 e = Eccentricity (in.)  
 Heavy Spot = Actual Unbalance Location  
 High Spot = Point of Maximum Rotor Deflection



**FIGURE 6.05C**  
**CHANGE OF VIBRATION DISPLACEMENT AND PHASE LAG WITH RPM**  
**ABOVE, BELOW, AND AT ROTOR RESONANCE (BODE' PLOTS)**



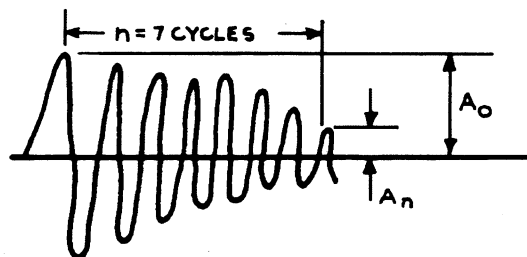
HALF POWER POINTS AT  $(.707)(A_n)$  DEFINE  $f_L$  AND  $f_H$

$$\frac{x}{x_0} = Q = \frac{f_n}{f_H - f_L} = \frac{3000}{3200 - 2750} = 6.67$$

$$Q = \frac{1}{\sqrt{[1 - (\frac{\omega}{\omega_n})^2]^2 + (2\frac{c}{c_c} \frac{\omega}{\omega_n})^2}} \quad \text{AT CRITICAL} = \frac{1}{2\frac{c}{c_c} \frac{\omega}{\omega_n}}$$

$$6.67 = \frac{1}{2\frac{c}{c_c}} \quad \frac{c}{c_c} = \frac{1}{(2)(6.67)} = .075$$

DAMPING FROM AN IMPULSE TEST :



$$\text{LOG DECREMENT } \delta = \frac{1}{n} L_n \left( \frac{A_0}{A_n} \right) \leftarrow \text{KNOWN FROM IMPULSE TEST}$$

$$\frac{c}{c_c} = \frac{\delta}{\sqrt{4\pi^2 + \delta^2}}$$

FIGURE 6.05D  
CALCULATING DAMPING FROM A COASTDOWN AND IMPULSE TEST

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Please note one other important fact shown in Figure 6.05B. When this machine passed through resonance, the vibration phase changed exactly  $90^\circ$  right at the natural frequency as predicted and continued to change almost  $180^\circ$  until it suffered no more resonant amplification. Note that this happened at both resonant events in this machine example. The second event might be a little confusing, but note that the phase changed from about  $90^\circ$  to  $0^\circ$  at resonance and then continued from there to about  $270^\circ$  when it passed out of trouble ( $360^\circ$  phase is the same as  $0^\circ$ ).

Figure 6.05C is likewise a Bode' Plot and shows how a rotor responds in each of three regions (A, B and C), each of which show a unique relationship of the rotor heavy spot (actual unbalance location) relative to its high spot (point of maximum rotor deflection). In region A, where stiffness controls dynamic resistance, there is little phase difference between the heavy spot and high spot. That is, if it were possible to measure rotor displacement with a dial indicator in region A, the greatest displacement would occur almost instantly at the heavy spot pass by the dial indicator during each shaft revolution. Also in region A, unbalance vibration response will increase with the square of the speed (tripling the speed will result in about 9 times higher displacement).

In Figure 6.05C, as the rotor increases speed and enters region B, damping causes the high spot to begin to lag the heavy spot of the shaft by a predictable and repeatable angular amount. This angular lag (phase) is caused by delay in the time it takes for the heavy spot to create shaft displacement. In fact, as the rotor approaches the first balance critical speed, Figure 6.05C shows that the maximum displacement (high spot) does not occur until the heavy spot is already passed by one-fourth of a revolution (or  $90^\circ$ ). When this  $90^\circ$  phase lag occurs at resonance, the only parameter limiting infinite vibration response is the amount of damping ( $\zeta$ ).

Figure 6.05C then shows what happens when the rotor begins to pass through the amplitude magnification region and out of resonance (into region C). Phase lag continues to change dramatically, finally approaching  $180^\circ$  when it has passed completely through the critical. Now, what this means is that the high spot (point of maximum rotor deflection) now will actually lag the heavy spot a full  $180^\circ$ ; that is, the heavy spot will complete one-half a revolution before the maximum rotor deflection even occurs. In this region C, the dynamic resistance is largely controlled by mass with almost no resistance applied from either spring stiffness or damping as the rotor system responds almost as a pure mass. Also interesting is the fact that in region C well beyond the first critical (and well before the vicinity of the second critical), the displacement becomes nearly constant even though centrifugal force ( $F_c$ ) continues to increase with the square of speed. The reason for this is that the resistance to vibration in region C is governed by the product of mass times acceleration ( $m \omega^2$ ) where  $\omega = 2 \pi f/60$ . Therefore, even though centrifugal force increases with the square of speed, dynamic resistance to motion likewise increases with the square of speed counteracting centrifugal force, and resulting in a displacement which is near constant.

Figure 6.05D illustrates how far away one must move a natural frequency to avoid resonant amplification. Here, a figure is shown graphing vibration amplitude versus frequency. When at resonance, this machine has amplitude  $A_n$  at frequency  $f_n$ . Resonant amplification occurs at so-called half-power points which are defined as points below and above natural frequency at 70.7% of the amplitude at resonance (see  $f_L$  and  $f_H$ ). The equation below the diagram shows how the magnification factor (Q) can be calculated if both the natural frequency and half-power points are known. In the example given, resonant amplification is 6.67; that is, resonance amplified vibration by a factor of almost 7X. Then, a calculation of damping factor was made by solving the equation given in Figure 6.05A. This results in a damping factor of about .075 which is a little higher than many common machine materials.

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Finally, Figure 6.05D also shows how damping can be calculated from an impulse test when something such as a rubber-coated hammer is used to impact a machine to excite system natural frequencies. The time waveform in Figure 6.05D shows the response of the machine from the time of impact until damping can dissipate the vibration. Here, one would look at the time waveform and take the number of cycles ( $n$ ) that best show the system response. In the example shown,  $A_0$  is the amplitude at impact where amplitude  $A_n$  is the amplitude at the  $n$ th cycle.

Therefore, it is most important that plants recognize a resonance when it occurs and make immediate attempts to resolve the problem rather than continuing to subject their expensive machinery to excessive vibration unnecessarily. With the instrumentation available today, there is no reason why this cannot easily be done, even by rather inexperienced vibration personnel. If this is accomplished, these machines will have significantly enhanced lifetimes and, during this life, will have significantly lower costs associated with required repairs.

#### **6.051 Identifying Characteristics of Natural Frequencies that Help "Give Them Away":**

Resonant vibration is indicated by the following characteristics:

##### **1. Lack of Response to Balance Attempts:**

Examination of Figure 6.05C explains why it is almost impossible to balance a machine which is at or near resonance. The machine will show a dramatic phase change for even a small change of speed in the vicinity of resonance which can approach  $180^\circ$ . Thus, to balance this rotor, it will be necessary either to change the natural frequency, operate it at a different speed, or remove the rotor from the machine and balance it on a fixed balancing stand.

##### **2. Highly Directional Vibration:**

Resonant vibration will cause much higher vibration in one direction as compared to the other two triaxial directions (for example, horizontal might be 10X higher than either vertical or axial levels). If resonant, it is common for vibration in the resonant direction to be 5 to 15X higher than that in the other two triaxial directions. This fact is now used in many expert diagnostic software systems when looking for a possible resonance. This is also why it is important that readings be taken in all 3 directions at each bearing during routine Predictive Maintenance Surveys.

##### **3. Phase Behavior in the Resonant Measuring Direction:**

Resonant frequencies will show a great change of phase with RPM in the direction in which the machine is resonant since phase will change  $90^\circ$  right at a natural frequency, and almost  $180^\circ$  when passing completely through resonance, depending on the amount of damping present. On the other hand, phase changes in non-resonant measurement directions may be small at the same time since they are not undergoing a natural frequency event.

##### **4. Probable Phase Difference in Measuring Direction Perpendicular to Resonant Measuring Direction:**

If one radial direction is resonant, the phase difference when the transducer is moved  $90^\circ$  to measure in the other radial direction will likely approach either  $0^\circ$  or  $180^\circ$ , depending on which side of the bearing that the pickup is placed (instead of about a  $90^\circ$  difference as in the case of dominant unbalance). That is, if the horizontal direction is resonant, the horizontal phase will likely either equal the vertical or differ from it by almost  $180^\circ$ . This

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is due to the fact that an additional 90° phase change is introduced by operating at a natural frequency. In either case, either a 0° or 180° horizontal-vertical phase difference represents highly directional motion characteristic of resonance (or an eccentricity).

**5. Characteristic Shape of a Resonant Peak:**

Normally, a resonant peak will have a rather wide "apron" at its base as opposed to the much narrower aprons for non-resonant peaks. That is, the base of a resonant peak will usually be wider than those of surrounding non-resonant peaks.

**6. Frequencies at Which Resonances Can Occur:**

Resonance does not occur only at 1X RPM. It can respond to any forcing frequency that happens to coincide with a natural frequency. In these cases, it is meaningful to compare the amplitude of this frequency in this direction with the same frequency in the other two triaxial directions. If resonant, the frequency should be much higher in one of these three directions. This frequency might be a peak at 4X, 5X or 6X RPM (or even higher) corresponding to a blade pass frequency (BPF), bearing defect frequency, gear mesh frequency (GMF), or even to a mechanical looseness condition. If action is taken on the source of this exciting frequency resulting in lowering of the amplitude of the forcing frequency itself, it may also lower the response of this natural frequency to the forcing frequency. Remember that Resonant Vibration = Forcing Frequency Vibration (f) X Magnification Factor (Q).

**7. Excessive Motion and Dynamic Stress of any Resonant Member:**

Not only must one be concerned about resonance of the machine rotor (rotating assembly), but also with exciting natural frequencies of the support frame, foundation and even connected piping and/or duct work. Very often, fatigue failures will occur in the connected frame and/or piping due to the fact that they happen to be resonant to the forcing frequency coming from the machine. Resolution will require either lowering the amplitude source of the forcing frequency in the machine, isolating the resonant frame member from the machine, changing the rotor speed, or changing the natural frequency of the frame member itself.

**8. Resonance Can Suddenly Occur on a Long-Running Machine Never Before Subjected to It:**

Resonance can suddenly occur without warning in a machine running for years with little or no problems. For example, bearing wear can reduce stiffness of a shaft/bearing system lowering the natural frequency which might put it in resonance with a forcing frequency. Also, simple journal bearing replacement can cause a change in natural frequency which could put the rotor in resonance if the bearings are not properly blued and scraped to establish good, continuous contact with the shaft. After this has been done, one can simply resolve the problem after installing the bearing and checking for required clearance specifications and proper bearing alignment. The reason that each of these items affect resonance is that they each can have a direct effect on stiffness.

**9. Possible Effect of Coupling Changes:**

Coupling changes can sometimes result in a different torsional natural frequency, particularly if a different diameter coupling is used, or a different coupling type or construction is employed. This can introduce significant vibration in bearings on either side of the coupling.

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## **10. Summary of Actions Needed to Resolve Resonance Problems:**

### **a. Change Stiffness:**

Resonance can sometimes be resolved by increasing the stiffness of pedestals or frames, clamping of feet, removing distortion induced by shimming, or using reduced clearance bearings (which will affect stiffness, and therefore natural frequency).

### **b. Adding or Removing Weight (Mass):**

Although it is normally simpler and less expensive to change stiffness, natural frequency can likewise be changed by adding or removing weight. However, it is important to point out this effect will only occur on the part that is resonant.

### **c. Adding Isolation Mechanisms:**

Vibration isolation can be effective on a machine which is forced into resonance at one of its natural frequencies due to an incoming vibration from external equipment. For example, a properly designed pump operating at 1780 RPM and having a 750 CPM natural frequency may have a problem if a large fan operating nearby at about 750 RPM is out of balance. In this case, it would be necessary to either isolate the fan from the pump, balance the fan, or isolate the pump itself from the floor.

### **d. Adding Damping Materials:**

Resonant amplitude amplification can be lowered by increasing damping of the part that is resonant (however, it is important to point out that damping itself will be of little help if the part on which the damping materials are placed is not resonant). Damping is the only parameter that prevents catastrophic failure when a part is resonant.

### **e. Installing Dynamic Vibration Absorber:**

A "tuned" dynamic vibration absorber can sometimes be installed on equipment on which other reduction techniques either have not proven practical, or have not sufficiently reduced vibration to acceptable levels. However, this device is only effective in those situations where only one dominant frequency is causing excessive vibration and this frequency is a natural frequency of the part on which the tuned damper is to be mounted. In these cases, the tuned damper can be designed with the natural frequency equal to the troublesome frequency and will respond 180° out of phase with the resonant motion which will in effect "cancel out" the problem natural frequency.

### **f. Precision Dynamic Balancing:**

Since resonant vibration equals forcing frequency vibration X magnification factor, if one cannot effectively move the natural frequency out of resonance or lower the magnification (by addition of damping materials), one can still lower resonant vibration by reducing the sources of the forcing frequencies (for example, unbalance). In this case, if the balancing amplitude can be significantly reduced by several orders, the result can be acceptable vibration, even though at resonance. However, it is important to point out that it may be necessary to remove the rotor in order to even attempt to balance it because of the tremendous phase changes at resonance previously mentioned.

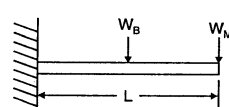
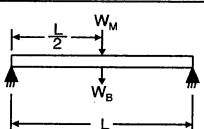
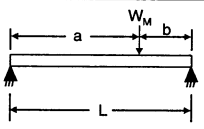
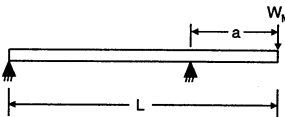
## 6.052 How Natural Frequencies Can Be Approximated For Overhung Rotors and Machines with Loads Supported Between Bearings:

Following below is Equation 6.05B which applies to calculation of natural frequency for a single degree-of-freedom system like those shown in Table 1:

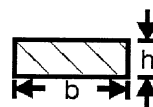
$$f_n = \frac{60}{2\pi} \sqrt{\frac{K}{m}} = \frac{60}{2\pi} \sqrt{\frac{g_c}{\Delta}} \quad (\text{Eqn. 6.05B})$$

where:  $f_n$  = Natural Frequency (CPM)  
 $g_c$  = Standard Gravitational Constant  
 $= 32.2 \text{ ft/sec}^2 = 386 \text{ in/sec}^2$  (SEA LEVEL)  
 $K$  = Stiffness (lb/in)  
 $m$  = Mass ( $\text{lb}_f \cdot \text{sec}^2/\text{in}$ ) = Weight/ $g_c$   
 $\Delta$  = Deflection (in)

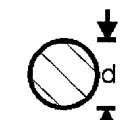
Table 6.05A provides formulas for approximating natural frequency for cantilevered, simply supported and overhung rotors. In the simply supported case, the applied load ( $W_m$ ) can act either at the center of this shaft span (Case B) or at any point along the shaft between the bearings (Case C).

| CASE | DIAGRAM                                                                             | FORMULA FOR $f_n$                                     |
|------|-------------------------------------------------------------------------------------|-------------------------------------------------------|
| A.   |   | $f_n = 324.96 \sqrt{\frac{EI}{L^3 (W_M + 0.24 W_B)}}$ |
| B.   |  | $f_n = 1302 \sqrt{\frac{EI}{L^3 (W_M + 0.49 W_B)}}$   |
| C.   |  | $f_n = 324.96 \sqrt{\frac{EIL}{a^2 b^2 W_M}}$         |
| D.   |  | $f_n = 324.96 \sqrt{\frac{EI}{a^2 L W_M}}$            |

$f_n$  = Natural Frequency (CPM)  
 $E$  = Modulus of Elasticity of Beam or Shaft Material ( $\text{lb/in}^2$ )  
 $I$  = Principal Moment of Inertia of Beam Section ( $\text{in}^4$ )  
 $W_M$  = Applied Load (lb)  
 $W_B$  = Weight of Beam or Shaft (lb)  
 $L$  = Length of Beam or Shaft (in)  
 $a, b$  = Dimensions as Shown (in)



$$I = \frac{bh^3}{12} (\text{in}^4)$$

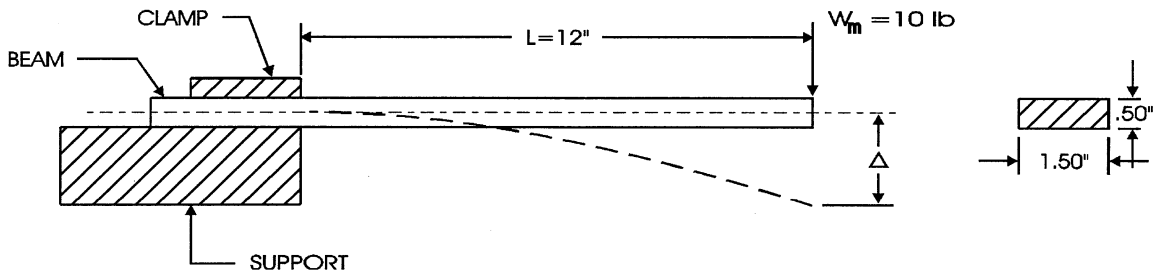


$$I = \frac{\pi d^4}{64} (\text{in}^4)$$

**TABLE 6.05A**  
**NATURAL FREQUENCY FORMULAS FOR OVERHUNG ROTORS,**  
**CANTILEVER BEAMS AND SHAFTS MOUNTED BETWEEN BEARINGS**

In teaching seminars, the author has found that there is often a lack of understanding if formulas and tables are simply presented without examples. Therefore, several examples applying the above equations and showing how they can be used to calculate natural frequency will now be provided:

**Example 1 - Applying Cantilever Beam Natural Frequency Formulas:**



Given: Rectangular Steel Beam ( $E = 29,600,000 \text{ lb/in}^2$ ), 12 inch cantilevered portion, 1.50" wide by .50" deep; Applied Load ( $W_m$ ) of 10 lb at End.

**Part A - What is system natural frequency if you neglect the beam weight?**

Moment of Inertia  $I$ , for a rectangular beam,

$$I = \frac{bh^3}{12} = \frac{(1.50)(.50)^3}{12}$$

$$I = .01563 \text{ in}^4$$

From Table 6.05A, Case A (but neglecting beam weight  $W_B$ ):

$$f_n = 324.96 \sqrt{\frac{EI}{W_m L^3}} = 324.96 \sqrt{\frac{(29,600,000)(.01563)}{(10)(12)^3}}$$

$$f_n = 1681 \text{ CPM} = 28.0 \text{ Hz} \quad (\text{neglecting beam weight})$$

**Part B - What is natural frequency if the Beam Weight is included?**

Beam  $W_t$  = Density X Volume =  $(.283 \text{ lb/in}^3) [(12)(1.50)(.50) \text{ in}^3]$

Beam Weight = 2.547 lb (12" Long, 1.5" Wide, .5" Deep)

Now including beam weight ( $W_B$ ) and using formula from Table 6.05A, Case A:

$$f_{nB} = 324.96 \sqrt{\frac{EI}{L^3 (W_m + 0.24 W_B)}} = 324.96 \sqrt{\frac{(29,600,000)(.01563)}{(12)^3 [10 + (0.24)(2.547)]}}$$

$$f_n = 1632 \text{ CPM} = 27.2 \text{ Hz} \quad (3.0\% \text{ lower when including Beam Weight})$$

Part B of example 1 shows that including the beam weight did drop the calculated natural frequency 3.0% from 1681 to 1632 CPM. Therefore, in this case, including the beam weight made only a slight difference.

**Part C - What is the beam deflection ( $\Delta$ ) if you neglect the beam weight?**

From Equation 6.05B,

$$f_n = \frac{60}{2\pi} \sqrt{\frac{g_c}{\Delta}}$$

Solving Equation 6.05B for Deflection ( $\Delta$ ),

$$\Delta = \frac{3600g_c}{(2\pi f_n)^2} \quad (\text{Eqn. 6.05C})$$

Substituting:

$$\Delta = \frac{(3600)(386)}{(2\pi \times 1681)^2}$$

$$\Delta = .0125 \text{ in} = 12.5 \text{ mils} \quad (\text{neglecting beam weight})$$

NOTE: If you include beam weight,  $\Delta = .0132 \text{ in} = 13.2 \text{ mils}$  (including beam weight)

Checking Answer C with Cantilever Beam Deflection Formula (neglecting beam weight):

$$\Delta = \frac{PL^3}{3EI} \quad (\text{Eqn. 6.05D})$$

$$\Delta = \frac{(10)(12)^3}{(3)(29,600,000)(.0156)} = .0125 \text{ in} \quad (\text{OK})$$

Similarly, natural frequencies can be approximated for machines with shafts mounted between bearings or those with overhung rotors using the formulas provided in Table 6.05A.

---

## 6.06 MECHANICAL LOOSENESS

Many texts and seminars today simply refer to a general term called "Mechanical Looseness" and that it either can be detected by high vibration at 2X RPM or multiple running speed harmonics. However, a comprehensive study of a great amount of available literature as well as review of a number of case histories accomplished by Technical Associates actually has revealed that there are at least 3 different types of Mechanical Looseness, each of which has its own characteristic vibration spectra as well as vibration phase behavior. A discussion will follow on each of these which are as follows:

- 6.061 Type A - Structural Frame/Base Looseness (Primarily 1X RPM);
- 6.062 Type B - Looseness Due To Rocking Motion Or Cracked Structure/Bearing Pedestal (Primarily 2X RPM);
- 6.063 Type C - Loose Bearing In Housing or Improper Fit Between Component Parts (Multiple Harmonics Due To Nonlinearity Often Induced By Impulse Events)

One of the important facts about each type of mechanical looseness is that it alone is not a cause of vibration. Instead, looseness is a reaction to other problems which are present such as unbalance, misalignment, eccentricity, bearing problems, etc. Resolution of these other problems often will remove many of the symptoms, and therefore the response of looseness. However, the problem is that only minute amounts of such problems as misalignment or unbalance can cause vibration if a looseness condition exists. Looseness aggravates the situation. Therefore, mechanical looseness allows much more vibration than would otherwise occur from these other problems alone. Resolution of the other problems themselves will often remove most of the symptoms due to looseness. However, this is often virtually impossible in reality because such steps would require extraordinary levels of precision of alignment or balancing. Therefore, in these cases, the looseness condition will first have to be resolved. Then, if remaining vibration is still high, other steps such as alignment and balancing can be accomplished with much greater ease than before the looseness conditions were resolved. Following below is a discussion on each of the 3 types of looseness which were listed previously:

### 6.061 Type A - Structural Frame/Base Looseness (1X RPM):

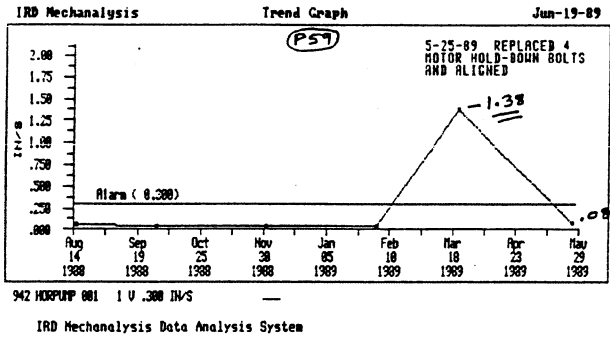
This type of looseness includes each of the following problems:

- Structural Looseness/Weakness of Machine Feet, Baseplate & Concrete Base;
- Deteriorated or Crumbled Grouting;
- Distortion of Frame or Base (Soft Foot);
- Loose Hold-Down Bolts.

Type A Looseness Problems are often misdiagnosed as unbalance or misalignment problems since they have almost identical vibration spectra. Therefore, it is important to look beyond vibration spectra and compare relative amplitudes between directions, look closely at phase behavior which departs radically from such things as unbalance, and to examine other characteristics which are listed as follows:

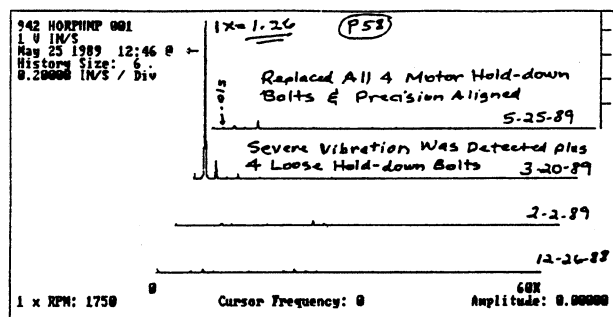
1. Type A Looseness spectra are dominated by high 1X RPM vibration and appear identical to that for an unbalance or eccentric rotor condition. A spectrum illustrating this type signature is shown in Figure 6.06A.

- 
2. Often, high vibration is pretty well confined to only one rotor (that is, the driver or driven component or the gearbox alone). This is unlike unbalance or misalignment in which rather high vibration levels due to these other problems are not confined to just one of the rotors.
  3. Two different phase behaviors can occur with Type A Looseness:
    - a. When comparing vertical and horizontal phase on each of the bearing housings, the vibration will sometimes be found to be highly directional with phase differences of either  $0^\circ$  or  $180^\circ$  depending on whether or not the horizontal reading was taken on one side or the other (either a phase difference of  $0^\circ$  or  $180^\circ$  means that the motion is directly up and down or side to side). This does not normally occur with simple unbalance in which horizontal and vertical phase usually differs approximately  $90^\circ$  ( $\pm 30^\circ$ ).
    - b. When this first phase behavior occurs ( $0^\circ$  or  $180^\circ$  phase difference in horizontal and vertical), the analyst should not confine his measurements to the bearing housings alone, but move on down to the machine foot, baseplate, concrete base and surrounding floor. This is illustrated in Figure 6.06B. Here, comparative amplitude and phase measurements should show relatively identical amplitude and phase at 1X RPM at each location. If there is a great difference in amplitude and phase, this will suggest relative motion. Using the point where this great phase change occurs, one can locate where the problem exists. For example, the measurements in Figure 6.06B show a problem between the baseplate and concrete base indicated by the great difference in phase (note  $180^\circ$  out of phase with the other two measurements). This indicates structural looseness/weakness allowing relative movement in machine components which may be due to a problem with the grouting between the baseplate and concrete base; or broken or cracked foundations, etc. On the other hand, if a great phase difference occurred between machine foot and baseplate, this might suggest looseness of the mounting bolt and/or possible stripped mounting bolt threads. Either of these two problem conditions can cause a great vibration at 1X RPM like that shown in Figure 6.06A on the machine component where this occurs, particularly on the bearing housing directly above this base location.



Machine 942 HORPUMP 001 Pos 1 Dir V Units IN/S Alarm .300

| Ampl | Insp     | Date     | Time  | Ampl | Insp     | Date | Time |
|------|----------|----------|-------|------|----------|------|------|
| Code | mm/dd/yy |          |       | Code | mm/dd/yy |      |      |
| .062 | 0        | 08-14-88 | 10:01 |      |          |      |      |
| .054 | 0        | 08-15-88 | 13:00 |      |          |      |      |
| .042 | 0        | 09-29-88 | 11:03 |      |          |      |      |
| .038 | 0        | 12-01-88 | 9:49  |      |          |      |      |
| .041 | 0        | 02-02-89 | 15:00 |      |          |      |      |
| 1.38 | 0        | 03-20-89 | 13:43 |      |          |      |      |
| .080 | 0        | 05-25-89 | 12:46 |      |          |      |      |



001X x 2.0000 = 0.02000 1 x RPM: 1750.00

Machine 942 HORPUMP 001 Pos 1 Dir V Units IN/S Date May 25 1989 12:46

| Frequency | 1 x RPM | Amplitude |
|-----------|---------|-----------|
| 3600      | 2.06    | 0.02478   |
| 7200      | 4.11    | 0.05782   |

3X x 2.0000 = 0.06000 1 x RPM: 1800.00

Machine 942 HORPUMP 001 Pos 1 Dir V Units IN/S Date Mar 20 1989 13:43

| Frequency | 1 x RPM | Amplitude |
|-----------|---------|-----------|
| 1800      | 1.00    | 1.26005   |
| 3450      | 1.92    | 0.12994   |

001X x 2.0000 = 0.02000 1 x RPM: 1750.00

Machine 942 HORPUMP 001 Pos 1 Dir V Units IN/S Date Feb 02 1989 13:00

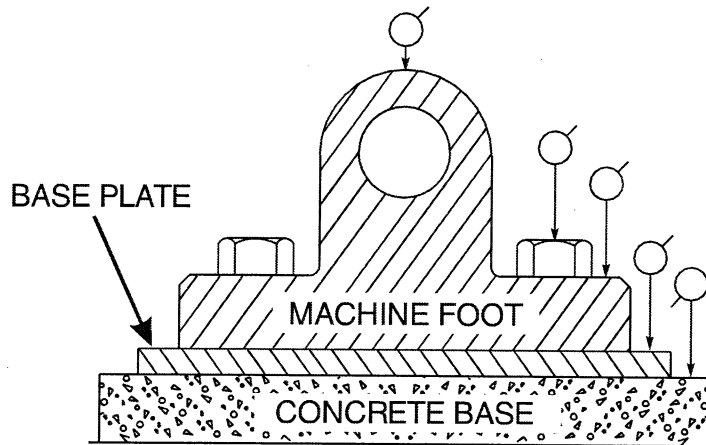
| Frequency | 1 x RPM | Amplitude |
|-----------|---------|-----------|
| 21600     | 12.34   | 0.03940   |

1X x 2.0000 = 0.02000 1 x RPM: 1750.00

Machine 942 HORPUMP 001 Pos 1 Dir V Units IN/S Date Dec 26 1988 10:01

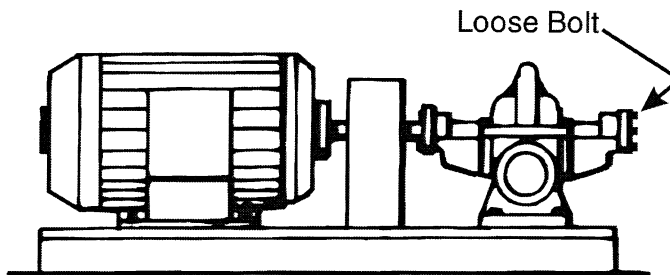
| Frequency | 1 x RPM | Amplitude |
|-----------|---------|-----------|
| 150       | 0.09    | 0.02542   |
| 21600     | 12.34   | 0.02478   |

FIGURE 6.06A  
SEVERE LOOSENESS INTRODUCED BY LOOSE HOLD-DOWN BOLTS



**FIGURE 6.06B**  
**PHASE ANALYSIS USED TO PINPOINT LOOSENESS**

4. Figure 6.06C illustrates another case involving this high 1X RPM looseness vibration. In this case, bolts that are parallel to the pump shaft are mounted on each pump bearing housing acting in the axial direction. In this case, if these bolts become loose, it will cause a high vibration in the axial direction at 1X RPM which will closely resemble a misalignment problem. However, simply tightening these bolts will greatly reduce the vibration.



**FIGURE 6.06C**  
**EFFECT OF LOOSE BOLT ON PUMP OUTBOARD BEARING HOUSING**

5. Distortion induced either by a soft foot or piping strain shows another situation which would result in high 1X RPM vibration signatures looking like unbalance. However, in these cases, when phase readings are taken, they will show highly directional vibration with the difference in horizontal and vertical phase approaching either  $0^\circ$  or  $180^\circ$  ( $\pm 30^\circ$ ) rather than  $90^\circ$  in the case of simple unbalance. If the problem were distortion rather than looseness, amplitude and phase measurements would show the machine foot, baseplate and concrete base pretty much vibrating in the same direction (equal phase readings). However, they may show that the amplitude on either one of the foot bolts alone is much higher than that on any of the other 3 bolts; or, for example, the right front foot and left rear foot may be significantly higher than the left front and right rear feet. In this second case, the motor would be vibrating diagonally. Great amplitude differences on these foot bolts would suggest a soft foot which must be corrected to reduce the high vibration levels. (In fact, the analyst may find that if he backs off slightly on those bolts having high vibration, he may see much lower vibration than before).

---

#### **6.062 Looseness Due To Rocking Motion or Cracked Structure/Bearing Pedestal (2X RPM):**

The 2X RPM "looseness symptom" referenced in many vibration texts seems only to occur for the following looseness problems:

- Crack in Structure or Bearing Pedestal;
- Rocking Motion Sometimes Induced By Differential Length Support Legs;
- Occasionally on Some Loose Bearing Housing Bolts;
- When Loose Bearing or Improper Component Fit Problems Are Only of Minor Severity (No Impulse or Impact Events).

Figure 6.06D shows a spectrum typically showing these problems. These problems exhibit the following characteristics:

1. Typically, these problems are suggested when the amplitude at 2X RPM exceeds about 50% of that at 1X RPM in the radial direction.
2. Amplitudes are somewhat erratic.
3. If phase readings are taken with a strobe light, it will often show 2 reference marks which are slightly erratic.
4. These looseness symptoms will not normally occur unless there is some other exciting force such as unbalance or misalignment. However, if this looseness condition exists, it will be extremely difficult to balance or align the unit sufficiently to bring down the final vibration sufficiently.
5. If the looseness problem is a bearing loose in the housing or a loose component on the shaft, the vibration will pretty well remain at 1X and 2X RPM until it worsens allowing an impulse or impact event. When this occurs, these impulses cause nonlinearities in the time waveform which will begin exciting many harmonics advancing to Type C Looseness.

#### **6.063 Loose Bearing In Housing or Improper Fit Between Component Parts (Multiple Harmonics Due To Nonlinearity Often Induced By Impulse Events):**

Each of the following problems occur in Type C Looseness:

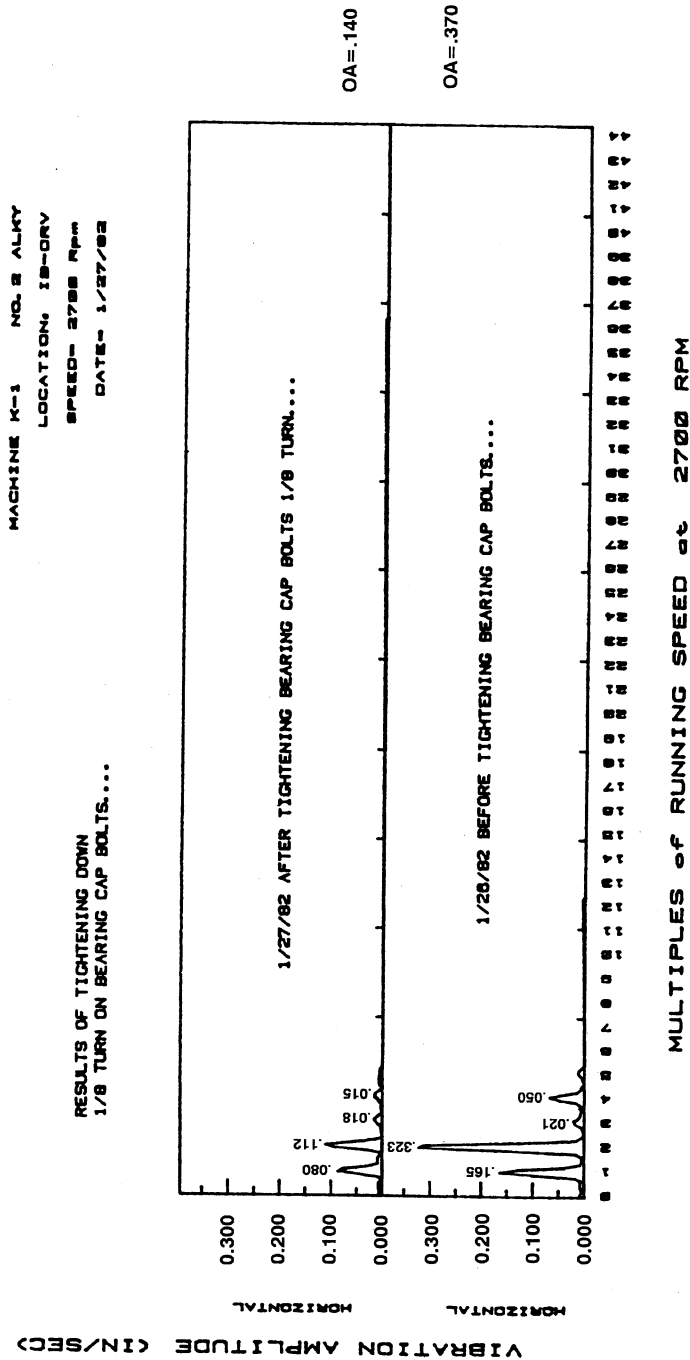
- Bearing Loose in Housing;
- Excessive Internal Bearing Clearances;
- Bearing Liner Loose In Its Cap;
- Loose Rotor;
- Bearing Loose and Turning on Shaft.

Figures 6.06E thru 6.06G illustrate typical spectra indicating Type C Looseness. Note the presence of multiple running speed harmonics in both spectra. Also note in Figure 6.06G that one of the running speed harmonics lies close to a natural frequency causing a resonant response at this frequency (if the looseness condition is resolved, the resonance condition will likely be as well). **Type C is the most common mechanical looseness problem** and exhibits the following characteristics:

# CASE VI

This case is on our K-1 refrigeration compressor located at our Alkylation Plant. The turbine and compressor both run between 2,700-3,100 RPM requiring 6,000 HP. In this signature (Figure 13) there is a high two times running speed frequency showing up on the turbine inboard. The vibration in this case was suspected to be caused by a

looseness problem between the bearing and the bearing housing. The bearing housing nuts were found to be slightly loose and they were tightened. The signature also shows the level had dropped significantly after the housing was tightened; a slight misalignment problem might still be indicated.

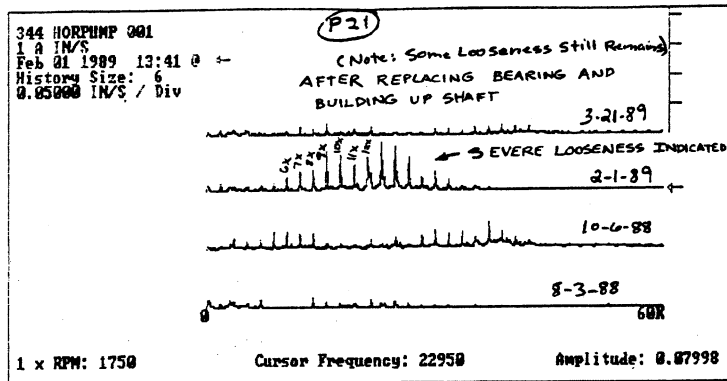


Vibration signatures of Turbine IB bearing showing highest level at two times running speed. Bottom signature is before bearing cap is tightened down and top is after.

FIGURE 6.06D  
EXAMPLE OF LOOSENESS OF FASTENERS OCCURRING AT 2X RPM  
(Ref. 17)

- 
1. Multiple running speed harmonics sometimes up to 10X or even 20X RPM are clearly present in this spectrum. These harmonics are a result of impulses and truncation (limiting) in the machine response. This impulse event causes a nonlinearity in the time waveform. When this occurs, multiple harmonics will appear in the resulting FFT spectrum (see Figure 6.06E).
  2. This looseness tends to produce vibration that is directional which differs from unbalance. It normally will be highest in the direction and vicinity of the looseness problem. For example, this may show that the highest vibration is not either horizontal or vertical, but somewhere in between the two.
  3. If the amplitude of harmonics becomes significant, this can also generate frequencies spaced at 1/2 times RPM (that is, .50X, 1.50X, 2.50X, etc.) or even sometimes at 1/3 times RPM.
  4. The analyst is cautioned that amplitudes of these 1/2 times RPM "harmonics" may appear deceptively low when compared with those at 1X RPM and running speed harmonics. However, he should remember that no peaks at 1/2 times RPM intervals should be present whatsoever. If the peaks are clearly evident, they do indicate a more advanced looseness problem (or possibly, presence of a rub).
  5. One half times RPM "harmonics" usually are accompanied by other problem sources such as unbalance and misalignment.
  6. Phase measurements of Type C looseness problems are normally somewhat erratic, but can approach differences of 0° and 180° between horizontal and vertical directions if the vibration itself becomes highly directional. It normally acts in a radial direction, but can occur in the axial, dependent on the exact type of looseness.
  7. In the case of a loose rotor such as a loose pump impeller, phase will vary from one startup to the next. The amplitude itself may be steady for a given run, but likewise will vary from startup to startup. Such a loose rotor is impossible to balance since the heavy spot itself is constantly changing directions. This shift in amplitude and phase is likely caused by center of gravity shifts.
  8. **CAUTION:** Vibration spectra simply appearing to be Type C looseness (many 1X RPM harmonics) can, in fact, signal a problem of much greater severity - a bearing loose and turning on a shaft. This can be the case even if amplitude of 1X RPM and its harmonics are fairly low, on the order of .05 in/sec or less. In these cases, the turning of the bearing on the shaft can actually cause great damage, actually removing material from the shaft diameter. In these cases, it can cause catastrophic failure of the machine as the bearing finally locks, and can do so without even generating any bearing defect frequencies. Figure 6.06E below is an example of just such a catastrophic failure where over .25 inch of the shaft diameter was removed from a 3.50 inch shaft before the bearing locked up.





$$2\% \times 0.5000 = 0.01000 \quad 1 \times \text{RPM:} \quad 1750.00$$

| Machine        | Pos     | Dir       | Units | Date              |
|----------------|---------|-----------|-------|-------------------|
| 344 HOPUMP 001 | 1       | A         | IN/S  | Mar 21 1989 12:57 |
| Frequency      | 1 x RPM | Amplitude |       |                   |
| 12300          | 7.03    | 0.01247   |       |                   |
| 15900          | 9.09    | 0.01843   |       |                   |
| 19500          | 11.14   | 0.01048   |       |                   |
| 21600          | 12.34   | 0.01096   |       |                   |
| 35250          | 20.14   | 0.01144   |       |                   |
| 37050          | 21.17   | 0.01191   |       |                   |
| 38850          | 22.20   | 0.01493   |       |                   |
| 40650          | 23.23   | 0.01295   |       |                   |
| 42450          | 24.26   | 0.01096   |       |                   |

$$006\% \times 0.5000 = 0.03000 \quad 1 \times \text{RPM:} \quad 1750.00$$

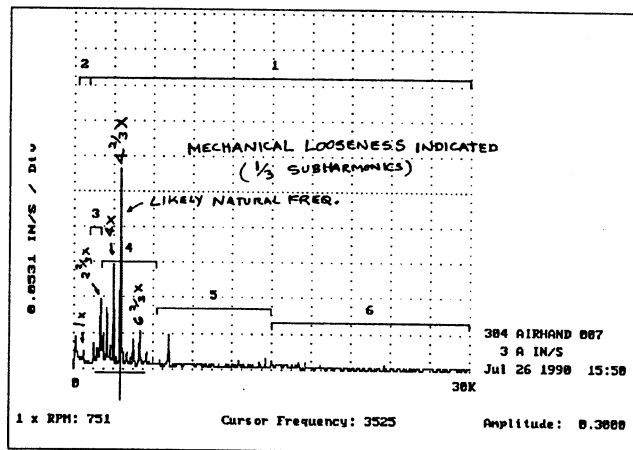
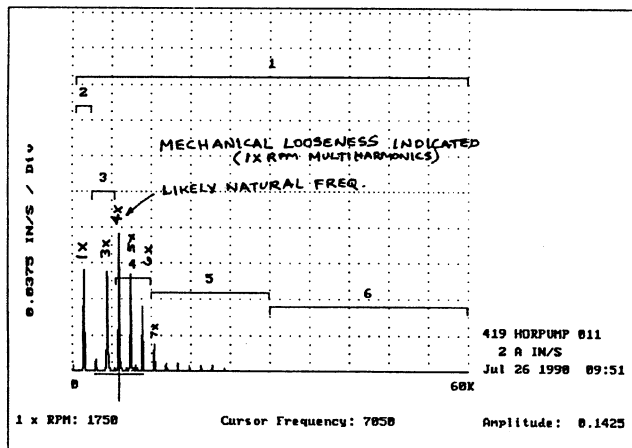
| Machine        | Pos     | Dir       | Units | Date              |
|----------------|---------|-----------|-------|-------------------|
| 344 HOPUMP 001 | 1       | A         | IN/S  | Feb 01 1989 13:41 |
| Frequency      | 1 x RPM | Amplitude |       |                   |
| 12300          | 7.03    | 0.03002   |       |                   |
| 14100          | 8.06    | 0.03503   |       |                   |
| 15900          | 9.09    | 0.06497   |       |                   |
| 17700          | 10.11   | 0.05997   |       |                   |
| 19500          | 11.14   | 0.04249   |       |                   |
| 21150          | 12.09   | 0.05496   |       |                   |
| 22950          | 13.11   | 0.07998   |       |                   |
| 24750          | 14.14   | 0.07498   |       |                   |
| 26550          | 15.17   | 0.05496   |       |                   |
| 30000          | 17.14   | 0.03002   |       |                   |

2-1-89

$$002\% \times 0.5000 = 0.01000 \quad 1 \times \text{RPM:} \quad 1750.00$$

| Machine        | Pos     | Dir       | Units | Date              |
|----------------|---------|-----------|-------|-------------------|
| 344 HOPUMP 001 | 1       | A         | IN/S  | Aug 03 1988 07:27 |
| Frequency      | 1 x RPM | Amplitude |       |                   |
| 300            | 0.17    | 0.01398   |       |                   |
| 3000           | 1.71    | 0.01096   |       |                   |
| 7200           | 4.11    | 0.01096   |       |                   |
| 14100          | 8.06    | 0.01597   |       |                   |
| 21600          | 12.34   | 0.01247   |       |                   |

**FIGURE 6.06F**  
**ADVANCING PROBLEM WITH BEARING LOOSE ON SHAFT**  
**(PARTIALLY RESOLVED)**



002% x 0.500 = 0.0100      1 x RPM: 751.00

| Machine         | Pos     | Dir          | Units     | Date              |              |
|-----------------|---------|--------------|-----------|-------------------|--------------|
| 304 AIRHAND 007 | 3       | A            | IN/S      | Jul 26 1990 15:50 |              |
| Frequency       | 1 x RPM | Amplitude    | Frequency | 1 x RPM           | Amplitude    |
| 150             | 0.20    | 0.0450       | 5550      | 7.39              | 0.0225 = 1½x |
| 750             | 1.00    | 0.0275 = 1x  | 5700      | 7.59              | 0.0100       |
| 975             | 1.30    | 0.0100 = 1½x | 6000      | 7.99              | 0.0150 = 2x  |
| 1125            | 1.50    | 0.0100 = 1½x | 6525      | 8.69              | 0.0125 = 1¾x |
| 1500            | 2.00    | 0.0350 = 2x  | 7200      | 9.59              | 0.0500 = 4½x |
| 1800            | 2.40    | 0.0300       | 12525     | 16.68             | 0.0125 = 1½x |
| 2025            | 2.70    | 0.1025 = 2¾x | 14025     | 18.68             | 0.0100 = 1½x |
| 2250            | 3.00    | 0.0525 = 3x  | 14550     | 19.37             | 0.0150 = 1¾x |
| 2475            | 3.30    | 0.0900 = 3¾x | 15000     | 19.97             | 0.0100 = 2x  |
| 2775            | 3.70    | 0.0325 = 3¾x | 17025     | 22.67             | 0.0100       |
| 3000            | 3.99    | 0.1550 = 4x  | 17550     | 23.37             | 0.0100       |
| 3225            | 4.29    | 0.0150 = 4½x |           |                   |              |
| 3525            | 4.69    | 0.3000 = 4¾x |           |                   |              |
| 4050            | 5.39    | 0.0225 = 5½x |           |                   |              |
| 4275            | 5.69    | 0.0150 = 5¾x |           |                   |              |
| 4500            | 5.99    | 0.0425 = 6x  |           |                   |              |
| 5025            | 6.69    | 0.0575 = 6¾x |           |                   |              |

T - Threshold adjust      Q - Quit

**FIGURE 6.06G**  
**2 DIFFERENT TYPES OF SPECTRA INDICATING**  
**TYPE C MECHANICAL LOOSENESS**

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## 6.07 ROTOR RUB

Rotor rub occurs when the rotating parts of a machine contact stationary parts and can be catastrophic, depending on specifically which parts contact one another, the material construction of each, rotational speed-induced friction, structural rigidity of both the rotating and stationary parts and the impact velocity itself. Technical papers, including reference 6, have classified two major types - (1) Partial Rub, and (2) Full Annular Rub. Each of these cases will be separately discussed in Sections 6.071 and 6.072, respectively. Rotor rub encompasses a variety of events, some of which are catastrophic, and others which are less catastrophic (but still potentially serious). These include the following examples:

### CATASTROPHIC RUB EVENTS

Journal Contacting Bearing Babbitt

Rotor Contacting Stator in Electric Motor

Impeller Vanes Contacting Diffuser Vanes

Turbine Blades Contacting Diffuser Vanes

### LESS CATASTROPHIC RUB EVENTS

Shaft Rubbing a Seal

Coupling Guard Rubbing a Shaft

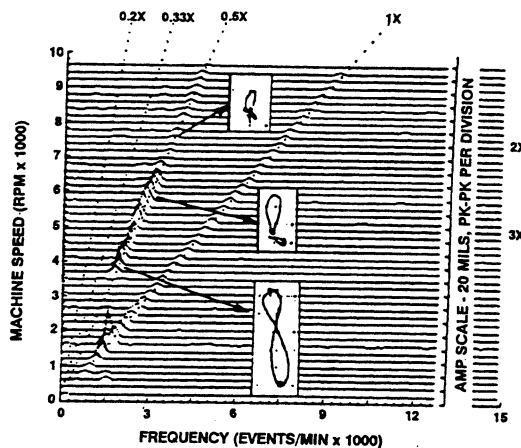
Belt Rubbing Belt Guard

Fan Blades Contacting Shroud

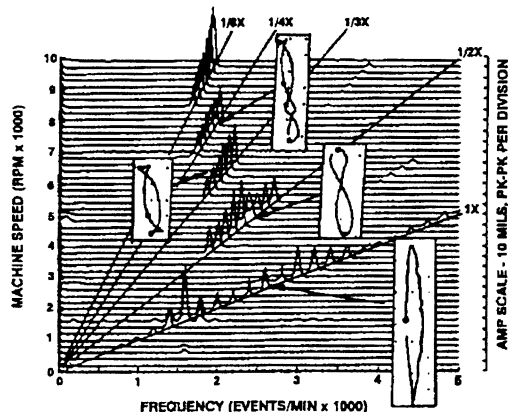
Characteristics which describe rotor rub in general are as follows:

1. When rub occurs, it can actually change the stiffness of the rotor which will therefore change the rotor natural frequency during the rub event. This can become a real problem if the rotor was formerly operating in a nonresonant condition, but now is thrust into resonant operation due to the natural frequency change.
2. The impact can cause local wear and deformation depending on the angle of attack and impact velocity.
3. Not only can high forces be generated due to impact, but also high torques which can almost instantaneously affect electrical characteristics, sometimes causing a significant increase in amperage. Of course, if the rub is continuous, this can bring about failure, not only of the contact area, but also potentially of the driver itself.
4. When rub occurs, friction will oppose rotation depending on the amount of force and surface properties such as coefficient of friction. This high friction force can even cause a backwards whirling motion, sometimes known as "backward precession". This does not mean the shaft changes rotational direction, but backward whirling implies that the shaft can actually begin to orbit in a direction opposite shaft rotation at a frequency independent of rotational speed.

5. A Partial Rub usually precedes a Full Annular Rub.
6. Waterfall plots (also called Cascades) are especially helpful in diagnosing radial rubs. Example Cascade plots are shown in Figure 6.07A which illustrates a partial rub. These cascade plots show how vibration amplitude and frequency changes with machine RPM. Looking at Figure 6.07A, this figure shows a series of single spectra plotting amplitude versus frequency, and it also shows how they change as the machine is brought up to speed (note that the 1X RPM goes through a natural frequency at about 1500 to 1600 RPM. Later, the amplitude at 1/2 times RPM goes through the same natural frequency, at which point the amplitude 1/2 times RPM is much greater than that at 1X RPM when the speed of the machine is approximately 3600 to 3800 RPM. This is very unusual and, when 1/2 times RPM is much greater than that at 1X, rub is one of the primary suspects.
7. Shaft orbit displays, which are also shown in Figure 6.07A, are also very helpful in diagnosing rubs. These orbit displays show the actual path the shaft itself follows inside the bearing. The shape of the orbit is helpful in determining phase as well as distinguishing the nature of the rub, its severity, and probable frequency content.



**FIGURE 6.07A**  
**CASCADE DIAGRAM INDICATING A PARTIAL RUB EVENT**  
**UNDER HIGH FORCE (Ref. 6, pg. 3)**



**FIGURE 6.07B**  
**PARTIAL RUB EVENT WITH LOW RUB FORCE GENERATING MANY**  
**INTEGER SUBHARMONICS (Ref. 6, pg. 3)**

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8. Again, two distinct types of radial rubs will occur including partial rub and full annular rub:

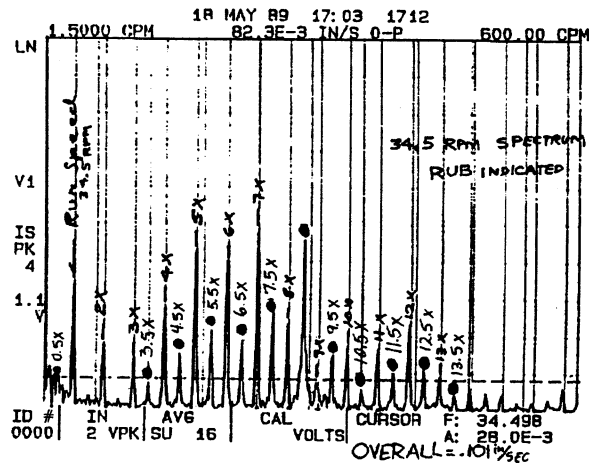
- a. Both types can take place separately or simultaneously;
- b. Both types originate either from excessive shaft vibration, from a smaller vibration if the shaft is not rotating in its geometric centerline, or if insufficient clearances have been established between rotating and stationary parts.

While many things are shared in common between partial and full annular rubs, each type has its own unique characteristics which are summarized in Sections 6.071 and 6.072, respectively.

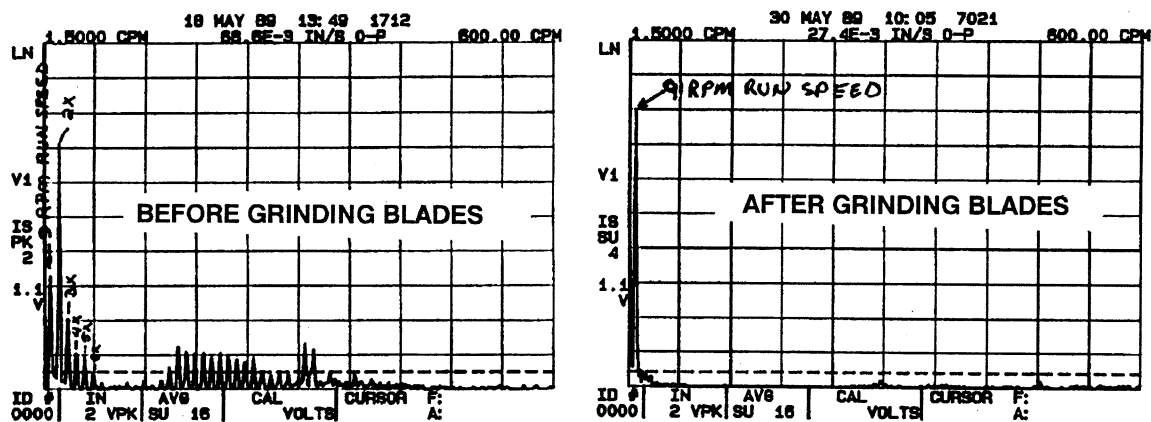
#### **6.071 Partial Rub:**

Partial Rubs usually begin only as very short-lived, transient events which often grow in severity and in duration as time passes. Because most of these rub events are only transient, the backwards whirling or precession motion does not normally occur. Left uncorrected, localized heating will occur which can actually bow the shaft and cause even greater severity rubs and cause rotor instability. Of course, as more severe rubs occur, even greater heat is generated in the shaft bowing it further which can lead to eventual catastrophic failure. Therefore, it is important to recognize partial rubs and act on them before they are given the chance to become even more serious. Following below are some of the characteristics exhibited by a partial rub which will help the analyst recognize a partial rub event:

1. Partial rubs will generate noticeable subsynchronous vibration (vibration below 1X RPM) which can sometimes be of significant amplitude and often will excite high frequencies as well if any noticeable impact occurs. However, an important fact is that subsynchronous vibration itself should not occur. Just the fact that subsynchronous vibration is occurring is important, and its source needs to be identified. If noticeable impact does occur, it can excite the system natural frequencies (independent of speed) in the mid to high frequency ranges.
2. If the subsynchronous vibration is caused by partial rub, it will normally show integer subharmonics (1/2X or 1/3X or 1/4X, etc.). Which integer sub-harmonic is excited depends on the shaft speed relative to rotor system natural frequency. In general, with higher shaft speed, lower fractions of subharmonic vibration can occur.
3. Several different integer subharmonics can simultaneously occur when a light rub with lower radial force occurs. Such a response is shown in Figure 6.07B which shows both the spectra and the orbit responses for 1X, 1/2X, 1/3X, 1/4X, and 1/5X RPM.
4. Often, when a heavier radial force partial rub occurs, only one subharmonic is present - predominantly, 1/2X RPM will remain.
5. If sharp impacts occur, multiples of these subharmonic frequencies can also appear (.5X, 1X, 1.5X, 2X, 2.5X, etc.) as well as high frequency resonant response. This is due to inherent nonlinearities which occur in an FFT spectrum when sharp impact takes place (instead of the time waveform being a smooth sinusoidal shape, it is truncated during an impact event which can generate almost a square waveform, resulting in a series of running speed peaks on an FFT spectrum). Such a spectrum is seen in Figure 6.07C with a whole series of 1/2 X RPM subharmonics which occurred when blading of a reactor began rubbing the stainless steel tank wall. Note that Figure 6.07D then compares "before" and "after" spectra taken prior to and after the repair actions at 9 RPM speed, respectively.



**FIGURE 6.07C**  
**SPECTRUM CAPTURED DURING THE CONFIRMED RUB OF A CHEMICAL REACTOR BLADE AGAINST ITS STAINLESS STEEL WALL WITH BLADE SPEED OF 34.5 RPM (Notice 1/2X RPM Harmonics Identified By Dots)**



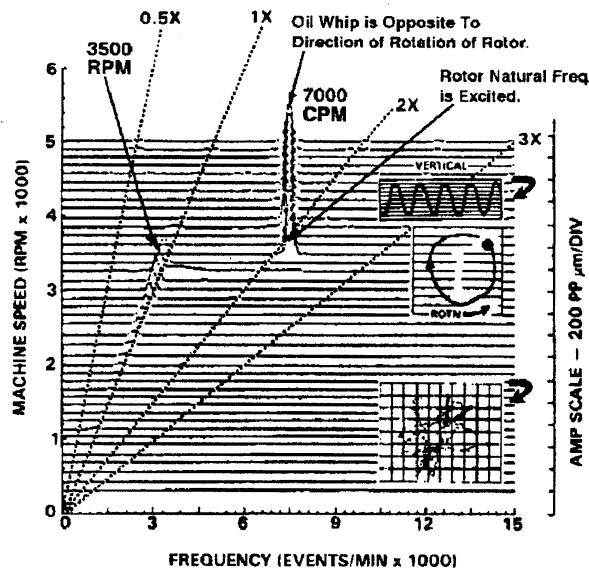
**FIGURE 6.07D**  
**COMPARISON OF SPECTRA BEFORE AND AFTER REPAIR RESOLVING RUB PROBLEM (BOTH SPECTRA CAPTURED AT 9 RPM REACTOR SPEED)**

- During a partial rub, the response is highly directional which results in horizontal and vertical phase difference of subsynchronous vibrations approaching 180° between horizontal and vertical. This is seen in Figure 6.07B. Note the elongated shape of the orbit at 1/2X RPM as well as that at 1/4X RPM, each of which approach straight line motion with 180° phase difference between the horizontal and vertical proximity probes.

#### 6.072 Full Annular Rub:

As previously mentioned, partial rub events can eventually lengthen in duration and impact leading to full annular rub in which case virtually a continuous rub event occurs. A full rub may cause counter-rotation, also known as backwards precession during which the shaft will continue to rotate in the same direction, but will orbit in a direction opposite this. This backward precession is caused by friction applied to the rotor generating a force in the opposite direction

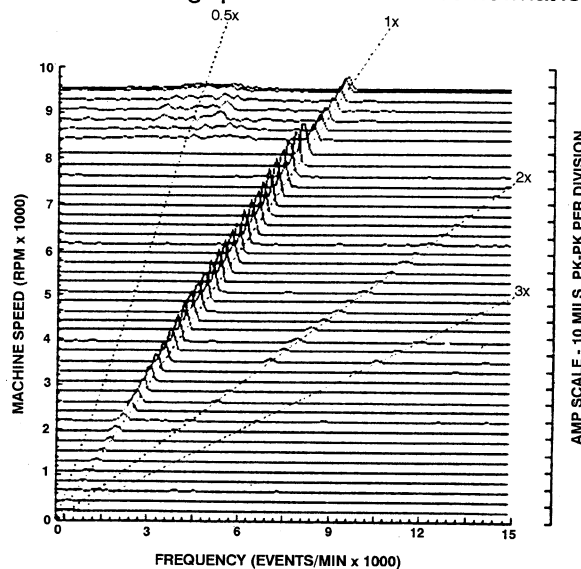
1. Figure 6.07E shows a spectrum taken during a full annular rub along with its shaft orbit displays when a seal was rubbing on a shaft. This shows how the response changed as the machine was brought up to speed. Note that at lower speeds little vibration was noticed until the rotor was up to approximately 2500 to 3500 RPM. At these speeds, the rotor began bouncing around the seal as seen by the orbit display. At higher speeds on the order of 3500 RPM, a response at about 7000 CPM began which remained at this same frequency even when the machine was brought all the way up to 5000 RPM. This 7000 CPM peak corresponded to the rotor/seal system natural frequency which was higher than the natural frequency of the rotor alone before it contacted the seal due to the additional stiffness supplied by the seal during the rub event. This 7000 CPM frequency then was the “speed” of the backward precession. That is, this backward precession will easily initiate at resonance and will remain at the same frequency location, independent of shaft rotational speed, whether or not the speed is increased or decreased.



**FIGURE 6.07E**  
**MULTIPLE IMPACTING SEAL RUB WHICH GENERATED REVERSE**  
**PRECESSION AT APPROX. 3500 RPM EXCITING A 7000 CPM**  
**ROTOR/SEAL SYSTEM NATURAL FREQUENCY (Ref. 17, Pg. 19)**

2. Backward precession is inherently unstable and can be very violent causing catastrophic rotor destruction.
3. Therefore, a full annular rub is characterized by a forward precession at 1X RPM corresponding to the unbalance in the system which can lead to backward precession at the natural frequency of the rotor/contact surface system (in which case both 1X RPM and the rotor/contact surface natural frequency will appear, often with 1X RPM at much lower amplitude).

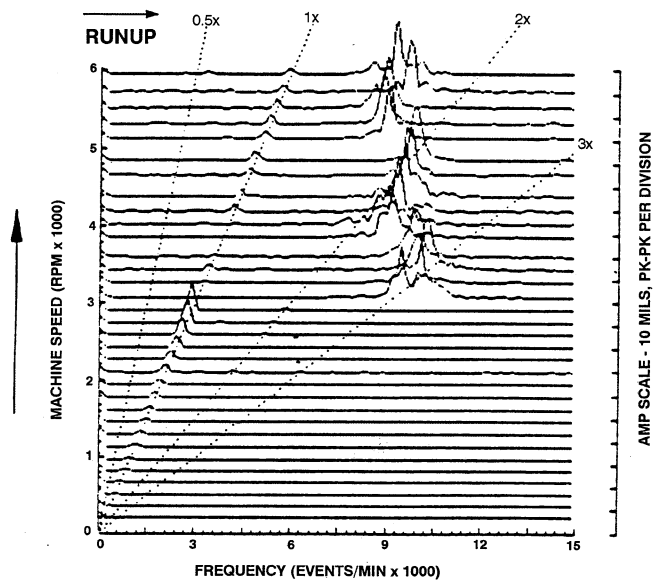
4. The catalyst which causes self-excited backward precession is friction between the rotating and stationary components. Since friction itself is nonlinear, it also can generate a series of running speed harmonics. Here again, higher harmonics are generated in an FFT spectrum by time waveform nonlinearities.
5. Figure 6.07F shows another type of full annular rub which is dominated by the response at 1X RPM along with some running speed harmonics at somewhat lower amplitudes.



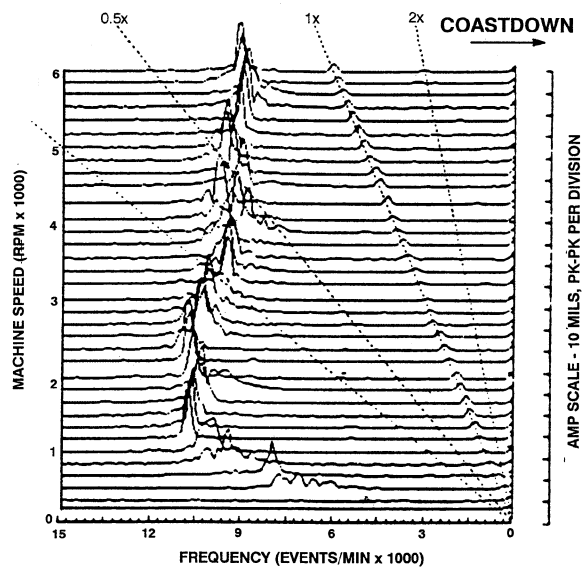
**FIGURE 6.07F**  
**FULL ANNULAR RUB DOMINATED BY FORWARD PRECESSION**  
**SYNCHRONOUS SPEED (Ref. 6, Pg. 4)**

6. Figures 6.07G and 6.07H are examples of a full annular rub where self-excited backward precession has occurred at a speed just below the first balance resonance. Note that the backward precession remained throughout the runup and even continued to occur as the machine coasted back down in Figure 6.07H. Note that when the backward precession was occurring, the comparative amplitude at 1X RPM was very low.
7. Note that since friction remains continually throughout a full annular rub, it will often introduce nonlinearities generating many running speed harmonics as well.
8. The vibration amplitude which will result in a full annular rub depends on clearances and damping of the system.

Permanent monitors with horizontal and vertical probes set 90° apart are recommended on all critical machinery in which rub can occur. This normally is thought of to be very large turbo-machinery and compressors outfitted with sleeve bearings. However, rotor rub events are not confined to these machines. They also can occur on smaller machinery and on units outfitted with antifriction bearings. Therefore, if the machine is truly critical, and there is a real possibility of a rub event occurring, these machines should also be outfitted with permanent vibration monitoring. This will allow not only capture of spectra during normal operation, but will allow the analyst to capture both waterfall spectra and shaft orbit displays similar to that shown in Figure 6.07A during startups and coastdowns. Not only will this go a long way in improving chances for detecting rubs when they do occur, but also permanent monitors will provide protection against undetected, catastrophic failure when no measurements are being made, particularly in those situations where the whole rub and catastrophic failure event can occur within minutes, or even seconds.



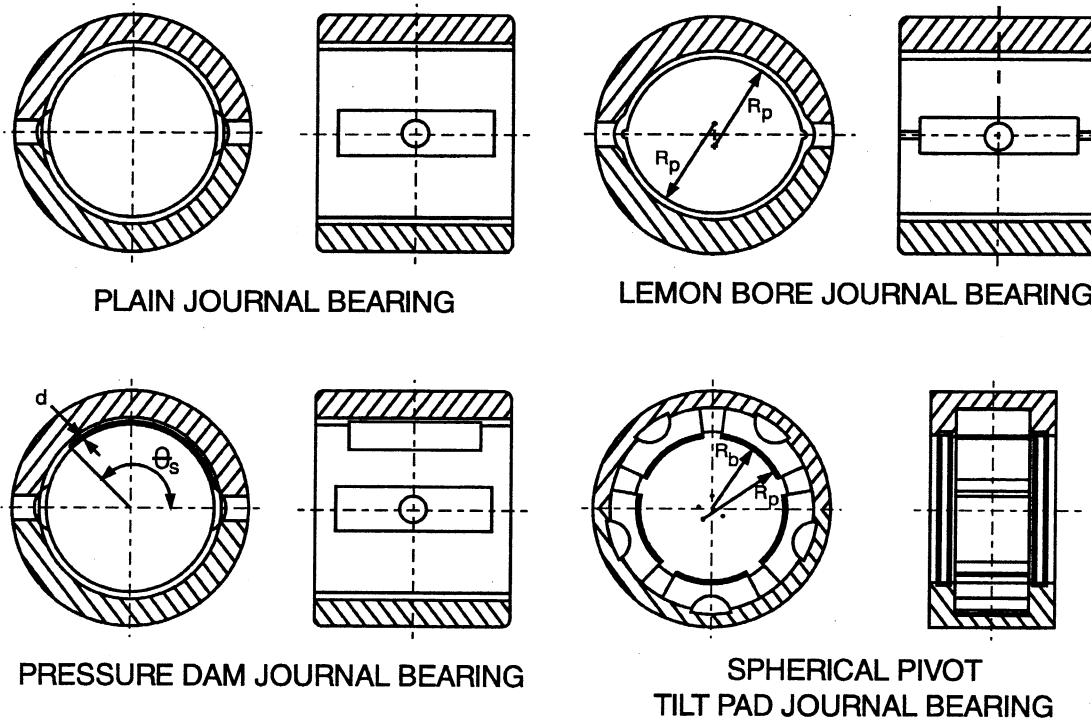
**FIGURE 6.07G**  
**FULL ANNULAR RUB DURING RUNUP WITH SELF-EXCITED BACKWARD**  
**PRECESSION (Ref. 6, Pg. 5)**



**FIGURE 6.07H**  
**FULL ANNULAR RUB DURING COASTDOWN (NOTE CONTINUED**  
**BACKWARDS PRECESSION UNTIL VERY LOW SPEED) (Ref. 6, Pg. 5)**

## 6.08 JOURNAL BEARINGS

Journal Bearings are also sometimes called Hydrodynamic Journal Bearings (technically probably a better name), sleeve bearings, Fluid Film Bearings or Plain Bearings. There are a variety of these bearings, some of which are listed in Figure 6.08A. As per Reference 10, "Hydrodynamic Bearings operate by developing a fluid film between the rotating shaft journal and the stationary bearing bore". Figure 6.08B shows the hydrodynamic bearing pressure profile and how it is distributed relative to the rotating shaft. Note in this figure that the shaft is rotating counterclockwise and that the maximum pressure is located roughly 15° to 20° counterclockwise of vertical. This is common in fluid film bearings.



PLAIN JOURNAL BEARING

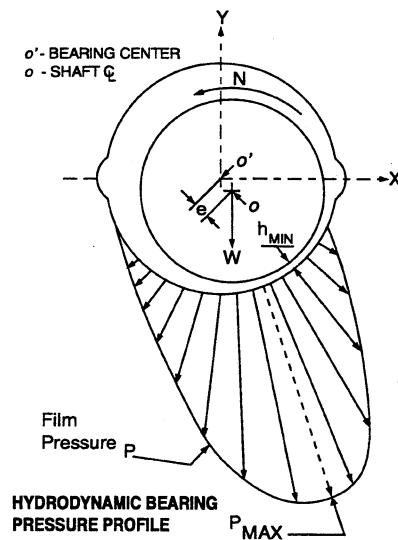
LEMON BORE JOURNAL BEARING

PRESSURE DAM JOURNAL BEARING

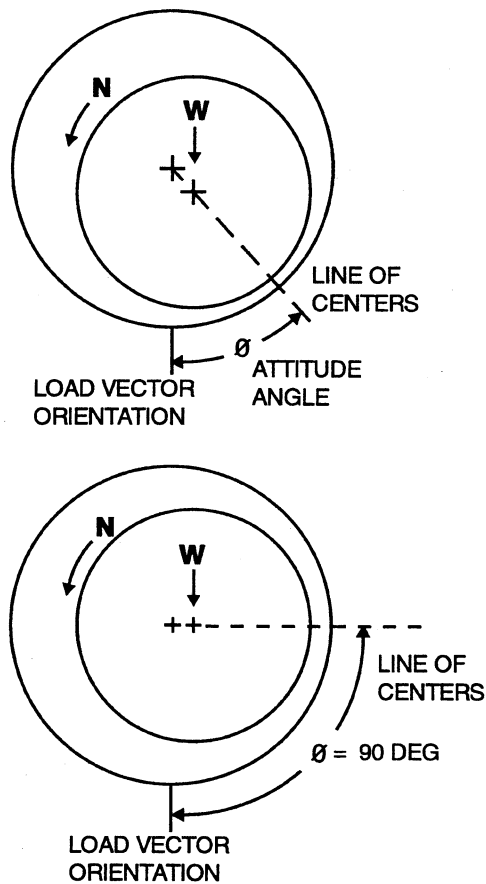
SPHERICAL PIVOT  
TILT PAD JOURNAL BEARING

**FIGURE 6.08A**  
**VARIOUS TYPES OF HYDRODYNAMIC JOURNAL BEARINGS**  
(Reference 10, Pages 48, 49, 50 & 53)

Figure 6.08C helps demonstrate one of the key items in a journal bearing - that of stability. The offset between the centerline of the bearing and that of the journal (shaft) is known as the bearing eccentricity. Importantly, eccentricity decreases as load decreases as oil viscosity ( $\mu$ ) increases and/or as speed increases. Note the dashed line running through each of the centerlines. Note on the drawing that this is referred to as the "line of centers" which is used to determine the "attitude angle" which is the angle between the line of centers and the applied load vector. These two quantities, eccentricity and attitude angle, are key indicators of bearing stability. Note that as the bearing load drops and/or the rotor speed increases, the attitude angle increases in plain cylindrical journal bearings. As this attitude angle increases, the bearing stability will decrease. Rotor instability occurs when a bearing is unable to exert sufficient preload to keep the rotating shaft in a stable position or as the rotor-bearing system loses its damping capabilities. When a rotor bearing system is prone to becoming unstable, any outside force which acts to upset the

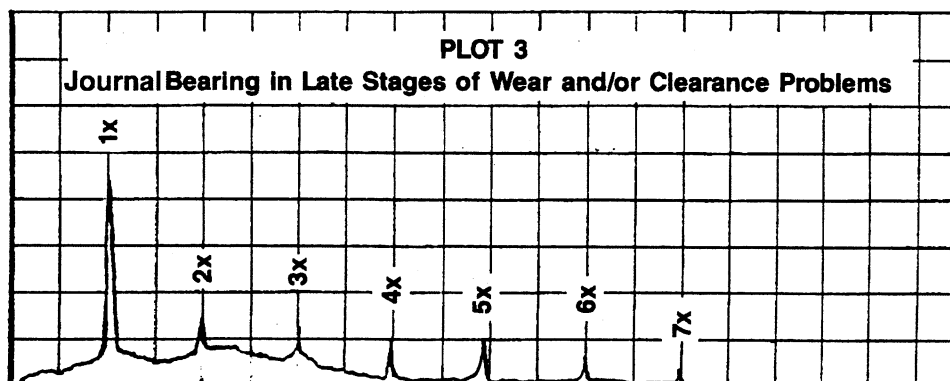
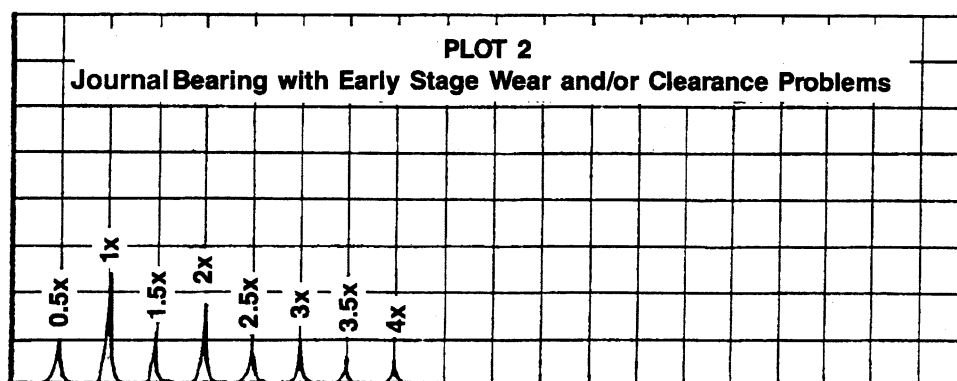
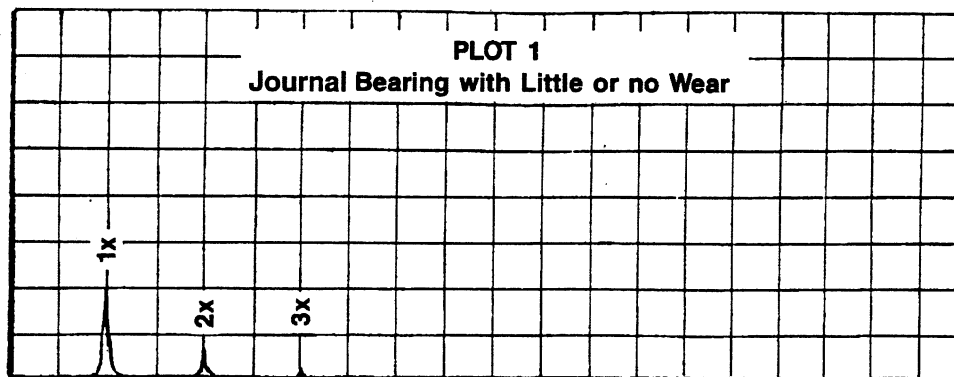


**FIGURE 6.08B**  
(Reference 10, Page 47)



**FIGURE 6.08C**  
**KEY JOURNAL BEARING DESIGN PARAMETERS** (Reference 10, Page 47)

**FIGURE 6.08D**  
**TYPICAL SPECTRA SHOWING JOURNAL BEARING PROBLEMS**



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bearing load may provide the condition necessary to cause instability (in some cases, a rotor-bearing system may suddenly become unstable even if there is no outside disturbance). Instability is a condition of rotor operation in which various elements in the system combine to induce self-excited vibration that can remain even after the original stimulus has been removed (self-excited vibrations are due to an internal feedback mechanism transferring the rotational energy into vibration). Normally, only a significant reduction in operating speed will again stabilize the rotor. In these cases, the rotor will often stabilize allowing it to be returned to its normal operating speed, but will often remain stable only until some disturbing force again upsets the system. When the forces restraining instability are reduced or lost, the vibration amplitude may suddenly increase dramatically, but yet stabilize at a value much higher than the amplitude at running speed. At this point, the dominant vibration itself actually is normally occurring subsynchronous to the operating speed (that is, lower than rotating speed). Therefore, one of the key items to watch in vibration signatures on journal bearings will be subsynchronous vibration below running speed.

In general, the greatest concern in the subsynchronous region for machines outfitted with journal bearings is concentrated between 35% and 55% of operating speed. However, particular problems can cause subsynchronous vibration as low as 20% up to as high as 80% of operating speed, depending on the particular problem. Following in Sections 6.081 through 6.084 will be a discussion on some of the more common journal bearing problems which can be detected by vibration analysis including journal bearing wear and clearance problems, oil whirl, instability and oil whip instability.

#### **6.081 Journal Bearing Wear and Clearance Problems:**

Worn journal bearings can be detected either by placing a velocity pickup or an accelerometer on the bearing cap. However, probably the best journal bearing condition data at lower frequencies up to approximately 5X RPM is captured from non-contact probes reading relative shaft vibration (also known as proximity probes). These proximity probes are placed just off the shaft surface (approximately 40 to 60 mils depending on shaft diameters and materials and always must be placed within the linear range of the non-contact pickup) and sense shaft motion relative to the probe itself. Therefore, they measure vibration displacement which is probably a good indicator of condition out to about the 4th or 5th running speed harmonic. The key point is that the proximity probe is looking directly at shaft motion, whereas both the accelerometer and velocity pickup on the bearing cap must sense a signal which must pass from the shaft through the oil film, and then through several metal interfaces to the bearing cap surface.

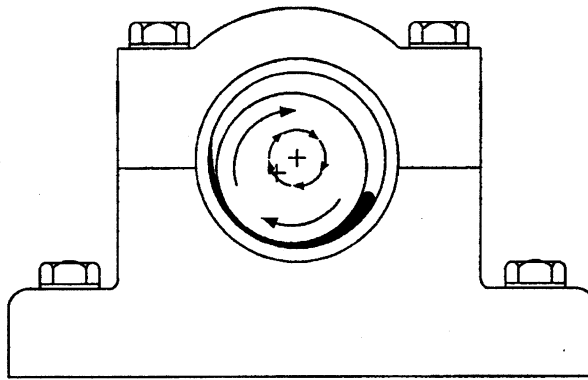
It is important to point out that some journal bearing failure incidents take place in only a matter of minutes, or even seconds, depending on the particular problem and the type of instability to which the system is subjected. However, in those cases where the deterioration does take place over a longer period of time, the oil film characteristics themselves begin to change and eventually may bring about rotor instability. In these cases, worn journal bearings can be detected either directly from the shaft or on the bearing cap.

When taking data from a proximity probe, it is important to point out it is quite normal to see several running speed harmonics. This is unlike velocity spectra taken from bearing caps in which case normally only the first 2 or 3 harmonics are seen, and each succeeding harmonic normally is only about 1/3 the height of the former (if no problems are present). Still, even with proximity probe shaft vibration data, the harmonics should also disappear into the spectral base.

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Following below are characteristics displayed by journal bearings with clearance and/or wear problems:

1. When high vibration levels are generated by faulty journal bearings, they are generally the result of lubrication problems, improper bearing load, looseness (babbitt loose in the housing), or excessive bearing clearance (caused by wiping or chemical erosion).
2. Figure 6.08D shows typical spectra for a journal bearing with little wear and/or clearance problems, one with early stage problems, and one with significant problems evident. Note the presence of subsynchronous vibration during early stages which may sometimes be at 1/2X or even 1/3X RPM subharmonics. In the latter stages of journal bearing wear, note that the spectrum will typically show a high vibration at 1X RPM, along with harmonics superimposed on an elevated baseline, particularly below 3X RPM.
3. A wiped journal bearing can often be detected by comparing horizontal and vertical vibration amplitudes. Normally, the horizontal vibration should be somewhat higher than that in the vertical direction due to more support in the vertical direction. A wiped journal bearing will often allow abnormally high vibration in the vertical direction as compared to that in the horizontal.
4. A wiped bearing with excessive clearances can allow the shaft to actually change position within the bearing which may result in misalignment which often will generate a vibration at 2X RPM. Sometimes, a bearing with excessive clearance will finally allow the shaft to rub the bearing or possibly another component such as a seal. In these cases, it can be critical in that this may be just the right disturbing force necessary to set the shaft into violent motion causing rotor instability. However, there have been cases reported where misalignment has actually had a stabilizing effect on the journal bearing. This may have been due to the additional load caused by the misalignment acting on a shaft and having a bearing which might have been too lightly loaded to maintain stability.
5. A journal bearing having excessive clearance may allow a small unbalance, misalignment or other related force to result in mechanical looseness, thereby generating the lower vibration spectra of Figure 6.08D. In these cases, the bearing is not the source of the problem. However, the vibration amplitude would be much lower if the bearing clearances were to spec.
6. Some oil film bearings act as thrust bearings. In these cases, many of them have pads or shoes. Of course, when running properly, the shaft should not contact the thrust bearing, but ride on an oil film instead. When problems develop with these thrust bearings, much higher vibration will result. This vibration sometimes will occur at 1X RPM, predominantly in the axial direction. However, if the thrust bearing is outfitted with pads, the vibration will be at the #pads X RPM. Surprisingly, this so-called pad passing frequency (#pads X RPM) can sometimes generate higher radial vibration than that in the axial direction. In any case, when excessive levels do occur at either the pad passing frequency or at 1X RPM, thrust bearing problems should be suspected. Of course, if this occurs at 1X RPM, the question may arise as to whether the problem is unbalance or thrust bearing wear. If this thrust bearing is placed in a compressor, one test that can be performed to detect which problem dominates is simply taking data with the compressor fully loaded and comparing that with the machine unloaded. If the problem is with the thrust bearing, the vibration should drop significantly as the load drops. However, if unbalance dominates, there should be little or no difference in 1X RPM amplitude since you have only dropped the load and not the machine RPM.



**FIGURE 6.08E**  
**OIL FILM WITHIN A JOURNAL BEARING**

**6.082 Oil Whirl Instability:**

Oil Whirl is probably the most common cause of subsynchronous instability in hydrodynamic journal bearings. Normally, the oil film itself will flow around the journal to lubricate and cool the bearing. In so doing, it will develop an average speed slightly less than 50% of the journal surface speed. This is pictured in Figure 6.08E. Normally, the shaft rides on the crest of an oil pressure gradient, rising slightly up the side of the bearing slightly off vertical at a given, stable attitude angle and eccentricity. The amount of rise depends on the rotor speed, rotor weight and oil pressure. With the shaft operating eccentrically relative to the bearing center, it draws the oil into a "wedge" to produce this pressurized load-carrying film.

If the shaft receives a disturbing force such as a sudden surge or external shock, it can momentarily increase the eccentricity from its equilibrium position. When this happens, additional oil is immediately pumped into the space vacated by the shaft. This results in an increased pressure of the load-carrying film which creates additional force between the film and shaft. In this case, the oil film can actually drive the shaft ahead of it in a forward circular motion and into a whirling path around the bearing within the bearing clearance. If there is sufficient damping within the system, the shaft can be returned to its normal position and stability. Otherwise, the shaft will continue in its whirling motion which can become violent depending on several parameters.

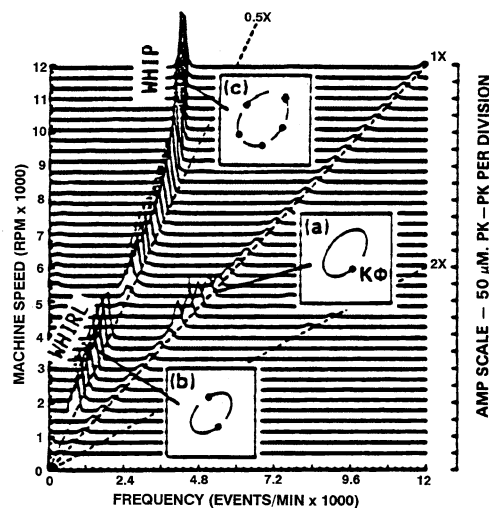
*Oil whirl will demonstrate the following characteristics:*

**1. The oil whirl condition can be induced by several conditions including:**

- light dynamic and preload forces;
- excessive bearing wear or clearance;
- a change in oil properties (primarily shear viscosity);
- an increase or decrease in oil pressure or oil temperature; improper bearing design (sometimes an over design for the actual shaft loading);
- fluid leakage in the shroud of blades and shaft labyrinth seals (so-called "Alford force" or "aerodynamic force");
- change in internal damping (hysteretic, or material damping; or dry (coulomb) friction);
- gyroscopic effects - especially on overhung rotors having much overhang.

Any of the above conditions can induce oil whirl after a disturbing force induces an initial rotor deflection.

2. Sometimes machines can exhibit oil whirl intermittently that have nothing to do with the condition of the journal bearing, but rather to external vibratory forces transmitting into the unit or from sources within the machinery itself. In these cases, these vibratory forces just happen to have the same frequency as the oil whirl frequency of that bearing and can be just the amount of disturbing force at just the right tuned frequency to set it into the whirl motion. This vibration can transmit from other machinery through attached structures such as piping and braces, or even through the floor and foundation. If this does occur, it may be necessary to either isolate this machine from surrounding machinery, or to isolate the offending machine itself.
3. Oil whirl can be easily recognized by its unusual vibration frequency which is generally on the order of 40% to 48% of shaft RPM (Reference 2 states that pure oil whirl occurs at 43% of shaft speed, but that the instability may occur at the 1st critical speed).
4. Figure 6.08F shows the development of oil whirl just after the shaft is brought up to speed. Note that the shaft definitely went into whirl at a machine speed of about 1800 RPM and remained in whirl until about 4000 RPM. At this point, note that the shaft 1X RPM begin entering resonance which actually caused enough force to overcome whirl. However, once the machine has passed through resonance, whirl once again occurred just above 5200 RPM.



**FIGURE 6.08F**  
**DEVELOPMENT OF OIL WHIRL JUST AFTER STARTUP**  
**(FOLLOWED BY OIL WHIP FROM 9200-12,000 RPM)**  
**(Reference 17, Page 20)**

5. Oil whirl is considered severe when vibration amplitudes reach 50% of the normal bearing clearance. At this point, corrective action must be taken.
6. Temporary corrective measures include changing the temperature of the oil (and therefore, the oil viscosity), purposely introducing a slight unbalance or misalignment to increase the loading, "temporarily" shifting the alignment by heating or cooling support legs, scraping the sides or grooving the bearing surface to disrupt the lubricant "wedge" or changing the oil pressure.

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7. More permanent corrective steps to solve oil whirl problem include installing a new bearing shell with proper clearances, preloading the bearing by an internal oil pressure dam, or completely changing the bearing type to oil film bearings less susceptible to oil whirl (including axial-groove bearings, lobed bearings or tilting pad bearings). The tilting pad bearing is possibly one of the best choices since each segment or pad develops a pressurized oil wedge tending to center the shaft in the bearing, thereby increasing the system damping and overall stability.

#### **6.083 Oil Whip Instability:**

Oil Whip can occur on those machines subject to oil whirl when the oil whirl frequency coincides with and becomes locked into a system natural frequency (often a rotor "balance" or lateral natural frequency). For example, please refer again to Figure 6.08F. When the rotor speed increased to just above 9200 RPM, this brought its speed to 2X its first balance natural frequency. At this time, the oil whirl which was approximately 43% of RPM was brought into coincidence with this natural frequency. At this point, the oil whirl was suddenly replaced by oil whip- a lateral forward precessional subharmonic vibration of the rotor. At this point, the oil whip frequency remains the same, independent of the rotor RPM. Note that the oil whip frequency never changed even though the machine continued on up to 12,000 RPM. When a shaft goes into oil whip, its dominant dynamic factors become mass and stiffness in particular and its amplitude is limited only by the bearing clearance. Left uncorrected, oil whip may cause destructive vibration which can cause catastrophic failure.

#### **6.084 Dry Whip:**

Dry Whip occurs in journal bearing machines subject either to a lack of lubrication or the use of the wrong lubricant. When this occurs, excessive friction is generated between the stationary bearing and rotating journal. This friction can excite vibration in the bearing and other components. This kind of vibration is called Dry Whip. Dry Whip can also be caused by journal bearings having excessive clearance as well as those having insufficient clearance.

The Dry Whip condition is similar to taking a moistened finger and rubbing it over a dry pane of glass. It will generate a frequency specifically dependent upon the shaft and construction materials, geometries and lubricant properties. Normally, this frequency will be quite high producing something similar to a squealing noise similar to that produced by dry rolling element bearings. The frequency content itself will not be an integer multiple of the machine speed. When Dry Whip is expected, it is important that it be taken care of quickly in order to prevent potentially catastrophic failure. When Dry Whip is suspected, both the lubricant itself and the lubrication system should be closely inspected and the bearings should be checked to ensure that they have proper clearances.

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## 6.09 TRACKING OF ROLLING ELEMENT BEARING FAILURE STAGES USING VIBRATION AND HIGH FREQUENCY ENVELOPING AND DEMODULATED SPECTRAL TECHNIQUES

### ABSTRACT:

Throughout the last decade, much research along with in-plant testing has been conducted in an attempt to accurately evaluate the condition of rolling element bearings. This has included studies to approximate remaining bearing life. This life is very dependent upon the vibration to which the bearing is subjected. The following paper will present a number of failure scenarios which have been identified to date for tracking rolling element bearing failure stages using both vibration and spike energy spectral analysis. These results have been directly correlated with studies which have been rigorously conducted in laboratories. Such studies have included evaluation of many types of rolling element bearings including deep groove ball, angular contact ball, needle, cylindrical roller, spherical roller and tapered roller bearings. It is the expressed purpose of this paper to provide the reader with solid tools with which he can not only evaluate the current health of specific rolling element bearings, but also can assist him in predicting remaining life and/or taking proactive steps immediately required to noticeably extend the life of the bearings.

### INTRODUCTION:

A tremendous cross section of today's process and utility machinery is outfitted with rolling element bearings. In most all cases, these bearings are the most precise components within the machine, generally held to tolerances only 1/10th those of many of the remaining machine components. Yet, only about 10 to 20% of bearings achieve their design life due to a variety of factors. These primarily include lubrication inadequacies, use of the wrong lubricant, contamination with dirt and other foreign particles, improper storage outside their shipping packages, introduction of moisture, false brinelling during shipment or when standing idle, misapplication of the wrong bearing for the job, improper installation of bearings, etc.

Figure 6.09A shows the components of a rolling element bearing. Note the location of the accelerometer relative to the bearing's outer race, inner race, rolling elements and cage. When rolling element bearings wear, the vibration signal most readily travels from defects on the outer race to the accelerometer. Such flaws will normally appear on two or more of these components prior to eventual failure. Figure 6.09B illustrates various types of rolling element bearings.

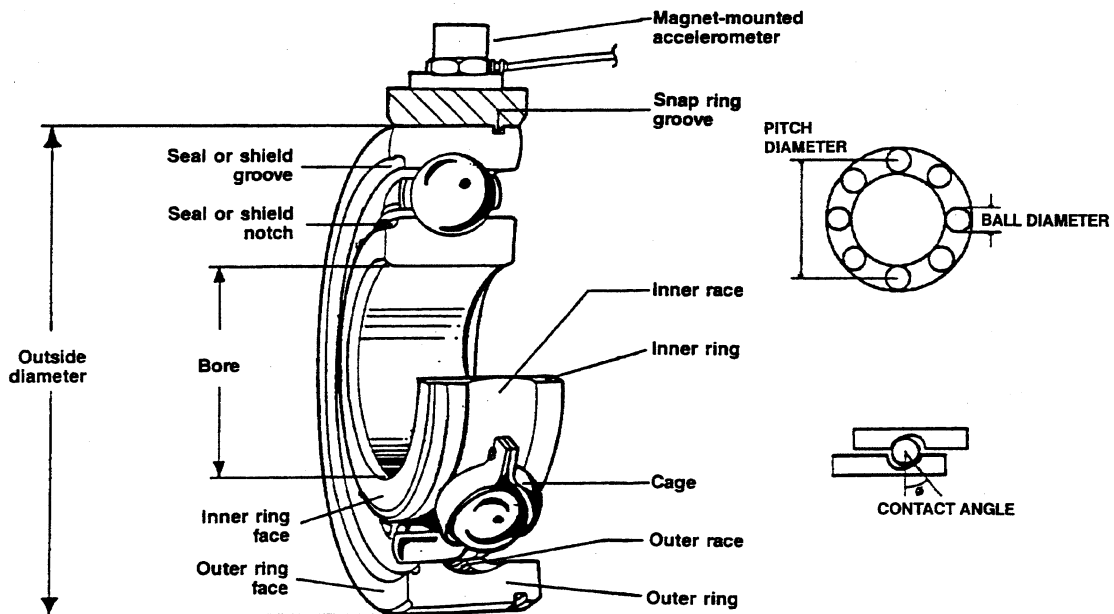
One of the leading contributors to premature rolling element bearing failure is excessive vibration and the high dynamic loads that it can transmit into bearings. Following below is the design formula used in calculating theoretical ball bearing life which will show why it is so critical to bearing life to minimize the dynamic loads imposed upon them from vibration:

$$L_{10} \text{ Life} = \left( \frac{16,666}{\text{RPM}} \right) \left( \frac{\text{RATING}_B}{\text{LOAD}_E} \right)^3 \text{ HOURS}$$

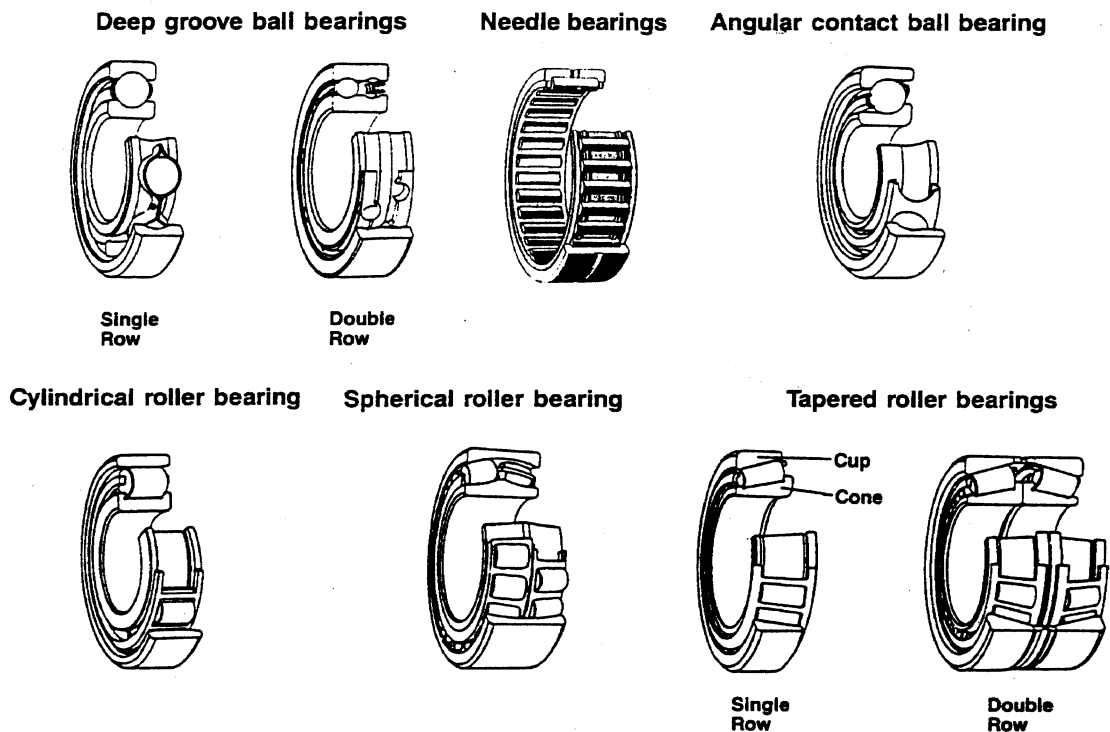
where:

- $L_{10} \text{ Life}$  = No. of Hours that 90% of a group of bearings should attain or exceed prior to onset of fatigue failure.
- $\text{RATING}_B$  = Basic Dynamic Load Rating for a given bearing (lb)
- $\text{LOAD}_E$  = Equivalent Radial Load taken by a bearing - including Radial and Axial Loads (lb)

**FIGURE 6.09A**  
**ROLLING ELEMENT BEARING TERMINOLOGY**



**FIGURE 6.09B**  
**ROLLING ELEMENT BEARING TYPES**



This formula shows that the greater the speed, the lower the anticipated life. However, of even greater significance is that the theoretical ball bearing life varies with the 3rd power of the load to which it is subjected (and to the 3.33 power in the case of roller bearings). Therefore, if the designer only considered the static loads on the bearing as well as those from other components such as belt tension, he may be surprised to learn of the magnitude of the dynamic forces which can be introduced from vibration.

The key point is that rolling element bearings in truth rarely fail due to defective workmanship themselves. In most all cases, outside influences act on the bearing to bring about its premature failure. If outside influences such as unbalance, misalignment, belt drive problems, soft foot, inadequate lubrication and improper installation can be taken care of, the bearings themselves should have satisfactory life.

To bring home the point, if one considers only the load from unbalance, this alone can generate significant dynamic loads. For example, consider a 2000 lb rotor turning at 6000 RPM with 1 oz of unbalance on a 3 foot diameter (18" radius). The amount of centrifugal force from the unbalance alone can be calculated as follows:

$$F_c = \frac{mrw^2}{g_c} = \left( \frac{Wr}{(386)(16)} \right) \left( \frac{2\pi n}{60} \right)^2$$

$$F_c = .000001775 U n^2 = .00002841 W r n^2$$

where:

- $F_c$  = Centrifugal Force (lb)
- $U$  = Unbalance of Rotating Part (oz-in)
- $W$  = Weight of Rotating Part (lb)
- $r$  = eccentricity of the rotor (in)
- $n$  = Rotating Speed (RPM)

Now substituting for the sample rotor with a 1 oz unbalance at an 18" radius ( $U = 18$  oz-in) turning 6000 RPM,

$$F_c = (.000001775)(18 \text{ oz-in})(6000 \text{ RPM})^2$$

$$F_c = \underline{1150 \text{ lbs}} \text{ (from centrifugal force due to unbalance alone)}$$

That is, only a 1 oz unbalance on a 3 foot diameter wheel turning 6000 RPM would introduce a centrifugal force of 1150 lbs which would have to be supported by the bearings in addition to the 2000 lb static rotor weight. Therefore, if the designer had only anticipated supporting 2000 lbs by the bearings, but in fact had to withstand 3150 lbs, his design life calculation would be off by a factor of:

$$\text{Corrected } L_{10} \text{ Life} = (\text{Initial Life}) \left( \frac{2000}{3150} \right)^3$$

$$\text{Corrected } L_{10} \text{ Life} = \underline{.25 \text{ X Initial Life Calculation}} \text{ (only 25\% of design life)}$$

Therefore, the actual theoretical life would only be 25% of his initial design life if this wheel were subjected only to unbalance, not to mention other dynamic forces introduced from vibration due to misalignment, looseness, cavitation or any other problems.

Of great importance is the ability to "track" the condition of rolling element bearings and to know when they will need replacement, right from the beginning when initial baseline signatures are

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acquired. Much has been learned during the past decade on how vibration signature analysis can contribute to this capability. Following in Section III will be a separate illustrated discussion on how this can be accomplished using not only vibration signature analysis, but also some of the newer high frequency enveloping signal processing techniques as well. This discussion will follow sections on "Which vibration parameter should normally be used?" (Section 6.091) and "What types of vibration frequencies are generated by defective rolling element bearings?" (Section 6.092). Then, recommendations on when rolling element bearings should be replaced will follow in Section 6.095.

#### **6.091 OPTIMUM VIBRATION PARAMETER FOR ROLLING ELEMENT BEARING CONDITION EVALUATION (ACCELERATION, VELOCITY OR DISPLACEMENT)?**

Particularly during this last decade, we have learned that we certainly can no longer depend on overall vibration alone to accurately evaluate the health of rolling element bearings. Also, we have found we cannot depend only on the measurement of ultrasonic frequency broadband measurements (Spike Energy, HFD, Shock Pulse, etc.) which make measurements in approximately the 5000 Hz to 60,000 Hz region. What we have learned is that these ultrasonic measurements are only an indicator, not the indicator of bearing health. They likewise are most effectively used in conjunction with vibration signature analysis to best evaluate bearing condition.

Of course, when employing vibration signature analysis, either of 3 vibration parameters can be used - acceleration (g), velocity (in/sec) or displacement (mils). Figure 6.09C shows how each of these parameters vary with frequency in terms of severity. Following below are comments on the attributes of each of these vibration parameters when specifically evaluating rolling element bearing health:

- A. Displacement** - Unfortunately, displacement spectra miss a great deal of bearing health information. Since displacement is low frequency intensive, it tends to suppress or almost eliminate much of the spectral content available that indicates bearing defect problems. In fact, one of the real problems with employing displacement on low-speed machines less than 200 RPM is the fact that while the spectrum may successfully display 1X RPM, it most often will almost completely miss bearing frequencies until the problem is quite severe.
- B. Acceleration** - Unlike displacement, acceleration tends to overemphasize much of the high frequency content generated by the rolling element bearing defects. As a result, if one is not greatly familiar with working with it, acceleration spectra might "cry wolf" far too often. For plant programs trying to establish themselves, this can do great harm to credibility. Although acceleration itself is probably a better indicator in the very early stages of bearing problems, it quickly gives way to vibration velocity which more accurately and clearly tells the "true story" of current bearing health.
- C. Velocity** - Velocity spectra should be one of the best parameters for evaluating most bearing problems, even on low-speed machines (for example, even if the speed was only about 60 RPM, much of the bearing frequency content would be above 500 CPM). For common rotational speeds ranging from 1200 to 3600 RPM, most of the spectral vibration content containing bearing defect information will be below 2000 Hz (120,000 CPM). In general, depending on the type of transducer employed, velocity will remain somewhat "flat" in frequencies ranging from 600 CPM up to 120,000 CPM (10 Hz - 2000 Hz). This means that somewhat "equal weight" can be assessed to a bearing defect frequency occurring at either 6000 CPM or at 60,000 CPM, whereas this could not be done with acceleration or displacement which are highly frequency dependent (see Figure 6.09C). When rolling element bearing machines operate at speeds above approximately 10,000 RPM, acceleration would then likely be the best rolling element bearing health indicator.

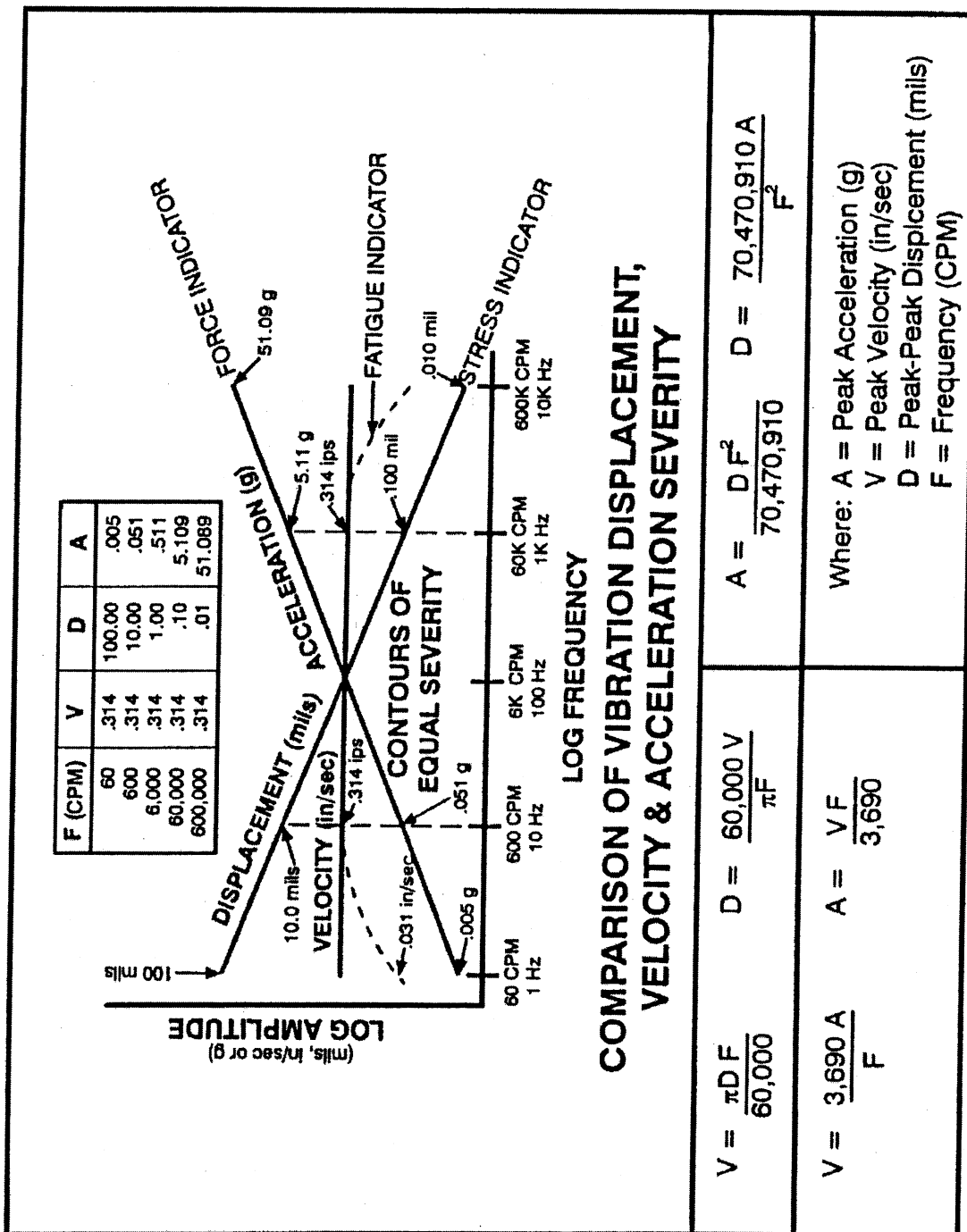


FIGURE 6.09C

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The remainder of this section on rolling element bearing health will assume velocity spectra. However, if one takes into account how the other parameters such as acceleration vary with frequency, he can apply many of these same techniques. Acceleration spectra may detect such bearing problems earlier than will velocity spectra, particularly on high-speed machinery. In addition, high frequency envelope demodulated spectra will likely provide even earlier warning of bearing wear and lubrication problems, and should be employed in addition to vibration spectra alone on critical machinery (this will be discussed in Section 6.093 of the paper).

#### **6.092 TYPES OF VIBRATION SPECTRA CAUSED BY DEFECTIVE ROLLING ELEMENT BEARINGS:**

Defective rolling element bearings generate each of 4 types of frequencies once they begin to develop defects. These frequencies include (a) random, ultrasonic frequencies; (b) natural frequencies of bearing components; (c) rotational defect frequencies; and (d) sum and difference frequencies. Following below will be a discussion on each of these types of frequencies and their significance:

##### **(a) Random, Ultrasonic Frequencies:**

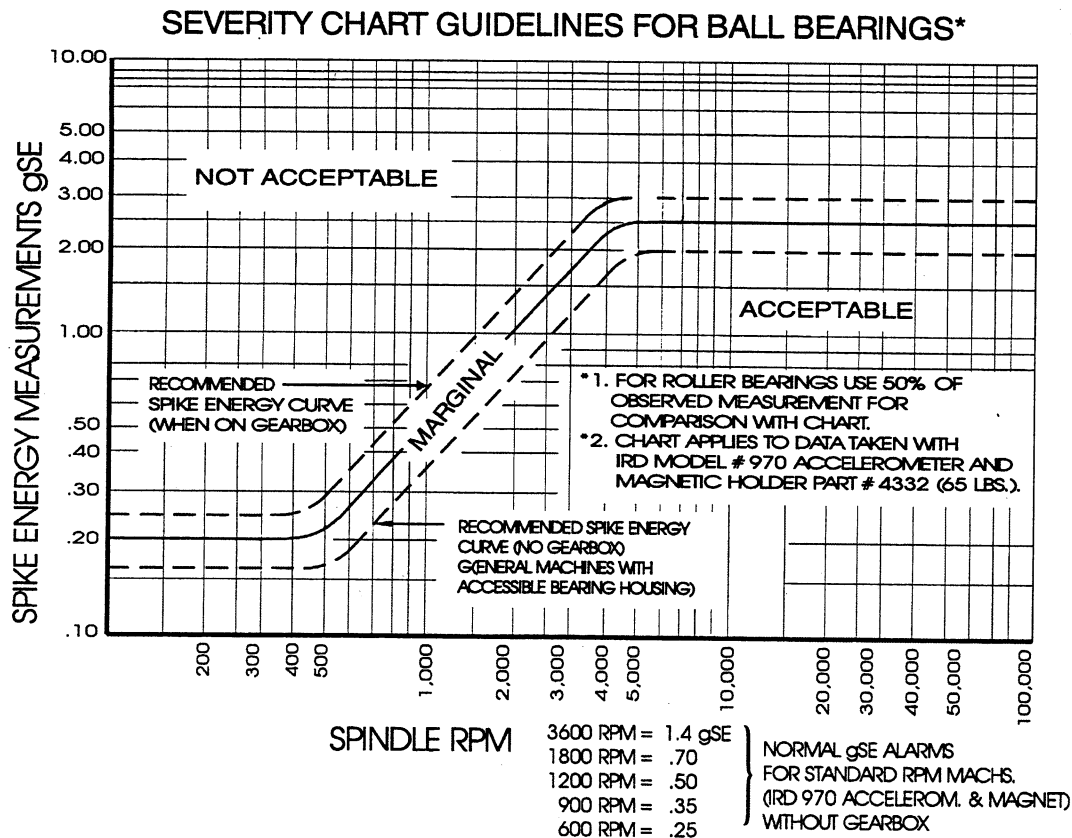
Measurements in this ultrasonic frequency region ranging from approximately 5000 Hz to 60,000 Hz are made by a variety of instruments and employ a similar variety of techniques. These include spike energy, HFD high frequency acceleration, shock pulse measurement and others. Each of these are meant to be incipient failure detection parameters that can track bearing health from its installation until just prior to or including eventual failure. Each of them have their own strengths and weaknesses, not only concerning their bearing health evaluation accuracy, but also how well each of them can be trended with time. In general, the overall number they provide gives just one more piece of information to be considered when evaluating bearing health. However, the information contributed by the vibration spectral data should be given significantly more weight.

Figure 6.09D provides a severity chart for specifying alarm levels of Spike Energy. This chart shows that machine speed must be taken into account when evaluating Spike Energy amplitudes. Similarly, Figure 6.09E provides a comparable severity chart for "HFD" and Shock Pulse (SPM), comparing the relative amplitudes of these parameters with Spike Energy levels in a study conducted by Mr. Charles Berggren (References 16 & 19).

##### **(b) Natural Frequencies of Bearing Components (when installed):**

Reference 1 documents that the natural frequencies of installed rolling element bearing components range from approximately 500 to 2000 Hz (30,000 to 120,000 CPM). Like every other member, these bearing components "ring" at these natural frequencies when they are impacted. In the case of rolling element bearings, intermittent impacts of the rolling elements striking flaws on the raceways ring their natural frequencies. Actually, there are several bearing component natural frequencies in the region of 30,000 to 120,000 CPM, but some are much more predominant than others. Therefore, when defects progress beyond microscopic size, they begin to excite these natural frequencies making them the "second line of detection" (Failure Stage 2, discussed in Section 6.093). As the defects worsen, they can cause greater impacts which cause greater response from the natural frequency peaks. Eventually, when wear progresses, more frequencies around these resonances appear, many of which will be 1X RPM sidebands of these natural frequencies (often, such modulating peaks will be spaced at bearing defect frequencies rather than 1X RPM sidebands).

**FIGURE 6.09D**  
**RECOMMENDED SPIKE ENERGY SEVERITY CHART (IRD SPIKE ENERGY)**



### SPIKE ENERGY™ MEASUREMENTS

Energy is generated by repetitive transient mechanical impacts. Such impacts typically occur as a result of surface flaws in rolling-element bearings or gear teeth.

This energy is conducted from its source through various paths to the outer surface of the machine structure, and is seen as a small-amplitude vibration at the surface. Accelerometers coupled to the surface generate corresponding electrical signal.

The accelerometer signals processed by unique filtering and detection circuitry to produce a single "figure of merit" related to the intensity of the original impacts. This figure of merit is expressed in "gSE" units.

SPIKE ENERGY™ gSE readings are measurements which can with experience, be correlated with the severity of the casual surface flaws. Even though gSE readings are affected by the nature of the conductive path between the impact source and the accelerometer, similar machine structures will provide a reasonable basis for comparison between the structures.

The gSE figure of merit has proven to be effective in detecting mechanical defects in meshing gears and rolling element bearings. The gSE measurement, when used in conjunction with conventional measurement of vibration velocity and acceleration, provides early indications of mechanical deterioration.

**\*\*** When used with magnetic holders, accelerometers must be installed with a light coating of silicone grease and tightened to 40 in-lb. torque.

**FIGURE 6.09E**  
**MAINTENANCE DIAGNOSTIC VIBRATION AND HIGH FREQUENCY**  
**GENERAL TOLERANCE CHART FOR PROCESS MACHINERY WITH**  
**ROLLING ELEMENT BEARINGS<sup>1</sup>**

| Unfiltered<br>Vibration<br>Velocity | Shock<br>Pulse &<br>Acoustic<br>Emission | Microlog<br>HFD | IRD<br>Spike<br>Energy <sup>②</sup> | Severity Level<br>and<br>Alarm Status |
|-------------------------------------|------------------------------------------|-----------------|-------------------------------------|---------------------------------------|
| Inches/Second<br>Peak               | Db                                       | g               | gSE                                 |                                       |
| 1.5+                                | 50+                                      | 5.0+            | 3.0+                                | Danger<br>(Shutdown)                  |
| .75 - 1.49                          | 40 - 49                                  | 3.0 - 4.99      | 1.50 - 2.99                         | Very Rough<br>(Alert)                 |
| .40 - .74                           | 30 - 39                                  | 1.50 - 2.99     | .80 - 1.49                          | Rough<br>(Alert)                      |
| .20 - .39                           | 20 - 29                                  | .75 - 1.49      | .40 - .79                           | Fair<br>(Acceptable)                  |
| .10 - .19                           | 10 - 19                                  | .30 - .74       | .20 - .39                           | Good<br>(Acceptable)                  |
| .01 - .09                           | 1 - 9                                    | .01 - .29       | .01 - .19                           | Smooth<br>(Acceptable)                |

① Provide by Charles Berggren of Monsanto

② Spike Energy Amplitudes measured using an IRD 970 accelerometer outfitted with IRD's 2-pole, 65 lb. magnet.

One important point about these bearing component natural frequencies is the fact that they are independent of running speed. That is, whether the shaft is turning at low or at very high speeds, the natural frequencies remain at the same frequency location. However, their response amplitude is proportional to the impact velocity, which means that with greater rotational speeds, they normally will respond at greater amplitudes.

**(c) Rotational Defect Frequencies:**

Through the years, a series of formulas have been developed which can help detect specific defects within rolling element bearings. They can separately detect faults on the inner race, outer race, cage or rolling elements themselves. They are based on the bearing geometry, the number of rolling elements and the bearing rotational speed. Figure 6.09F provides the formulas for each of these four rolling element bearing defect frequencies.

The power of these equations is that if one knows the design parameters of his bearings (pitch diameter, rolling element diameter, number of rolling elements & contact angle), he is able to detect problems which occur on the races, cage or rolling elements, and he is enabled to track these problems as deterioration continues. In many cases, the analyst may not know all the parameters to insert for a particular bearing in the equations, but he might know the bearing manufacturer and model number. In these cases, there are several publications and software offerings which tabulate each of the 4 defect frequencies for each of the bearing model numbers. Figure 6.09G is an example of one of the better known publications listing these defect frequencies (as per Reference 12). Note on Figure 6.09G that this sheet provides the number of rolling elements ( $N_b$ ), rolling element diameter ( $B_d$ ), bearing pitch diameter ( $P_d$ ), contact angle

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( $\emptyset$ ), outer race defect frequency (BPFO), inner race defect frequency (BPFI), cage defect frequency (FTF) and rolling element defect frequency (BSF). Note that Figure 6.09G also provides an example assuming the SKF N220 bearing. Note that each one of the bearing frequencies are given in terms of running speed orders. Therefore, the frequencies for the SKF N220 bearing show, for example, that the inner race frequency (BPFI) is 9.13X RPM. Thus, if this bearing were turning at 1000 RPM, and a frequency was detected at 9130 CPM, he could conclude that this peak is the bearing inner race defect frequency and that a fault is developing there. On the other hand, if a frequency of 6860 CPM occurred on this same unit, he would know this is the outer race defect frequency of this SKF N220 bearing (since BPFO = 6.86X RPM).

The great advantage of knowing these bearing defect frequencies is that, for example, an analyst can separately evaluate the outboard and inboard bearings on the same machine, particularly if they are different model numbers. For example, assume that the outboard bearing of our example was an SKF N220 and that its inboard bearing was an SKF N228 bearing. Note from Figure 6.09G that they would have an entirely different set of defect frequencies (for example, the inner race frequency on the N228 would be 10.19X RPM as compared to 9.13X RPM on the N220).

A number of interesting facts can be stated about these bearing defect frequencies as follows:

**1. How Bearing Frequencies Differ From Other Frequencies (Defect Frequencies):**

One thing setting rolling element bearing defect frequencies apart from other vibration sources is the fact that they are defect frequencies. In other words, bearing defect frequencies should not be present. When they are present, they signal at least an incipient problem. On the other hand, other common frequencies such as 1X RPM are always present whether or not there is satisfactory or unsatisfactory balance or alignment; pumps and vanes always show vibration at some amplitude for blade pass frequencies; gears cause vibration at the number of teeth X RPM. However, the presence of these other frequencies does not mean there is necessarily a defect or problem. The appearance of bearing defect frequencies sends a message to the analyst to "pay attention". However, it is also important to point out the presence of such defect frequencies do not necessarily mean there are defects within the bearing. They also will appear if there is insufficient lubricant allowing metal-to-metal contact, or if the bearings are improperly loaded (excessive press fit, excessive thrust on a bearing not necessarily designed to take thrust, or if a thrust bearing is installed backwards, etc.).

**2. Bearing Defect Frequencies are Noninteger Multiples of Operating Speed:**

Referring back to Figure 6.09G, note that each of the bearing defect frequencies are noninteger multiples. That is, they are one of the few machinery vibration sources that do not generate integer multiples of rotational speed ( $GMF = \# \text{teeth} \times \text{RPM}$ ;  $BPF = \# \text{blades} \times \text{RPM}$ ). This is helpful when the manufacturer and model number of the bearing is unknown. Look for real number RPM multiples (such as 5.78X, or 7.14X, etc.).

**3. Sum of Race Frequencies = "Bearing Ball Pass Frequency":**

Figure 6.09F shows an interesting relationship between the outer and inner race frequency multipliers and the number of rolling elements. Note that the product of the number of balls X RPM equals the sum of the outer and inner race frequencies (BPFO + BPFI). For years, many have looked for the appearance of frequencies at the number of balls X RPM similar to what they have experienced with gears ( $\# \text{teeth} \times \text{RPM}$ ), blade pass frequency ( $\# \text{Blades} \times \text{RPM}$ ) and so forth. However, the author has rarely seen the appearance of a frequency at "bearing ball pass frequency", but has commonly seen each of the race frequencies themselves which are on the other side of the equation.

## FIGURE 6.09F ROLLING ELEMENT BEARING DEFECT FREQUENCIES

These equations assume the inner race is rotating with the shaft while the outer race is fixed (stationary):

$$\begin{aligned} \text{INNER RACE} &= \text{BPFI} = \frac{N_b}{2} \left( 1 + \frac{B_d}{P_d} \cos \theta \right) \times \text{RPM} \\ \text{OUTER RACE} &= \text{BPFO} = \frac{N_b}{2} \left( 1 - \frac{B_d}{P_d} \cos \theta \right) \times \text{RPM} = N_b \times \text{FTF} \\ \text{BALL (OR ROLLER)} &= \text{BSF} = \frac{P_d}{2B_d} \left[ 1 - \left( \frac{B_d}{P_d} \cos \theta \right)^2 \right] \times \text{RPM} \\ \text{CAGE} &= \text{FTF} = \frac{1}{2} \left( 1 - \frac{B_d}{P_d} \cos \theta \right) \times \text{RPM} = .35 - .45 \times \text{RPM} \end{aligned}$$

**NOTE:**

If the inner ring is fixed (stationary) while the outer race rotates with the shaft, the minus sign must be changed to a plus sign within the parentheses of the Cage Frequency Equation. In this case,  $N_b \times \text{FTF}$  will now equal BPFI rather than BPFO; and  $\text{FTF} \approx .55 - .65 \times \text{RPM}$ . However, BPFO, BPFI and BSF will remain the same.

(NOTE THAT  $\text{BPFI} + \text{BPFO} = N_b$ )

where:

$$\begin{aligned} N_b &= \text{Number of Balls or Rollers} \\ B_d &= \text{Ball or Roller Diameter (in. or mm)} \\ P_d &= \text{Bearing Pitch Diameter (in. or mm)} \\ \theta &= \text{Contact Angle (degrees)} \end{aligned}$$

**APPROXIMATIONS** (If Bearing Dimensions are Unknown, but do know # of Balls or Rollers) (Reference 18):

$$\begin{aligned} \text{Approx. BPFI} &\approx \left( \frac{N_b}{2} + 1.2 \right) \times \text{RPM} \\ \text{Approx. BPFO} &\approx \left( \frac{N_b}{2} - 1.2 \right) \times \text{RPM} \\ \text{Approx. BSF} &\approx \frac{1}{2} \left( \frac{N_b}{2} - \frac{1.2}{N_b} \right) \times \text{RPM} \\ \text{Approx. FTF} &\approx \left( \frac{1}{2} - \frac{1.2}{N_b} \right) \times \text{RPM} \end{aligned}$$

**FIGURE 6.09G**  
**EXAMPLE TABULATION OF ROLLING ELEMENT BEARING DEFECT**  
**FREQUENCIES (REFERENCE 12)**

**SKF**

**SINGLE ROW CYLINDRICAL ROLLER BEARINGS**

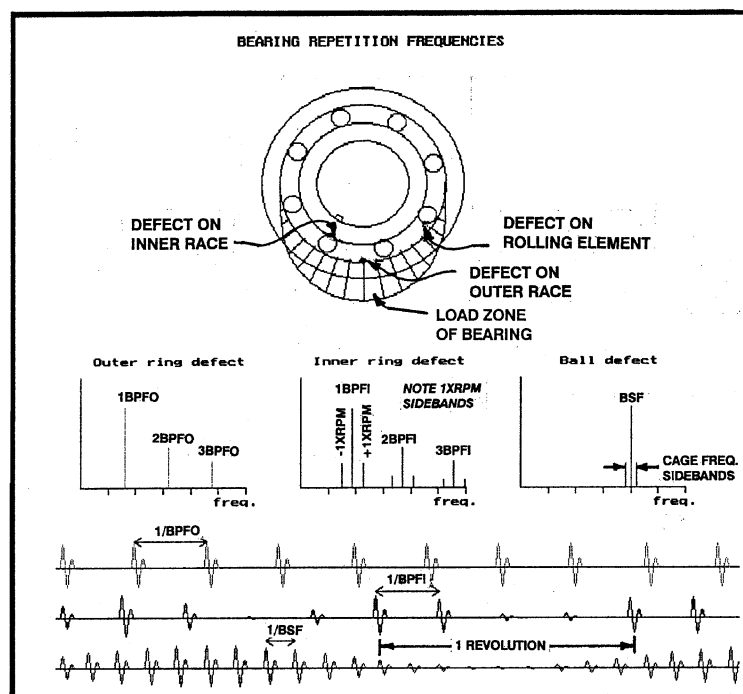
| <u>BEARING<br/>NUMBER</u> | <u>Nb</u> | <u>Bd</u> | <u>Pd</u> | <u>Ø</u> | <u>BPFO</u> | <u>BPFI</u> | <u>FTF</u> | <u>BSF</u> |
|---------------------------|-----------|-----------|-----------|----------|-------------|-------------|------------|------------|
| N205                      | 12        | .256      | 1.516     | 0        | 4.99        | 7.01        | .41        | 2.87       |
| N206                      | 12        | .295      | 1.811     | 0        | 5.00        | 6.94        | .41        | 2.99       |
| N207                      | 12        | .354      | 2.106     | 0        | 4.97        | 6.98        | .41        | 2.89       |
| N208                      | 13        | .394      | 2.362     | 0        | 5.42        | 7.58        | .41        | 2.92       |
| N209                      | 14        | .394      | 2.559     | 0        | 5.97        | 8.07        | .42        | 3.17       |
| N210                      | 15        | .394      | 2.968     | 0        | 6.50        | 8.48        | .43        | 3.70       |
| N211                      | 15        | .433      | 3.051     | 0        | 6.43        | 8.56        | .43        | 3.45       |
| N212                      | 16        | .472      | 3.476     | 0        | 6.92        | 9.08        | .43        | 3.61       |
| N213                      | 16        | .512      | 3.641     | 0        | 6.88        | 9.11        | .43        | 3.49       |
| N214                      | 16        | .512      | 3.738     | 0        | 6.93        | 9.06        | .43        | 3.68       |
| N215                      | 17        | .551      | 4.035     | 0        | 7.34        | 9.65        | .43        | 3.59       |
| N216                      | 16        | .590      | 4.331     | 0        | 6.91        | 9.08        | .43        | 3.60       |
| N217                      | 16        | .620      | 4.626     | 0        | 6.91        | 9.08        | .43        | 3.60       |
| N218                      | 16        | .709      | 4.921     | 0        | 6.84        | 9.15        | .43        | 3.40       |
| N219                      | 16        | .748      | 5.216     | 0        | 6.85        | 9.14        | .43        | 3.41       |
| N220                      | 16        | .787      | 5.512     | 0        | 6.86        | 9.13        | .43        | 3.43       |
| N222                      | 16        | .905      | 6.102     | 0        | 6.81        | 9.18        | .42        | 5.59       |
| N224                      | 16        | .945      | 6.594     | 0        | 6.85        | 9.14        | .42        | 3.41       |
| N225                      | 18        | .945      | 7.087     | 0        | 7.91        | 10.07       | .43        | 3.32       |
| N228                      | 18        | 1.024     | 7.677     | 0        | 7.76        | 10.19       | .43        | 3.68       |
|                           |           |           |           |          |             |             |            |            |
| N305                      | 10        | .354      | 1.713     | 0        | 4.01        | 5.99        | .40        | 2.32       |
| N306                      | 11        | .394      | 2.007     | 0        | 4.42        | 6.58        | .40        | 2.45       |
| N307                      | 11        | .433      | 2.264     | 0        | 4.45        | 6.55        | .40        | 2.51       |
| N308                      | 12        | .472      | 2.559     | 0        | 4.88        | 7.07        | .41        | 2.62       |
| N309                      | 11        | .551      | 2.854     | 0        | 4.43        | 6.56        | .40        | 2.49       |
| N310                      | 12        | .590      | 3.150     | 0        | 4.85        | 7.09        | .40        | 2.57       |
| N311                      | 11        | .669      | 3.445     | 0        | 4.42        | 6.56        | .40        | 2.48       |
| N312                      | 12        | .709      | 3.740     | 0        | 4.85        | 7.10        | .41        | 2.54       |
| N313                      | 12        | .748      | 4.035     | 0        | 4.87        | 7.08        | .41        | 2.60       |
| N314                      | 12        | .787      | 4.330     | 0        | 4.89        | 7.06        | .41        | 2.66       |
| N315                      | 12        | .866      | 4.626     | 0        | 4.85        | 7.09        | .41        | 2.57       |
| N316                      | 13        | .866      | 4.921     | 0        | 5.36        | 7.63        | .41        | 2.75       |
| N317                      | 12        | .945      | 5.216     | 0        | 4.89        | 7.06        | .41        | 2.67       |
| N318                      | 13        | .984      | 5.512     | 0        | 5.34        | 7.65        | .41        | 2.71       |
| N319                      | 14        | 1.024     | 5.807     | 0        | 5.77        | 8.22        | .41        | 2.74       |
| N320                      | 14        | 1.102     | 6.201     | 0        | 5.75        | 8.24        | .41        | 2.68       |
| N321                      | 14        | 1.181     | 6.496     | 0        | 5.72        | 8.27        | .41        | 2.66       |
| N322                      | 14        | 1.260     | 6.890     | 0        | 5.72        | 8.27        | .41        | 2.64       |

**EXAMPLE:**

FOR SKF N220 BEARING, BPFI = 9.13 X RPM (Inner Race)  
 BPFO = 6.86 X RPM (Outer Race)  
 BSF = 3.43 X RPM (Rolling Elements)  
 FTF = .43 X RPM (Cage Freq.)

#### 4. Description of How Defect Frequencies are Generated Within Rolling Element Bearings:

Figure 6.09F provides the formulas required to calculate the four defect frequencies for each rolling element bearing (BPFI, BPFO, BSF and FTF). Figure 6.09H illustrates how such defect frequencies are generated within the bearings. For example, if there is a defect on the outer race on the bottom of the bearing within the load zone as shown in Figure 6.09H, note that an impulse occurs in the time waveform at each instant when a rolling element passes over and impacts this defect. On the other hand, if the inner race had a defect, an impulse would occur in the time domain when the inner race rotated past each rolling element (assuming the inner race is press fit on the shaft). An important fact shown by Figure 6.09H is that the amount of response from the rolling elements striking the inner race defect will depend on where the inner race is positioned at that particular instant of time when the impact occurs (that is, if the inner race defect is positioned within the load zone, it will have significantly more response than it would if the impact occurred with the same inner race defect positioned  $180^\circ$  away, clearly out of the load zone). This explains why inner race defect frequencies are often surrounded by sidebands spaced at 1X RPM since their amplitude is modulated at the rate of once per revolution as shown by Figure 6.09H. On the other hand, since the outer race does not rotate in this instance, the amplitude response in the time domain should remain near constant. Therefore, 1X RPM sidebands surrounding outer race frequencies (BPFO) are much more serious than those surrounding the inner race frequency (BPFI), again assuming the inner race is press fit on the shaft (if the inner race is stationary while the outer race rotates, the reverse with respect to sidebands would occur). The presence of 1X RPM sidebands surrounding outer race (BPFO) frequencies normally means the problem is sufficiently serious to cause the bearing to actually impede the motion of the shaft. As Section 6.093 will later point out, when the inner race frequency (BPFI) becomes surrounded by several "families" of 1X RPM sidebands, this can likewise indicate a more serious problem.



**FIGURE 6.09H**  
**ILLUSTRATION OF HOW DEFECT FREQUENCIES ARE GENERATED WITH ROLLING ELEMENT BEARINGS**

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## **5. Relative Outer and Inner Race Amplitudes:**

Normally, the amplitude of the outer race frequency is higher than that of the inner race frequency. This is likely due to the fact that the transducer itself is much closer to the outer race (see Figure 6.09A). In addition, the vibration signal from the inner race frequency (or harmonics) must pass through several more interfaces including the constantly rotating rolling elements on its path out to the vibration transducer.

## **6. Usual Order of Appearance by Defect Frequencies:**

Normally, defects will first appear on the races before the problem propagates to the rolling elements and cage. Thus, outer and inner race frequencies are normally first established before the appearance of a ball spin frequency. Later, the cage frequency normally can appear either as a fundamental or as a sideband to another frequency. Likewise, the ball spin frequency will sometimes appear as a sideband above and below an outer or inner race defect frequency.

## **7. Where Cage Frequency Normally Appears:**

Although rolling element bearing problems are classically thought of as high frequency problems, fundamental cage frequencies will always be subsynchronous, ranging from approximately .33X RPM up to .48X RPM with the majority falling between .35X and .45X RPM. However, the cage frequency will not normally appear at its fundamental frequency. Instead, it most often will appear as a sideband around ball spin frequency (BSF), or around one of the race frequencies (BPFO or BPFI) with the sideband difference frequency equal to the cage frequency (FTF). For example, see Figure 6.09I which shows an outer race frequency BPFO at 10,260 CPM (5.78X RPM) with cage frequency sidebands (FTF) spaced at 720 CPM (.41 X running speed which is 1775 RPM). This is how a cage frequency most often will appear.

## **8. Frequencies Generated by Faults on Balls or Rollers:**

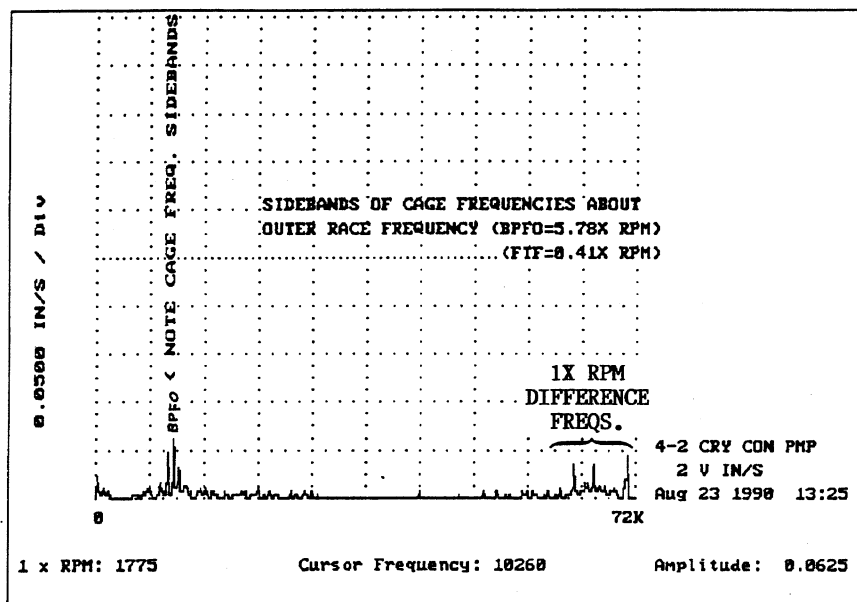
When defects occur on rolling elements themselves, they will often generate a frequency not only at the ball spin frequency (BSF), but also at the cage frequency (FTF, also known as fundamental train frequency).

## **9. Ball Spin Frequency May Appear if Cage is Broken:**

The appearance of a ball spin frequency does not always necessarily mean there is a defect on the rolling elements. However, it still means that there is a problem present. In this case, it can indicate that a cage is broken at a rivet and if the balls are thrusting hard against the cage (as per Reference 12).

## **10. Frequency Generated if More than One Rolling Element has Faults:**

If more than one rolling element has defects, a frequency equal to the number of balls having defects X the ball spin frequency will be generated. In other words, if defects are present on 5 balls or rollers, a frequency at 5X Ball Spin Frequency would most often appear.



002% x 0.500 = 0.0100 1 x RPM: 1775.00

| Machine         |         | Pos                  | Dir       | Units   | Date              |  |
|-----------------|---------|----------------------|-----------|---------|-------------------|--|
| 4-2 CRY CON PMP |         | 2                    | V         | IN/S    | Aug 23 1990 13:25 |  |
| Frequency       | 1 x RPM | Amplitude            | Frequency | 1 x RPM | Amplitude         |  |
| 180             | 0.10    | 0.0225               | 67140     | 37.83   | 0.0125            |  |
| 1080            | 0.61    | 0.0100               | 67860     | 38.23   | 0.0125            |  |
| 6840            | 3.85    | 0.0100               | 69120     | 38.94   | 0.0100            |  |
| 7200            | 4.06    | 0.0125               | 70920     | 39.95   | 0.0425            |  |
| 8640            | 4.87    | 0.0150               |           |         |                   |  |
| 9540            | 5.37    | 0.0475 ← -FTF (.41x) |           |         |                   |  |
| 10260           | 5.78    | 0.0625 ← BPFO        |           |         |                   |  |
| 10980           | 6.19    | 0.0325 ← +FTF (.41x) |           |         |                   |  |
| 11880           | 6.69    | 0.0125               |           |         |                   |  |
| 13860           | 7.81    | 0.0100               |           |         |                   |  |
| 14580           | 8.21    | 0.0100               |           |         |                   |  |
| 21600           | 12.17   | 0.0100               |           |         |                   |  |
| 61920           | 34.88   | 0.0100               |           |         |                   |  |
| 63720           | 35.90   | 0.0350               |           |         |                   |  |
| 65160           | 36.71   | 0.0150               |           |         |                   |  |
| 65520           | 36.91   | 0.0150               |           |         |                   |  |
| 66420           | 37.42   | 0.0350               |           |         |                   |  |

**FIGURE 6.09I**  
**HOW THE CAGE FREQUENCY (FTF) MOST OFTEN APPEARS**  
**WITHIN A VIBRATION SPECTRUM**

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## 11. Allowable Vibration at Bearing Defect Frequencies:

It is very difficult to assign definite vibration severity levels to bearing defect frequencies similar to that which is commonly done for unbalance amplitude at 1X RPM. For one, there are a variety of rolling element bearings in a variety of different machines, each of which may provide different paths for the vibration signal to the transducer. However, one can make a statement which will apply almost uniformly for all the various types and combinations - the allowable vibration for unbalance at 1X RPM is much greater than that allowed for a defect frequency for a rolling element bearing.

Much research has proven that no absolute answer can be given to allowable vibration amplitudes at bearing defect frequencies. Not only does it depend on which particular machine a bearing is installed and what the operating speed is, it also depends greatly on which bearing failure scenario path it will travel. Each of 6 bearing failure scenarios are identified in Section 6.093. For example, on rare occasions, the author has personally witnessed bearings which still did not have significant damage even with an amplitude of .30 in/sec at a fundamental BPFO when this damage was concentrated at one particular location on the outer race (as described by Bearing Failure Scenario B in Section 6.093). On the other hand, considerable damage has been discovered in other bearings when no one defect frequency had an amplitude greater than only .03 in/sec in everyday machinery like pumps and blowers running at common speeds such as 1780 RPM. In fact, extensive damage has been found in large dryer roll bearings on paper machines running less than 100 RPM with bearing frequency amplitudes that ranged from only .003 to .006 in/sec.

The key point common to each of the latter two scenarios having low vibration but significant bearing damage was that in each case, not just one bearing frequency was present in the spectra; instead, a number of bearing frequency harmonics were present (a "number of bearing frequencies present" means for example that either 3 to 5 BPFO or BPFI harmonics might simultaneously be present; or that 2 or more harmonics of BPFO might be present along with 2 or more harmonics of BPFI). In addition, further investigation has shown that when these bearing frequency harmonics were surrounded by sideband frequencies spaced at 1X RPM of the problem bearing, even more damage is indicated (particularly if these 1X RPM sidebands surround BPFO harmonics, assuming the bearing is press fit on the shaft). It should also be emphasized that these sidebands may be spaced at bearing frequencies themselves rather than at 1X RPM (i.e., 4 or 5 BPFO harmonics may all be present, each with sidebands of FTF or BSF above and below them which would likewise indicate a potentially serious problem).

**Therefore, the most important thing to look for indicating significant bearing wear is the presence of a number of bearing defect frequency harmonics, particularly if they are surrounded by sidebands spaced at either 1X RPM or sidebands spaced at other defect frequencies of the bearing - independent of amplitude. If these are present in a spectrum, replace the bearing as soon as possible.**

## 12. Evaluating Bearings on Low-Speed Machinery (less than 250 RPM):

Reference 9 points out special precautions must be taken when making measurements on low-speed machinery. Rolling element bearings have been successfully evaluated at speeds as low as 1.5 RPM. However, one must be particularly aware of the low frequency limitations of both his analyzer and his transducer. If he uses one of the computer-loaded data collectors popular today, he should be aware that many of these instruments are outfitted with high-pass filters that begin filtering out signals below fixed frequency ranges at rates of from approximately 12 to 24 dB/octave. This cutoff frequency may be close to DC (0 Hz) on some data collectors, but may be as high as 8 Hz (480 CPM) on other models. Thus, in the

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latter case, a machine may be generating 1.00 in/sec of vibration at 240 RPM, but the instrument may display only approximately .25 in/sec at this frequency if it has such a high-pass filter. Obviously, if one attempted to make measurements at frequencies below 120 CPM with this instrument, the data collector would detect and display only a small percentage of this peak. And, if one wanted to single or double integrate the signal from acceleration to velocity or displacement, respectively, there would likely be even greater suppression of the amplitude in the data collector (also, it is important to point out that some data collectors do not integrate acceleration signals to velocity at frequencies below approximately 120 CPM; data below this cutoff frequency is left nonintegrated). In addition to filtering, the low frequency response of the analyzer is also effected by its signal conditioning and input circuitry components which also needs to be considered.

Not only does one have to be concerned about the analyzer during low frequency measurements, he also must be aware of the frequency response of the transducer whether it is an accelerometer, a velocity pickup, or a proximity probe. Typically, most general purpose accelerometers in use with data collectors today are flat within 5% between approximately 5 Hz and 10,000 Hz (300 - 600,000 CPM). However, special seismic piezoelectric accelerometers can be obtained which will extend the  $\pm 5\%$  flat response down to as low as approximately 0.1 Hz (6 CPM). These transducers will typically have exceptionally high sensitivities ranging from 1000 to 10,000 mV/g (as compared to only 10 to 100 mV/g for the standard accelerometer) and will normally have much greater weight on the order of 400 to 1000 grams (as compared to only 10 to 50 grams for the standard unit). They also will typically have much longer discharge time constants of 20 seconds or more (compared to only 0.5 second for standard accelerometers) and will work best when connected to analyzers having input impedances of approximately 1,000,000 ohms. Finally, the low frequency performance of these transducers can be further enhanced by connecting them to special power supplies and signal conditioning equipment.

In addition to the instrument limitations discussed, it should be pointed out that both 1X RPM and bearing frequency amplitudes themselves will inherently be much lower on large, low-speed machinery. In the case of 1X RPM, there will be minimal unbalance forces since these vary with the square of speed. Thus, if one attempts to specify spectral alarm bands for this machinery, he will have to spec them at much lower alarm amplitudes than for machinery rotating above 1200 RPM. The best way to specify them will be to capture actual data and perform statistical analysis on the overall levels and those of individual frequency bands.

Fortunately, even though many of today's standard data collectors and transducers may not be capable of evaluating 1X RPM and 2X RPM vibration on much low-speed machinery, these same instruments might still be able to satisfactorily evaluate the health of their bearings. Please refer to Figure 6.09G which shows some typical rolling element bearing frequencies. Even though the fundamental cage (FTF) and ball spin frequencies (BSF) may still be below the "reach" of the system, the more common outer race (BPFO) and inner race frequencies (BPF1) will most often range from 4 to 12X RPM. Therefore, a measurement system which could not "see" vibration at, for example, 100 RPM speed very likely could detect fundamental outer and inner race frequencies which would probably range from about 500 to 1000 CPM on this machine with little or no loss of signals. And, of course, harmonics of these bearing frequencies could easily be evaluated.

Importantly, when evaluating bearings on machines particularly running below 100 RPM, it is highly recommended that both time waveform and FFT spectra be captured. Often, when speeds drop to these levels, impulses which occur when the rolling elements roll past defects on the races will not have sufficient energy to generate clearly detectable defect frequencies in FFT spectra, but still might clearly be seen in the time waveform. One such example is given in Figure 6.09J in which case the time waveform is shown directly below the FFT

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spectrum. In this particular case, the analyst would likely have seen the inner race defect frequencies present in the FFT spectrum. However, note they were likewise clearly present at much greater amplitude in the time waveform. For example, the time waveform amplitude of these pulses spaced at the inverse of the inner race frequency (1/BPFI) had amplitudes of about .25 in/sec, compared to amplitudes of about only .01 in/sec in the FFT spectrum (or about 25 times higher amplitude in the waveform). This is very common with defects associated with rolling element bearings and gears which cause short-lived impacts in the waveform. When the FFT mathematics are performed, these waveform transient pulse amplitudes are often heavily suppressed when converted into the spectrum, and often will not even be noticed, particularly if there are frequencies present associated with other mechanical problems of much higher amplitude (such as unbalance at 1X RPM, misalignment at 2X RPM, blade pass frequency, etc.).

When speeds drop to very low levels on the order of 1 to 20 RPM, FFT spectra alone will almost never detect rolling element bearing problems. However, the time waveform may still reveal the bearing problems as demonstrated by Figure 6.09K which from Reference 13. In this example, taken on a 200 ton BOF (basic oxygen furnace) vessel trunnion bearing in a steel mill turning at a nominal speed of 1.5 RPM (40 sec/rev), the time waveform clearly showed an outer race defect frequency (BPFO) when the spectrum did not. Here, BPFO equaled 20.4 CPM (0.34 Hz) as compared to the operating speed of 1.5 RPM (0.025 Hz). Notice each of the pronounced spikes produced when the trunnion bearing rollers passed over and impacted the defects on the outer race (also note that the machine speed actually varied from about 1.3 to 1.7 RPM during the analysis). The frequency was calculated by measuring the time between impacts (sec/cycle) and inverting the result. For example, referring to Figure 6.09K, taking the first annotated times (3.367 sec - 0.427 sec = 2.940 sec), these would correspond to a frequency of 1/2.940 sec or 0.34 Hz (20.4 CPM), which equalled BPFO for this bearing.

Reference 26 points out that when trying to decide upon an optimum sampling time ( $t_{MAX}$ ) which should be specified for low-speed machines, one should normally specify a sampling time equaling approximately 2 to 4 shaft revolutions. That is, if the machine were rotating at 60 RPM (1.0 rev/sec), this would correspond to 1 sec/rev. Therefore, a  $t_{MAX}$  of about 2 to 4 seconds should be specified to capture such bearing problems, in addition to capturing the FFT spectrum as well. On the other hand, if the machine were running at only 10 RPM (6 sec/rev), a sampling time ( $t_{MAX}$ ) of about 12 to 24 seconds should be employed. It should be pointed out for very long sampling times such as these, it would be better to specify at least 2048 samples (corresponding to an 800 line FFT) rather than the usual 1024 samples (400 line FFT). Doing so will provide a sufficient number of samples so that the full amplitudes will be traversed in the time waveform, besides drawing a much truer picture of the actual vibration response of the machine in the waveform. Note that such time waveforms should be non-synchronous waveforms rather than synchronous time waveforms (since synchronous waveforms actually remove bearing defect frequencies from both the waveform and FFT since they are not integer multiples of running speed).

For now, the most important point is that one can successfully evaluate bearing health on low-speed machinery if he takes into account the frequency response performance characteristics of his instrumentation system and takes the necessary provisions outlined in this section.

RECOMMENDED SAMPLING TIME ( $t_{MAX}$ ) AND NUMBER OF SAMPLES  
FOR TIME WAVEFORMS ON VARIOUS LOW-SPEED MACHINES

| RPM | SEC<br>REV | RECOM.<br>$t_{MAX}$ | RECOM.<br># SAMPLES | RESULTANT<br>$F_{MAX}$ |
|-----|------------|---------------------|---------------------|------------------------|
| 600 | .100       | .400 sec            | 1024                | 1000 Hz (60,000 CPM)   |
| 500 | .120       | .400                |                     | 1000 Hz (60,000 CPM)   |
| 450 | .133       | .400                |                     | 1000 Hz (60,000 CPM)   |
| 400 | .150       | .500                |                     | 800 Hz (48,000 CPM)    |
| 350 | .171       | .500                |                     | 800 Hz (48,000 CPM)    |
| 300 | .200       | .800                |                     | 500 Hz (30,000 CPM)    |
| 240 | .250       | .800                |                     | 500 Hz (30,000 CPM)    |
| 180 | .333       | 1.000               |                     | 400 Hz (24,000 CPM)    |
| 150 | .400       | 1.000               |                     | 400 Hz (24,000 CPM)    |
| 120 | .500       | 1.333               |                     | 300 Hz (18,000 CPM)    |
| 100 | .600       | 2.000               | 2048                | 200 Hz (12,000 CPM)    |
| 80  | .750       | 2.667               |                     | 150 Hz (9,000 CPM)     |
| 60  | 1.000      | 4.000               |                     | 200 Hz (12,000 CPM)    |
| 50  | 1.167      | 4.000               |                     | 200 Hz (12,000 CPM)    |
| 40  | 1.500      | 4.000               |                     | 200 Hz (12,000 CPM)    |
| 30  | 2.000      | 8.000               |                     | 100 Hz (6,000 CPM)     |
| 20  | 3.000      | 8.000               |                     | 100 Hz (6,000 CPM)     |
| 15  | 4.000      | 16.000              |                     | 50 Hz (3,000 CPM)      |
| 12  | 5.000      | 16.000              |                     | 50 Hz (3,000 CPM)      |
| 10  | 6.000      | 16.000              |                     | 50 Hz (3,000 CPM)      |
| 6   | 10.000     | 32.000              |                     | 25 Hz (1,500 CPM)      |
| 3   | 20.000     | 80.000              |                     | 10 Hz (600 CPM)        |

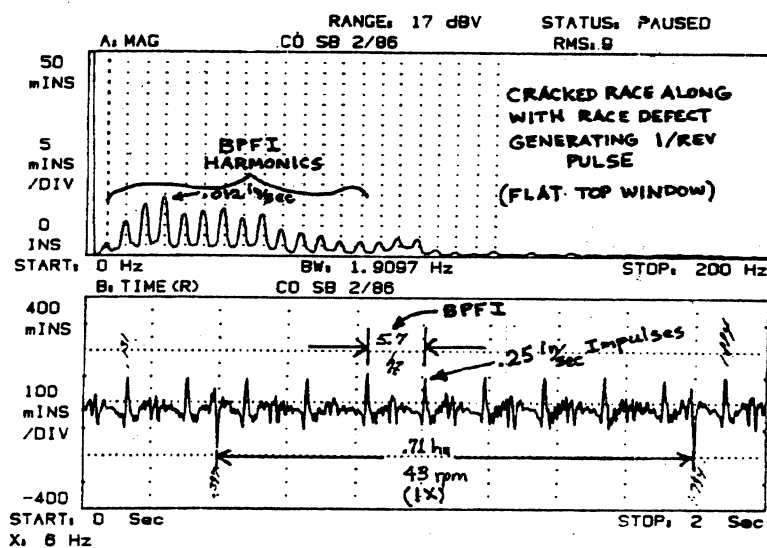
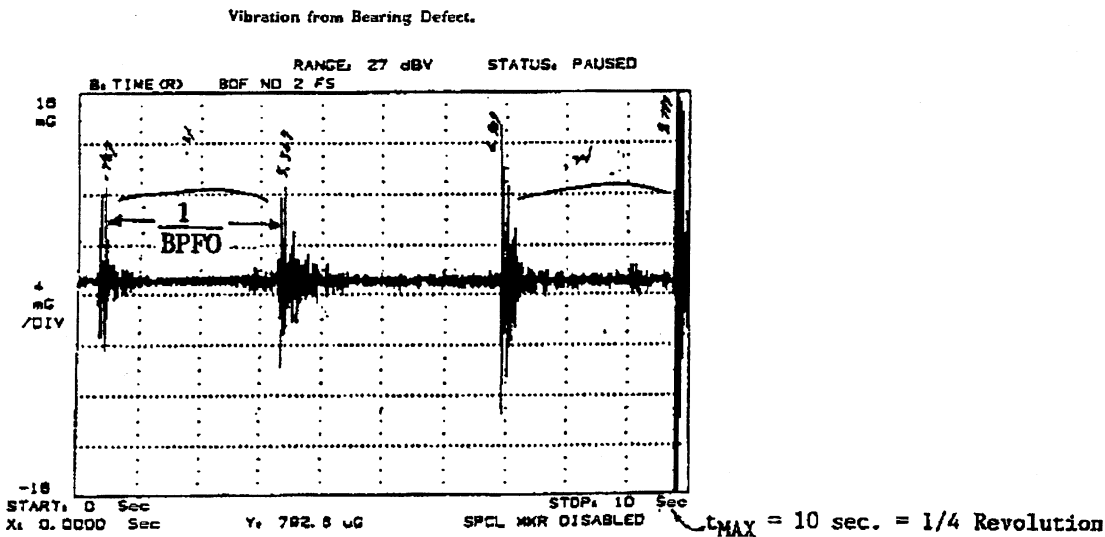


FIGURE 6.09J  
CRACKED INNER RACE WAVEFORM & SPECTRUM FOR A 43 RPM ROLL  
(Note the Pronounced Positive & Negative Pulses in the Time Waveform)  
(Ref. 13)



**FIGURE 6.09K**  
**DETECTION OF BEARING FAULTS USING THE TIME DOMAIN ON A**  
**TRUNNION BEARING OPERATING AT ABOUT 1.5 RPM (As Per Ref. 13)**

**13. Where Transducer Should be Placed to Properly Detect Bearing Frequencies:**

It is most important to place the vibration transducer as close as possible to the load zone of the bearing, particularly if the bearing only supports radial load. Reference 13 points out that this is of critical importance for a spherical roller bearing in which he claims the vibration signal strength can be affected by as much as 100% if the transducer is not placed in the load zone, giving the analyst a false sense of security when he actually does have a problem.

**14. Approximate Bearing Defect Frequencies if only the Number of Rolling Elements is Known:**

Figure 6.09F provides 4 formulas which approximate the bearing defect frequencies if only the number of rolling elements ( $N_b$ ) is known (as per Reference 18). These equations have been proven to be considerably more accurate than older approximations which assumed BPFO to equal approximately  $.4 \times N_b \times \text{RPM}$  and BPFI to equal approximately  $.6 \times N_b \times \text{RPM}$ . Still, optimum precision can only be realized if one knows each of the other bearing design parameters ( $B_d$ ,  $P_d$  and  $\emptyset$ ), particularly in the case of the ball spin frequency (BSF).

**15. Sensing of Improper Bearing Load or Installation:**

Not only can bearing frequencies be used to detect defects within the bearings, but also they can be used to detect when bearings are improperly loaded or installed. For example, even when a new bearing is installed, if there is excessive interference of the bearing seat on the bearing housing causing it to be "jammed into" the seat, it can result in take-up of all the internal clearances forcing the rolling elements against the races. If this occurs, the bearing will immediately generate ball pass frequencies of the outer and/or inner race upon startup. In addition, the author has taken data several times when thrust bearings were installed backwards. When this has occurred, the improperly installed thrust bearings have generated excessively high amplitudes at race defect frequencies, sometimes on the order of 1.0 in/sec, or greater. Excessive press fit of a bearing onto a shaft can also immediately generate a pronounced outer race or inner race defect frequency (BPFO or BPFI), letting the analyst

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know excessive and/or improper load has been placed on the bearing. And, even though no real wear might yet have occurred, if the problem is not detected and corrected, the bearing will likely fail long before its predicted life. In all improperly loaded situations, if the same bearings were reinstalled properly or loaded properly, the defect frequencies were significantly reduced or disappeared altogether. This has happened on a number of occasions. In fact, one client manufacturing textile machinery uses this technique to detect assembly problems in its quality assurance program.

#### **16. Frequencies Generated by Inadequate Bearing Lubrication:**

Reference 12 states that "unique signatures generated by inadequately lubricated bearings are characterized by 3 or 4 peaks in the frequency range of 900 to 1600 Hz (54,000 to 96,000 CPM). The difference frequency between the peaks ranges from 80 to 130 Hz (4800 to 7800 CPM). Some signatures of properly lubricated bearings contain these frequency components; however, the amplitude is very low - about 0.05 in/sec or less. The amplitude increases to as much as 0.10 or 0.20 in/sec when the lubrication is inadequate". He adds that "empirical evidence indicates that frequencies from 900 to 1600 Hz are natural frequencies of the installed bearing." With this in mind, if frequencies within this range (approximately 50,000 to 100,000 CPM) do occur with difference frequencies on the order of 5000 to 8000 CPM, it may be a good idea to check lubrication, particularly if high spike energy (or equivalent) levels are also measured on this same bearing housing. Also, even greater evidence of lubrication problems would occur if neither the high spike energy nor a spectrum having these components was present on the other bearing that is supporting this same shaft.

In addition, some research has shown that inadequately lubricated bearings which allow metal-to-metal contact can also generate bearing defect frequencies. In these cases, one might go ahead and add lubricant while monitoring both overall spike energy (or equivalent) and vibration FFT spectra to determine if the ultrasonic levels drop considerably, and to find if the bearing defect frequencies disappear from the spectrum. If they do, he should return to the machine 12 to 24 hours later and see if either have reoccurred. If not, lubrication was likely the problem. If they do return, the bearing is likely within one of the failure scenarios described in Section 6.093.

#### **17. Defect Frequencies for So-Called "Equivalent Bearings":**

The APPENDIX includes tables which allow one to determine "equivalent bearings" if he knows the AFBMA numbers of bearings within this machine (AFBMA numbers are often listed on the nameplates of many machines). As these tables show, this AFBMA number allows one to know several facts about the bearings including their bore, type (deep groove, angular contact, cylindrical roller, etc.), dimensional series (extra light, light, medium, etc.), type of cage or ball retainer, recommended lubricant and bearing tolerance/clearance. However, simply knowing the AFBMA number and corresponding "equivalent bearing" does not verify what bearing defect frequencies should be expected. Research has shown that so-called "equivalent bearings" have defect frequencies which can vary by as much as 22% due to differing numbers of rolling elements, different rolling element diameters, etc. That is, knowing that the AFBMA number is 65BC02JPP0 might mean that the bearing is either an SKF 6213, MRC 213S, Fafnir 213K, Norma-Hoffmann 213 or New Departure 3213 bearing. However, if one compared the actual bearing defect frequencies for these or other bearings, he will often find bearing frequencies which vary by a noticeable amount. Still, if only the AFBMA number is known, it is usually a good idea to at least record this number and to determine the equivalent bearings from the various manufacturers. Then, the analyst should list the bearing defect frequencies for 5 of 6 of these "equivalent bearings" so that he might have some idea where he might find the defect frequencies for these bearings, and how much the defect frequencies differ from one another.

## 18. Electrically-Induced Fluting Problems Within Motor Bearings:

Within the last few years, it has been discovered that vibration signatures can be used to detect electrically-induced fluting within rolling element bearings. This has been found to be a widespread problem, particularly with the outboard (or opposite drive-end) bearings of DC motors. Reference 27 provides information from one author on how he has used vibration analysis to detect the presence of such electrical fluting and also relates he was surprised to find how little voltage can be present, and yet, do so much damage to the bearing surfaces. In fact, in one of his experiments, he found a voltage of only .4 to .5 volt passing through the bearing causing considerable damage. He also found the following when he had electrical fluting,

"a spike that did not match any bearing frequency occurred between 40,000 - 60,000 CPM. Most often the spike was accompanied by the sidebands of whatever race had the biggest defect. It also could occur for both races. This was the one way we could routinely identify the electrical fluting problems. Another way to identify electrical fluting was listening to the motor itself. We noticed the motors sounded like they were under a heavy strain, yet the motors were under no load or under their normal load. As the bearing defect worsens, the sound intensifies and becomes deeper."

Figure 6.09L shows a velocity spectrum taken by this Reference 27 article taken on a DC motor bearing with electrically-induced fluting. His point here was that no real defect frequencies showed up at either BPFO or BPFI multiples. Instead, difference frequencies surrounded a significant spike which occurred at roughly 54,000 CPM (900 Hz). He reported the difference frequencies equalled the outer race and inner race frequencies in this spectrum.

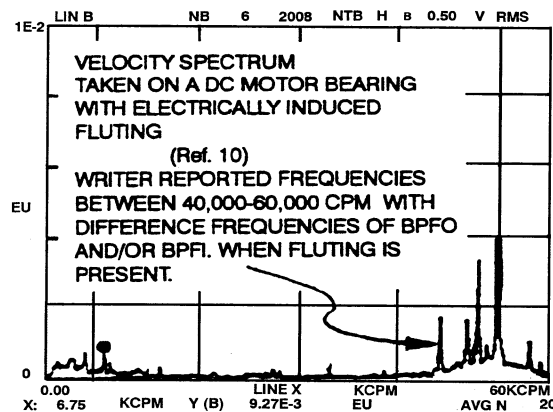


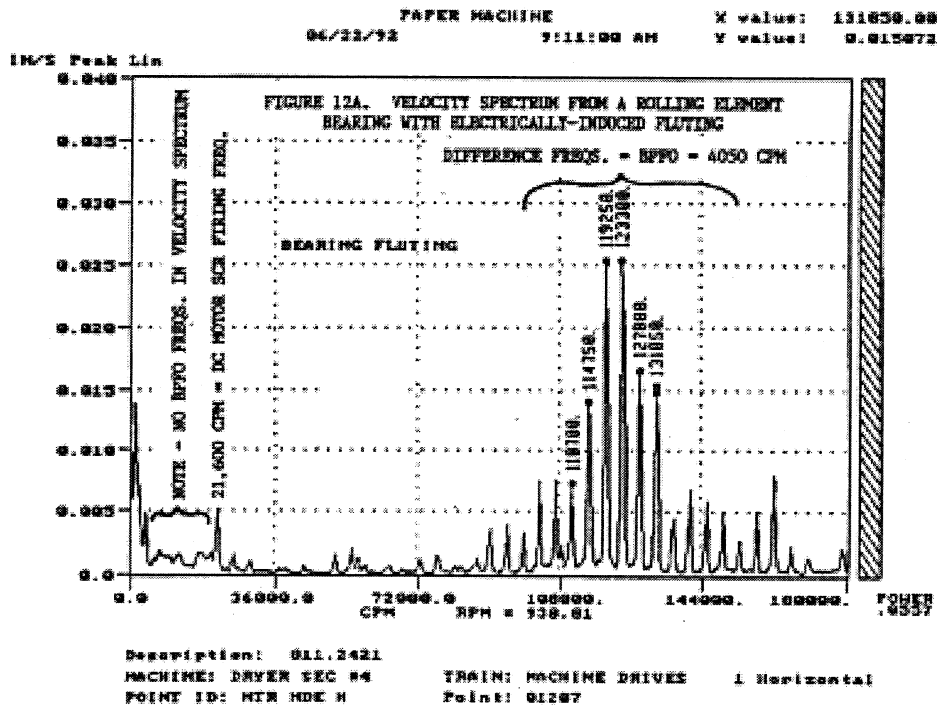
FIGURE 6.09L

Similar data was found by another analyst as shown in Figure 6.09M. Here, both velocity and spike energy spectra were taken on the outboard bearing of a DC motor having electrically induced fluting on its outer race. However, notice the great difference in the velocity and spike energy spectra. The velocity spectrum did not show any frequencies at the outer race frequency (BPFO), or BPFO harmonics. Instead, it showed a series of difference frequencies spaced at BPFO (approx. 4050 CPM) which were concentrated around peaks at 119,150 and 123,300 CPM. On the other hand, note the spike energy spectrum shown in Figure 6.09M-B which did show, in this case, 7 harmonics of outer race frequency (BPFO), with no other discernible frequencies in

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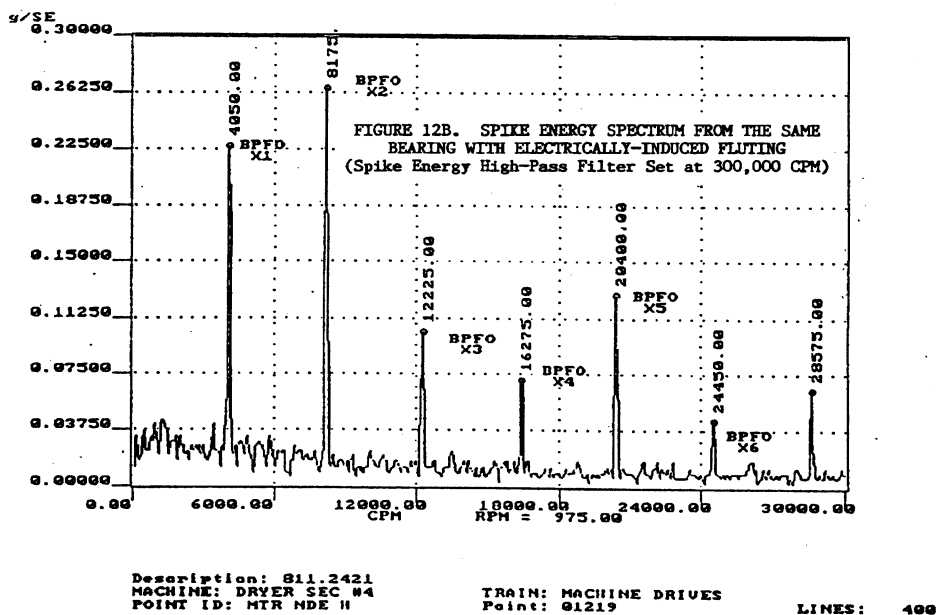
the spike energy spectra. One of the most important findings of these and other tests conducted by the author show that an analyst should capture at least one high frequency velocity measurement out to a frequency of approximately 180,000 CPM (3000 Hz) in order to detect possible electrically-induced fluting on DC motors. If he takes this data only in one location of a DC motor, it would likely best be taken on the outboard bearing. However, the author strongly recommends a similar high frequency measurement be taken on the inboard bearing (coupling side) as well (since higher frequency vibration does not often travel). Both measurements should be taken in the horizontal direction.

The writer of Reference 27 provided two solutions to the undesirable passage of electrical current through his motor bearings. In one case, the rear bearing on the motor was insulated by boring out the end bell of the motor and inserting a phenolic sleeve. Then the bearing was pressed back into the sleeve, thereby interrupting the path for current flow between the rear of the motor and the end bell. He reported this eliminated the electrical fluting problems in these bearings (however, the author has been told by several analysts that this solution works only about 80% to 90% of the time - not always). The Reference 27 analyst also tried installing carbon brushes on the end of the rear bearings where the shaft goes through the bearing. This resolved this particular problem, but he added that when the brushes wear, electrical current could once again pass through the bearings and do considerable damage if not detected soon enough.



PAPER MACHINE 06/29/92 01:04:00 PM X value: 28375.900  
Y value: 0.0650583

PAPER MACHINE 06/29/92 01:04:00 PM X value: 28375.900  
Y value: 0.0650583



**FIGURE 6.09M**  
**COMPARISON OF VELOCITY & SPIKE ENERGY SPECTRA TAKEN ON THE OUTBOARD BEARING OF A DC MOTOR HAVING FLUTING ON ITS OUTER RACE DUE TO PASSAGE OF ELECTRICAL CURRENT THROUGH THE BEARING**

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#### (d) Sum and Difference Frequencies:

When a single defect has developed on a bearing, it will generate a defect frequency and will begin deteriorating. When the defect grows, it can contribute to the development of other defects in the bearing. In so doing, other frequencies will be generated and a number of patterns might develop. Some frequencies will add to and subtract from others. In fact, the fundamental frequency for a particular defect might never occur. When it appears, this defect frequency might act as a sideband to other frequencies which are already present. For example, a cage frequency itself will not normally occur at its fundamental frequency of approximately .35X to .45X RPM. Instead, the cage frequency itself normally will sideband the race frequencies (BPFO or BPFI) or the ball spin frequency of the bearing (see Figure 6.09I). In some cases, bearing defect frequencies can even modulate frequencies generated by sources other than bearings. Following below are some of the more important facts concerning sum and difference frequencies and their behavior:

1. It is not uncommon for the fundamental ball spin frequency or harmonics never to appear even if significant faults are present on the rolling elements. In this case, BSF will appear, but as a sideband of other frequencies.
2. If only a single fault is present on either the outer or inner race, only a single race frequency will appear. However, when faults begin to appear around the periphery, a number of harmonics of these race frequencies will appear.
3. When defects grow on raceways, the amplitude of the race frequencies themselves will often increase somewhat. However, even greater indicators of deterioration are the number of bearing frequency harmonics, as well as the appearance of 1X RPM sideband frequencies above and below the race frequencies (particularly BPFO). In effect, the unbalance forces at shaft speed will tend to modulate the frequency components of the races, generating sidebands. In the case of BPFI, 1X RPM sidebands are often created around it since its amplitude is modulated with much greater response when the inner race defect impacts rolling elements within the load zone than those generated outside the load zone. The rate of this modulation in amplitude is at 1X RPM, therefore generating the 1X RPM sidebands (see Figure 6.09H).
4. In rolling element bearing frequency analysis, the emphasis is not on amplitude, but on content of the spectra. In fact, amplitudes of bearing frequencies themselves often begin to drop as condition worsens, particularly in the case where faults begin to propagate around the periphery of the outer or inner race. In this case, serious weight should be placed on the fact that a multiple number of fault frequencies are appearing and that many of these fault frequencies are sidebanded by vibration at bearing RPM.
5. As deterioration continues, modulation effects can continue to have more and more influence until eventually, the spectrum can become a series of nothing but 1X RPM harmonics.
6. Reference 14 shows that double row bearings having staggered rolling elements can generate 2 sets of bearing defect frequencies - (a) If a defect is confined to only one side of the raceway, the bearing defect frequencies generated should be calculated using the number of rolling elements only in that single row; however, (b) if a defect occurs on both sides of the raceway, the frequencies should be calculated using the total number of rolling elements. Here again, either of these frequencies can act as a sideband around other frequencies with the sideband difference depending on whether the fault has occurred on one or both raceways. Importantly, this number of balls (either those for one row or the total number) must be entered into the outer race, inner race and ball spin frequencies.

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7. A single defect on the inner race of a rolling element can be difficult to diagnose from frequency spectra alone since amplitudes will most often be low, and discrete spectral lines might not be seen at calculated defect frequencies. This particularly applies to bearings with inner race rotation mounted with an interference fit between the inner race and the shaft. In these cases, the load zone might be noticeably less than 180° which explains why discrete peaks at defect frequencies may not appear. Reference 15 points out the real problem occurs if a crack appears on the inner race. In these cases, the shaft fit can be relieved and internal clearances of the bearing lost. He points out that such a situation is extremely dangerous which can bring about rapid seizure of the bearing causing the inner race to spin on the shaft or the outer race to spin in the housing resulting in catastrophic failure. In these cases, since the inner race frequency itself or harmonics may not show, but act as a sideband on other frequencies, he recommends referring to the time domain (or time waveform) in addition to the vibration spectra themselves.
  8. In rolling element bearings, when multiple running speed harmonics are present, they can signal either looseness of the bearing on the shaft or within its seat or, much more importantly, they can likewise signal a bearing turning on the shaft or in the housing. The author has been involved in several situations where a number of very low amplitude running speed harmonics (less than .04 in/sec) were present which resulted in catastrophic failures, even with only the low amplitudes. In many of these cases, there were in fact no real defects within the bearings themselves, but the bearing turning on the shaft resulted in severe shaft damage (in one case, over .25 inch diameter was lost before eventual failure).

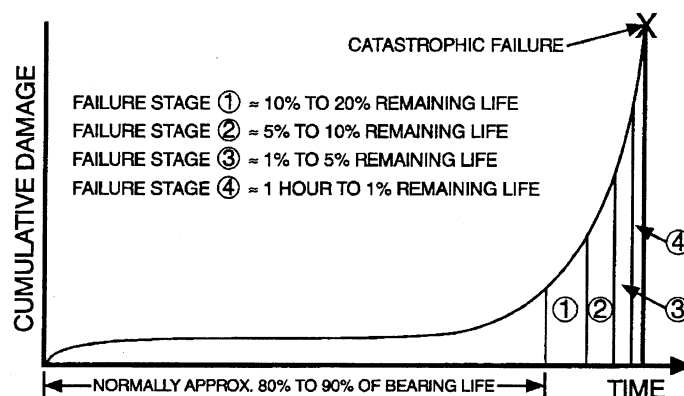
Thus, there is great information present within the signatures of rolling element bearings that will help identify their current condition and assist the analyst in recommending possible corrective actions. Following in Section 6.093 will be a presentation of a series of typical spectra which can be used by the analyst to track the condition of rolling element bearings under some of the most common failure scenarios which have occurred in the experience of the author.

### 6.093 TYPICAL SPECTRA FOR TRACKING FAILURE STAGES THROUGH WHICH ROLLING ELEMENT BEARINGS PASS:

During the last decade, concentrated research and experimental investigations have been conducted on how to best evaluate rolling element bearing condition using vibration analysis and high frequency enveloping techniques. Some of the better papers which have been written include References 2, 12 and 19. In addition, Technical Associates has been deeply involved in further developing rolling element bearing diagnostic capabilities. We have consulted with a variety of clientele on a wide array of machinery ranging from massive, low-speed machines outfitted with large, expensive rolling element bearings to very small, high speed rolling element bearing machinery.

Experiments have been conducted purposely running rolling element bearings to failure while carefully tracking them with vibration signatures (Reference 11). At various intervals, the bearings were broken open and closely inspected for the extent of damage, the type of deterioration and the probable cause for the onset of faults within the bearings. These results would then be closely correlated with the vibration spectra, along with various ultrasonic frequency measurements (i.e., Spike Energy, Shock Pulse and HFD). The bearings were purposely opened up and examined when such data indicated them to be in a wide range of condition (from those with supposedly no faults to those indicated to have extensive damage). Some only had increases in ultrasonic measurements (not in vibration); others showed increases only in high frequency regions; while clearly defined bearing defect frequencies were within vibration spectra of others (some were allowed to catastrophically fail while vibration and ultrasonic responses were closely tracked). Such experiments were performed on a variety of rolling element bearings at different speeds and conditions. This work has enabled the development of fairly well proven diagnostic methods for the tracking of rolling element bearing condition using vibration signature analysis, along with time waveform and ultrasonic analysis.

It was found that the majority of rolling element bearings followed a fairly predictable failure path from the very onset of deterioration through eventual catastrophic failure. This failure path is graphically portrayed in Figure 6.09N which plots bearing damage versus time. Note importantly that bearing damage typically will increase exponentially during the final 10% to 20% of its life. It is here where intensive research and field investigations have found how to use vibration analysis and high frequency enveloping tools to identify "failure stages". From this, a classic 4 stage failure scenario has been developed which will apply to approximately 80% of rolling element bearing failures. This will be presented in Figure 6.09P as Scenario A entitled "4 Primary Failure Stages Through Which Most Rolling Element Bearings Pass". Text describing each of the 4 failure stages will accompany Figure 6.09P.



**FIGURE 6.09N**  
**TYPICAL FAILURE PATH TAKEN BY ROLLING ELEMENT BEARINGS**

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As was expected, not all rolling element bearings followed the classic failure path (Figures 6.09N and 6.09P). Therefore, this paper will attempt to both illustrate and document the other failure paths which were taken by a much smaller percentage of rolling element bearings which will likewise be illustrated and documented in Scenarios B through F, respectively.

It is difficult to categorically state the exact percentage of total failures represented by each of the six failure scenarios presented. However, at this time, it appears that Scenario A alone probably represents approximately 80% of all rolling element failure paths. Therefore, that for all the remaining five failure scenarios represents only approximately 20% of the total population.

#### **SCENARIO A. 4 PRIMARY FAILURE STAGES THROUGH WHICH MOST ROLLING ELEMENT BEARINGS PASS (FIGURE 6.09P):**

Figure 6.09P presents typical velocity spectra for each of the 4 failure stages for most rolling element bearings. These spectra follow the bearing from the very onset of bearing problems in Stage 1 through imminent failure of the bearing in Stage 4 (see Figure 6.09N). Note that the overall spike energy (or HFD) amplitudes given to the right of each spectrum are meant to be rough approximations only. Also, note that documentation below the title of each one of the stages refers to an approximate " $L_{10}$  Life Remaining" which corresponds to the approximate remaining anticipated life of the bearing based on a 90% confidence level ( $L_{10}$  Life is documented and its formula is given in the introductory discussion on page 1). Today, in general, the machine designer normally will attempt to provide a design life of approximately 5 to 10 years for most common machinery. Therefore, when a percentage on the order of 10% of  $L_{10}$  Life is quoted below, this will generally mean from 6 months to 1 year remaining life anticipated (depending on the type of machine and its intended application). Of course, this can vary widely with the machine type, with particular design parameters and whether or not the bearings are provided with the proper lubrication, operating temperature and subjected to acceptable vibration levels.

It is important to point out that bearings do not follow a linear deterioration path, but instead tend to fail exponentially; that is, when the bearing for example enters Stage 3 outlined below, it may still have a fair remaining life, or it may fail rapidly. Once the bearing enters Stage 3, the failure can progress rapidly, particularly towards the end of this stage. Generally, low speed bearings may still have fairly predictable remaining life (unless exposed to high dynamic loads - see  $L_{10}$  Life Formula discussion on page 1 which shows that bearing life is inversely proportional to the 3rd power of imposed loading).

Table 6.09A compiled by Charles Berggren in Reference 16 (reprinted here with his permission) roughly adheres to the 4 stages illustrated in Figure 6.09P and documented below:

##### **Stage 1 (Approximately 10% to 20% $L_{10}$ Life Remaining):**

Spike energy (or equivalent) normally first appears before any frequencies do so in the vibration spectrum (velocity or acceleration). For example, Stage 1 shows a normal spectrum indicating a healthy bearing and has only the normal first 3 running speed harmonics in the velocity spectrum. The only evidence of possible bearing problems is that Spike Energy has grown from near 0 to approximately .25 gSE (example amplitude only; actual levels depend on the particular bearing and how close the measurement is to the bearing housing). Note that the Spike Energy reading itself is not meant to be part of the spectrum in Figure 6.09P, but only an overall level. During Stage 1, no sound will be detectable by the human ear indicating bearing damage, and no change in bearing temperature would be anticipated. Table 6.09A shows the defects that would be expected in this stage, many of which would be near microscopic at this stage.

**STAGE ①**

1x  
2x  
3x

**STAGE ②**

1x  
2x  
3x

1x RPM Sidebands  
Bearing Component  
Natural Frequency (fn)

**STAGE ③**

A. Lone Bearing Defect Frequency appears (without 1x Sidebands)  
B. Harmonics of Bearing Defect Frequencies then appear as slight wear progresses around periphery of Raceways.  
C. 1x RPM Sidebands then surround Bearing Frequencies when wear becomes more pronounced; Other Defect Frequencies appear.

1x  
2x  
3x

RPM BPFO RPM  
C A C

BPFI RPM  
C C

RPM 2 BPFO RPM  
C B C

RPM 3 BPFO RPM  
C B C

2 BPFI C

2x RPM  
RPM  
fn  
RPM  
2x RPM

**STAGE ④**

1x  
2x  
3x  
4x

2x RPM  
RPM  
BPFO  
RPM  
2x RPM

Pronounced Wear Indicated  
(Remaining Life Questionable)

Random Higher Frequency

Example  
Overall  
gSE  
or  
HFD  
Level

.20 gSE

.50 gSE

1.0 gSE

30 to 50 gSE

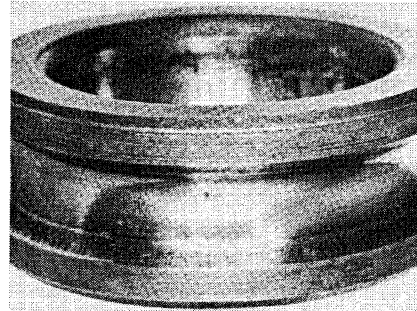
Actually drops at start Stage 4

Later can increase dramatically  
just prior to seizure

**TABLE 6.09A**  
**4 ROLLING ELEMENT FAILURE STAGES (Reference 16)**  
**Typical Stages Of Bearing Degradation**

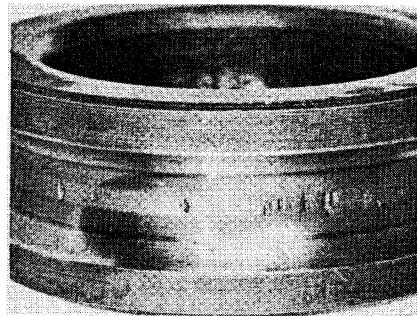
***a. Initial Phase.***

- noise level normal
- temperature normal
- measurable increase in ultrasonic sound, acoustic emission, spike energy, and outer race deflection
- overall vibration low; no discrete spikes at bearing
- frequencies remaining life more than 10% of B-10 rating\*



***b. Second Phase.***

- slight increase in noise level
- temperature normal
- large increase in ultrasonic sound, acoustic emission, spike energy, and outer race deflection
- slight increase in overall vibration acceleration and velocity
- bearing frequencies clearly visible on log scale, barely visible on linear scale of vibration spectrum; noticeable rise in noise floor
- remaining life less than 5% B-10 rating



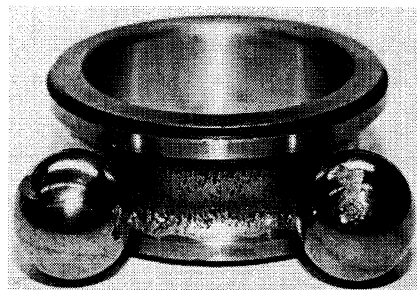
***c. Third Phase.***

- noise level quite audible
- slight increase in temperature
- very high ultrasonic sound, acoustic emission, spike energy, and outer race deflection
- large increase in overall vibration acceleration and velocity
- bearing frequencies with harmonics and sidebands clearly visible on linear scale
- of vibration spectrum; noticeable rise in noise floor
- remaining life less than 1% B-10 rating\*



***d. Final Phase.***

- change in pitch of noise level
- significant temperature increase
- gradual decline followed by rapid increase in ultrasonic sound, acoustic emission, spike energy, and outer race deflection immediately prior to failure
- significant increase in overall vibration displacement and velocity; decrease in acceleration
- vibration spikes predominant at lower bearing frequencies; very high noise floor in spectrum remaining life less than 0.2% B-10 rating\*



\* based on 90% confidence level

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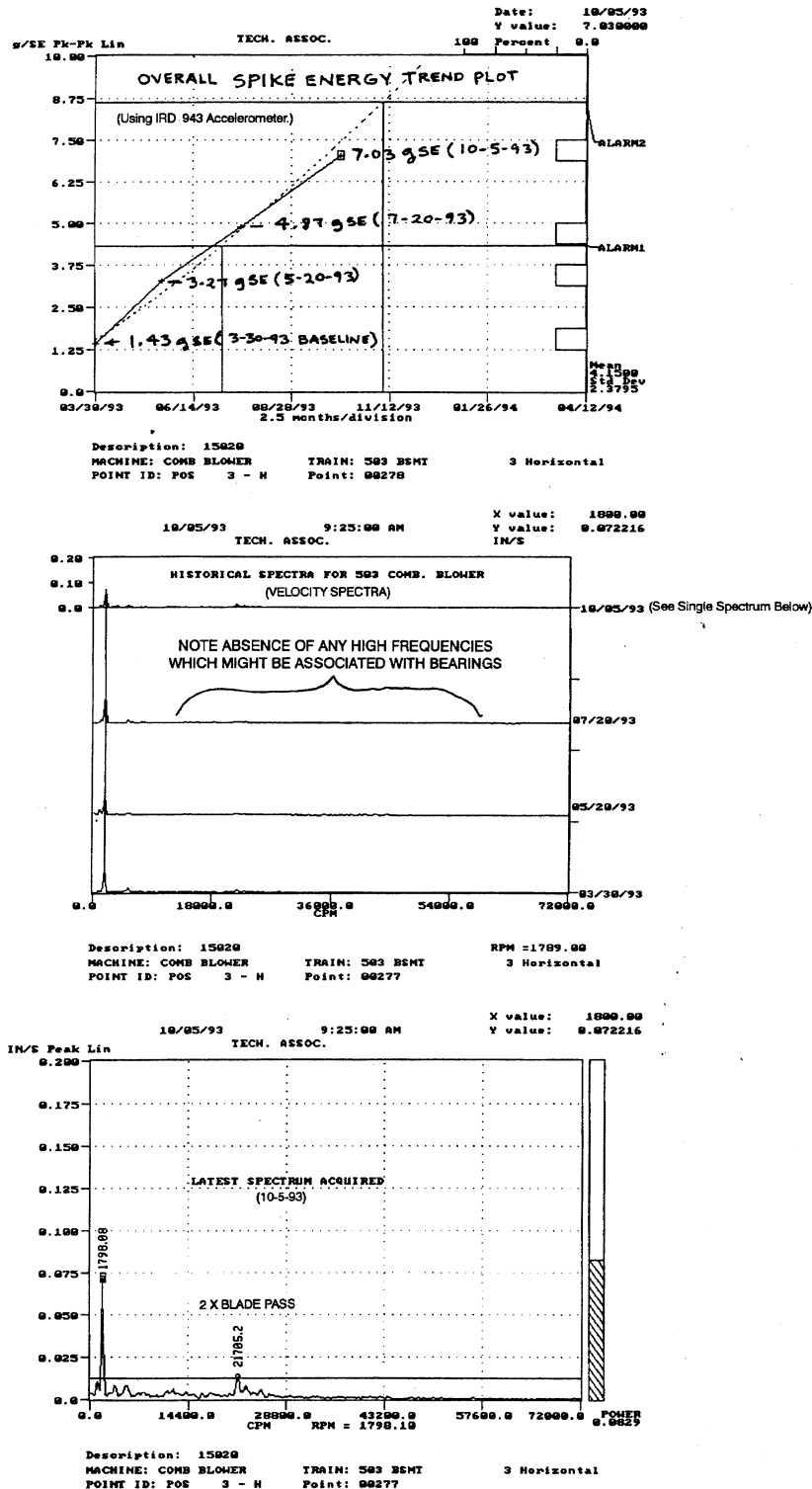
Figure 6.09Q shows what would be referred to as a classic "Failure Stage 1" scenario. This data was taken on a 1790 RPM combustion blower at a fiberglass plant where it was one of the most critical machines. First, note the trend of overall spike energy which increased from 1.43 up to 7.03 gSE on this inboard blower bearing (note that this data was taken with an IRD 943 accelerometer connected to an IRD FAST TRACK/FS analyzer; therefore, the spike energy values tend to be higher with the model 943 than they formerly did with the older IRD 970 accelerometer; in any case, the significant spike energy trend signaled a problem). Examination of the spectral history and the latest spectrum did not reveal any frequencies associated with the bearings within this machine. Therefore, referring to failure Scenario A in Figure 6.09P, this would place this bearing within Stage 1 if the increase in spike energy was due to bearing damage (since increased spike energy levels can not only be due to bearing problems, but also to lubrication, gear problems, rub, cavitation and many other "friction-related" events which can generate very high frequencies).

It is exactly at this point that vibration analysts were stymied for a number of years. Ever since the development of Spike Energy (and related parameters such as Shock Pulse and Acoustic Emission), analysts had wondered if the bearing truly had a problem, why was nothing apparently showing up in either vibration velocity or acceleration spectra. And another question was asked after microscopic evaluations of the bearing component surfaces were conducted after measurements showed an increase in one of these ultrasonic quantities (when it was confirmed the increase was indeed due to bearing problems). Finally, only a few years ago, metallurgists found the answer. Even though the component surfaces themselves showed little or no apparent damage, the metallurgists discovered an important fact - if the bearing is failing properly in fatigue, the damage most often will begin approximately 4 to 6 mils (.004 - .006 inch) beneath the surface and will work its way out to the surface. Initially, there will only be damage to the actual crystalline structure of the bearing materials themselves beneath the surface. Later, a microcrack will work its way to the surface, but will still be microscopic in size. However, the important fact is that the damage mechanism has begun, and cumulative damage from this point onwards will likely be exponential. When the rolling elements roll past and impact this microscopic flaw, they will generate a very short-lived transient impulse, normally generating a response less than only .001 g. Therefore, the simple fact is that vibration due to mechanical and electrical deviants such as unbalance, misalignment, blade pass, rotor bar problems, etc., will easily mask (or cover up) these responses in the lower frequency ranges (particularly below 120,000 CPM or 2000 Hz), making it impossible to see the actual bearing defect frequencies which indeed are there, but are "invisible" to the vibration instruments and spectra.

However, it is well known that very short-lived transients generate numerous frequency harmonics (resembling "white noise" causing the harmonics to virtually "propagate forever", but still of very small amplitude. However, since they do propagate all the way into very high frequencies on the order of 20,000 to 40,000 Hz (2,400,000 - 4,800,000 CPM), they eventually will excite the resonance either of the transducer itself measuring the ultrasonic response, or of the surrounding structure or some other member. It is within these frequencies that such parameters as Spike Energy, Shock Pulse and Acoustic Emission do their detection work. As a matter of fact, each of these parameters purposely employs high pass filters to essentially remove the contribution in the spectrum from the mechanical and electrical sources mentioned above which were masking the bearing problem.

Some of the new data collection systems now employ high frequency enveloping and demodulation techniques which can now work with the remaining data in the high frequency ranges. In essence, they use the time waveform (which might include the numerous bearing frequency harmonics as well as the natural frequency of one of the transducer or structural components); full-wave rectify the modulated waveform; determine the spacing of the exciting frequencies (bearing frequencies in this case); and then pass the remaining waveform data through a low-pass filter to remove the high frequency resonant response or so-called "carrier

**FIGURE 6.09Q**  
**CLASSIC STAGE 1 APPEARANCE OF SPIKE ENERGY TREND AND**  
**ACCOMPANYING VELOCITY SPECTRA**



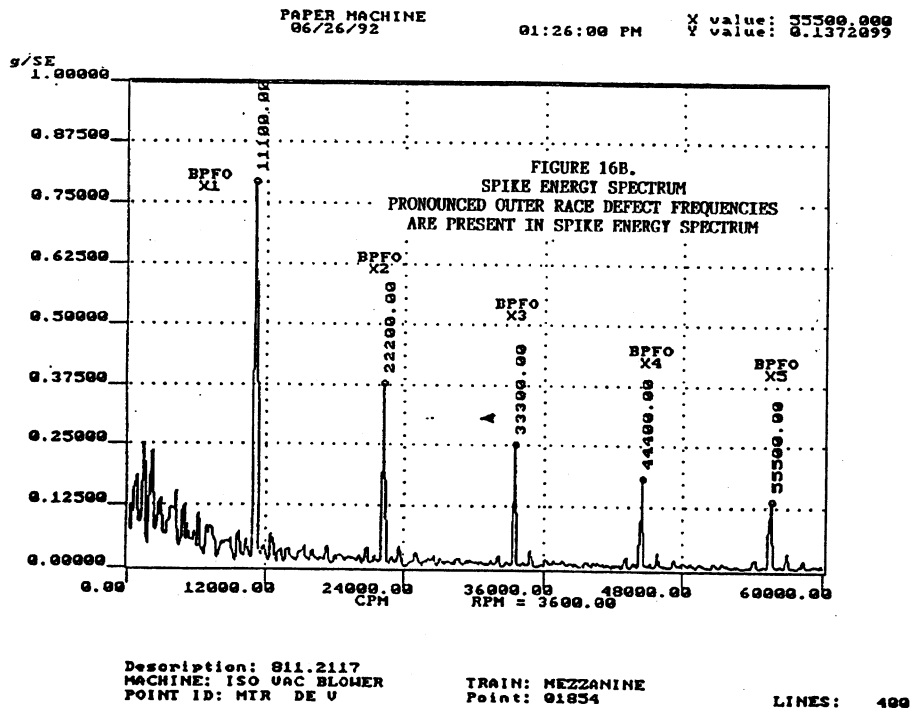
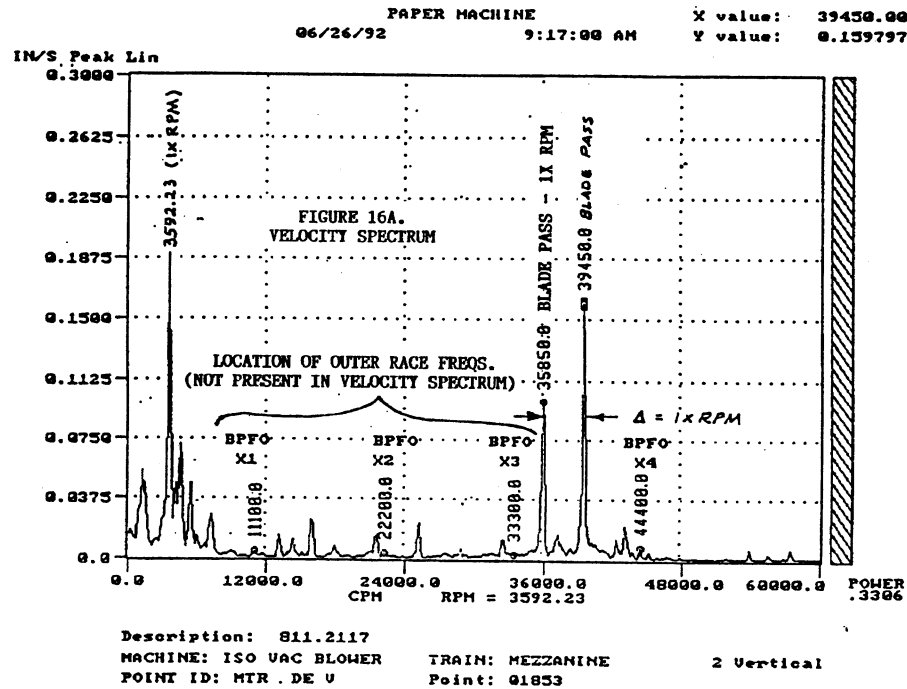
frequency". The purpose here will be to see what frequencies were exciting the resonance and, using the low-pass filter, to fold them back into a lower frequency span to identify these frequencies. If in fact they were bearing frequencies, it would point to the bearing as a problem. Now, the problem might not necessarily be wear at this time. Instead, it might only be a lack of lubricant for example. Therefore, at this point, the analyst should add lubricant to the bearing and repeat these ultrasonic spectra as well as measurements of the overall levels. If they drop considerably or even disappear, he should then repeat the measurements within 12 to 24 hours to see if they return. If they remain of little or no amplitude, the problem was likely lubrication and the bearing has now been given a chance to hopefully enjoy a significantly increased life had the lubrication problem not been caught. On the other hand, if the bearing defect frequencies have reoccurred, we are likely within Stage 1.

Figures 6.09R through 6.09T give examples of such so-called ultrasonic spectra with data acquired from each of three vendors on various machines. Figure 6.09R illustrates detection of a Stage 1 bearing problem by spike energy spectra which were not present within the vibration spectrum alone (this data was taken with an IRD FAST TRACK/FS analyzer). Looking at the velocity spectrum in Figure 6.09R-A, note where outer race frequencies (BPFO) would be present if they were in the velocity spectrum. You will note that although cursors have been placed in these locations, there seems to be no real response due to the masking problem previously discussed. However, examination of the spike energy spectrum in Figure 6.09R-B clearly showed the presence of 5 BPFO harmonics on this 3590 RPM vacuum blower which would have totally been missed had vibration spectra alone been taken. When the same frequencies remained after lubrication, this confirmed the onset of failure Stage 1 for this bearing.

Similarly, Figure 6.09S used another analyzer (SKF CMVA10 analyzer) to detect bearing problems not visible within the vibration spectrum. In this case, it used not only what are known as acceleration envelope techniques, but also the new SEE spectral technology to confirm the bearing problems ("SEE" stands for "Spectral Emitted Energy" and was developed by SKF). This data was taken on a 1250 RPM belt-driven fan bearing. Figure 6.09S-A is the acceleration spectrum on which are overlaid a series of 8 outer race frequency markers which showed where such outer race frequencies would be if they were in the acceleration spectrum (which they were not). On the other hand, Figure 6.09S-B used a bandpass filter of about 5000 to 40,000 Hz and employed acceleration enveloping with the SKF CMVA10 analyzer and did show that both outer race and inner race frequencies were located after using the bandpass filter of 5,000 to 40,000 Hz (300,000 - 240,000 CPM) to filter out the lower frequency mechanical and electrical problems. Similarly, Figure 6.09S-C employed SKF's new SEE technology to confirm the bearing problems seen in the acceleration envelope. Currently, the SEE spectrum uses a much higher bandpass filter of approximately 250,000 to 350,000 Hz (15,000,000 - 21,000,000 CPM). In this case, note that the bearing problems were even clearer in the SEE spectrum with higher signal to noise ratio than was the case with the acceleration envelope. Note that further discussion of the SEE spectrum and these other related high frequency enveloping and modulation technologies is provided in Reference 28.

Figure 6.09T demonstrated the fact that even these high frequency demodulation technologies can be employed on very low speed machines. In fact, they will often detect problems which cannot even be detected by either the FFT spectrum nor the time waveform when using vibration measurements alone. In this case, the measurements shown in Figure 6.09T were taken on a 22 RPM (.366 Hz) dryer roll which had a bearing frequency of 3.125 Hz (187.5 CPM). In this case, a CSI 2115 analyzer was outfitted with a Model 750 Amplitude Demodulator which was attached to the 2115 providing it with this amplitude demodulation capability. First, vibration data shown in Figures 6.09T-A and 6.09T-B did not show any evidence of a bearing problem. Figure 6.09T-A was the time waveform with a time span of 4.0 sec (corresponding to almost 1.5 roll revolutions). It showed the presence not only of the running speed, but evidence of higher frequencies as well. However, it would have been extremely difficult to pick out the inner race frequency from this time waveform. Figure 6.09T-B showed the FFT spectrum corresponding to the time waveform

**FIGURE 6.09R**  
**DETECTION OF STAGE 1 BEARING PROBLEM BY A SPIKE ENERGY**  
**SPECTRUM WHICH WAS NOT INDICATED BY THE VIBRATION**  
**SPECTRUM ALONE (Used IRD FAST TRACK/FS Analyzer)**



**FIGURE 6.09S**  
**DETECTION BEARING PROBLEMS BY ACCELERATION ENVELOPE AND**  
**SEE SPECTRAL TECHNIQUES WHICH WERE NOT EVIDENT WHATSOEVER**  
**IN THE VIBRATION SPECTRUM (Used an SKF CMVA10 Analyzer)**

SET: MULTILOG 1 TYPE: FFT DATE: 12-NOV-92 16:30:58  
 POINT ID: POINT 1 ACC LABEL 2 DESC: Acceleration on Bearing A  
 WINDOW: HANNING LINES: 400 AVER: 1 FREQ: 0 - 1000 Hz  
 DETECT: RMS RPM: 1250 THRESHOLD: 0.1000 UNITS: Gs

FREQ: 20.00 AMP: 0.004 ORDER: 0.960 DEG: ---

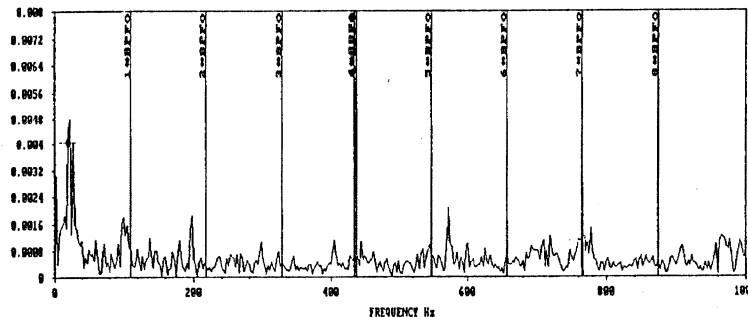


FIGURE A. ACCELERATION SPECTRUM TAKEN ON A BELT-DRIVEN FAN BEARING  
 (No Evidence of Bearing Problems)

SET: MULTILOG 1 TYPE: FFT DATE: 12-NOV-92 16:33:09  
 POINT ID: POINT 1 ENV LABEL 2 DESC: Envel Acceleration on Bearing  
 WINDOW: HANNING LINES: 400 AVER: 8 FREQ: 0 - 1000 Hz  
 DETECT: PEAK TO PEAK RPM: 1250 THRESHOLD: 0.4000 UNITS: Gs

FREQ: 17.50 AMP: 0.2565 ORDER: 0.840 DEG: ---

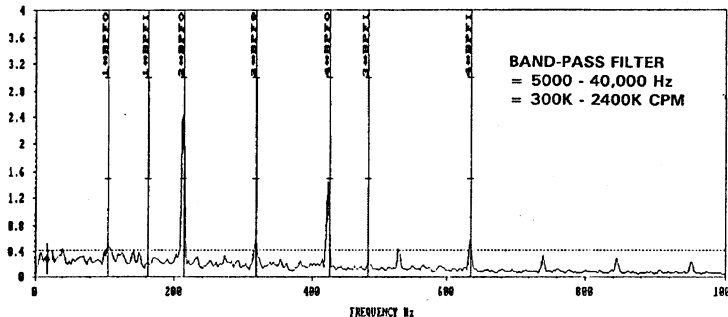


FIGURE B. ACCELERATION ENVELOPED SPECTRUM SHOWING OUTER RACE (BPFO)  
 AND INNER RACE (BPFI) PROBLEMS

SET: MULTILOG 1 TYPE: FFT DATE: 12-NOV-92 16:32:44  
 POINT ID: POINT 1 SEE LABEL 1 DESC: See Meas on Bearing A  
 WINDOW: HANNING LINES: 400 AVER: 8 FREQ: 0 - 2000 Hz  
 DETECT: PEAK TO PEAK RPM: 1263 THRESHOLD: 6.0000 UNITS: SEES

FREQ: 20.00 AMP: 10.87 ORDER: 0.950 DEG: ---

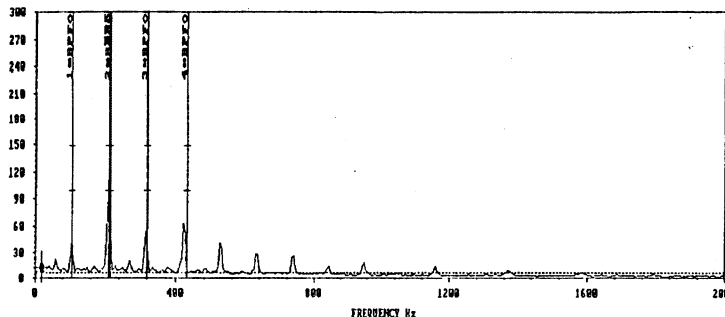
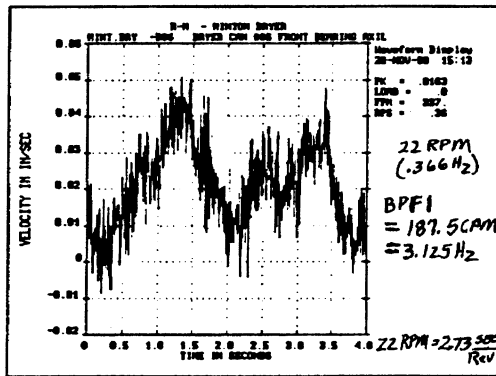
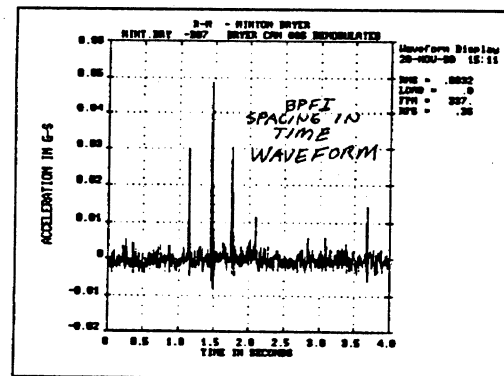


FIGURE C. SKF'S SEE SPECTRUM CONFIRMING THE BEARING PROBLEMS  
 SEEN IN THE ACCELERATION ENVELOPED SPECTRUM

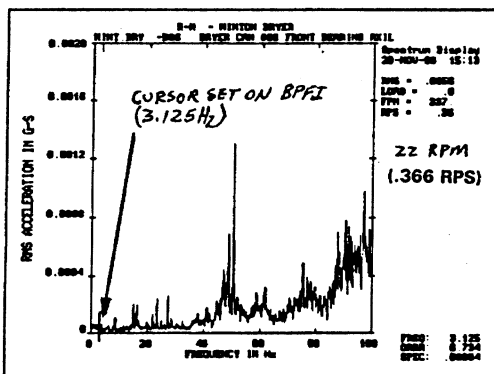
**FIGURE 6.09T**  
**DETECTION OF INNER RACE PROBLEMS ON A 22 RPM DRYER ROLL BEARING**  
**BY AMPLITUDE DEMODULATED SPECTRA WHICH WERE COMPLETELY MASKED**  
**BY BOTH VIBRATION TIME WAVEFORM AND FFT SPECTRA (Ref. 29)**  
**(Used a CSI 2115 Analyzer and CSI Model 750 Amplitude Demodulator)**



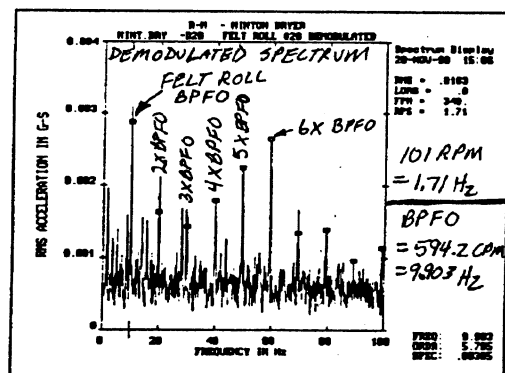
**FIGURE A**  
**RAW VIBRATION VELOCITY WAVEFORM**  
**FROM A DRYER BEARING**



**FIGURE C**  
**DEMODULATED VIBRATION**  
**ACCELERATION WAVEFORM FROM**  
**WHICH THE SPECTRUM OF**  
**FIGURE D IS DERIVED**



**FIGURE B**  
**VIBRATION ACCELERATION**  
**SPECTRUM DERIVED FROM**  
**WAVEFORM IN FIGURE A**



**FIGURE D**  
**DEMOMULATED VIBRATION**  
**ACCELERATION SPECTRUM OF**  
**THE SAME BEARING SHOWN IN**  
**FIGURES A AND B**

captured in Figure 6.09T-A. In this case, the cursor was purposely positioned on the inner race frequency (BPFI) of 3.125 Hz and showed almost no response. The only major peak showing within the spectrum was at approximately 60 Hz which corresponded to the gear mesh frequency, and even it had an amplitude of only .0012 g. Figures 6.09T-C and 6.09T-D showed the resulting spectral and waveform data which resulted after the Model 750 Amplitude Demodulator was invoked. First, both 1X and 2X BPFI were evident in the FFT spectrum (note that the amplitude at BPFI was only .0005 g). In addition, note the clear BPFI spacing in the amplitude demodulated time waveform in Figure 6.09T-D. It has been the experience of the author that in many cases, neither the FFT spectrum nor the time waveform showed any evidence of bearing problems. However, employing high frequency spectral techniques such as this amplitude demodulated spectrum and waveform provided just the evidence needed to prove there was indeed a problem. In a few rare cases, particularly on machines running below 20 RPM, nothing ever did show up in either the vibration waveform nor spectrum, but did show in the high frequency enveloped and demodulated spectra, thereby preventing the failure of bearings costing tens of thousands of dollars. Further information is provided on high frequency enveloping and demodulation spectral techniques in Reference 11. The important point to be made here is that these technologies will not replace vibration, but serve as invaluable tools to confirm the presence of bearing problems; and in some cases, serve as the only detection tools available on very low speed machines with bearing problems.

#### **Stage 2 (Approximately 5% to 10% $L_{10}$ Life Remaining):**

Slight bearing defects begin to "ring" natural frequencies of the installed bearing components. These natural frequencies are concentrated in the 30,000 to 120,000 CPM range (500 to 2000 Hz). Fundamental natural frequencies of most bearings in industrial machinery are approximately 30,000 to 90,000 CPM (500 - 1500 Hz) while those for high precision bearings normally range between 75,000 to 120,000 CPM (1250 - 2000 Hz); and on such high precision machine tooling, may occasionally range as high as 150,000 to 180,000 CPM (2500 - 3000 Hz). Remember, this is not one individual natural frequency, but a set of natural frequencies for the bearing outer race/inner race/rolling element/housing system. These are natural frequencies of the assembled rolling element bearings themselves which do not change in frequency with a change in operating speed (however, they normally will show higher amplitudes with increasing speed due to greater impact velocity). These natural frequencies are excited by the momentary impact between the rolling elements and bearing races which not only excite the bearing natural frequencies, but also increases ultrasonic response (for example, roughly doubling the overall ultrasonic level in many cases). It has been the experience of the author that during initial Stage 2, only one or more discrete frequencies appear in these regions. Later, towards the end of Stage 2, these frequencies will not only grow in amplitude, but also will become modulated with the running speed when wear progresses (that is, 1X RPM sidebands will later appear above and below these natural frequencies). On the other hand, these sidebands can often be spaced at bearing defect frequencies themselves (BPFO, BPFI, BSF, etc.).

Note that the defects themselves in Stage 2 may not yet be readily visible to the naked eye. There should be only a slight increase in bearing noise and its temperature should still be roughly normal. Notice that bearing defect frequencies will not likely yet be visible in the velocity spectrum. However, acceleration spectra may now begin to pick up harmonics of defect frequencies, particularly if a log amplitude scale is employed. Still, at this stage, bearing defect frequency response will normally be erratic.

Figure 6.09U shows a classic machine which has advanced into Stage 2 of the failure scenario. In this case, Figure 6.09U pictured spectra taken from the inboard pump bearing of a 1790 RPM chilled water pump. On Figure 6.09U are shown trends of the overall spike energy, a cascade (or waterfall) diagram showing the historical spectra which have been taken on this machine, as well as the latest spectrum itself which vividly shows the problem. In the case of the spike energy trend, note that the overall levels have advanced from as low as .163 up to 5.14 gSE. In the case

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of the cascade diagram, note that no bearing component natural frequencies were evident on spectra taken up through March 29, 1993. However, they made their initial appearance during the May 20, 1993 survey and were surrounded by 1X RPM sidebands with the amplitude of the probable natural frequency at about .023 in/sec. Note that on July 21, the apparent natural frequency located at about 54,500 CPM grew from .023 up to .036 in/sec. Then, on the survey on October 7, the bearing advanced late into Stage 2 as evidenced by the growth of the probable 54,500 CPM natural frequency up to .071 in/sec; combined with the growth and the number of 1X RPM sideband families surrounding this frequency. Also, possible bearing frequencies were beginning to show as evidenced by noninteger running speed multiples at approximately 12.2X RPM (21,800 CPM) and 15.3X RPM (27,250 CPM) (although the bearing model number could not be confirmed at this time).

### **Stage 3 (Approximately 1% to 5% $L_{10}$ Life Remaining):**

Note that each of 3 progressive events are documented for Stage 3 in Figure 6.09P (shown as A, B and C). The letters shown below the horizontal axis of the Stage 3 spectrum in Figure 6.09P correspond to peaks which appear during these 3 progressive events. For the first time, bearing defect frequencies associated with faults in the inner race (BPFI), outer race (BPFO), rolling elements (BSF) and/or cage (FTF) appear in the velocity spectrum (Event A). Later on in Stage 3, harmonics of these bearing defect frequencies then will appear as slight wear progresses around the periphery of the raceways and/or faults appear on more than one rolling element (Event B). Normally, the defect frequencies themselves will not be modulated by running speed when they first appear (unless the bearing is fairly heavily loaded in which case BPFI will often be surrounded by 1X RPM sidebands when it first appears since the inner race defect will rotate in and out of the load zone). When deterioration progresses, 1X RPM sidebands will surround the bearing defect frequencies. Spike energy (or equivalent) will continue to grow, doubling or tripling in amplitude. At the end of Stage 3, not only will 1X RPM sidebands appear around bearing defect frequencies, but more sideband families will appear around the bearing component natural frequencies (Event C).

A word of caution should be taken at this point. When the bearings approach the conclusion of Stage 3, the rate of wear becomes highly unpredictable. In any case, it will be well into the exponential part of the failure curve (Figure 6.09N). How much longer the bearing lasts will largely depend on its lubrication, temperature, cleanliness and dynamic loads being imposed upon it by vibration forces from unbalance, misalignment and so forth. At this point, there will be a noticeable change in sound level and frequency and a slight increase in bearing housing temperature. Addition of lubricant at this point may temporarily lower spike energy and possibly have some effect in reducing vibration. However, since wear is the problem at this point, both vibration and spike energy will return, normally within 12 to 24 hours (the lubricant addition at this point could hasten failure if the bearings already were over-lubricated).

Figures 6.09V and 6.09W show a spectrum and spike energy trend plot, respectively, taken from a pump bearing with a classic Stage 3 bearing failure underway. Notice that the spike energy increased from .271 to .944 gSE between the May 24 and August 23 measurements exceeding the alarm of .750 gSE (Figure 6.09W). Then, looking at the velocity spectrum of Figure 6.09V, each of 4 harmonics of inner race frequency are present (BPFI = 7.44 X RPM = 13,200 CPM). Note the harmonics at 26400, 39600 and 52800 CPM. Also note the 1X RPM sidebands which have appeared around 2BPFI and 3BPFI. Looking at both the velocity spectral data and spike energy data, the plant would be well advised to replace this bearing right away even though amplitudes of all bearing frequencies are below .10 in/sec.

# **FIGURE 6.09U** **COMMENCEMENT AND DETERIORATION OF FAILURE STAGE 2** **ON A PUMP BEARING**

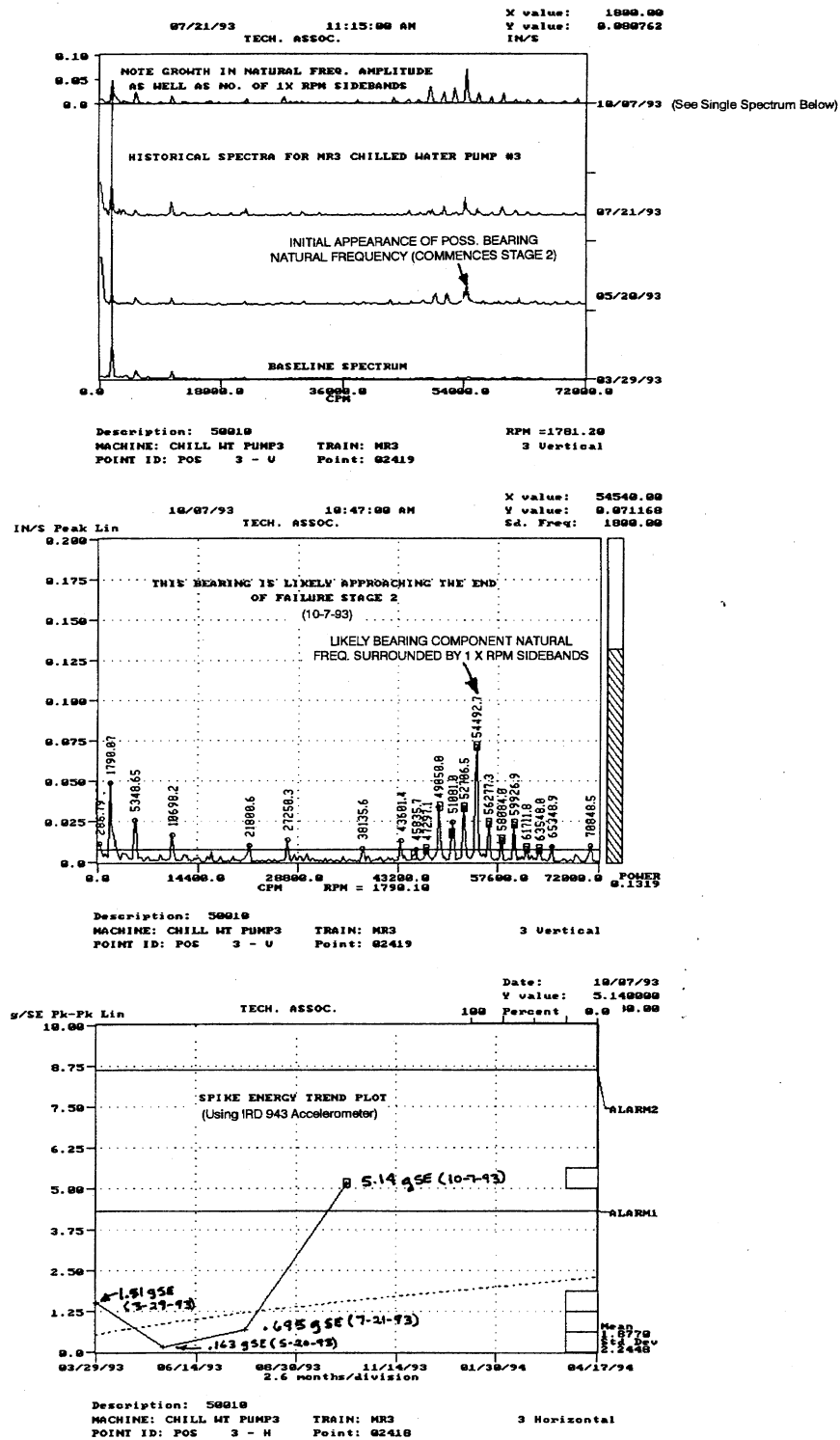
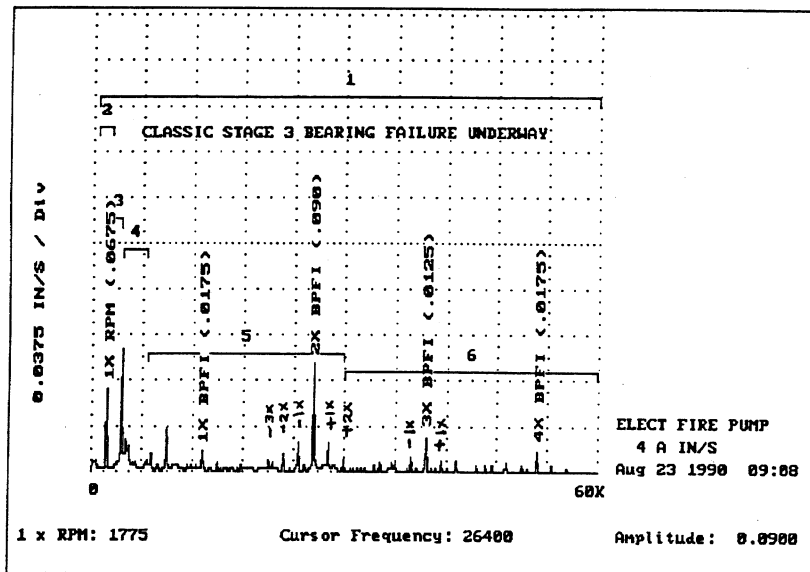


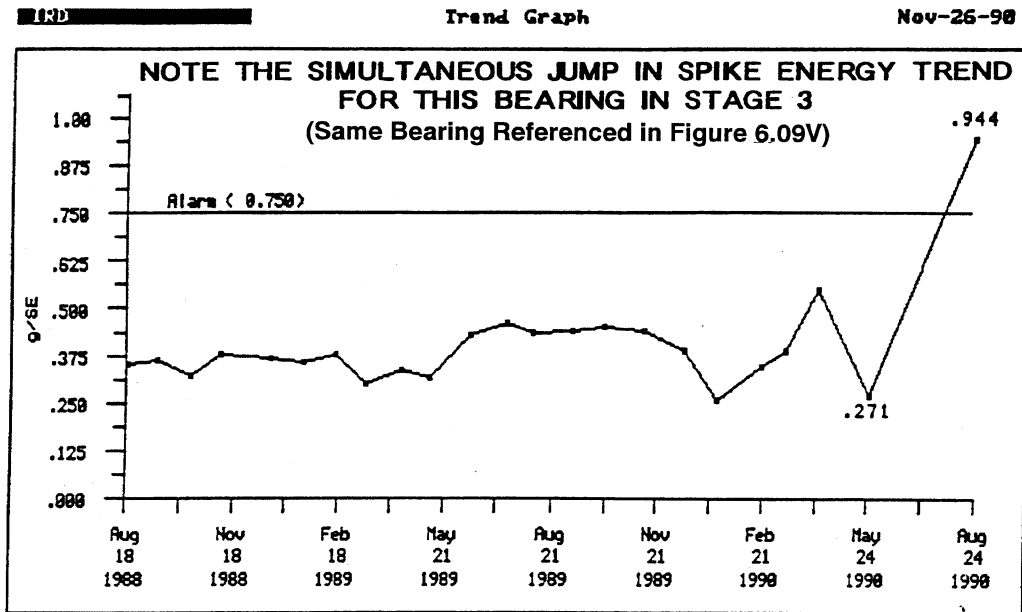
FIGURE 6.09V



2% x 0.500 = 0.0100      1 x RPM: 1775.00

| Machine         |         | Pos       | Dir       | Units   | Date              |         |
|-----------------|---------|-----------|-----------|---------|-------------------|---------|
| ELECT FIRE PUMP |         | 4         | A         | IN/S    | Aug 23 1990 09:08 |         |
| Frequency       | 1 x RPM | Amplitude | Frequency | 1 x RPM | Amplitude         |         |
| 150             | 0.08    | 0.0100    | 39600     | 22.31   | 0.0275            | 3X BPFI |
| 600             | 0.34    | 0.0100    | 41400     | 23.32   | 0.0100            |         |
| 1800            | 1.01    | 0.0675    | 43200     | 24.34   | 0.0100            |         |
| 3600            | 2.03    | 0.1000    | 52800     | 29.75   | 0.0175            | 4X BPFI |
| 4050            | 2.28    | 0.0250    |           |         |                   |         |
| 4500            | 2.54    | 0.0200    |           |         |                   |         |
| 6600            | 3.72    | 0.0100    |           |         |                   |         |
| 7200            | 4.06    | 0.0150    |           |         |                   |         |
| 8850            | 4.99    | 0.0350    |           |         |                   |         |
| 13200           | 7.44    | 0.0175    |           |         |                   | BPFI    |
| 21000           | 11.83   | 0.0100    |           |         |                   | 3X RPM  |
| 22800           | 12.85   | 0.0150    |           |         |                   | 2X RPM  |
| 24600           | 13.86   | 0.0225    |           |         |                   | 1X RPM  |
| 26400           | 14.87   | 0.0900    |           |         |                   | 2X BPFI |
| 28200           | 15.89   | 0.0225    |           |         |                   | 1X RPM  |
| 30000           | 16.90   | 0.0125    |           |         |                   | 2X RPM  |
| 37800           | 21.30   | 0.0125    |           |         |                   |         |

FIGURE 6.09W



ELECT FIRE PUMP 4 H .750 g/SE

IRD Mechanalysis Data Analysis System

| Machine | ELECT FIRE PUMP | Pos      | 4     | Dir  | H    | Units    | g/SE  | Alarm | .750 |
|---------|-----------------|----------|-------|------|------|----------|-------|-------|------|
| Ampl    | Insp            | Date     | Time  | Ampl | Insp | Date     | Time  |       |      |
| Code    | Code            | mm/dd/yy |       | Code | Code | mm/dd/yy |       |       |      |
| .353    | 0               | 08-18-88 | 9:25  | .442 | 0    | 09-07-89 | 8:51  |       |      |
| .361    | 0               | 09-13-88 | 0:07  | .449 | 0    | 10-05-89 | 8:50  |       |      |
| .320    | 0               | 10-11-88 | 9:18  | .442 | 0    | 11-09-89 | 10:00 |       |      |
| .378    | 0               | 11-07-88 | 9:19  | .390 | 0    | 12-14-89 | 15:38 |       |      |
| .368    | 0               | 12-20-88 | 10:34 | .261 | 0    | 01-11-90 | 8:52  |       |      |
| .355    | 0               | 01-17-89 | 9:10  | .347 | 0    | 02-19-90 | 8:40  |       |      |
| .380    | 0               | 02-14-89 | 8:51  | .388 | 0    | 03-12-90 | 8:58  |       |      |
| .302    | 0               | 03-14-89 | 8:45  | .549 | 0    | 04-09-90 | 7:54  |       |      |
| .338    | 0               | 04-13-89 | 8:03  | .271 | 0    | 05-24-90 | 7:51  |       |      |
| .316    | 0               | 05-09-89 | 9:18  | .944 | 0    | 08-23-90 | 9:08  |       |      |
| .429    | 0               | 06-13-89 | 9:17  |      |      |          |       |       |      |
| .460    | 0               | 07-13-89 | 9:20  |      |      |          |       |       |      |
| .434    | 0               | 08-03-89 | 9:25  |      |      |          |       |       |      |

Page 1

Machine Report (Latest Data) for: ELECT FIRE PUMP

| Date      | Time  | Pos | Dir | Ampl  | Units | Change | Alarm | Inspection |
|-----------|-------|-----|-----|-------|-------|--------|-------|------------|
| Aug-23-90 | 09:06 | 1   | A   | .150  | IN/S  | -6%    | .300  |            |
| Aug-23-90 | 09:06 | 1   | H   | .129  | IN/S  | 48%    | .300  |            |
| Aug-23-90 | 09:06 | 1   | H   | .329  | g/SE  | 186%   | .600  |            |
| Aug-23-90 | 09:06 | 1   | O   | .117  | IN/S  | -----  | .300  |            |
| Aug-23-90 | 09:06 | 1   | V   | .077  | IN/S  | 8%     | .300  |            |
| Aug-23-90 | 09:07 | 2   | A   | .157  | IN/S  | -16%   | .300  |            |
| Aug-23-90 | 09:07 | 2   | H   | .127  | IN/S  | 14%    | .300  |            |
| Aug-23-90 | 09:07 | 2   | H   | .332  | g/SE  | -1%    | .600  |            |
| Aug-23-90 | 09:07 | 2   | O   | .129  | IN/S  | -----  | .300  |            |
| Aug-23-90 | 09:07 | 2   | V   | .061  | IN/S  | -65%   | .300  |            |
| Aug-23-90 | 09:07 | 3   | A   | .211  | IN/S  | 41%    | .300  |            |
| Aug-23-90 | 09:07 | 3   | H   | .180  | IN/S  | 10%    | .300  |            |
| Aug-23-90 | 09:07 | 3   | H   | .337  | g/SE  | 42%    | .600  |            |
| Aug-23-90 | 09:07 | 3   | V   | .189  | IN/S  | 26%    | .300  |            |
| Aug-23-90 | 09:08 | 4   | A   | .204  | IN/S  | -19%   | .300  |            |
| Aug-23-90 | 09:08 | 4   | H   | .204  | IN/S  | 2%     | .300  |            |
| Aug-23-90 | 09:08 | 4   | H   | *.944 | g/SE  | 248%   | .750  |            |
| Aug-23-90 | 09:09 | 4   | V   | .121  | IN/S  | -15%   | .300  |            |

18 lines this report

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Figure 6.09X shows evidence of another machine which has advanced well into Stage 3 as of its July 21, 1993 spectrum. An important point which should be made here is that this bearing passed completely through Stage 3 during the time interval between surveys of only 8 weeks (another reason why surveys should not be scheduled this far apart - the client elected to do this to "save money"). This data was taken on the inboard bearing of the motor which is a 6-pole induction motor operating at 1192 RPM and outfitted with NTN 2313 bearings. In this case, the outer race frequency was at about 5.19X RPM while the cage frequency (FTF) was at about .41X RPM. First, notice the great number of outer race frequency harmonics showing at the 2nd, 3rd, 5th, 7th, 8th and 9th harmonic outer race frequencies. In addition, what is probably of much greater concern are the great number of multiples of cage frequency sidebands surrounding these outer race frequencies, in addition to the appearance of the cage frequency itself (FTF). This bearing is in great jeopardy and should not be guaranteed to operate any length of time. Once the cage fails, catastrophic failure can occur quickly.

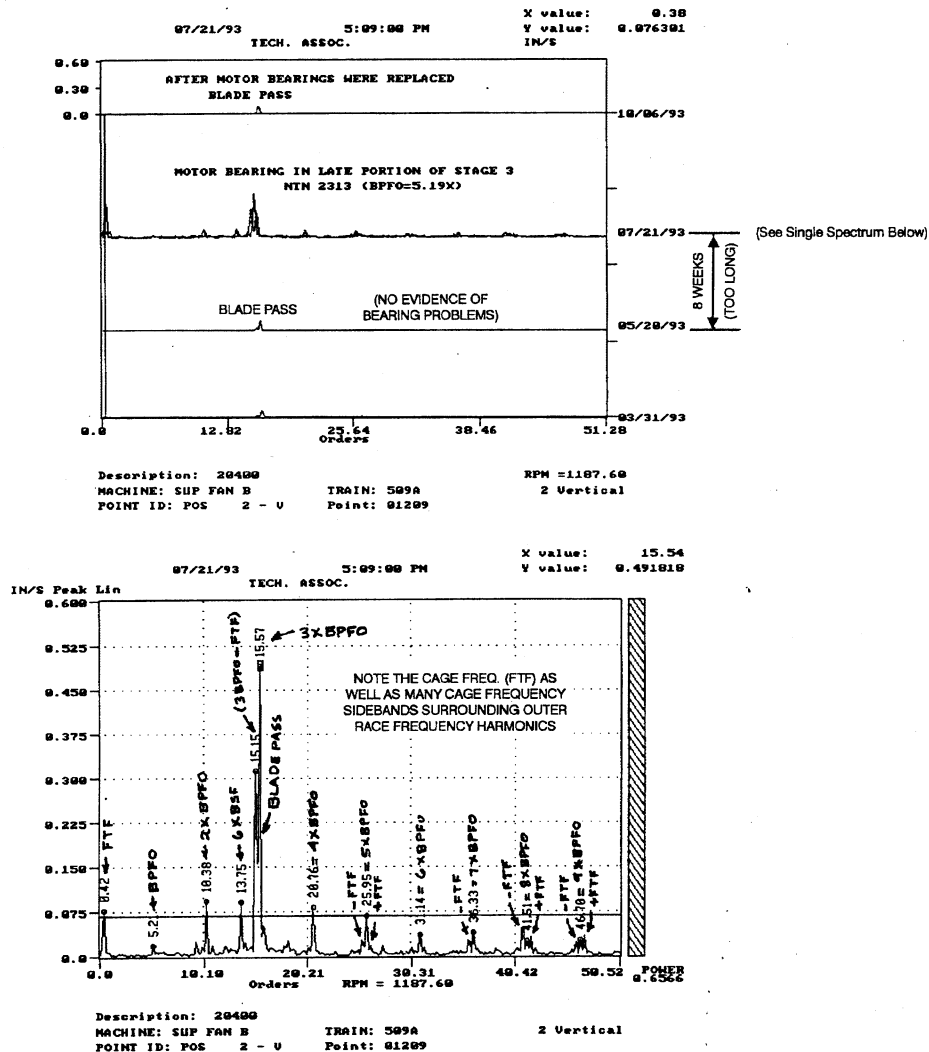
Figure 6.09Y is an example of a bearing where 4X BPFI initiated Stage 3 (rather than 1X BPFI). In fact, the fundamental inner race frequency (1X BPFI = 7.14X RPM) never did grow to significant amplitude. However, this figure showed the development of very serious Stage 3 bearing problems in this vertical pump operating at about 1787 RPM. Note especially the cascade plot which showed no real problems with the machine as of May 11, 1992. Then, the predominant bearing frequency which initially appeared was at 4X inner race frequency (4X BPFI) on June 9, 1992. Later, even more harmonics of inner race frequency appeared and grew to excessively high amplitudes (for bearing defect frequencies). In fact, on December 8, the amplitude at 4X BPFI was .267 in/sec and had a 1X RPM sideband up to .342 in/sec. Interestingly, note that the middle plot on Figure 6.09Y showed that during the next survey on January 13, 1993, the amplitudes at bearing frequencies actually dropped to .100 in/sec at 4X BPFI, and down to .081 in/sec at the 1X RPM sideband to the left of 4X BPFI which was excessively high on December 8. Note that it is not uncommon for amplitudes at bearing frequencies to actually decrease as the condition is truly getting worse. Note also the spike energy trend in Figure 6.09Y which finally showed about 10.5 gSE (excessive for measurements with an IRD 970 accelerometer). Although the plant had been strongly recommended to replace the bearing for several months, they were now told if they did not replace the bearing soon, it would soon take care of itself (catastrophically). The cascade spectrum in Figure 6.09Y shows the plant did finally replace the bearing and all vibration at bearing defect frequencies disappeared; and overall spike energy levels significantly dropped and have remained low since that time.

#### **Stage 4 - Approaching Catastrophic Failure At Any Time (Approx. 1 Hour to 1% $L_{10}$ Life Remaining):**

The level of 1X RPM normally begins to grow for the first time throughout the bearing failure process in Stage 4, along with harmonics at 2X and 3X RPM. Spike energy levels typically actually drop and amplitudes of the higher bearing frequency harmonics and natural frequencies also normally drop. Many 1X RPM sidebands appear around bearing defect frequencies (indicating pronounced wear throughout the periphery of the bearing). There will now be a noticeable change in pitch of the bearing noise, and likely a significant increase in bearing housing temperature.

Later on in Stage 4, discernible bearing defect and component natural frequencies actually begin to disappear and are replaced by a random broadband high frequency "noise floor" which can extend far down into the spectrum, obliterating discrete frequency peaks. Finally, at the end of Stage 4, spike energy normally will decline again (as the bearing itself "disappears"); but, just prior to failure, the spike energy levels can grow dramatically up to 50 to 100 gSE just before final seizure. Thus, a bearing should never be allowed to operate in Stage 4 for no one knows when it will catastrophically fail.

**FIGURE 6.09X**  
**EXAMPLE OF A BLOWER BEARING WHICH APPARENTLY SEEMED TO**  
**ADVANCE NEAR THE END OF FAILURE STAGE 3**  
**(Due to too long a time interval between surveys to "Save Money")**



07/21/93  
 5:09:00 PM

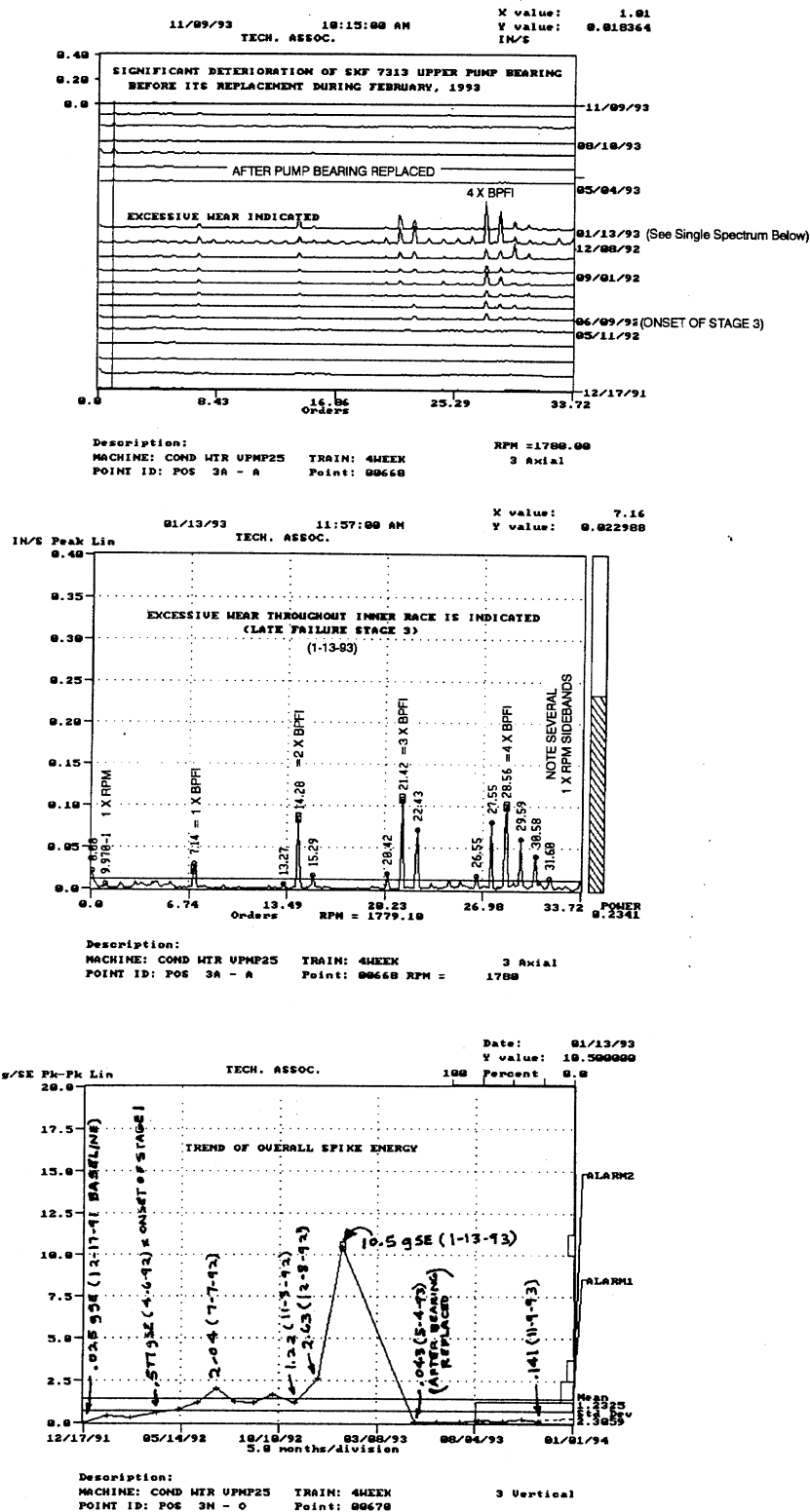
### Threshold Report

PLANT: PPG  
 Description: 20400  
 MACHINE: SUP FAN B  
 POINT ID: POS 2 - V  
 Units: IN/S Peak Lin  
 Threshold: 0.067000

TRAIN: 509A  
 Point: 01209  
 RPM: 1187.60

| Frequency Hz | CPM      | Order    | Spect Amp | % Exceeded |
|--------------|----------|----------|-----------|------------|
| 8.34         | 500.6    | 0.42     | 0.076301  | 13.88      |
| 205.45       | 12327.2  | 10.38    | 0.092370  | 37.87      |
| 272.10       | 16326.3  | 13.75    | 0.090190  | 34.61      |
| 299.94       | 17996.3  | 15.15    | 0.312373  | 366.23     |
| 308.26       | 18495.3  | 15.57    | 0.491818  | 634.06     |
| 410.91       | 24654.4  | 20.76    | 0.081970  | 22.34      |
| 513.60       | 3.08E+4  | 25.95    | 0.067710  | 1.06       |
| Power        | Sync     | Subsync  | Non-Sync  |            |
| 0.656627     | 0.327053 | 0.081327 | 0.563543  |            |

**FIGURE 6.09Y**  
**EXAMPLE OF A BEARING WHERE 4X BPFI INITIATED STAGE 3 (RATHER**  
**THAN 1X BPFI) AND LATER GENERATED MORE BPFI HARMONICS AS**  
**WELL AS 1X RPM SIDEBANDS WHEN ITS CONDITION DETERIORATED**



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While most bearings track through the failure stage presented in Scenario A, some will take other paths. These will be outlined in Scenarios B through F which will follow.

**SCENARIO B. CONTINUED DETERIORATION OF ONE PRONOUNCED FAULT ON A RACEWAY (Figure 6.09Z):**

Normally, faults appear on one bearing component and then spread to the other components through the rolling elements due to the impact vibration and dynamic forces they generate. However, occasionally a single fault will appear on one or the other raceways and the bearing will continue to failure with very little effect on the other components. This may be due to possible metallurgical problems with the material of a raceway or other related problems. In these cases, the bearing will normally pass through the first 2 failure stages outlined in Scenario A. However, when in Stage 3, the single fault will predominantly excite only one race defect frequency which itself will generate little or no harmonics. Sidebands at 1X RPM will then appear around this singular defect frequency as shown in Figure 6.09Z (note sidebands about BPFI).

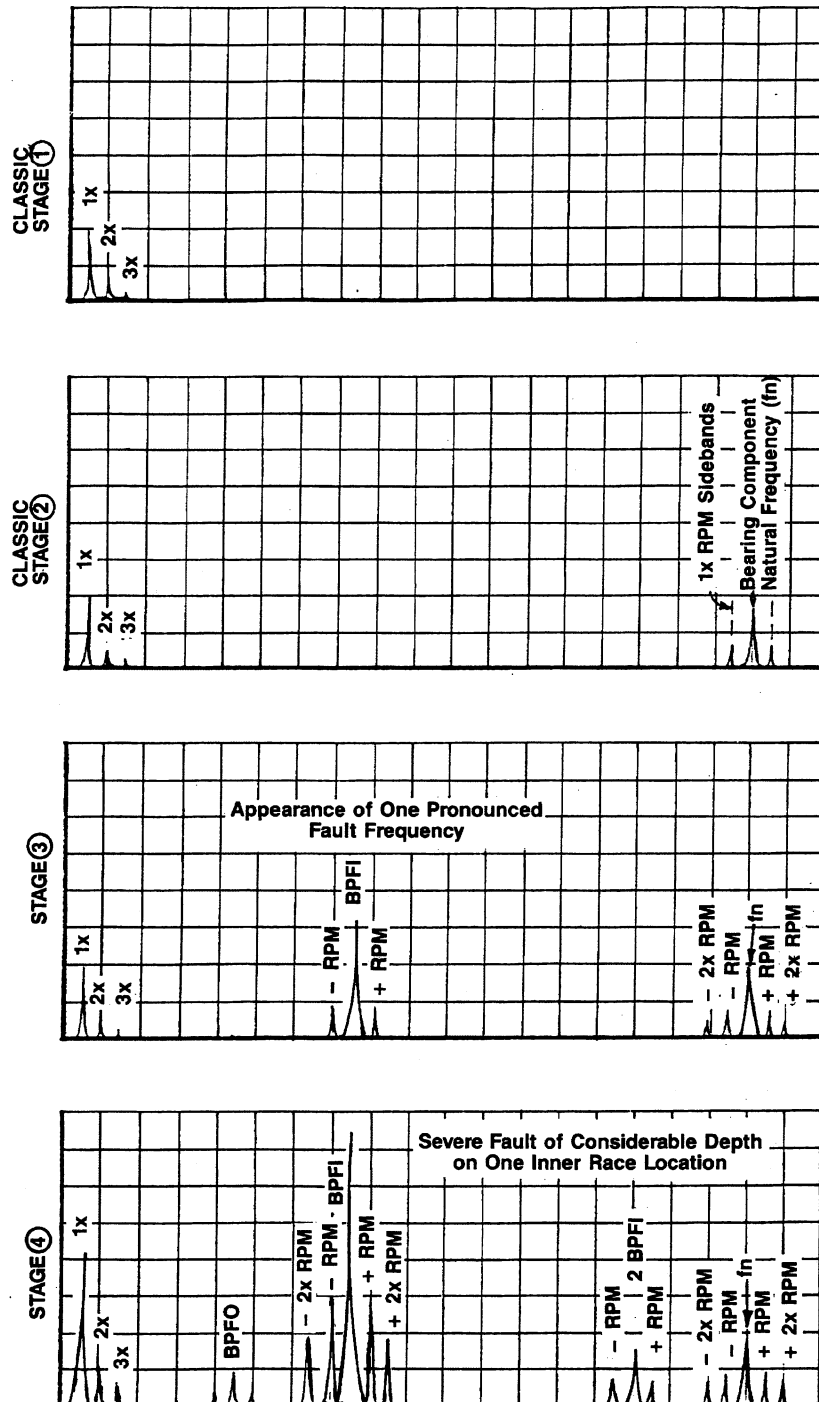
Later on, in Stage 4, the single defect frequency itself will typically increase in amplitude dramatically. In the author's experience, these frequencies have been seen to grow to 1.0 in/sec or more if the bearing has a localized fault as is the case of Scenario B (this rarely happens to the amplitude of any bearing frequency in Scenario A). Then, 1X RPM sidebands will grow to amplitudes much higher than normal. Sometimes, more than one family of sidebands will appear around this frequency (as shown in Stage 4 of Figure 6.09Z). Late in Stage 4, harmonics of this defect frequency will appear and finally, defect frequencies for the remaining bearing components do also as seen in the figure. At this point, the amplitude at 1X RPM and lower harmonics is affected. If the bearing is not replaced, it often will almost literally fall apart when removed, and there will be one pronounced defect of considerable depth on at least one raceway, surrounded by general wear throughout the remainder of the bearing.

**SCENARIO C. CONTINUAL WEAR THROUGHOUT THE PERIPHERY OR LOAD ZONE OF ONE RACEWAY (Figure 6.09AA):**

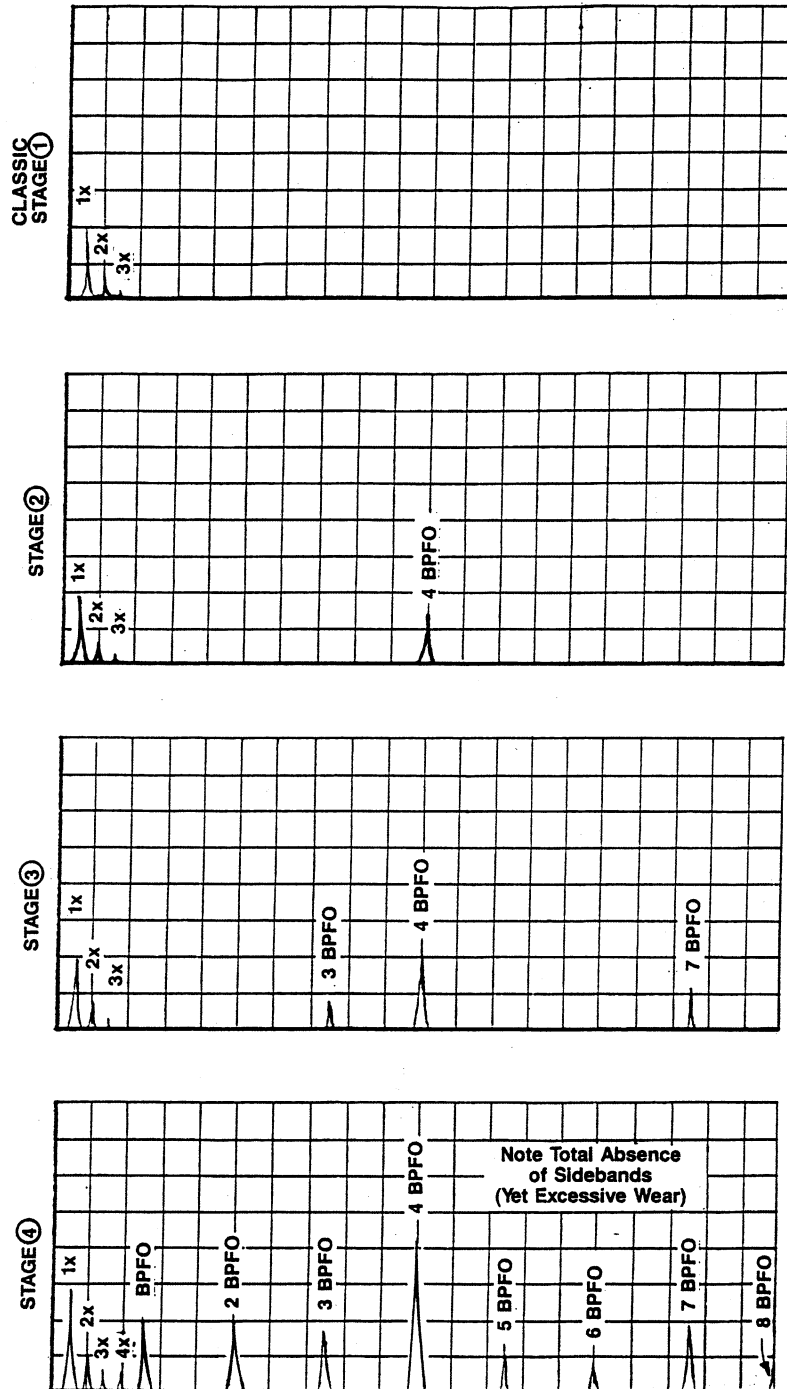
Figure 6.09AA illustrates this failure mode. Note that it passes through Stage 1 just as in Scenario A with the appearance only of increased amplitudes of overall ultrasonic parameters (Spike Energy, HFD, etc.). However, then one of the first frequencies to appear will be some harmonic of a race frequency (for example, the 4th or 5th harmonic outer or inner race frequency similar to that shown in Figure 6.09AA). In these cases, it is not yet clear why a harmonic defect frequency first responds before the bearing component natural frequencies. One possibility is that a system natural frequency (rotor, frame, etc.) may be located nearby a particular defect frequency harmonic (such as 4BPFO as shown in Figure 6.09AA). In this case, any excitation of this resonant frequency would generate much more response from this peak located close to the natural frequency than that from the other harmonics of the defect frequencies.

Later on in this failure mode, Stage 3 shows the appearance of more race frequency harmonics (in this case, 3BPFO and 7BPFO, in addition to 4BPFO). Another interesting thing happens in this particular failure mode. 1X RPM sidebands often do not appear. Still, the wear is becoming serious and is progressing around the periphery of the raceway. Later, in the final stages (shown in Stage 4), even more race frequency harmonics appear. They grow in amplitude, but still often are not surrounded by 1X RPM sidebands. Even though the height of any singular race defect frequency harmonic may only be .03 to .05 in/sec, a bearing generating this many harmonics of defect frequencies will normally show excessive wear which borders on bearing seizure.

**FIGURE 6.09Z**  
**CONTINUED DETERIORATION OF ONE PRONOUNCED FAULT ON A**  
**RACEWAY (Scenario B)**



**FIGURE 6.09AA**  
**CONTINUAL WEAR THROUGHOUT THE PERIPHERY OR LOAD ZONE OF**  
**ONE RACEWAY (Scenario C)**



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**SCENARIO D. DEVELOPMENT OF A SERIOUS FAULT FREQUENCY ACTING AS A SIDEBAND RATHER THAN AS A FUNDAMENTAL (Figure 6.09BB):**

Scenario D illustrates another way impending failures can be detected by signature analysis. Not only can the rolling element bearing defect frequencies appear as fundamentals or harmonics, they also can act as sidebands around other forcing frequencies. For example, when faults appear on balls or rollers, they often will generate a cage frequency (FTF) since the rolling elements impact the cage regardless of the rotation. However, the cage frequency itself seldom appears at its fundamental of about .35 to .45X RPM. Instead, it will more often modulate other frequencies causing sidebands spaced at cage frequency (FTF). When several defects are present, some will be identified from their fundamental frequency, but others will act at these Sum and Difference Frequencies. This is illustrated in Figure 6.09BB. Note the appearance of the cage frequency (FTF) acting as a sideband about a race frequency (BPFI in this case) in Stage 3. Note that when deterioration continues as shown in Stage 4, advanced faults will likely be present throughout the bearing. Note the multiple families of cage frequency sidebands around the 1st and 2nd harmonic inner race frequency and the 1X RPM sidebands around the ball spin frequency (BSF) and the bearing component natural frequency ( $f_n$ ) in this example. When this occurs, the life of the bearing is in great jeopardy.

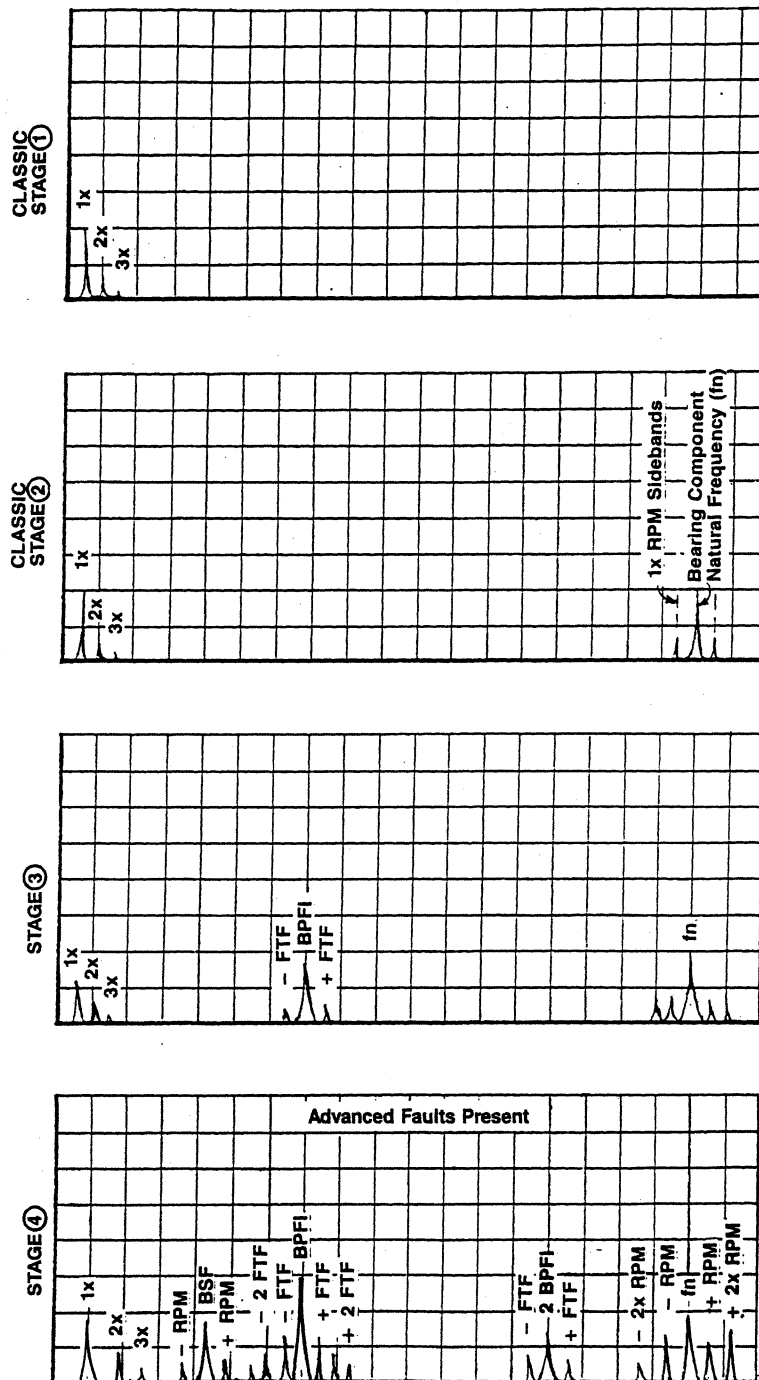
**SCENARIO E. CONDITION DETERIORATION ENDING EITHER WITH SEVERE MECHANICAL LOOSENESS OR THE BEARING TURNING ON THE SHAFT (Figure 6.09CC):**

Figure 6.09CC tracks the onset of a much more subtle bearing/shaft failure evidenced by the appearance of more and more running speed harmonics. This can eventually end up with either severe mechanical looseness or the bearing turning on the shaft (it is not possible at this point to differentiate between the two as they end up with almost identical spectra). Note the additional running speed harmonics appearing in Stages 2 and 3, with even more generated in Stage 4. However, note the absence of any bearing defect frequencies. In this case, the problem can be mechanical looseness, or it can be a problem of much greater severity - that is, a bearing turning on the shaft. In neither case would significant wear be expected in the bearing itself. However, the danger is the fact that significant damage can be taking place on the shaft due to a bearing turning on it and yet, very little vibration amplitudes may be occurring (commonly less than .05 in/sec). Therefore, if a signature approaching spectra shown in Stages 3 or 4 is captured as a baseline, it could be a moderate problem (mechanical looseness), or it could be a very serious problem (bearing turning on shaft). In these cases, it would be a good idea to lift up on the shaft (with the machine stopped) to determine the amount of play to help guide which problem is dominant and what steps should be taken.

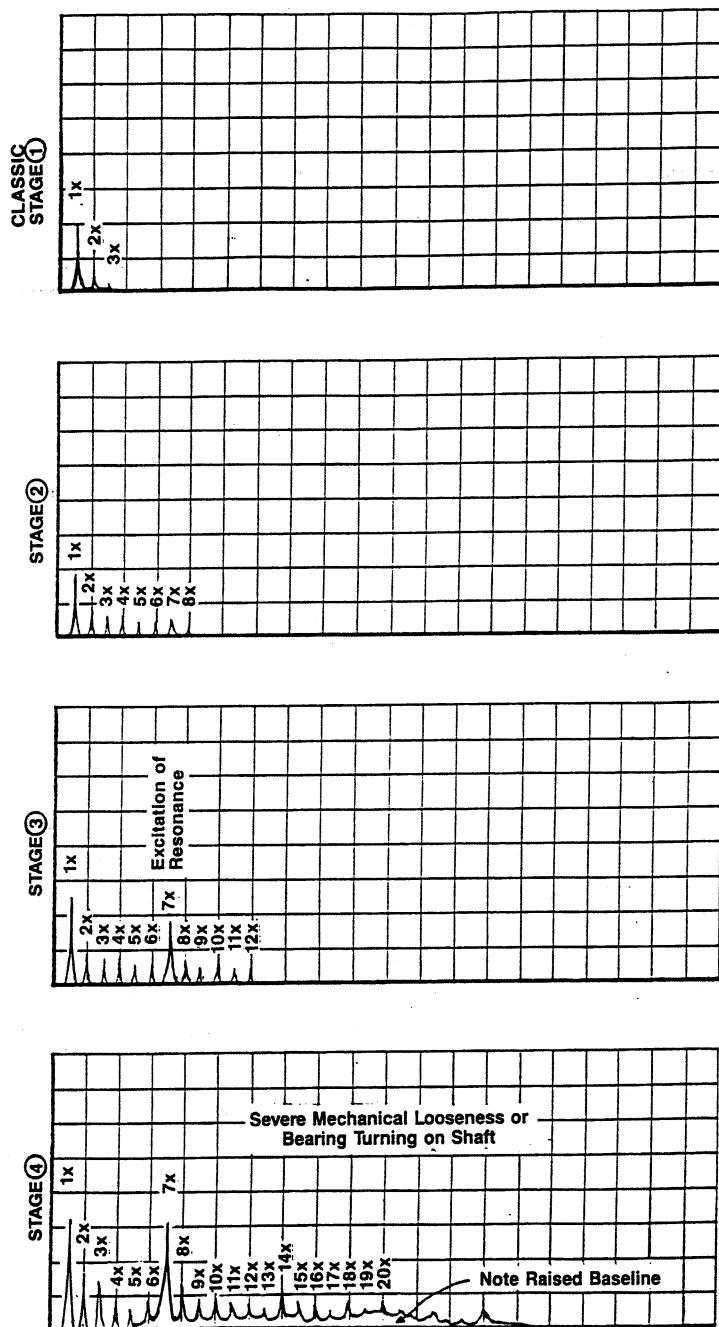
**SCENARIO F. DEVELOPMENT OF EXCESSIVE 1X RPM MODULATION ABOUT RACE FREQUENCIES ENDING UP WITH A WHOLE SERIES OF RUNNING SPEED HARMONICS CAUSED BY EXCESSIVE FATIGUE SPALLING ON RACEWAYS (Figure 6.09DD):**

Figure 6.09DD illustrates this problem involving excessive fatigue spalling on raceways. When this occurs, the bearing will normally pass through the first 2 stages normally, but will generate considerably more running speed sidebands about race defect frequencies in Stage 3. When this occurs, it can eventually cause a spectrum like that in Stage 4 - that is, a fairly high once per revolution vibration, followed by a whole series of running speed harmonics as the 1X RPM sidebands pretty well take over this spectrum. In this case, the noise floor will also normally lift up during the final stages before failure as shown in Stage 4 of Figure 6.09DD. At this point, even though 1X RPM might be of considerable magnitude, balancing will not really help the problem (and may not be even possible at this time).

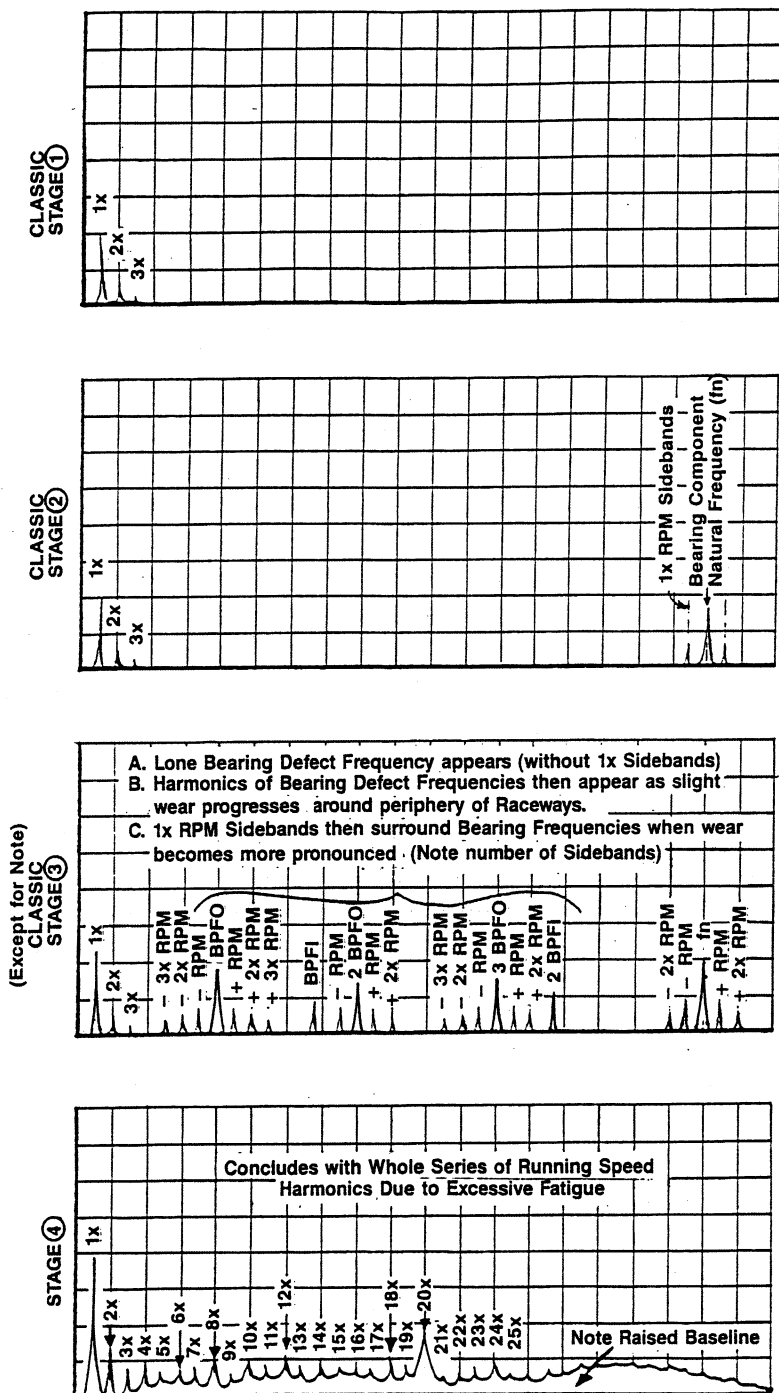
**FIGURE 6.09BB**  
**DEVELOPMENT OF A SERIOUS FAULT FREQUENCY ACTING AS A**  
**SIDE BAND RATHER THAN A FUNDAMENTAL (Scenario D)**



**FIGURE 6.09CC**  
**CONDITION DETERIORATION ENDING EITHER WITH SEVERE**  
**MECHANICAL LOOSENESS OR THE BEARING TURNING ON THE SHAFT**  
**(Scenario E)**



**FIGURE 6.09DD**  
**DEVELOPMENT OF EXCESSIVE 1X RPM MODULATION ABOUT RACE**  
**FREQUENCIES ENDING UP WITH RUNNING SPEED HARMONICS CAUSED**  
**BY EXCESSIVE FATIGUE SPALLING ON RACEWAYS (Scenario F)**



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## 6.094 WORD OF WARNING CONCERNING INSTRUMENTS AND TRANSDUCER MOUNTINGS

### A. HOW 8 BIT DATA COLLECTORS CAN MISS POTENTIALLY SERIOUS BEARING PROBLEMS:

The chart on Figure 6.09EE helps demonstrate the difference between an 8 bit and a 12 bit data collector. Importantly, the dynamic range of an 8 bit instrument is about 48 dB versus about 72 dB for a 12 bit instrument as shown in Figure 6.09EE. This 24 dB difference translates into a sensitivity difference of almost 16 times - that is, 12 bit data collectors are 16 times as amplitude sensitive as are their 8 bit counterparts. Experience has shown that 8 bit data collectors are not sufficiently sensitive to detect many rolling element bearing problems until the very final stages. Figure 6.09FF and the following anecdote help illustrate why this is so significant:

Charlie and Joe were two mechanics who decided to go into the vibration analysis business together. One day, opportunity struck when they travelled down to the local K-Mart there in Hertzville where they laid their eyes on some fancy, new "vibration rulers". They noticed two types were available. Each ruler ranged in length from 0 to 6 inches, but the "12 bit" ruler was delineated in 1/16 inch increments while the "8 bit" ruler was divided only into full 1 inch increments (see Figure 6.09FF). Now, Charlie thought he knew a good bargain when he saw one and he could see no reason why he should pay \$99 for that fancy "12 bit" ruler when he reasoned he could get by just fine with the "8 bit" model at only \$10. In fact, Charlie thought "Why, with all those little marks, they would probably just confuse me anyhow".

On the other hand, Joe went ahead and invested the extra \$89 since he thought he just might need that "little extra accuracy" sometime, and also reasoned that this "12 bit" ruler was 16 times as sensitive as was the "8 bit" model so it "might" be worth it sometime.

The next day, Charlie and Joe went on their first job with their shiny new "vibration rulers". On the first machine, they noticed it was "shaking and baking" quite a bit, so they paid close attention. Figure 6.09FF shows the vibration peaks that were present, each of which was measured by Charlie's "8 bit" and Joe's "12 bit" vibration rulers. Figure 6.09FF shows that Charlie measured a big spike at 1X RPM and claimed it was 6 inches high. Joe said "No, not quite, it's only 5 3/4 inches high". Charlie retorted, "Big deal! What's the difference anyhow?" Joe had to reply "Well, very little I guess", and Joe began to worry whether or not the extra "5 Bucks" was really worth it.

However, Charlie then said "Well, that's it. I guess we're ready to go to the next machine". But Joe replied, "Wait a minute, Charlie. We need to write down the measurements on those other 3 peaks, too". Charlie answered "What peaks?" Joe replied, "You know, those other 3 peaks out there with the 'whatcha-ma-call-its' surrounding that bigger peak on each side". Charlie looked at his ruler again and said, "Joe, I'm telling ya - there just ain't nothing out there!" Joe abruptly answered "Well Charlie, I'm telling you - there's a bearing frequency out there 5/16 inch high and there's 2 sidebands on each side of it that are 1/8 inch high each - this machine has a bearing problem!"

About this time, Charlie's heart began to sink as he realized he just blew "10 Bucks" and must now spend \$89 more when he recalled the little sign accompanying the vibration ruler display saying "Blue Light Specials On Sale Today - No Returns". On the other hand, Joe felt pretty good when he left for home that day since his first customer gave him an extra \$250 bonus for uncovering what could have been a serious problem had they not caught it in time on this "No.1 Widget Fabricator" which was their most critical machine in the plant!

Now that Joe and Charlie have helped illustrate the difference between an 8 bit and 12 bit instrument, we can examine what happened on a forming fan at a major industrial client when measurements were simultaneously taken with both an 8 bit instrument (IRD 818) and a 12 bit instrument (IRD 890) on the same bearing housing, using the same IRD 970 accelerometer connected to each instrument (initially connected it to the 890; then the 970 was left in place, but the connector at the data collector was removed from the 890 and attached to the 818). Note that the cascade spectrum shown in Figure 6.09GG shows that measurements had been taken for several surveys with the IRD 818. Then, when Technical Associates visited the plant with an IRD 890, an abruptly different signature was noticed by our analyst as per the 12/3/90 spectrum in Figure 6.09GG. This was reported to the plant and a decision was made to conduct the simultaneous measurement test at first opportunity.

On 12/5/90, this simultaneous measurement test was performed. First the IRD 890 captured the data at 13:51; then the transducer cable connector was removed from the 890 and connected to the 818 (leaving the accelerometer in place). The 818 captured the spectrum at 13:52 which is also shown in Figure 6.09GG. The important difference between the spectra captured by the 8 bit and 12 bit instruments is seen by referring to Figure 6.09GG and the discussion on "Failure Stage 3". Note that when a pair of sideband frequencies appear around a bearing defect frequency, they indicate further bearing deterioration. In the case of the 12 bit spectrum on Figure 6.09GG, note the presence not only of a single pair of sidebands, but 2 fully developed sideband families and a third on the way. Although this 12 bit spectrum does not indicate imminent failure, particularly due to the absence of any other bearing frequencies or bearing frequency harmonics, the multiple families of sidebands that were totally missed by the 8 bit instrument do suggest that this machine deserves close attention.

This is just one example of important information that has been detected by 12 bit instruments that has been missed by 8 bit models. This strongly points out the need to move up from 8 bit data collector models as soon as possible.

DECIBEL CONVERSION TABLE

$$dB = 20 \log (V_i/V_o)$$

$$V_i = V_o \log^{-1} \left( \frac{dB}{20} \right)$$

| dB  | Ratio     | A/D<br># Bits<br>Used | # Of Bits<br>Value | Dynamic Range<br>Largest<br>Number | dB<br>Range |
|-----|-----------|-----------------------|--------------------|------------------------------------|-------------|
| 100 | 100,000:1 | 1                     | 1                  | 2                                  | 6           |
| 90  | 31,623:1  | 2                     | 2                  | 4                                  | 12          |
| 80  | 10,000:1  | 3                     | 4                  | 8                                  | 18          |
| 70  | 3,162:1   | 4                     | 8                  | 16                                 | 24          |
| 60  | 1,000:1   | 5                     | 16                 | 32                                 | 30          |
| 50  | 316:1     | 6                     | 32                 | 64                                 | 36          |
| 48  | 251:1     | 7                     | 64                 | 128                                | 42          |
| 46  | 200:1     | 8                     | 128                | 256                                | 48          |
| 40  | 100:1     | 9                     | 256                | 512                                | 54          |
| 34  | 50:1      | 10                    | 512                | 1024                               | 60          |
| 26  | 20:1      | 11                    | 1024               | 2048                               | 66          |
| 20  | 10:1      | 12                    | 2048               | 4096                               | 72          |
| 14  | 5:1       | 13                    | 4096               | 8192                               | 78          |
| 12  | 4:1       | 14                    | 8192               | 16,384                             | 84          |
| 10  | 3.16:1    | 15                    | 16,384             | 32,768                             | 90          |
| 8   | 2.51:1    | 16                    | 32,768             | 65,536                             | 96          |
| 6   | 2:1       |                       |                    |                                    |             |
| 5   | 1.78:1    |                       |                    |                                    |             |
| 4   | 1.58:1    |                       |                    |                                    |             |
| 3   | 1.41:1    |                       |                    |                                    |             |
| 2   | 1.26:1    |                       |                    |                                    |             |
| 1   | 1.12:1    |                       |                    |                                    |             |
| 0   | 1:1       |                       |                    |                                    |             |

4096 = 16 (24dB)  
256

12 BIT A/D  
INSTRUMENTS  
ARE 16 TIMES  
AS SENSITIVE  
AS ARE 8 BIT  
INSTRUMENTS!

1 Part in 100 = 40dB  
1 Part in 200 = 46dB (8 BIT)  
1 Part in 1000 = 60dB  
1 Part in 2000 = 66dB  
1 Part in 4000 = 72dB (12 BIT)

FIGURE 6.09EE  
8 BIT VERSUS 12 BIT INSTRUMENTS

**FIGURE 6.09FF**  
**COMPARISON OF AMPLITUDE MEASUREMENTS DETECTED BY**  
**CHARLIE'S "8-BIT" AND JOE'S "12-BIT" VIBRATION RULER**

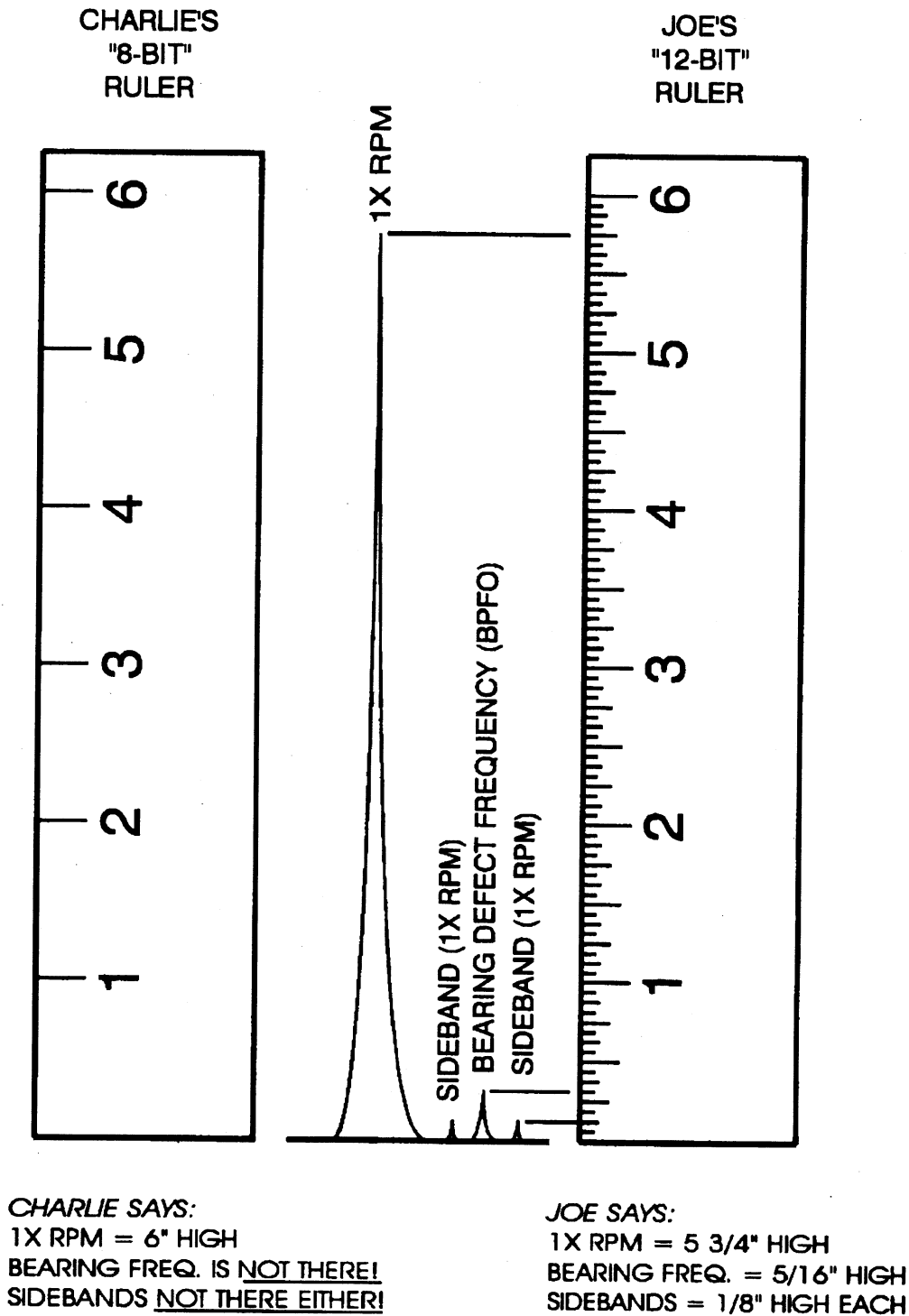
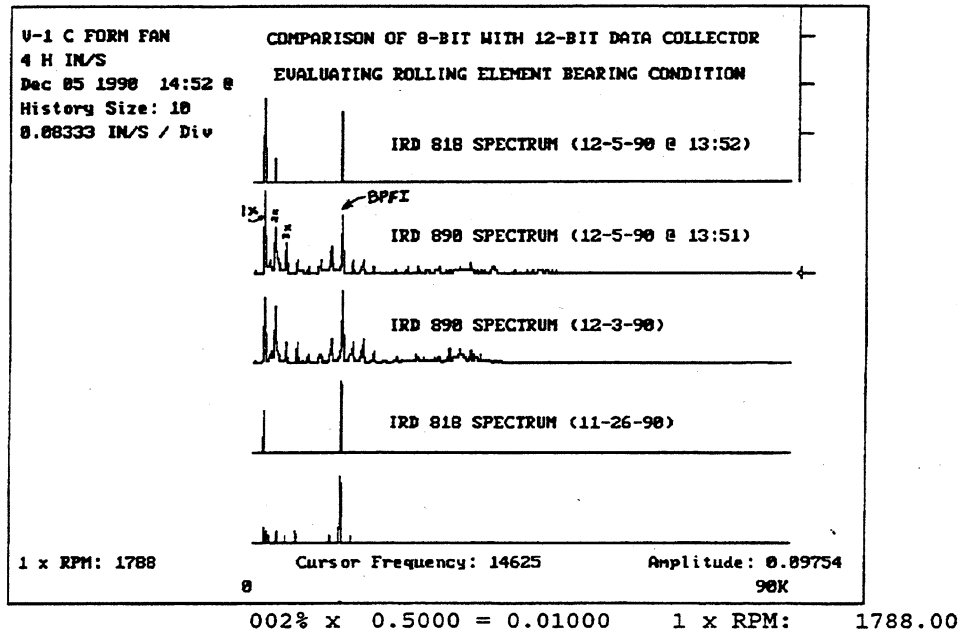


FIGURE 6.09GG



| Machine        |         |           | Pos        | Dir     | Units                            | Date        |  |  |
|----------------|---------|-----------|------------|---------|----------------------------------|-------------|--|--|
| V-1 C FORM FAN |         |           | 4          | H       | IN/S                             | Dec 05 1990 |  |  |
| Frequency      | 1 x RPM | Amplitude | Frequency  | 1 x RPM | Amplitude                        |             |  |  |
| 1800           | 1.01    | 0.14249   | 40050      | 22.40   | 0.01001                          |             |  |  |
| 2700           | 1.51    | 0.02002   |            |         |                                  |             |  |  |
| 3600           | 2.01    | 0.07752   |            |         |                                  |             |  |  |
| 5400           | 3.02    | 0.04996   |            |         |                                  |             |  |  |
| 7200           | 4.03    | 0.02248   |            |         |                                  |             |  |  |
| 9000           | 5.03    | 0.01001   |            |         |                                  |             |  |  |
| 11025          | 6.17    | 0.02002   | ←BPFI - 2X | }       | DETECTED<br>BY 12-BIT<br>IRD 890 |             |  |  |
| 12825          | 7.17    | 0.04750   | ←BPFI - 1X |         |                                  |             |  |  |
| 14625          | 8.18    | 0.09754   | ←BPFI      |         |                                  |             |  |  |
| 16425          | 9.19    | 0.02248   | ←BPFI + 1X |         |                                  |             |  |  |
| 18225          | 10.19   | 0.02248   | ←BPFI + 2X |         |                                  |             |  |  |
| 20025          | 11.20   | 0.01247   | ←BPFI + 3X |         |                                  |             |  |  |
| 25650          | 14.35   | 0.01001   |            |         |                                  |             |  |  |
| 27450          | 15.35   | 0.01001   |            |         |                                  |             |  |  |
| 30825          | 17.24   | 0.01247   |            |         |                                  |             |  |  |
| 34425          | 19.25   | 0.01247   |            |         |                                  |             |  |  |
| 36225          | 20.26   | 0.01501   |            |         |                                  |             |  |  |

DETECTED  
BY 12-BIT  
IRD 890

T - Threshold adjust      Q - Quit

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## B. IMPACT OF TRANSDUCER MOUNTING ON DETECTING ROLLING ELEMENT BEARING PROBLEMS:

Possibly, even more important and prevalent a problem than the "8 bit/12 bit" instrument performance question is the mounting of transducers when periodic vibration surveys are taken. Figure 6.09HH (provided by Bruel & Kjaer) illustrates this key point by comparing the frequency response for seven different mountings of a B&K 4367 accelerometer (whose characteristics closely resemble the standard accelerometers in use today by most data collector vendors). Notice how much the "maximum useful frequency" drops with different mountings as one moves from Type 1 through Type 7. For example, when the accelerometer is stud mounted on a thin film of silicon grease (Type 1), the mounting natural frequency is approximately 28,000 Hz (1,680,000 CPM) and the maximum useful frequency is about 10,000 Hz (600,000 CPM). Stud mounting always provides the best possible frequency response characteristics for either an accelerometer or velocity pickup.

Moving down to Type 4 with the accelerometer on an adhesive mounting, notice the difference when a good material such as Cyanoacrylate cement was used as opposed to when a soft glue with too great a thickness was employed. In the case of the Cyanoacrylate cement, it gave almost as good a frequency response as that obtained with Type 1 ending up with a mounting natural frequency of 26,000 Hz (1,560,000 CPM) and a flat response out to approximately 9000 Hz (540,000 CPM). However, the soft glue provided a flat response out to only 6000 Hz (360,000 CPM).

The spectra for the Type 5 mounting point out an even greater difference if too thick a double sided adhesive tape is used. In one case, the thin double sided tape provide a 19,000 Hz natural frequency and 5000 Hz useful frequency (300,000 CPM), whereas the flat response drops all the way down to only 500 Hz (30,000 CPM) when too thick a double sided adhesive tape is employed.

Mounting Types 6 and 7 should be carefully reviewed by all those taking periodic vibration surveys on key machines. A magnet is used under the accelerometer in Type 6 whereas a probe is attached directly to the accelerometer in the case of Type 7. Both mountings result in a drop in mounting natural frequency and in flat response. However, the drop off for the hand-held probe is considerably worse, particularly for those wanting to evaluate the condition of rolling element bearings.

In the case of the frequency response for the magnet mount in Type 6, this response is fairly typical of that for most of the accelerometer/ magnet systems offered by data collector vendors today. Its flat response is out to about 2000 Hz (120,000 CPM), but it can be successfully used out to as high as 3000 Hz (180,000 CPM) if the analyst takes careful precautions when mounting his magnet (making sure it is securely fastened to the machine with no rocking motion). Remember that a rocking motion of only .001 inch at 1800 RPM will produce an equivalent "false velocity" of almost .10 in/sec; and only .006 inch rocking at 1800 CPM will produce nearly .60 in/sec! Thus, magnet mounted accelerometers can produce good, repeatable measurements if they are mounted securely.

On the other hand, the frequency response for the Type 7 probe mount shown in Figure 6.09HH is considerably lower than either the Type 1 stud mount (or the Type 6 magnet mount for that matter). Note that the mounting natural frequency drops from 28,000 Hz (1,680,000 CPM) in the case of Type 1 down to only 1650 Hz (99,000 CPM) for this hand-held probe. Even more dramatic, its flat response drops from 10,000 Hz (600,000 CPM) for Type 1 down to only 450 Hz (27,000 CPM) when the probe is attached to the accelerometer. Another concern is seen by looking at what happens in Type 7 when actual forcing frequencies are present higher than the 1650 Hz natural frequency. Note that the response "drops off the table". That is, if there are

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frequencies related to bearing health beyond about 2500 Hz (150,000 CPM), the hand-held probe may not even detect their presence.

Finally, one of the other problems with hand-held probes in general is the fact that they will display vibration frequencies that appear to be of real concern right in the middle of the zone where so many bearing component natural frequencies and defect frequencies will occur between approximately 30,000 and 120,000 CPM. The reason for this is the great resonant amplification caused by the probe itself when any of these frequencies are present in this wide resonant amplification zone. The Type 7 spectrum shows that it can amplify frequency levels by as much as 26 dB (or a factor of 20 times higher). Therefore, this can lead to continually deceiving the analyst into believing bearing problems are present on one machine after another when, in fact, there may actually be little or nothing wrong with most all of the bearings.

Therefore, when any measurements are desired requiring a maximum frequency ( $F_{MAX}$ ) greater than 30,000 CPM, do not use a hand-held probe (unless there is no other way the bearing housing can safely be reached. If measurements up to 240,000 CPM are required, use at least a magnet mount. And, if measurements higher than 300,000 CPM are desired, a stud-mount (or at least a "Quick-Connect Mount") will have to be used to obtain meaningful, repeatable data, and to detect potentially significant problems which might be indicated at frequencies above approximately 1500 - 2000 Hz (90,000 - 120,000 CPM) which include not only bearing frequencies, but also gear mesh frequencies (and harmonics), rotor bar pass frequencies (and harmonics), lubrication-related frequencies, synchronous motor electrical problems at the coil passing frequency (and harmonics), dry whip in a plain bearing, etc.

## FIGURE 6.09HH EFFECT ON USEFUL FREQUENCY LIMIT ON MOUNTING OF ACCELEROMETER

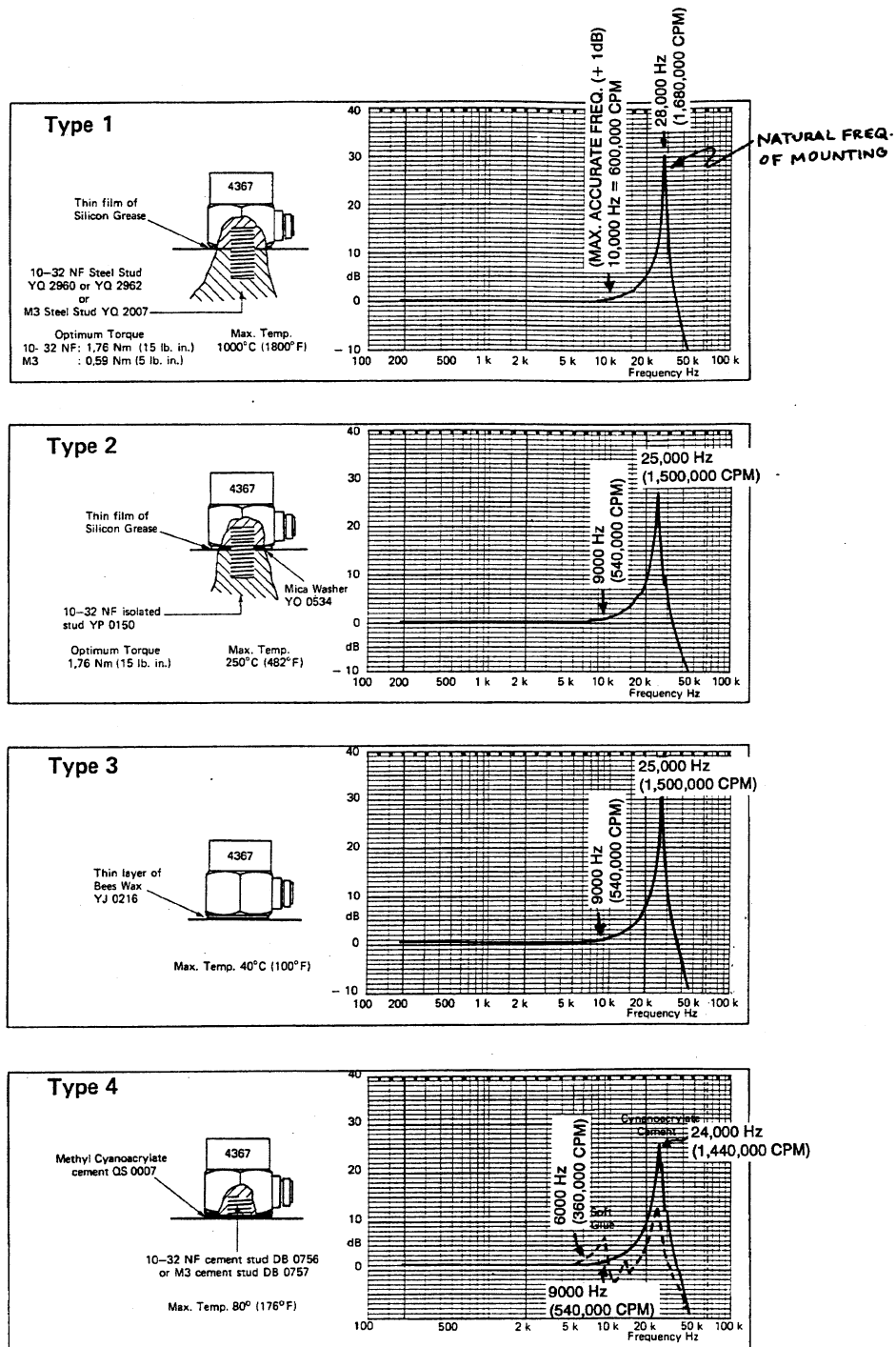
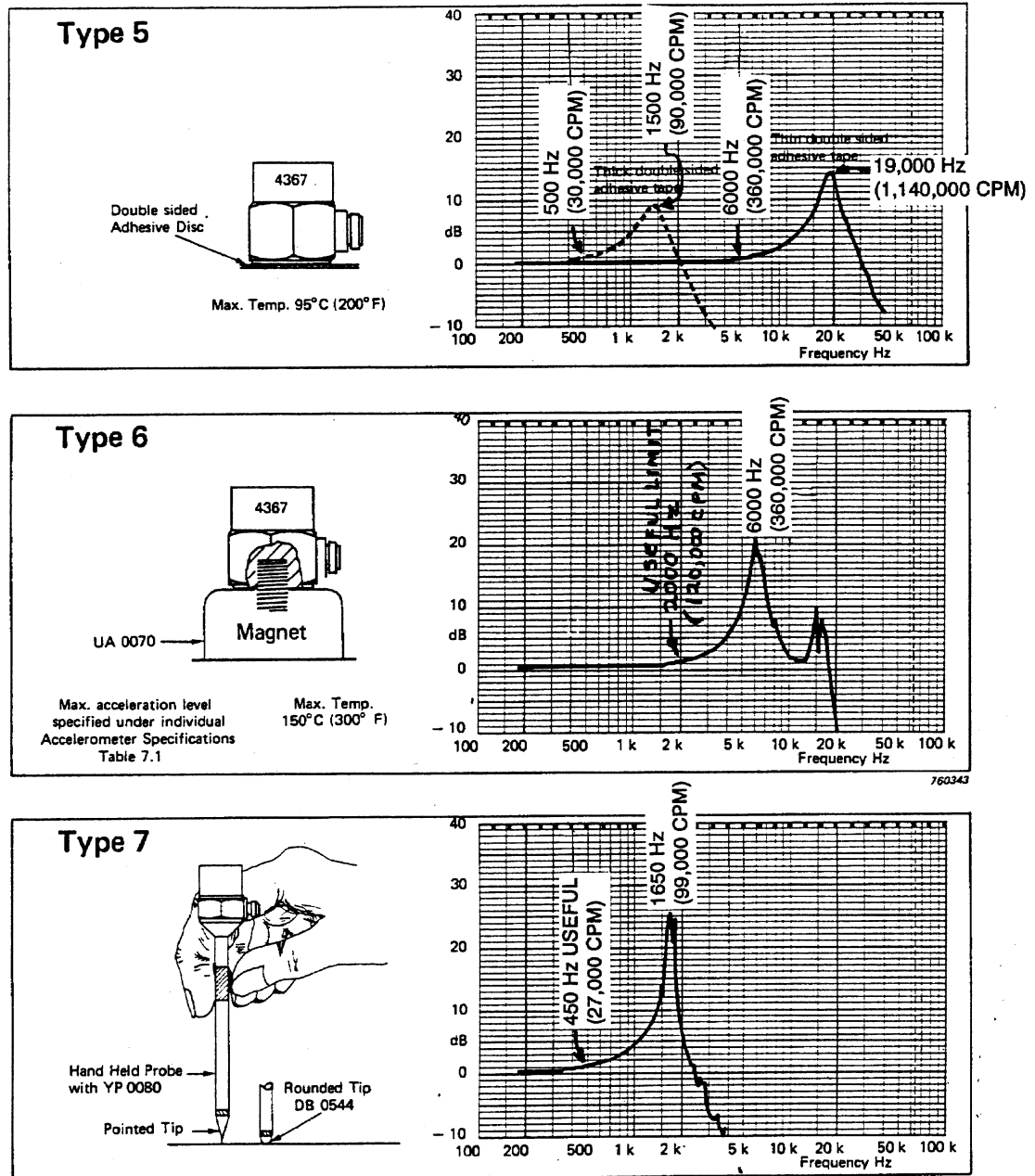


Fig. 4.2. Methods of mounting B & K piezoelectric accelerometers and examples of typical frequency responses obtained

(Reference: Piezoelectric Accelerometer & Vibration Preamplifier Handbook; Bruel & Kjaer; Revision March 1978; pages 52-53.)

**FIGURE 6.09HH**  
(Continued)



*Fig. 4.3. Methods of mounting B & K piezoelectric accelerometers and examples of typical frequency responses obtained*

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**6.095 RECOMMENDATIONS ON WHEN ROLLING ELEMENT BEARINGS SHOULD BE REPLACED:**

With a knowledge now of the various paths which can be taken by rolling element bearings to failure and how these paths affect the spectral content, this can go far in answering the age-old question of "When do we need to replace the bearings?" First, the analyst must have several questions answered for him such as:

1. How critical is this machine?
2. What is the cost per hour for it to be down?
3. What are the costs of replacement bearings (and how long will it require for the replacement bearing to arrive once an order has been placed)?
4. Is bearing deterioration affecting the machine productivity/quality performance?
5. If we decide not to replace it now, what would be the consequences were it to fail catastrophically?
6. What exactly is causing the bearing to fail (unbalance, misalignment, lubrication problems, excessive temperature, etc.)?
7. Is there any backup for this machine should it fail before predicted?
8. What is the production work load now? Is there a letup in the near future?
9. Will there be a scheduled downtime soon (if so, will this bearing "survive" until then)?
10. Am I positive this bearing has a problem?

These are just some of the questions that must be answered before a decision can be made. Again, note that the analyst himself should not attempt to answer all these questions himself.

With these questions and answers in mind, analysis on when to replace the bearings should likely be broken down into - (a) general machines of moderate criticality having reasonably priced replacement components with nominal lead times; or (b) large, highly critical machines, with expensive replacement components with long lead times.

Since this paper has shown there are a number of different failure scenarios through which rolling element bearings can pass, each of these must be considered separately for both the Type A (Noncritical, General Machinery) and Type B (Critical, Expensive Machinery). Table 6.09B on the following page has taken each of these into account and is offered to guide the analyst in his decision of whether he must take action and when he should do so.

**TABLE 6.09B**  
**WHEN TO REPLACE ROLLING ELEMENT BEARINGS ON NONCRITICAL GENERAL MACHINERY VERSUS**  
**CRITICAL, EXPENSIVE MACHINERY**

| FAILURE SCENARIO                                                                                                                    | TYPE A MACHINES<br>(NONCRITICAL, GENERAL MACHINERY<br>WITH NOMINAL LEAD TIMES)                                                                                                                                                                                                                | TYPE B MACHINES<br>(CRITICAL, EXPENSIVE MACHINERY<br>WITH LONG LEAD TIMES)                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A. Primary Failure Stage Scenario<br>(Most Rolling Element Bearings)                                                                | (A-1) If any defect frequency is present $\geq 30\%$ OA Alarm<br>OR<br>(A-2) If 3 or more defect frequencies are present each of which are $\geq 15\%$ OA Alarm<br>OR<br>(A-3) If 1X sidebands $\geq 10\%$ OA Alarm are present around any defect frequency which itself $\geq 20\%$ OA Alarm | (B-1) If any defect frequency is present $\geq 20\%$ OA Alarm<br>OR<br>(B-2) If 2 or more defect frequencies are present each of which are $\geq 10\%$ OA Alarm<br>OR<br>(B-3) If 1X sidebands $\geq 7.5\%$ OA Alarm are present around any defect frequency which itself $\geq 15\%$ OA Alarm |
| B. One Pronounced Fault Exciting Dominant Defect Frequency<br>(Metallurgical Problem)                                               | Rule (A-1)<br>OR<br>Rule (A-2)<br>OR<br>(A-4) If 1X sidebands $\geq 10\%$ OA Alarm are present around any defect frequency which itself $\geq 25\%$ OA Alarm (Since only 1 dominant defect frequency exists)                                                                                  | Rule (B-1)<br>OR<br>Rule (B-2)<br>OR<br>(B-4) If 1X sidebands $\geq 7.5\%$ OA Alarm are present around any defect freq. which itself $\geq 15\%$ OA Alarm (Since only 1 dominant defect frequency exists)                                                                                      |
| C. Continual Wear Throughout Periphery of Raceway<br>(No Sidebands Developed)<br>(This Should Not Be Present At All)                | Rule (A-1)<br>OR<br>Rule (A-2)<br>OR<br>Rule (A-3)                                                                                                                                                                                                                                            | Rule (B-1)<br>OR<br>Rule (B-2)<br>OR<br>Rule (B-3)                                                                                                                                                                                                                                             |
| D. When Defect Frequency Appears As Sideband Frequency Itself<br>(Roller/Ball Or Cage Problem)                                      | (A-5) When any difference frequency $\geq 30\%$ OA Alarm equal any defect frequency<br>OR<br>(A-6) If 3 or more difference frequencies $\geq 15\%$ OA Alarm equal any defect frequency<br>OR<br>Rule (A-3)                                                                                    | (B-5) When any difference frequency $\geq 20\%$ OA Alarm equal any defect frequency<br>OR<br>(B-6) If 2 or more difference freq. $\geq 10\%$ OA Alarm equals any defect frequency<br>OR<br>Rule (B-3)                                                                                          |
| E. 1X RPM Multi-harmonics Ending With Severe Mechanical Looseness or Bearing Turning on Shaft<br>(Undersized Shaft)                 | (A-7) If any 5 running speed harmonics exists each of which are $\geq 15\%$ OA Alarm                                                                                                                                                                                                          | (B-7) If any 4 running speed harmonics exists each of which are $\geq 10\%$ OA Alarm                                                                                                                                                                                                           |
| F. Excessive 1X RPM Sidebanding About Race Frequencies Ending Up With 1X RPM Multi-harmonics<br>(This Should Not Be Present At All) | Rule (A-1)<br>OR<br>Rule (A-2)<br>OR<br>Rule (A-3)<br>OR<br>Rule (A-7)                                                                                                                                                                                                                        | Rule (B-1)<br>OR<br>Rule (B-2)<br>OR<br>Rule (B-3)<br>OR<br>Rule (B-7)                                                                                                                                                                                                                         |

**NOTES:**

1. "Defect Frequencies" = Refers to any defect frequency whether it is a fundamental defect frequency or a harmonic of a defect frequency.
2. "Difference Frequency" = The mathematical difference between any 2 frequencies in the spectrum.
3. "OA Acceptance = Acceptance Level of Overall Velocity (in/sec) specified for particular machine type (rms or peak).
4. Table XX is offered as a guide to help the analyst decide when bearing replacement is required. It can be adapted by analysts within a specific plant (as required) to further enhance this important decision which will affect not only Maintenance, but also Production.

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## 6.10 FLOW-INDUCED VIBRATION

Unlike many of the other mechanical vibration problems such as unbalance, misalignment, resonance and so forth, flow-induced vibration is often very dependent upon operating conditions. That is, it can be very affected by load depending on the type of machine and the work being performed by the machine. Flow-induced vibration includes each of the following:

- Hydraulic and Aerodynamic Forces
- Cavitation and Starvation
- Recirculation
- Flow Turbulence
- Surge
- Choking

Each of these problems will be discussed in Section 6.101 through 6.106. Each of these phenomena are capable of generating extremely high energy levels and must be avoided by proper design and operation of the machines involved. In addition, each of them generate particular frequencies of vibration which again are often dependent upon the load and other operating conditions. Following below will be a discussion on each type of flow-induced vibration.

### 6.101 Hydraulic and Aerodynamic Forces:

Various types of machines including pumps, blowers, turbines, vacuum pumps and so forth inherently produce hydraulic or aerodynamic forces as their impellers impart work on the fluid (liquid or gaseous) they are handling. In most cases, the vibration generated from hydraulic and aerodynamic forces is not a problem unless it happens to excite resonant frequencies, if a potentially destructive phenomenon known as cavitation occurs, or if vibration generated by the impellers themselves becomes excessive subjecting expensive machine components to undesirably high vibration. Also, when vibration at blade pass frequency suddenly jumps, it can indicate something has gone wrong with the driven part of the machine. This will be covered below.

In general, hydraulic and aerodynamic forces are generated as per the following equation:

$$\text{Blade (or Vane) Pass Frequency} = \boxed{\text{BPF} = \# \text{Vanes} \times \text{Impeller RPM (CPM or Hz)}}$$

These forces are generated by a pressure variation or pulse each time a blade (or vane) loads or unloads as it passes nearby stationary components (such as diffuser vanes or discharge volutes). That is, if one were at a fixed point in space and he observed a six vane pump impeller, he would "feel" each of 6 pulses as the impeller made one revolution (in this case, 6 events per revolution).

Hydraulic and aerodynamic force vibration exhibits each of the following characteristics:

1. Significant vibration both at blade pass frequency and blade pass harmonics will be generated if an impeller is not centrally located within its housing and properly aligned with diffusers. For example, if the gap between impeller and diffuser vanes is .25" on one side, but .50" on the other a whole series of blade pass frequencies can be generated, likely at high vibration levels. Clearances all throughout the periphery should be identical unless there are specific design reasons to depart from this (as is the case with some types of pumps and fans).

2. It is important to ensure that neither the blade pass frequency nor blade pass harmonics coincide with natural frequencies of the rotor or support structure. Excessive vibration can result if allowed to do so.
3. A great increase in blade pass frequency will occur if welds securing diffuser vanes experience failure allowing the diffusers to shift position slightly relative to the impeller. This will affect hydraulic/aerodynamic flow which should be obvious from a waterfall spectrum trending vibration spectra from one survey to the next. Such an example is given in Figure 6.10A. Note from the lower plot that Blade Pass vibration at 14,310 CPM (16X RPM) significantly increased 10X higher from .010 to .100 in/sec between one survey and the next; then almost doubled again up to .193 in/sec by the next survey due to diffuser weld fatigue.
4. Similarly, a great increase in blade pass vibration will be created if a pump impeller wear ring seizes to the shaft.
5. Blade pass frequency in some particular fan types is particularly sensitive to damper settings. In other words, with these fans, just altering the damper setting alone can greatly affect blade pass vibration. Here again, this vibration alone is not particularly destructive to any machine component unless it should exceed approximately .30 to .40 in/sec peak velocity for most blower types.
6. Another frequency which can sometimes be generated on centrifugal machinery outfitted with rotating vanes and stationary diffusers is known as the Blade Rate Frequency (BRF). The blade rate frequency has to do with the number of times and rate at which rotating and stationary vanes coincide with one another. When they do coincide, there can be a pronounced pulse due to compression of the fluid (gas or liquid) between the rotating and stationary vanes. These pulses from simultaneous coincidence of two or more sets of vanes can be much stronger than if only one rotating and stationary vane were directly in line with one another at a given instant of time. This is why it is not a good idea to have numbers of rotating vanes and diffusers which have common denominators of one another.

The Blade Rate Frequency (BRF) is as follows:

$$\text{BRF} = \frac{(\# \text{Impeller Vanes})(\# \text{Diffuser Vanes})(\text{RPM})}{K}$$

where:

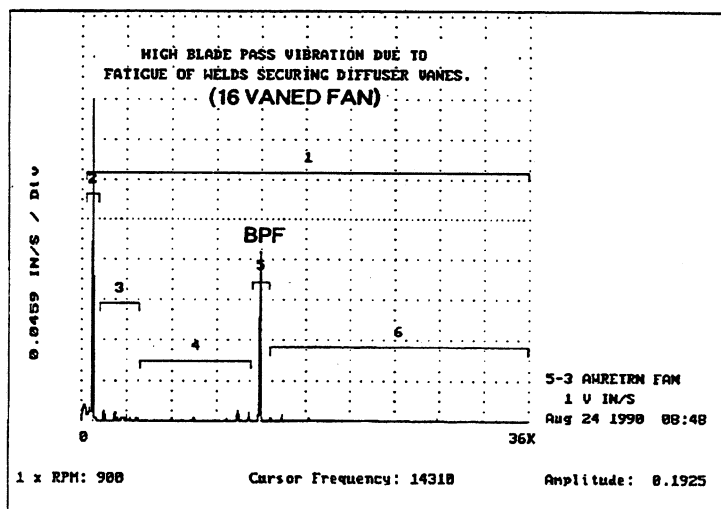
BRF = Blade Rate Frequency (CPM or Hz)

K = Highest Common Factor of #Impeller Vanes and #Diffuser Vanes

For example, if a machine had 18 impeller vanes and 24 diffuser vanes, K would equal to 6 so that:

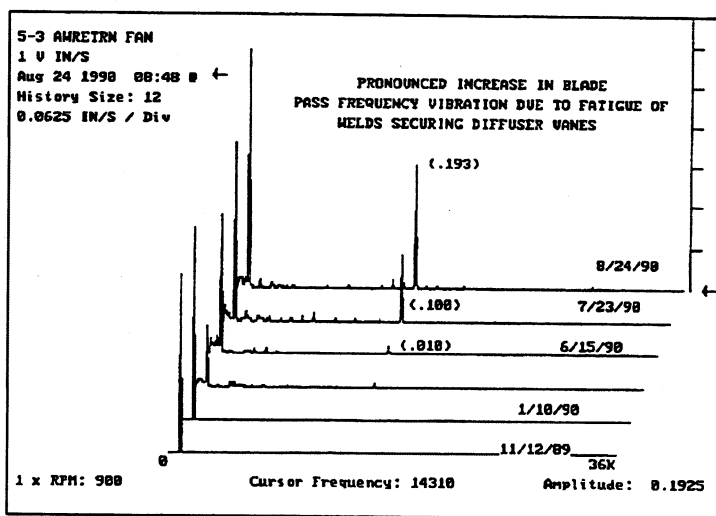
$$\begin{aligned} \text{BRF} &= \frac{(18 \text{ Impeller Vanes})(24 \text{ Diffuser Vanes})(\text{RPM})}{6} \\ &= \underline{72 \times \text{RPM}} \end{aligned}$$

Thus, BRF (72 X RPM) would be 4X higher than BPF (18 X RPM) in this case. And, as was pointed out above, this machine would likely suffer much higher pulsations due to the fact that more than one set of impeller and diffuser vanes would line up with one another (in this case, 6 impeller vanes would simultaneously be directly opposite diffuser vanes at angles of 0°, 60°, 120°, 180°, 240° and 300°), resulting in pronounced pulsations at Blade Rate Frequency (BRF). If instead there were either 17 impeller vanes or 25 diffuser vanes, at no instant in time would more than one set of impeller and diffuser vanes line up with one another, and therefore, the machine would likely experience lower vibration.



001 1/2 x 0.500 = 0.0050 1 x RPM: 900.00

| Machine         | Pos     | Dir       | Units | Date              |
|-----------------|---------|-----------|-------|-------------------|
| 5-3 AWRETRN FAN | 1       | V         | IN/S  | Aug 24 1990 08:48 |
| Frequency       | 1 x RPM | Amplitude |       |                   |
| 270             | 0.30    | 0.0175    |       |                   |
| 630             | 0.70    | 0.0150    |       |                   |
| 900             | 1.00    | 0.3675    |       |                   |
| 1800            | 2.00    | 0.0125    |       |                   |
| 2700            | 3.00    | 0.0075    |       |                   |
| 3600            | 4.00    | 0.0050    |       |                   |
| 12510           | 13.90   | 0.0125    |       |                   |
| 13410           | 14.90   | 0.0075    |       |                   |
| 14310           | 15.90   | 0.1925    |       |                   |
| 16110           | 17.90   | 0.0050    |       |                   |



**FIGURE 6.10A**  
**HIGH BLADE PASS VIBRATION DUE TO FATIGUE OF WELDS**  
**SECURING DIFFUSER VANES (16 VANED FAN)**

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### 6.102 Cavitation and Starvation:

Cavitation is a common problem with centrifugal pumps and can be quite destructive to internal pump components. Often, pumps subjected to continuing cavitation will have badly pitted and/or eroded impellers. In some cases, cavitation can completely erode away impeller vanes.

Cavitation most often occurs when a pump is operating with excess capacity or low suction pressure. In essence, since the pump is "starved", the fluid entering the pump is almost literally "pulled apart" as it attempts to fill the voids which exists. This creates pockets of nearly perfect vacuum which are prone to collapse or implode quickly. As they do so, these implosions cause impacts which excite local natural frequencies of the impeller and nearby components. Since these implosions may occur at random intervals all throughout the pump or connected piping, the vibration will be highly random both in amplitude and in frequency.

Cavitation exhibits each of the following characteristics:

1. Classic spectra indicating cavitation have random, broad band energy most often between approximately 20,000 CPM up through approximately 120,000 CPM. It can appear as "white noise" with no discernible frequency content, or at times with vane pass frequencies superimposed. That is, several blade pass frequency harmonics sometimes will appear along with the random, higher frequency vibration.
2. When cavitation occurs it will increase ultrasonic measurements (spike energy, HFD, shock pulse, etc.). Figure 6.10B shows a spectrum indicative of cavitation problems. Note not only the random broad band vibration, but also the high levels of spike energy on both pump bearings (positions 3H and 4H which were both in alarm on spike energy). Normally, cavitation will induce high spike energies on both pump bearings, whereas lubrication or wear problems of one bearing usually will show high spike energy only on the one bearing with the problem.
3. Cavitation will most often generate unique noise. Milder cavitation normally sounds like sand being pumped whereas more severe cavitation will actually sound as if gravel is being passed through the pump (therefore, if the analyst hears this type of sound during his vibration survey when evaluating a pump, he should make a note of this during his route and/or enter an inspection code indicating this observation if his data collector has this capability; then, when back at the office, if the analyst sees both the random, high frequency vibration and notes the abnormal noise, it will greatly improve the validity of his diagnosis).
4. Starvation is the aerodynamic counterpart to cavitation. Like cavitation, it involves insufficient air flow relative to fan capacity. It normally involves damper settings and at times, improper application of equipment. Typical starvation spectra are almost identical to those of cavitation (higher frequency, broad band energy) which sometimes can be excessive causing great vibration in suction and discharge ducts in addition to the fan itself.

### 6.103 Recirculation:

In the case of a pump, recirculation is just the opposite of cavitation. Recirculation can occur when a pump is operated at too low a capacity or too high suction pressure. When it occurs, a portion of the fluid returns from the discharge to the impeller as the pump attempts to move an excessive amount of fluid through the pump. This brings about a reverse flow and results in 2 or more fluids moving in opposite directions causing noise and vibration. Recirculation exhibits the following characteristics:

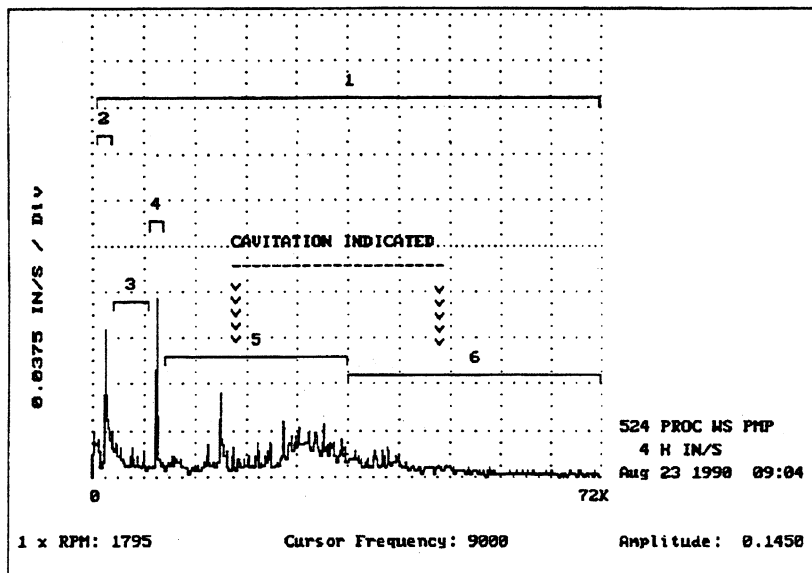
- 
1. Recirculation generates vibration spectra very similar to those caused by cavitation. That is, it will often have random, higher frequency vibration, sometimes superimposed with vane pass frequencies.
  2. Unlike cavitation, recirculation will not likely cause wear or erosion of pump components. However, if the vibration becomes excessive, it can damage bearings, wear rings, valves and other related components.

#### **6.104 Flow Turbulence:**

Flow turbulence occurs when something interferes with or resists the normal flow of liquids or gases through pumps, fans, compressors or vacuum pumps. Such flow disturbances might be caused by obstructions in the duct or pipe, sharp right angle turns, abrupt changes in diameter, etc. Correction of these problems would include removal of obstructions, insertion of turning vanes if a duct presents too sharp a right angle turn, and provision of a longer, stepped change in diameter where necessary to reduce duct or pipe cross section.

Figure 6.10C shows a typical spectrum indicating flow turbulence. Flow turbulence problems exhibit the following characteristics:

1. Although flow turbulence can generate random, higher frequency vibration, it normally will cause random, lower frequency vibration below 1X RPM. In the case of fans and blowers, reference 8 reports that flow turbulence will create random, low frequency vibration concentrated in a range from approximately 50 CPM up to 2000 CPM which will represent the natural frequencies of the machine, structure and duct work. In the case of pumps, Reference 9 reports that frequencies ranging from .55X RPM up to .78X RPM will be excited in the case of hydraulic instability, problems with wear rings, seal trouble, or difficulties with the balancing disc or drum.
2. Amplitudes and frequencies of flow turbulent vibration will be erratic, sometimes pulsating widely.
3. Sometimes noise produced by flow turbulence will be very high although the vibration accompanying it will be relatively low on the machine itself. The reason for this is that the turbulence condition itself exists outside the machine.



005% x 0.500 = 0.0250      1 x RPM: 1795.00

| Machine         | Pos | Dir | Units | Date              |
|-----------------|-----|-----|-------|-------------------|
| 524 PROC WS PMP | 4   | H   | IN/S  | Aug 23 1990 09:04 |

| Frequency | 1 x RPM | Amplitude | Frequency | 1 x RPM | Amplitude |
|-----------|---------|-----------|-----------|---------|-----------|
| 180       | 0.10    | 0.0375    | 32580     | 18.15   | 0.0425    |
| 900       | 0.50    | 0.0300    | 33120     | 18.45   | 0.0250    |
| 1800      | 1.00    | 0.1200    | 33480     | 18.65   | 0.0275    |
| 2700      | 1.50    | 0.0375    | 34020     | 18.95   | 0.0275    |
| 3240      | 1.81    | 0.0275    | 35100     | 19.55   | 0.0300    |
| 9000      | 5.01    | 0.1450    |           |         |           |
| 16200     | 9.03    | 0.0250    |           |         |           |
| 18000     | 10.03   | 0.0675    |           |         |           |
| 23220     | 12.94   | 0.0275    |           |         |           |
| 25020     | 13.94   | 0.0275    |           |         |           |
| 26820     | 14.94   | 0.0450    |           |         |           |
| 27720     | 15.44   | 0.0275    |           |         |           |
| 28080     | 15.64   | 0.0325    |           |         |           |
| 28620     | 15.94   | 0.0325    |           |         |           |
| 29160     | 16.25   | 0.0400    |           |         |           |
| 30600     | 17.05   | 0.0350    |           |         |           |
| 31500     | 17.55   | 0.0350    |           |         |           |

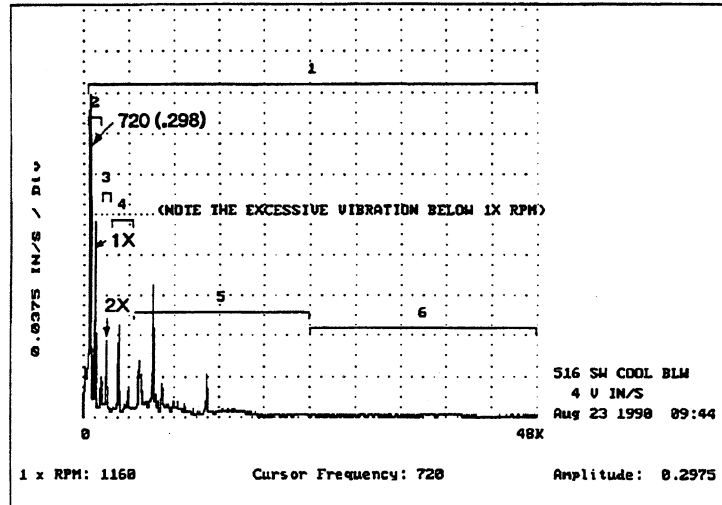
T - Threshold adjust      Q - Quit

#### Machine Report (Latest Data) for: 524 PROC WS PMP

| Date      | Time  | Pos | Dir | Ampl  | Units | Change | Alarm | Inspection                                 |
|-----------|-------|-----|-----|-------|-------|--------|-------|--------------------------------------------|
| Aug-23-90 | 09:03 | 1   | A   | .048  | IN/S  | -33%   | .300  |                                            |
| Aug-23-90 | 09:03 | 1   | H   | .096  | IN/S  | -20%   | .300  |                                            |
| Aug-23-90 | 09:03 | 1   | H   | .016  | g/SE  | -16%   | .600  |                                            |
| Aug-23-90 | 09:03 | 1   | O   | .087  | IN/S  | -----  | .300  |                                            |
| Aug-23-90 | 09:03 | 1   | V   | .054  | IN/S  | -2%    | .300  |                                            |
| Aug-23-90 | 09:03 | 2   | A   | .101  | IN/S  | 80%    | .300  |                                            |
| Aug-23-90 | 09:03 | 2   | H   | .123  | IN/S  | 15%    | .300  |                                            |
| Aug-23-90 | 09:03 | 2   | H   | .037  | g/SE  | 23%    | .600  |                                            |
| Aug-23-90 | 09:03 | 2   | O   | .125  | IN/S  | -----  | .300  |                                            |
| Aug-23-90 | 09:04 | 2   | V   | .063  | IN/S  | 9%     | .300  |                                            |
| Aug-23-90 | 09:04 | 3   | A   | .255  | IN/S  | 18%    | .300  |                                            |
| Aug-23-90 | 09:04 | 3   | H   | .212  | IN/S  | -21%   | .300  |                                            |
| Aug-23-90 | 09:04 | 3   | H   | *.640 | g/SE  | 23%    | .600  | PUMP INBOARD (3H) & OUTBOARD (4H) BEARINGS |
| Aug-23-90 | 09:04 | 3   | V   | .195  | IN/S  | 17%    | .300  |                                            |
| Aug-23-90 | 09:04 | 4   | A   | *.307 | IN/S  | 76%    | .300  |                                            |
| Aug-23-90 | 09:04 | 4   | H   | *.348 | IN/S  | 7%     | .300  |                                            |
| Aug-23-90 | 09:04 | 4   | H   | *0.74 | g/SE  | 4%     | .600  |                                            |
| Aug-23-90 | 09:04 | 4   | V   | .288  | IN/S  | 27%    | .300  |                                            |

18 lines this report

**FIGURE 6.10B**  
**DEVELOPMENT OF CAVITATION IN A WATER SUPPLY PUMP**



003% x 0.500 = 0.0150 1 x RPM: 1160.00

| Machine         | Pos     | Dir       | Units     | Date              |           |
|-----------------|---------|-----------|-----------|-------------------|-----------|
| 516 SW COOL BLW | 4       | V         | IN/S      | Aug 23 1990 09:44 |           |
| Frequency       | 1 x RPM | Amplitude | Frequency | 1 x RPM           | Amplitude |
| 120             | 0.10    | 0.0475    |           |                   |           |
| 720             | 0.62    | 0.2975    |           |                   |           |
| 1200            | 1.03    | 0.1800    |           |                   |           |
| 1800            | 1.55    | 0.0375    |           |                   |           |
| 2400            | 2.07    | 0.0700    |           |                   |           |
| 3720            | 3.21    | 0.0850    |           |                   |           |
| 4800            | 4.14    | 0.0275    |           |                   |           |
| 5880            | 5.07    | 0.0525    |           |                   |           |
| 6120            | 5.28    | 0.0400    |           |                   |           |
| 6600            | 5.69    | 0.0150    |           |                   |           |
| 7320            | 6.31    | 0.1225    |           |                   |           |
| 7680            | 6.62    | 0.0200    |           |                   |           |
| 7920            | 6.83    | 0.0150    |           |                   |           |
| 8280            | 7.14    | 0.0300    |           |                   |           |
| 9600            | 8.28    | 0.0150    |           |                   |           |
| 9960            | 8.59    | 0.0150    |           |                   |           |
| 13080           | 11.28   | 0.0400    |           |                   |           |

**FIGURE 6.10C**  
**PROBABLE TURBULENT FLOW PROBLEM INDICATED ON THIS BLOWER**

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#### **6.105 Surge:**

One of the more important problems plaguing high speed centrifugal and axial-flow compressors is surge. Typically a compressor experiencing surge is operating outside design limits. Surge occurs when the discharge pressure is too high or if the volumetric mass flow rate is too low relative to design conditions for a particular operating speed. When this occurs, the gas flow will actually reverse directions in the compressors. When the surge is only minor, this flow reversal may occur only at the boundary layers of the impeller blades. However, when full surge develops, the entire fluid flow will reverse directions and will flow from the discharge to the inlet. Surge must be prevented in a compressor due to the fact that it can wreak extensive damage. Fortunately, many electronic controls are available today which can make almost instantaneous adjustments within the machinery to prevent surge. Surge exhibits the following characteristics:

1. Minor surge causes a noticeable increase at blade pass frequency and harmonics within only a time period of seconds. Such blade pass frequencies may double or even triple in amplitude.
2. Fully developed surge will most often lift up the entire spectrum. That is, the entire "noise floor" of the spectrum will be lifted, creating high amplitude, random response covering a broad frequency range. This is caused by turbulent flow within the compressor exciting a series of natural frequencies including those of the impeller wheels, rotor blades, diffuser blades, casing, shaft, gearing and other components. Left uncorrected, surge can wreck a compressor within a short period of time.

#### **6.106 Choking:**

Choking (sometimes called "stone walling") is essentially the opposite of surging in a compressor. Choking occurs when discharge pressures are too low creating high velocities in the diffuser section. When such flow velocities approach Mach 1, a turbulent flow will occur between the diffuser blades which will have the effect of blocking the fluid flow. When this occurs, not only will there be an increase in vibration due to the turbulent flow, but also there will be a significant drop in compressor efficiency and pressure ratio. Vibration spectral characteristics of choking will be essentially identical to those of surging. Therefore, one will have to check other operating parameters such as pressure, mass flow, and so forth in order to differentiate which problem is occurring.

## 6.11 GEAR PROBLEMS

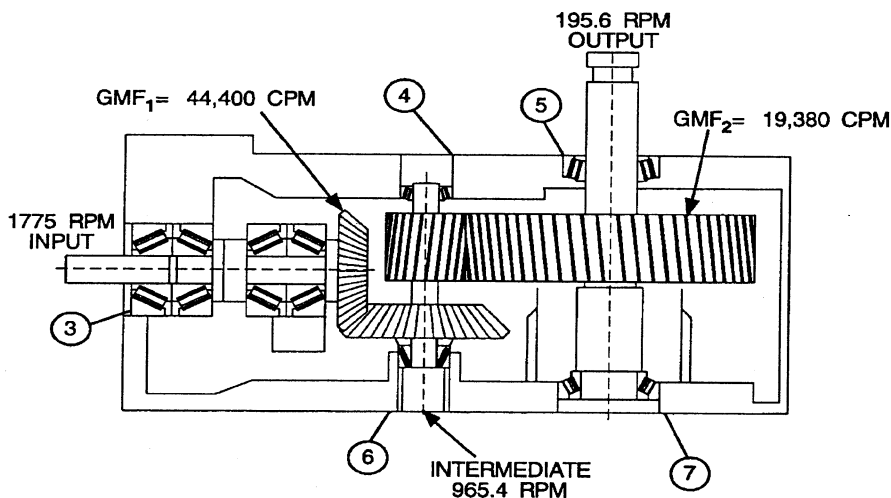
A variety of gear problems can be detected by vibration analysis. These include each of the following:

- Gear Tooth Wear
- Excessive Tooth Load
- Gear Eccentricity and/or Backlash
- Cracked, Chipped or Broken Gear Teeth
- Hunting Tooth Problems

Each of these problems will be discussed in Sections 6.111 through 6.116. First, several remarks in general should be made about vibration diagnostics on gears.

One of the key frequencies of interest when evaluating gear health will be gear mesh frequency ( $\# \text{teeth} \times \text{RPM}$ ). However, it is important to point out gear mesh frequency (GMF) is not a fault or defect frequency as is the case with bearing defect frequencies. All meshing gears generate gear mesh frequencies of some amplitude or another. In addition, all gear mesh frequencies will have sidebands of some amplitude spaced at the RPM of each mating gear in the mesh. However, if the gears are in good health and are properly aligned with one another (insignificant misalignment, backlash or gear eccentricity), amplitudes of GMF and its harmonics along with those of sidebands should be low, particularly those of the sidebands.

Figure 6.11A shows a standard setup for analyzing a right angle, double reduction gearbox outfitted in this case with bevel and helical gears. The example illustrated in Figure 6.11A will be used to make several comments:



**FIGURE 6.11A**  
**STANDARD SETUP FOR ANALYSIS OF A RIGHT ANGLE**  
**DOUBLE REDUCTION GEARBOX**

1. Vibration measurements should be made on each bearing housing which is accessible. Figure 6.11A shows measurements being taken on the double row bearing at position 3 as well as on each of the four tapered roller bearings in positions 4 through 7. The important point is that the transducer should be placed as near as possible to the bearings supporting the gears themselves. At times, this may involve measurements at a distance from the bearings. However, in this case, ensure that a frame or internal web that goes directly to the bearing housing is located and measurements made as close to these as possible.

- 
2. Measurements should be taken in all 3 orthogonal directions (horizontal, vertical and axial), particularly since some gears generate forces predominately in one direction or the other which may not be the same from one survey to the next, depending on load.
  3. In general, helical, herringbone and bevel gears generate significant vibration in the axial direction. Often, the best condition measurement on these gear types is in this axial direction.
  4. In general, spur gears are best evaluated in radial directions, but can sometimes have significant axial vibration as well, particularly if there is a problem with tooth alignment. At times, very high frequency measurements will have to be made to evaluate gear condition. One should always evaluate frequencies at least up to 3X gear mesh frequency (for regular PMP surveys, use an  $F_{MAX} = 3.25X \text{ GMF}$ ). If the number of teeth are unknown, set  $F_{MAX} = 200X \text{ RPM}$  using the speed of the shaft at each particular bearing positions being measured.

The author has often been involved on machines where little vibration was occurring at the fundamental gear mesh frequency (GMF), but where much higher vibration on the order of 10X those at the fundamental were occurring at either 2X gear mesh or 3X gear mesh frequency (2GMF or 3GMF). Therefore, potentially significant problems would have been overlooked had measurements not been made up in these frequency regions.

5. The analyst may sometimes have to employ more than one accelerometer on a gear box due to potentially high frequencies that might occur on one or more of the meshes. For example, if the fundamental gear mesh frequency was on the order of 1,200,000 CPM (20,000 Hz), he would evaluate balance, alignment, looseness, etc. Then, he would have to make a whole set separate measurements evaluating the gears with special accelerometers having much higher frequency capability. Of course, the converse could also be said if one were evaluating the output shaft which was at very low RPM (below 100 RPM) with a moderately high gear mesh frequency. In this case, one might have to employ a special, low frequency seismic accelerometer for the lower frequency problems and the general purpose accelerometer for the gear mesh measurements.
6. In most PMP programs using computer software, each of 2 measurements having 2 different frequency ranges must be made at each gear location because of the widely varying frequency between the lower speed harmonics and the gear mesh frequencies themselves. In these cases, he would use a lower  $F_{MAX}$  to evaluate such problems as unbalance, misalignment, looseness, electrical, etc. and a completely different set of measurements to evaluate gear health.
7. For a given mesh of gears having one common gear, the gear mesh frequency will always be the same no matter whether 2, or up to 5 or 6 gears happen to be in a common mesh. For example, in many of today's centrifugal air compressors, there is one bull gear which meshes with each of 4 pinions that are mounted on the first through fourth stage impellers. In this case of 5 meshing gears, there is only one gear mesh frequency.
8. Of course, the gear mesh frequency is different for each different mesh of gears (see Figure 6.11A). For example, one shaft may have one gear in a mesh on one end and another gear on its opposite end meshing with other gears. Each of these meshes will have individual gear mesh frequencies.
9. Referring to Figure 6.11A, an analyst should always make a drawing such as this showing each of the positions, the RPM of each shaft and the gear mesh frequency at each mesh. This will go far in helping him with his analysis.

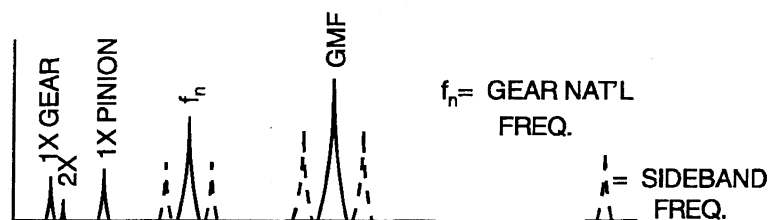
10. A multi-stage gear box should actually be treated as several individual vibration problems, each with its own unique set of operating speeds and gear mesh frequencies. For example, using Figure 6.11A, one problem should be analyzed on the first mesh of 44,400 CPM with the input and output shaft speeds being 1775 RPM and 965.4 RPM, respectively. In all likelihood, measurements on this input shaft will likewise contain vibration at the lower speed gear mesh and the output shaft speed. However, in most cases, he should neglect vibration being transmitted from the other meshes with the exception of resonances on this member being excited by vibration transmitted from the others.

Following below will be separate discussions on each of the gear problems previously listed:

#### 6.111 Gear Tooth Wear:

Figure 6.11B shows a spectrum indicating wear of gear teeth. In this case, wear does not refer to a chipped, broken or cracked tooth. Instead, it refers to surface damage across the tooth face. Worn gear teeth exhibit the following characteristics:

1. The key indicator of gear tooth wear is not the gear mesh frequency, but instead the gear natural frequency. In reality, there is of course more than one gear natural frequency including separate ones for the driver and driven gears as well as a set of those when the gears are meshing with one another. These natural frequencies can be identified by performance of impulse natural frequency tests outlined in Section 6.05. Like everything else in nature, when a member is impacted, it will respond at its natural frequency. In the case of gears, their natural frequencies respond each time a defective tooth hits or impacts as it goes into and out of mesh. The key here is that these natural frequencies will be modulated by the impact repetition rate which will correspond to the speed of the worn gear.
2. When significant tooth wear occurs, not only will sidebands appear about the gear mesh frequencies, but also about the gear natural frequencies. In the case of those around the gear mesh frequencies, the amplitude of the sidebands themselves is a better wear indicator than the amplitude of GMF frequencies.
3. With respect to the sidebands themselves, more than one set of sidebands may appear if the time waveform becomes less and less sinusoidal which may indicate a more serious gear wear problem.
4. If more than one gear in a mesh has worn teeth, sidebands will be established at each of the speeds of the gears having worn teeth.



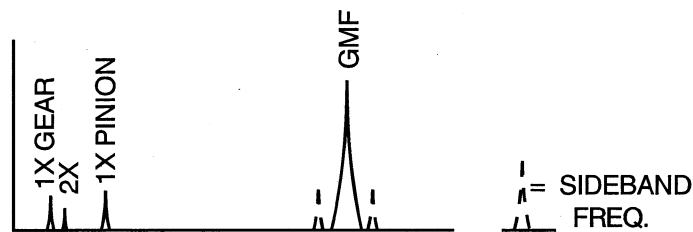
**FIGURE 6.11B**  
**SPECTRUM INDICATING GEAR TOOTH WEAR**

### 6.112 Significant Load Imposed on Gear Teeth:

Figure 6.11C provides a spectrum indicative of gear teeth subjected to significant load.

Gear mesh frequencies themselves are often very sensitive to the load imposed upon them. High GMF amplitudes do not necessarily indicate a problem with gear health. However, if the load is excessive and continues for a period of time, eventual fatigue of gear tooth surfaces will begin. Therefore, a gear mesh frequency which has substantially increased in amplitude between one survey and the next may not yet signify a problem (particularly if sideband amplitudes remain low and if no gear natural frequencies are excited).

Because gear mesh frequencies are sometimes so load sensitive, each vibration survey should be performed with the system under maximum operating load if this is possible.

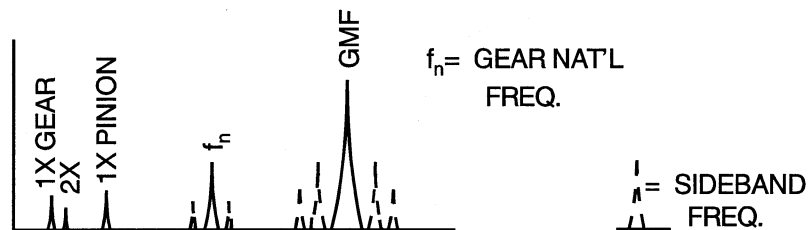


**FIGURE 6.11C**  
**SIGNIFICANT LOADING INDICATED ON GEARING**

### 6.113 Gear Eccentricity and/or Backlash:

Figure 6.11D shows an example spectrum indicating significant gear eccentricity and/or backlash. These problems display the following characteristics:

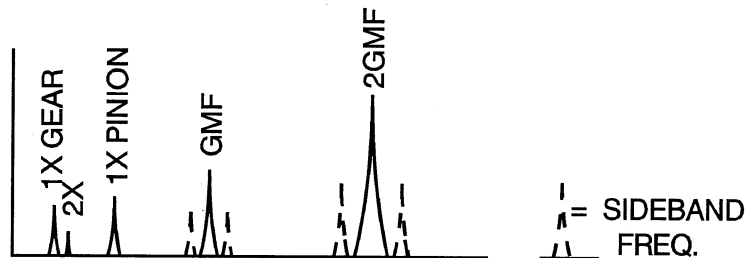
1. Both eccentricity and backlash excite the gear natural frequencies as well as gear mesh frequency. They also may generate a number of sidebands about both the natural and gear mesh frequencies.
2. If a gear is eccentric, it will modulate the natural frequency and gear mesh frequencies, both of which will be sidebanded at 1X RPM of the eccentric gear. An eccentric gear can generate significant forces, stresses and vibration if it is forced to bottom out with the meshing gears.



**FIGURE 6.11D**  
**GEAR ECCENTRICITY AND/OR BACKLASH INDICATED**

#### 6.114 Gear Misalignment:

Figure 6.11E is indicative of misaligned gears which almost always excites higher order gear mesh frequency harmonics. Often, only a small amplitude will be at the fundamental gear mesh frequency (GMF), but much higher levels will be at 2X and/or 3X GMF. Often, the sideband spacing about GMF frequencies might be 2X RPM, or even 3X RPM when gear misalignment problems are involved.



**FIGURE 6.11E**  
**SPECTRUM INDICATING MISALIGNMENT OF GEARS**

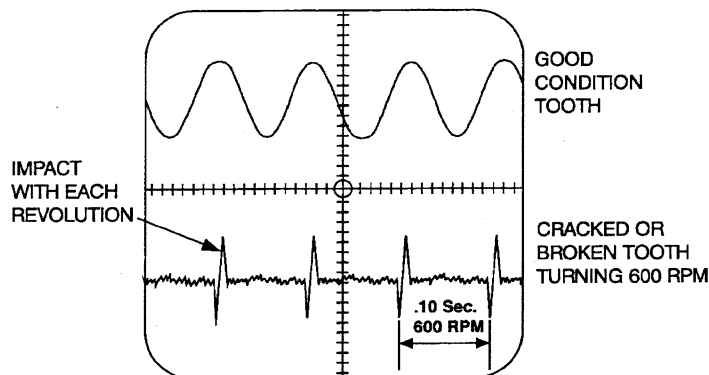
#### 6.115 Cracked, Chipped or Broken Gear Teeth:

A gear with a cracked, chipped or broken tooth will generate high vibration both at 1X RPM of this gear as well as the gear natural frequencies with sidebands around the natural frequency at gear RPM. This same behavior is exhibited by a gear tooth having a large, pronounced spall. Of course, an unbalanced gear would also cause high vibration at 1X RPM. Therefore, a time waveform like that shown in Figure 6.11F is of great assistance in determining whether the dominate problem is unbalance or gear tooth problems.

In many cases, a cracked or broken gear tooth will cause little vibration at 1X RPM in the FFT spectrum. However, it can have amplitudes 10 to 20 times higher in the time waveform with pronounced spikes spaced at the operating speed of the gear with the cracked or broken tooth.

Referring to Figure 6.11F, note that a good conditioned tooth will display a smooth, sinusoidal waveform (assuming there are no defective rolling element bearings supporting the shaft). However, cracked, chipped or broken gear teeth will generate a pronounced spike every time they go into and out of mesh. Looking at a time waveform, one can determine if the problem is with the gear teeth or from another impact event like a ball bearing problem. In the case of rolling element bearings, there would be a tremendous number of impacts within a short period of time. That is, high frequencies have correspondingly low periods. On the other hand, in the case of the gear tooth problem, if the distance in time between impact events corresponds to the RPM of the gear, this presents strong evidence of tooth problems. For example, referring to Figure 6.11F, if a 600 RPM gear showed a spike every .10 sec (.10 sec/cycle = 10 cycles/sec = 600 cycles/minute), there would be strong evidence of cracked, chipped or broken gear teeth.

If a spectrum is taken on a shaft which has more than one gear mounted on it and shows the pronounced time waveform plus natural frequencies in the spectrum sidebanded at RPM of the offending gear, it will not necessarily be clear which gear on the shaft has the problem. In this case, impulse natural frequency tests should be performed on each gear on the shaft as well as the gears to which they mesh to identify which of the gears has the problem.



**FIGURE 6.11F**  
**COMPARISON OF TIME WAVEFORM FOR A GOOD CONDITIONED**  
**VERSUS A CRACKED OR BROKEN GEAR TOOTH**

#### 6.116 Hunting Tooth Problem:

Figure 6.11G shows a spectrum which might indicate hunting tooth problems. Hunting tooth frequencies ( $f_{HT}$ ) appear when problems might have occurred during the gear manufacturing process or due to mishandling, as well as when problems occur in the field. For example, if faults appeared on both the gear and pinion, each time the defective teeth on each gear came in contact with one another, they would generate a pulse. Since most gears are not a 1:1 ratio, these two particular teeth would only come into contact periodically. For example, consider the case of a pump gerotor having a 6 tooth male and a 7 tooth female gear like that shown in Figure 6.11H. If each gear were numbered, tooth #1 on the gear would only be opposite tooth #1 on the pinion once every 7 revolutions. The actual formula for this hunting tooth frequency is given in Figure 6.11G, but is repeated here for clarity:

$$f_{HT} = \frac{(GMF)(N_a)}{(T_{GEAR})(T_{PINION})}$$

where:

$f_{HT}$  = Hunting Tooth Frequency (Hz or CPM)

GMF = Gear Mesh Frequency  
 = #teeth X RPM (Hz or CPM)

$N_a$  = Number of Unique Assembly Phases in a given Tooth Combination  
 (Product of Prime Factors Common to #teeth on each gear)

$T_{GEAR}$  = # Teeth on Gear

$T_{PINION}$  = # Teeth on Pinion

A gear set with this gear tooth repeat problem normally will generate a "growling" sound from the drive. It can cause quite high vibration, but since it occurs at low frequencies predominately below 600 CPM, it is often missed. It often requires the use of a seismic accelerometer to even detect its presence. However, left uncorrected, it can be quite destructive to gear teeth.

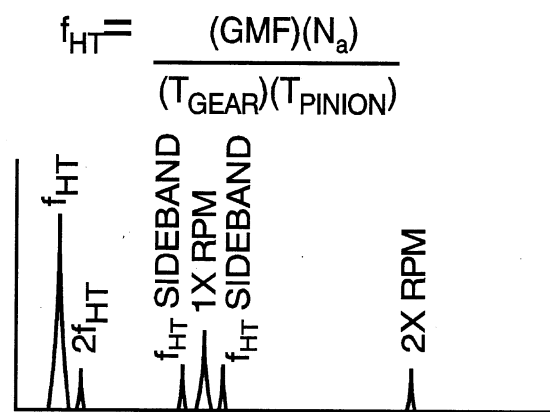
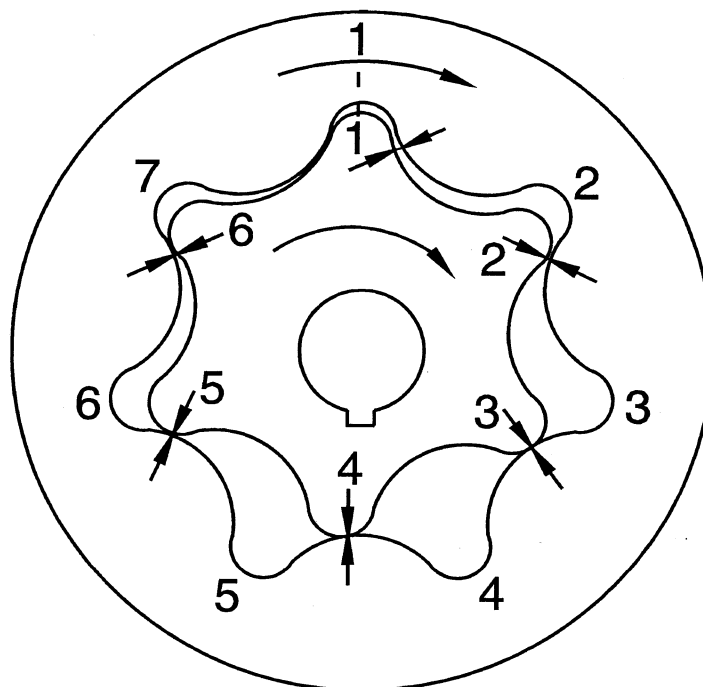


FIGURE 6.11G



*Courtesy Double A Products Co.*

FIGURE 6.11H  
SCHEMATIC DIAGRAM OF THE GEROTOR MECHANISM

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## 6.12 ELECTRICAL PROBLEMS

Many find it surprising that you can detect not only mechanical problems, but also electrical problems using vibration signature analysis. However, remember that the magnetic fields within a motor create flux which induces electromagnetic forces, and that these, along with forces from mechanically induced problems must all be supported by the bearings. These forces are then measured directly by force transducers placed on the bearing housings, or indirectly by vibration transducers such as accelerometers, velocity pickups or proximity probes. Accelerometers directly measure the quotient of force divided by mass ( $f/m$ ) which is acceleration (according to Newton's Law). Again, it does not matter whether these forces themselves are mechanically or electrically induced.

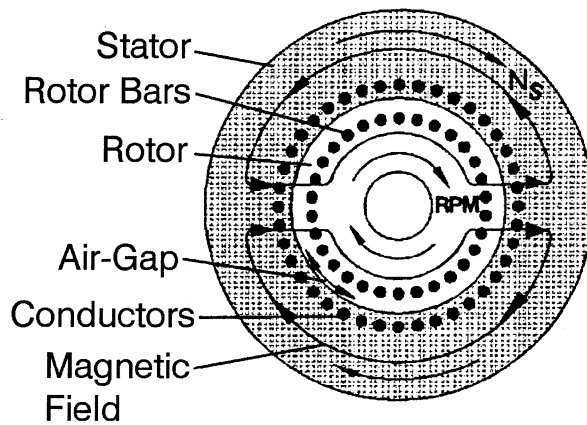
Figures 6.12A and 6.12B picture the stator and rotor of an induction motor, along with a close-up view of a rotor, rotor bars and shorting rings. Some of the problems which can be detected using vibration analysis include each of the following:

- Stator Eccentricity, Shorted Laminations and Loose Iron
- Eccentric Rotor (Variable Air Gap)
- Rotor Problems (Broken or Cracked Rotor Bars or Shorting Rings, Shorted Rotor Laminations, Loose Rotor Bars, etc.)
- Thermal Bow Induced By Uneven Localized Heating of a Rotor
- Electrical Phasing Problems Due to Loose or Broken Connectors
- Problems with Synchronous Motors
- Problems with DC Motors
- Torque Pulse Problems

Each of the above problems will be discussed in Sections 6.121 through 6.128 which will follow. However, a few things should be said about detection of electrical problems in general before getting into specifics.

First, it is important that if electrical problems are suspected, diagnostic tests to evaluate electrical condition should be run with the motor fully loaded. This is particularly due to the fact that the electromagnetic forces themselves vary with the square of stator current as per Reference 7. Very often, motors with proven electrical problems will not generate vibration signatures showing problem symptoms when operated "solo" (uncoupled from the driven machine), or even when operating a machine in an unloaded condition such as an air compressor. In these cases, definite symptoms appear when such machines are loaded, particularly at or near 100% load. Of course, this relates directly back to the fact that electromagnetic induced forces are so dependent on the current draw.

Probably a majority of all electrical problems detected involve "higher than normal" amplitudes at 2X electrical line frequency (also known as synchronous frequency). In the United States, the line frequency is set at 60 Hz whereas in Europe, it is at 50 Hz. Therefore, with respect to the United States, a frequency of great importance when detecting electrical problems will be that at 120 Hz (or 7200 CPM).



$$N_s = \frac{120 F_L}{P} = \text{SYNCHRON. SPEED}$$

$$F_s = N_s - \text{RPM} = \text{SLIP FREQ.}$$

$$F_p = (F_s)(P) = \text{POLE PASS FREQ.}$$

$$\text{RBPF} = \# \text{ ROTOR BARS} \times \text{RPM}$$

WHERE:

$F_L$  = ELECTRICAL LINE FREQUENCY (often 60 Hz)

RPM = ROTOR SPEED

$N_s$  = SYNCHRONOUS SPEED

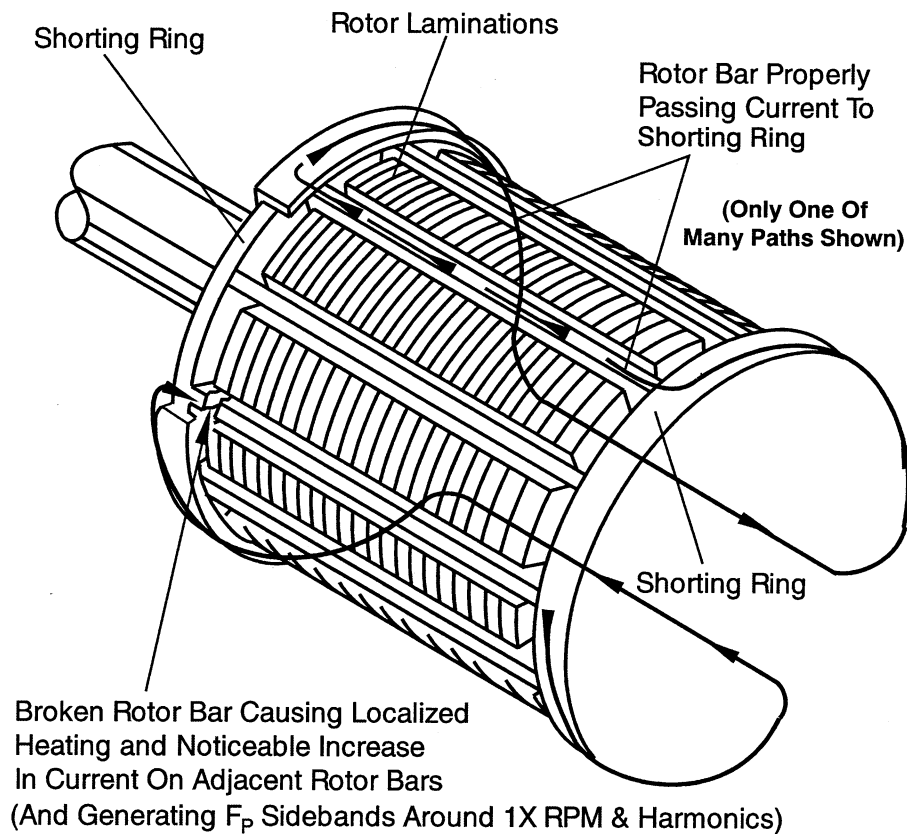
$F_s$  = SLIP FREQUENCY ( $N_s - \text{RPM}$ )

$F_p$  = POLE PASS FREQUENCY

$P$  = # POLES

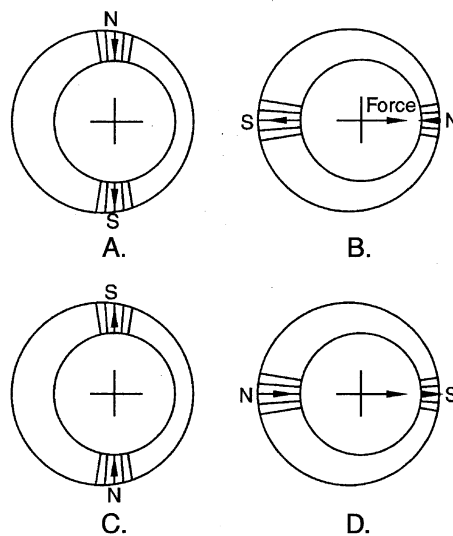
RBPF = ROTOR BAR PASS FREQUENCY

**FIGURE 6.12A**  
**DIAGRAM OF AN INDUCTION MOTOR ALONG WITH ITS STATOR, ROTOR, ROTOR BARS, AIR GAP AND MAGNETIC FIELD FOR 2-POLE MOTOR**

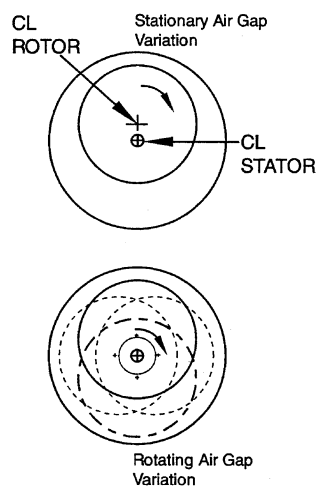


**FIGURE 6.12B**  
**ISOMETRIC VIEW OF A ROTOR INCLUDING ITS ROTOR BARS, SHORTING RINGS AND ROTOR LAMINATIONS FOR 2-POLE MOTOR**

Figure 6.12C helps explain why so many electrical problems involve twice line frequency rather than its fundamental. Referring to Figure 6.12C, note that during one rotation of the 3600 CPM stator field, the magnetic pull towards the closest pole rises from 0 to maximum twice in an eccentric rotor. Since the field itself revolves at 3600 revolutions per minute, the magnetic pull reaches a maximum 7200 times per minute (or 7200 CPM). Another way of looking at it is that because the close side of the rotor will first be attracted to the north, and then to the south pole, the force itself will vary at 2X the frequency of the magnetic field relative to the eccentricity. Therefore, when the rotor is not centered within the stator (either due to an eccentric rotor or stator), it will always affect 7200 CPM vibration. Figure 6.12D shows the difference between a stationary air gap problem where the gap itself remains at the same angular location and a rotating air gap variation which is caused by an eccentric rotor.



**FIGURE 6.12C**  
**UNEVEN AIR GAP BETWEEN MOTOR AND STATOR**



**FIGURE 6.12D**  
**STATIONARY & ROTATING AIR GAP VARIATIONS**

Before discussing diagnoses of specific electrical problems, several key items below will be defined which are common to many of the problems discussed:

(a)  $F_L$  = Electrical Line Frequency  
(60 Hz = 3600 CPM in USA; 50 Hz in Europe)

(b)  $N_s = \frac{120F_L}{P}$

where:

$N_s$  = Synchronous Speed with no Slip (RPM)  
 $P$  = Number of Poles

(c)  $F_s = N_s - \text{RPM}$   
where:

$F_s$  = Slip Frequency (CPM)

(d)  $F_p = (\# \text{Poles})(\text{Slip Frequency})$   
where:

$F_p$  = Pole Pass Frequency = a key sideband frequency which will signal the presence of many electrical problems

=  $2F_s$  for 2 Pole, nominal 3600 RPM motors

=  $4F_s$  for 4 Pole, nominal 1800 RPM motors

=  $6F_s$  for 6 Pole, nominal 1200 RPM motors

Equation (b) is used to calculate the Synchronous Speed of a motor driven by an Inverter which can be dialed into a range of electrical line frequencies ( $F_L$ ). See Table 6.12A below.

**TABLE 6.12A**  
**SYNCHRONOUS SPEEDS (RPM) FOR VARIOUS NUMBER OF POLES AT**  
**60 Hz LINE FREQUENCY**

| No. of Poles | SYNCH. SPEED |
|--------------|--------------|
| 2            | 3600         |
| 4            | 1800         |
| 6            | 1200         |
| 8            | 900          |
| 10           | 720          |
| 12           | 600          |

(e)  $\text{RBPF} = \# \text{Bars} \times \text{RPM}$   
where:

RBPF = Rotor Bar Pass Frequency

#Bars = The number of Rotor Bars

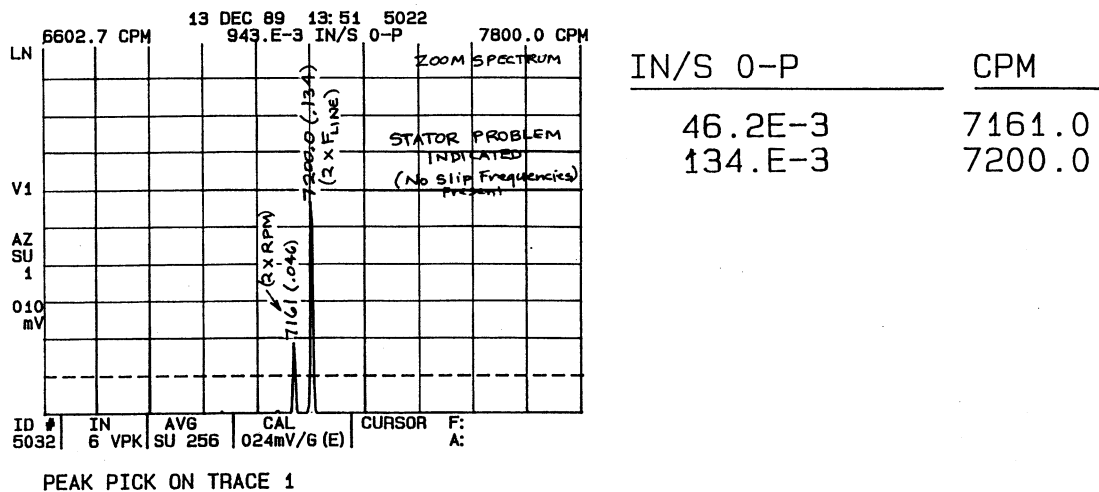
(f) Air Gap = Gap Between Rotor and Stator in an Electric Motor

(g) Magnetic Center = Point where Stator and Rotor Magnetic Fields are in balance and point to which rotor would be drawn.

Figure 6.12E shows a typical spectrum which would indicate stator problems. Note the higher than normal vibration of .134 in/sec at 2X line frequency ( $2F_L$ ). It is important to point out that when using most analyzers and data collectors, very good frequency resolution will have to be employed to separate 2X line frequency from motor running speed harmonics (2X RPM in the case of a 2 pole motor; 4X RPM in the case of a 4 pole, etc.). That is, for most frequency ranges chosen, the analyzer will not show 2 individual frequencies at 2X RPM and  $2F_L$ ; instead, it may display only 1 frequency with a so-called value of "7200 CPM" (depending on the maximum frequency chosen). In these cases, it will be necessary either to increase the number of FFT lines, lower  $F_{MAX}$ , or generate a "zoom spectrum" around 7200 CPM like that shown in Figure 6.12E in order to separate  $2F_L$  from running speed harmonics.

For example, on Figure 6.12E which was a 2 pole motor running at 3580.5 RPM, you would not be able to separate its 2X running speed (7161 CPM) from 7200 CPM if you employed 400 FFT lines and a maximum frequency of 60,000 CPM (since you could only resolve frequencies 150 CPM apart in this case). In this case, it would be a good idea either to use a factor of 10 zoom from 12,000 CPM (giving a 3 CPM resolution) or, for example, use 3200 FFT lines (which would give a resolution of 3.75 CPM). Then, one could evaluate whether the problem is of a mechanical nature (running speed harmonics) or of electrical ( $2F_L$  line frequency).

Importantly, it is a good idea to use log amplitude rather than linear amplitude when performing zoom spectra for any electrical problem when it will be necessary to zoom in around line frequency, 2X line frequency, rotor bar pass frequency, running speed harmonics or fundamental pole pass frequencies themselves ( $F_L$ ). If log amplitude is not employed, pole pass frequency sidebands around peaks such as 1X RPM, line frequency or 2X line frequency may easily be missed since such pole pass sidebands are normally so much lower in amplitude, and yet can still signal potential problems.



**FIGURE 6.12E**  
**ZOOM SPECTRUM INDICATING STATOR PROBLEM**

---

Following below will be a discussion of each of the electrical problems listed above in the introduction to the electrical problem section.

#### 6.121 Stator Problems (see Table 6.0, Section A):

Stator problems detectable by vibration analysis include each of the following:

- Stator Eccentricity (an eccentric stator producing a stationary differential air gap between the rotor and stator)
- Shorted Laminations (insulation problems with lamination layers which can cause localized heating)
- Loose Iron (any looseness or weakness in the stator)

These problems exhibit the following characteristics:

1. All such stator problems generate high vibration at 2X line frequency. However, they do not necessarily generate pole pass frequency sidebands since they originate within the stator, and are not therefore modulated by either running speed or slip frequency. Figure 6.12G indicates a serious electrical problem. Figure 6.12F is a spectrum that was captured by a data collector during a regularly scheduled PMP route. Note the high amplitude of .230 in/sec at 7200 CPM which exceeded Band 3. A real-time analyzer was then employed on the same day to better define this problem which is shown by the zoom spectrum of Figure 6.12G. This spectrum showed a high level of .228 in/sec at 7200 CPM ( $2F_L$ ) and only .0044 in/sec at 2X RPM. Please also note the absence of any pole pass sidebands about 2X line frequency which indicated a stator problem. Upon inspection, a stator eccentricity problem was found.
2. Concern should be given motors whose vibration exceeds .050 in/sec peak at 2X line frequency on new or rebuilt motors, or .100 in/sec peak on in-service motors having a stator problem (applies to motors in general ranging from 50 HP to 1000 HP). This amplitude applies specifically to the peak at 7200.0 CPM itself ( $2F_L$ ). However, if this motor is directly driving a precision machine tool spindle,  $2F_L$  levels will have to be much lower, on the order of .025 in/sec or less.
3. Stator eccentricity produces an uneven stationary air gap between the rotor and stator which results in highly directional vibration, depending on the largest gap differential. The largest magnetic forces occur at a minimum rotor/stator gap. Therefore, the electromagnetic forces themselves go from a minimum to a maximum each revolution producing vibration at twice line frequency (7200 CPM).
4. Differential air gaps should not exceed 5% for induction motors and 10% for synchronous motors. If the vibration amplitude at  $2F_L$  grows over time, the motor should be inspected (if physically possible) by marking a point both on the rotor and the stator. Then, measuring the air gap at the point where the marks align, rotate the rotor in 45° increments and measure the air gap at the point where the stator is marked. If the variation exceeds approximately 5%, the air gap difference is due to an eccentric rotor. The next step is to rotate the rotor again in 45° increments, this time measuring the gap at the point where the rotor is marked. If this gap varies more than 5% for these measurements, an eccentric stator is indicated.
5. Loose iron is due to localized stator support weakness or looseness.

6. Shorted stator laminations can cause uneven, localized heating which can actually distort the stator causing a stator eccentricity problem. This produces thermally-induced bow which can significantly grow with operating time. The thermally induced bow can sometimes actually cause the stator to bow and contact the rotor, which can be catastrophic.

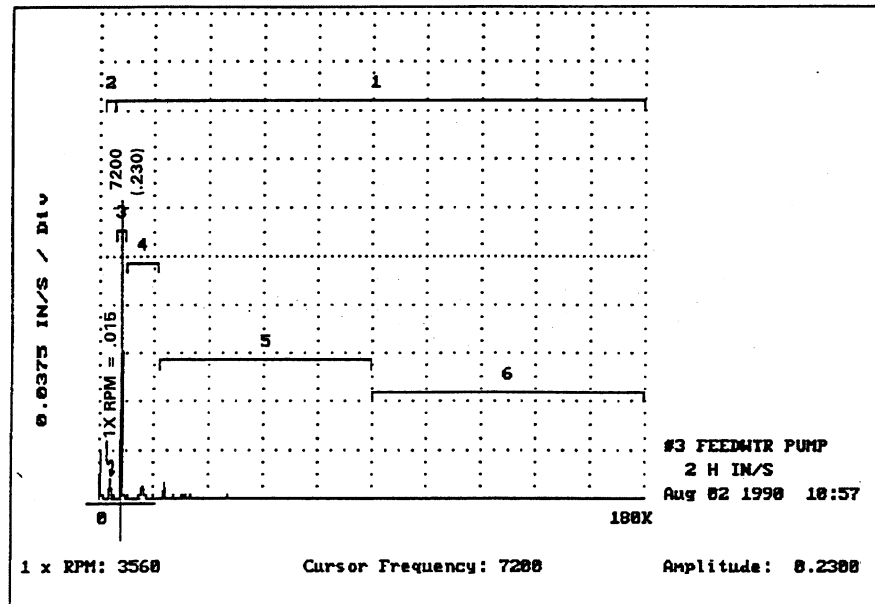


FIGURE 6.12F  
NORMAL PMP ROUTE SPECTRUM WITH FMAX = 50X RPM

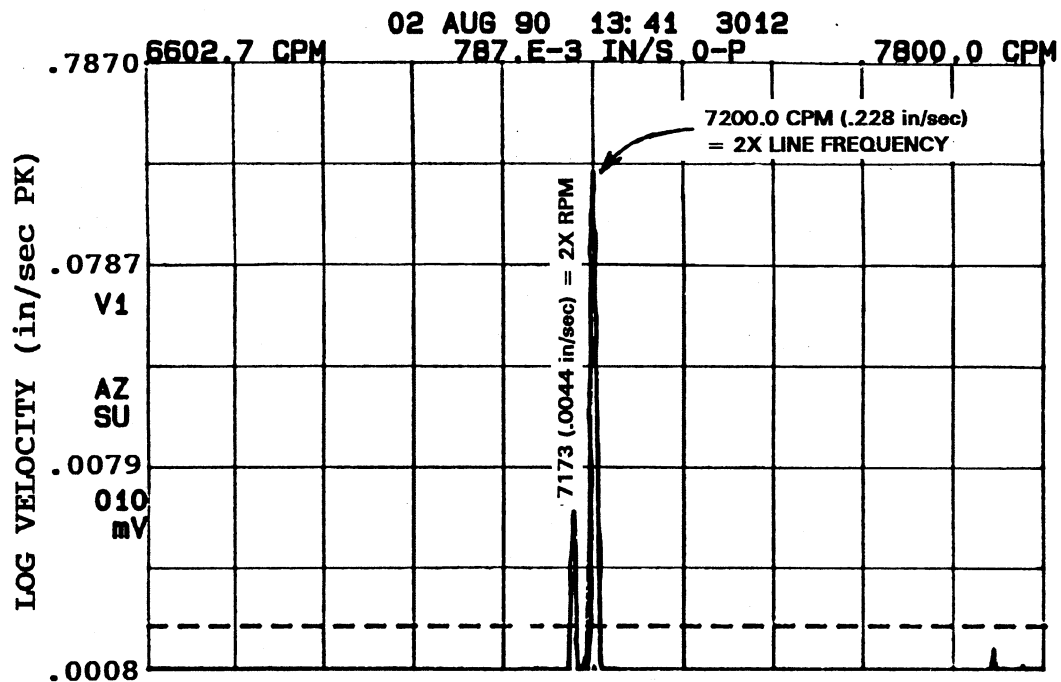


FIGURE 6.12G  
ZOOM SPECTRUM INDICATING PROBLEM WITHIN THE STATOR  
(NOTICE THE ABSENCE OF POLE PASS SIDEBANDS)

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### 6.122 Eccentric Rotor (see Table 6.0, Section B):

In the case of an eccentric rotor, the rotor itself is not concentric with its centerline (see Figure 6.12D). Therefore, an eccentric rotor produces an uneven air gap between the rotor and stator which will rotate with the rotor (as opposed to a stationary air gap in the case of an eccentric stator). An eccentric rotor can be caused by shorted rotor laminations resulting in localized heating and inducing a bowed rotor, or a simple out-of-round rotor.

1. An eccentric rotor most often will produce a high vibration at twice line frequency ( $2F_L$ ) accompanied by sidebands spaced at pole pass frequencies ( $F_p = \text{\#Poles} \times \text{slip frequency}$ ). That is, for a two-pole, 3600 RPM motor, the sidebands will be at twice slip frequency; while for a four-pole motor, sidebands would be at 4X slip frequency. Figure 6.12I shows a spectrum indicative of an eccentric rotor problem producing a variable air gap. First, the wideband spectrum of Figure 6.12H showed a high overall of .295 in/sec with .162 in/sec at a "so-called" 7200 CPM frequency. The zoom spectrum of Figure 6.12I revealed a high level of .166 in/sec at 7200 CPM, along with a well-formed pole pass sideband indicative of an eccentric rotor.
2. Concern should be given motors whose amplitude at twice line frequency (7200 CPM) exceeds approximately .050 in/sec for new or rebuilt motors; or .100 in/sec peak for in-service motors assuming these are general purpose motors. If serving a precision machine tool spindle, the allowable at  $2F_L$  would be only approximately .025 in/sec. This would apply to most induction motors ranging in size from approximately 50 HP to 1000 HP. It is important to clarify that this amplitude applies specifically to 2X line frequency itself.
3. An eccentric rotor may require adjustment of bearing housings themselves and/or machining of the rotor journals in order to provide an air gap within tolerance all around the periphery.
4. In a predictive maintenance program, when an eccentric rotor is indicated by pole pass frequency sidebands around  $2F_L$ , they should be closely trended in future surveys. For example, when a peak at  $2F_L$  exceeds roughly .100 in/sec, it should be closely trended in future surveys. If its amplitude noticeably increases, and if sideband amplitudes do likewise, much greater concern should be given, particularly if even more sidebands appear above and below  $2F_L$  (normally 7200 CPM). On the other hand, if the amplitudes at 7200 CPM and sidebands remain stable over several surveys, no further damage is likely being done to the motor - even if these amplitudes are as much as .175 in/sec at 7200 CPM. In these cases, it will likely be satisfactory just to continue trending. However, the motor itself may have a lowered life expectancy.
5. Note that a motor having an eccentric rotor will often experience higher and higher vibration as it comes up to temperature. For example, when such a motor is first started, it may have a level of only .10 in/sec. Then, after about 10 minutes operating time, the amplitude might increase to, for example, .14 in/sec. Next, after about 20 minutes, it may continue to grow up to about .18 in/sec. Finally, after 30 minutes, it may experience possibly .25 in/sec or more. This can be caused by nonuniform heating of the rotor on one side relative to the other. Left uncorrected, it can result in catastrophic failure if the rotor bows sufficiently to throw the rotor into the stator.

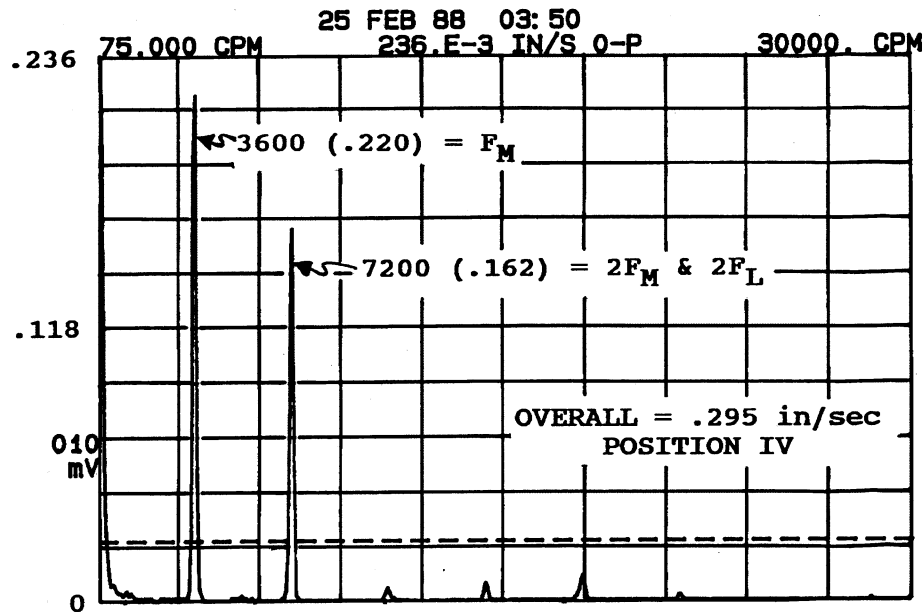


FIGURE 6.12H  
30,000 CPM SPECTRUM FOR A CIRCULATING WATER PUMP MOTOR

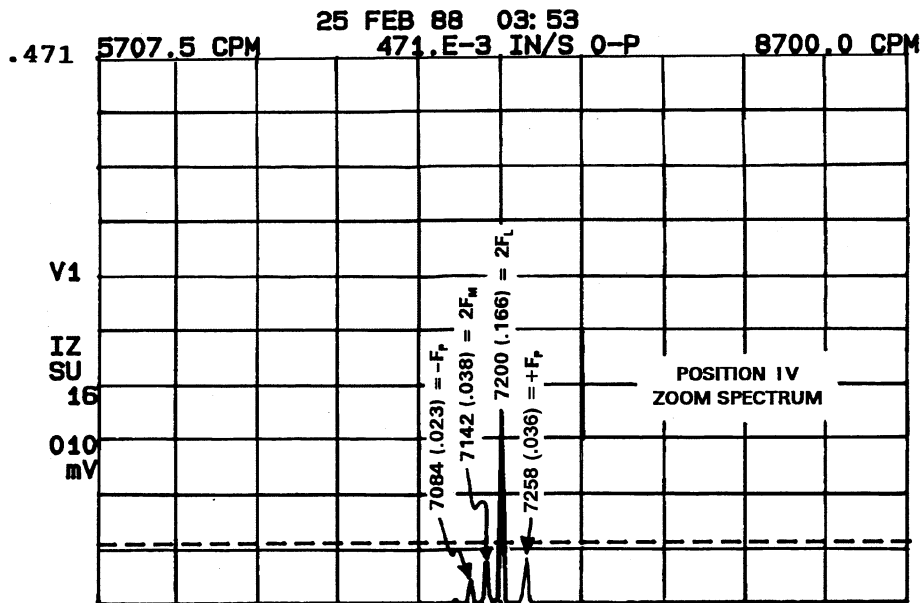


FIGURE 6.12I  
ZOOM SPECTRUM INDICATING AN ECCENTRIC ROTOR  
(NOT MECHANICAL LOOSENESS OR MISALIGNMENT)

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### 6.123 Rotor Problems (see Table 6.0, Section C):

Rotor problems detectable by vibration analysis include each of the following:

- Broken/Cracked Rotor Bars or Shorting Rings (see Figure 6.12B)
- Bad High Resistance Joints between Rotor Bars & Shorting Rings
- Shorted Rotor Laminations
- Loose/Open Rotor Bars not making good contact with End Rings

These problems exhibit the following characteristics:

1. Probably the key area of concern for broken or cracked rotor bars, is the presence of pole pass frequency sidebands around 1X RPM (in the case of 2-pole motors, these sidebands will be at 2X slip frequency while at 4X slip frequency for 4-pole motors). Figures 6.12J thru 6.12M show spectra typical for a 2-pole motor having serious rotor problems. Initially, the 30,000 CPM wideband spectrum gave little hint of a serious problem, seemingly showing mechanical looseness as evidenced by several running speed harmonics. However, upon closer examination, the zoom spectra of Figures 6.12K thru 6.12M revealed a series of well-formed pole pass sidebands around 1X, 2X and 3X RPM, respectively. These spectra all indicated cracked or broken rotor bars, shorting ring problems or shorted rotor laminations.
2. As suggested by Figures 6.12K thru 6.12M, in addition to the pole pass frequency sidebands around 1X RPM, broken or cracked rotor bars and/or high resistance joints can produce pole pass sidebands around higher running speed harmonics up to and including the 2nd, 3rd, 4th and 5th running speed harmonics. In this case, more than 1 rotor bar is often found cracked or broken since there is more than 1 pulse event per revolution. Figure 6.12N shows such a spectrum with multiple pole pass sidebands around 1X, 2X, 3X, and 4X RPM. In this case, the operating speed was about 1176 RPM (meaning that slip frequency =  $F_s = 24$  CPM). Since this was a 6-pole motor, the pole pass frequency ( $F_p$ ) equalled  $6 \times 24 \text{ CPM} = 144$  CPM. This was the approximate spacing of each of the multiple  $F_p$  sidebands about 1X RPM through 5X RPM as shown by the zoom spectrum of figures 6.12P. Later, this motor was found to have 4 cracked rotor bars.
3. The key area of concern for loose/open rotor bars is vibration at much higher frequencies at rotor bar pass frequency (RBPF) and also harmonics of this frequency (RBPF = # Rotor Bars X RPM).
4. Here, the concern is amplitudes exceeding approximately .06 in/sec at either rotor bar pass frequency (RBPF) or higher harmonics (2 RBPF or 3 RBPF). In addition, the sideband spacing around RBPF and its harmonics will be exactly twice line frequency ( $2F_L$ ). The reader is cautioned that while RBPF itself may be acceptable, if he extends the frequency range to encompass 2X RBPF, he may find amplitudes 10 or more times those at RBPF.

For example, Figure 6.12Q shows a spectrum for a motor confirmed to have two or more open rotor bars. This motor had 57 rotor bars and operated at a speed of 1793 RPM, giving the fundamental RBPF at about 102,200 CPM. The wideband spectrum of Figure 6.12Q showed an amplitude of only .008 in/sec at RBPF. However, the story out at 2X RBPF was completely different. The zoom spectrum of Figure 6.12Q showed an excessive .340 in/sec at 204,380 CPM, or 2X RBPF (over 28 times higher amplitude than at RBPF). Importantly, if a maximum frequency high enough only to capture the fundamental RBPF was taken, this problem would have been missed entirely. Again, the key indicators were the excessively high level at 2X RBPF, which was accompanied by sidebands at exactly 7200 CPM ( $2F_L$ ).

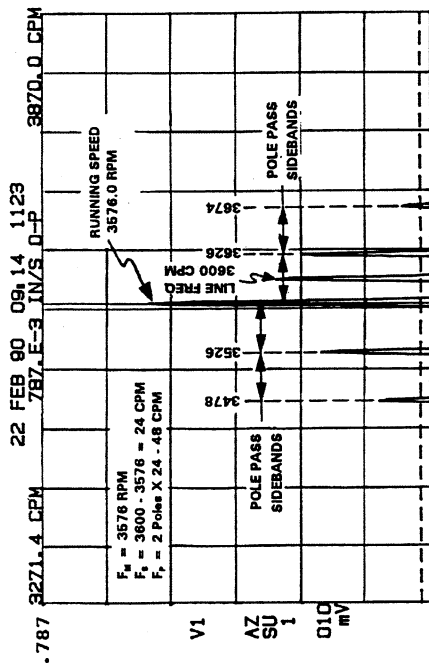


FIGURE 6.12K  
ZOOM SPECTRUM AROUND 1X RPM SHOWING MULTIPLE  
POLE PASS SIDE BANDS (BROKEN OR CRACKED ROTOR  
BARS/SHORTING RING PROBLEMS INDICATED)

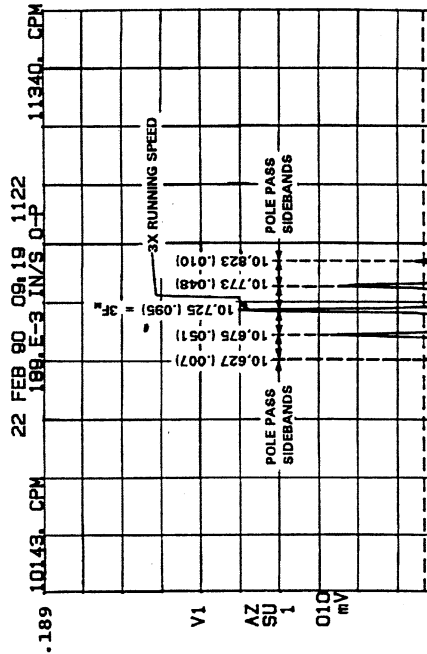


FIGURE 6.12M  
ZOOM SPECTRUM AROUND 3X RPM  
(NOTE POLE PASS SIDE BANDS HERE ALSO)

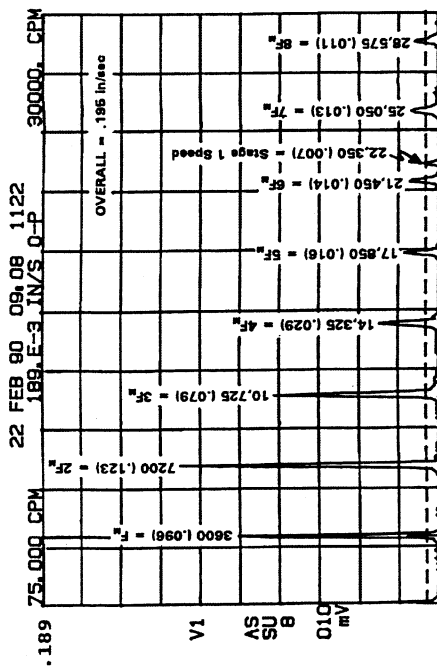


FIGURE 6.12J  
30,000 CPM SPECTRUM ON A CENTAC MOTOR (POS. 2H)

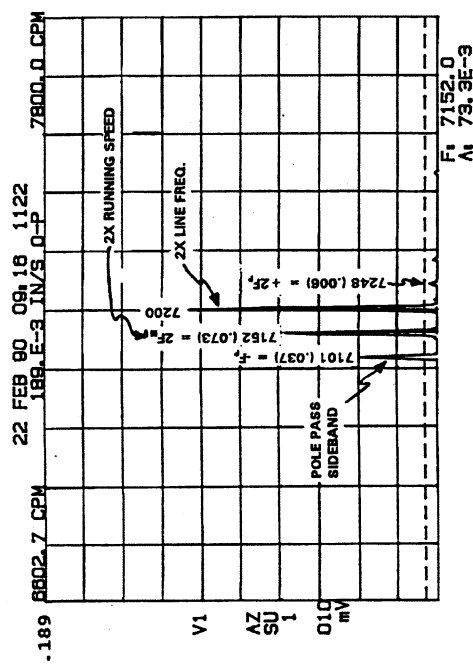
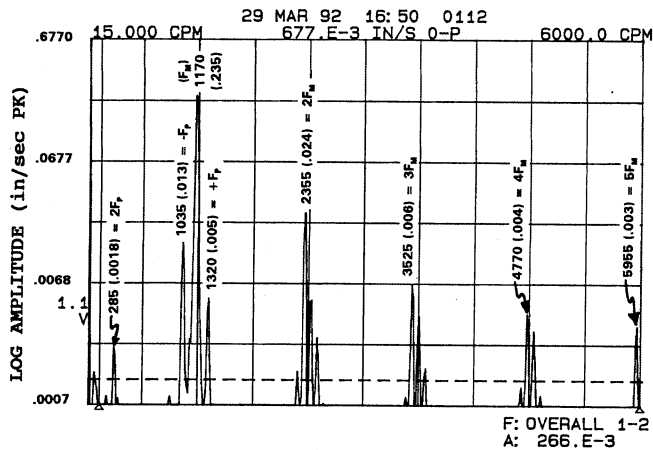
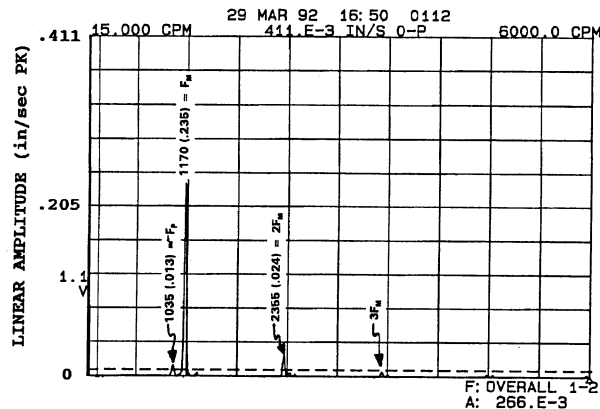


FIGURE 6.12L  
ZOOM SPECTRUM AROUND 2X RPM  
(POLE PASS SIDE BANDS ALSO HERE)

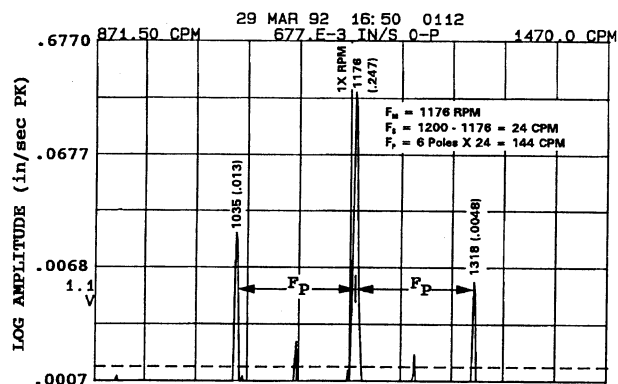
## WIDEBAND & ZOOM SPECTRA FOR A MOTOR HAVING CRACKED OR BROKEN ROTOR BARS OR SHORTING RING PROBLEMS



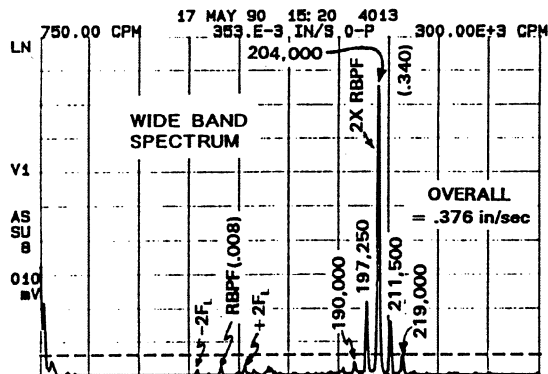
**FIGURE 6.12N**  
LOG AMPLITUDE SPECTRUM CLEARLY SHOWING POLE PASS  
FREQUENCY SIDEBANDS AROUND 1X RPM THRU 4X RPM  
(4 CRACKED ROTOR BARS WERE FOUND IN MOTOR)



**FIGURE 6.12O**  
LINEAR AMPLITUDE SPECTRUM DOES NOT ADEQUATELY SHOW  
MULTIPLE POLE PASS FREQUENCY SIDEBANDS



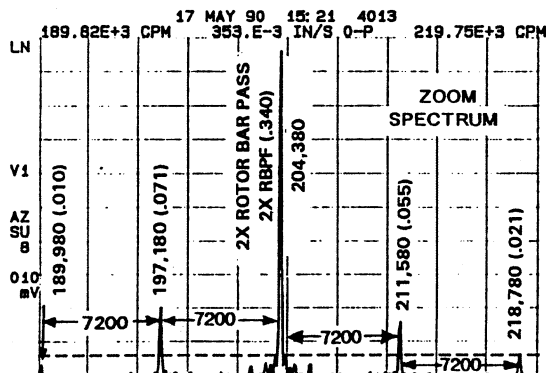
**FIGURE 6.12P**  
LOG AMPLITUDE ZOOM SPECTRUM CLEARLY DISPLAYS 1X RPM  
AND POLE PASS SIDEBANDS



| IN/S 0-P | CPM       |
|----------|-----------|
| 77.4E-3  | 3000.0    |
| 77.4E-3  | 197.24E+3 |
| 305.E-3  | 204.00E+3 |
| 56.4E-3  | 211.49E+3 |
| 21.0E-3  | 219.00E+3 |

SERIOUS ROTOR BAR PROBLEM DETECTED AT 2X RBPf,  
BUT COMPLETELY MISSED BY 1X ROTOR BAR PASS

FINISH FAN



| IN/S 0-P | CPM       |
|----------|-----------|
| 70.9E-3  | 197.18E+3 |
| 21.0E-3  | 204.07E+3 |
| 340.E-3  | 204.38E+3 |
| 21.0E-3  | 204.60E+3 |
| 54.8E-3  | 211.58E+3 |

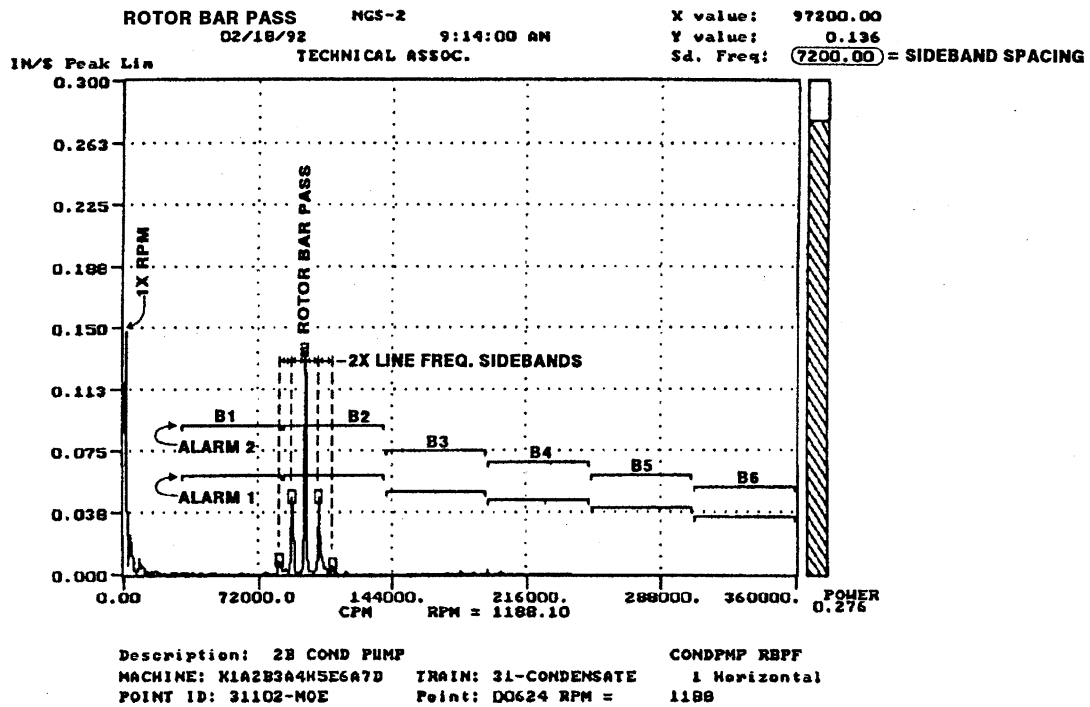
SERIOUS ROTOR BAR PROBLEM DETECTED  
AT 2X ROTOR BAR PASS FREQUENCY

NOTE: 6-10-90 G.E. Repair Shop phoned to confirm  
this motor had 2 or more open rotor bars which will  
likely result in motor replacement.

FIGURE 6.12Q  
SERIOUS ROTOR BAR PROBLEM DETECTED AT 2X  
ROTOR BAR PASS FREQUENCY

- 
5. Referring back to the lower spectrum on the Diagnostic Chart in Table 6.0 under "Rotor Problems", note that sometimes the amplitudes of RBPF or its harmonic will not be the highest amplitude. Instead, one of the sidebands spaced at difference frequencies of 2X line frequency (usually 7200 CPM with 60 Hz  $F_L$ ) might be highest. This "array" of frequencies will still include RBPF and exactly  $2F_L$  sidebands, and will still indicate loose or open rotor bars (assuming 60 Hz  $F_L$ ) and/or variable air gap.
  6. Even though the amplitude of concern in the area of rotor bar pass frequency and harmonics is about .06 in/sec for most motors, the important task will be first to detect these problems, and then to trend them before making any rash decisions about overhauling motors. If trending of several sets of spectra do not show any real increases, substantial damage is likely not continuing even with RBPF levels of approximately .10 to .15 in/sec. On the other hand, if the rate of change shows substantial increases from one survey to the next, it does indicate rapidly deteriorating condition which does warrant maintenance action.
  7. In Condition Monitoring programs, it is a good idea to specify each of the following two special points on a route for each motor to detect electrical problems:
    - a. Low Frequency Motor Electrical Point:  
 $F_{MAX} = 12,000$  CPM; 3200 FFT lines; 2 Ave. This will allow one to separate the true amplitude at  $2F_L$  and at motor running speed harmonics.
    - b. High Frequency Rotor Bar Pass Evaluation Point:  
 $F_{MAX} = 360,000$  CPM; 1600 FFT lines; 8 Ave. (For motors with more than two poles, an  $F_{MAX}$  of 240,000 CPM will likely be sufficient to pick up both 1X and 2X RBPF.) This will allow the detection of potential problems at RBPF and its multiples. Look for difference frequencies spaced at exactly  $2F_L$  (usually 7200 CPM), even if the number of rotor bars is unknown. Figure 6.12R is a good example. Note the high amplitude of .136 in/sec at RBPF which penetrated both "Alarm 1" (.060 in/sec) and "Alarm 2" (.100 in/sec) narrowband envelopes. Also, notice the 7200 CPM ( $2F_L$ ) sidebands surrounding RBPF. The reader is cautioned to carefully mount his transducer and to employ one with sufficient frequency response out to 360,000 CPM (6000 Hz) if he must make these higher frequency measurements on 2-pole motors.

Experience has shown that these measurements are normally best taken in the horizontal direction. Remember that these two spectral measurements are in addition to the standard condition monitoring route points on each motor. However, they will likely prove invaluable in detecting potentially significant electrical problems before they result in catastrophic motor failure.



**Threshold Report**

PLANT: NGS-2  
Description: 2B COND PUMP  
MACHINE: K1A2B3A4H5E6A7B  
POINT ID: 31102-MOE  
Units: IN/S      Peak Lin  
Threshold: 0.005

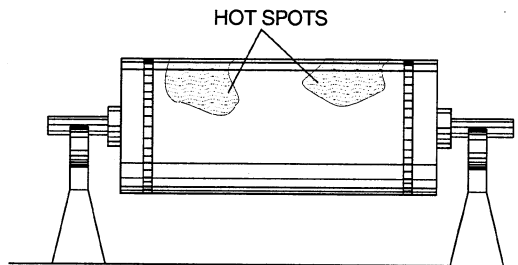
TRAIN: 31-CONDENSATE  
Point: 00624  
RPM: 1188.10

| Frequency Hz   | CPM      | Order   | Spect Amp | % Exceeded |
|----------------|----------|---------|-----------|------------|
| 18.95          | 1136.9   | 0.96    | 0.148     | 2861.08    |
| 61.43          | 3685.6   | 3.10    | 0.024     | 374.55     |
| 138.02         | 8281.4   | 6.97    | 0.010     | 95.67      |
| 1383.46        | 83007.6  | 69.87   | 0.008     | 68.81      |
| 1503.55        | 90213.2  | 75.93   | 0.047     | 835.97     |
| 1623.67        | 97420.2  | 82.00   | 0.136     | 2610.03    |
| 1743.41        | 104604.8 | 88.04   | 0.047     | 835.97     |
| 1863.95        | 111836.9 | 94.13   | 0.005     | 5.24       |
| ROTOR BAR PASS |          |         |           |            |
| Power          | Sync     | Subsync | Non-Sync  |            |
| 0.276          | 0.262    | 0.000   | 0.087     |            |

**FIGURE 6.12R**  
**TYPICAL SETUP OF SPECTRAL ALARM BANDS FOR A 6-POLE**  
**CONDENSATE PUMP MOTOR (1180 RPM NAMEPLATE SPEED)**

#### 6.124 Thermal Bow Induced By Uneven Localized Heating of a Rotor:

Figure 6.12S illustrates a rotor subjected to localized heating which can result in thermal bow. Grossly uneven localized heating of a rotor can occur if several adjacent rotor laminations are shorted together. This heat is generated by noticeably higher eddy current and hysteresis losses in the shorted laminations. Likewise, broken rotor bars can generate significant heat resulting in such localized heating of the rotor.



**FIGURE 6.12S**  
**DEVELOPMENT OF HOT SPOTS ON A ROTOR DUE TO BROKEN ROTOR BARS OR SHORTED ROTOR LAMINATIONS**

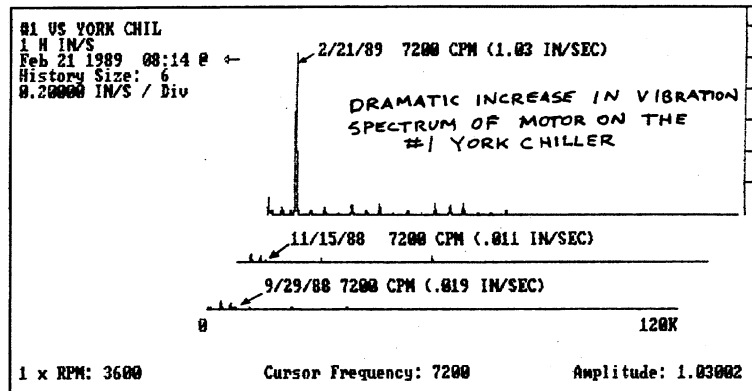
This localized heating can actually bow the rotor which can eventually contact the stator. One of the problems with thermal bow is that it can tend to "feed on itself". That is, once it begins, the shaft will bow slightly. As the shaft bows, it will create more and more electromagnetic, unbalance forces which will create more heat. This will then cause more bow in the rotor which will again generate more heat, etc. Therefore, this problem can be catastrophic. Thermal bow is indicated by each of the following characteristics:

1. When thermal bow occurs in a rotor, one will notice that after first starting up the machine, the amplitude at 1X RPM will continue to increase with operating time as the thermal bow itself increases. It will resemble a spectrum of unbalance. However, simple balancing will have no effect.
2. As the thermal bow increases, vibration spectra will sometimes indicate eccentric rotor with high 7200 CPM vibration accompanied by slip frequency sidebands (assuming 60 Hz  $F_L$ ).
3. As the thermal bow problem becomes more pronounced, vibration phase in the axial direction will approach 180° difference between that on the outboard and inboard bearings. In other words, it will simulate a bent shaft behavior.
4. An infrared camera can sometimes be used to locate the hot spots on a rotor subjected to localized heating which will be of use in determining which areas need repair.

#### 6.125 Electrical Phasing Problems (Loose Connectors):

A spectrum indicating electrical phasing problems is included in the Table 6.0 on page 3 of the Illustrated Vibration Diagnostic Chart. Phasing problems due to loose or broken connectors can cause highly excessive vibration sometimes exceeding 1.0 in/sec at 2X line frequency ( $2F_L$ ) which will have sidebands surrounding it spaced at one third line frequency ( $1/3F_L$ ). This is sometimes particularly a problem if the defective connector is only sporadically making contact.

Figures 6.12T through 6.12V illustrate a severe electrical phasing problem. Figure 6.12T was captured from the outboard bearing of a motor driving a refrigeration chiller. This motor was served by an inverter which allowed it to change speed according to the incoming electrical line frequency and voltage. Figure 6.12T was captured during a normal Predictive Maintenance



3% x 2.0000 = 0.06000 1 x RPM: 3600.00

| Machine         | Pos     | Dir               | Units | Date              |
|-----------------|---------|-------------------|-------|-------------------|
| #1 VS YORK CHIL | 1       | H                 | IN/S  | Feb 21 1989 08:14 |
| Frequency       | 1 x RPM | Amplitude         |       |                   |
| 300             | 0.08    | 0.10008           |       |                   |
| 7200            | 2.00    | 1.03002 ← 2/21/89 |       |                   |
| 21600           | 6.00    | 0.06005           |       |                   |
| 28500           | 7.92    | 0.06990           |       |                   |
| 42900           | 11.92   | 0.06990           |       |                   |
| 46800           | 13.00   | 0.06005           |       |                   |
| 50100           | 13.92   | 0.06990           |       |                   |

001% x 2.0000 = 0.02000 1 x RPM: 3600.00

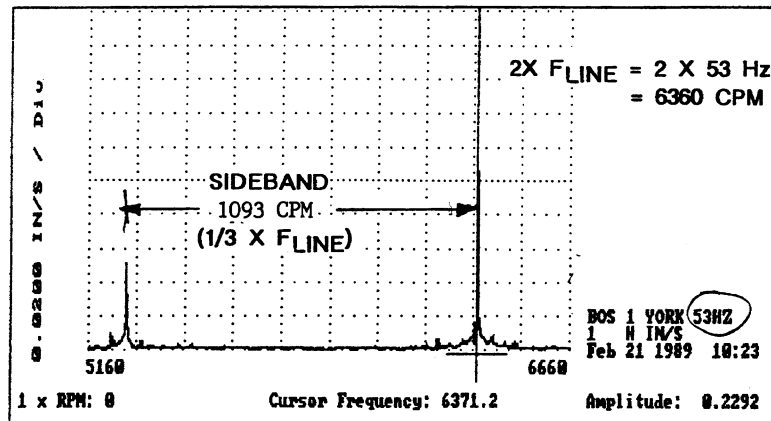
| Machine         | Pos     | Dir       | Units | Date              |
|-----------------|---------|-----------|-------|-------------------|
| #1 VS YORK CHIL | 1       | H         | IN/S  | Nov 15 1988 13:09 |
| Frequency       | 1 x RPM | Amplitude |       |                   |
| 3600            | 1.00    | 0.05274   |       |                   |
| 6000            | 1.67    | 0.03272   |       |                   |
| 21300           | 5.92    | 0.02097   |       |                   |
| 49800           | 13.83   | 0.03781   |       |                   |

000% x 2.0000 = 0.00000 1 x RPM: 3600.00

| Machine         | Pos     | Dir               | Units | Date              |
|-----------------|---------|-------------------|-------|-------------------|
| #1 VS YORK CHIL | 1       | H                 | IN/S  | Sep 20 1988 09:35 |
| Frequency       | 1 x RPM | Amplitude         |       |                   |
| 300             | 0.08    | 0.00731           |       |                   |
| 900             | 0.25    | 0.00699           |       |                   |
| 3600            | 1.00    | 0.04988           |       |                   |
| 6000            | 1.67    | 0.03590           |       |                   |
| 7200            | 2.00    | 0.01875 ← 9/20/88 |       |                   |
| 10800           | 3.00    | 0.01334           |       |                   |
| 21300           | 5.92    | 0.01843           |       |                   |
| 35700           | 9.92    | 0.01176           |       |                   |

FIGURE 6.12T

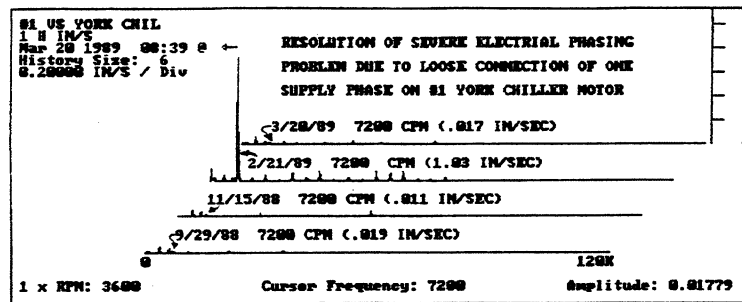
Survey by technicians from our company. Note the dramatic change that occurred at 2X line frequency on the survey of 2/21/89 increasing from .011 in/sec before up to 1.03 in/sec on 2/21/89. After consultation with engineers at our office, it was decided to lower the inverter frequency from 60 Hz to 53 Hz to see if the problem would remain, and what effect this would have on its spectral characteristics. Figure 6.12U shows a 3200 line FFT "zoom" around 2X line frequency which was now at 6371 CPM (106.2 Hz = 2X 53.1 Hz line frequency). Here, the important thing was the spacing of the sideband which was at about 1093 CPM (18.2 Hz) that placed it at 1/3 line frequency. A diagnostics of electrical phasing problems was made and recommendations were suggested to closely inspect all the supply lines and connectors, not only from the substation into the main supply, but also from the main supply into the inverter, and from the inverter into the motor itself. Subsequent checks found that one of the 3 connectors carrying one of the phases from the inverter directly into the motor itself was loose, making only sporadic contact from one instant to the next. Figure 6.12V shows the resultant effect on vibration after this loose connector was repaired. Note that the level at twice line frequency (7200 CPM) dropped dramatically from 1.03 in/sec down to .017 in/sec. It has since remained at this level during all subsequent Predictive Maintenance Surveys.



004% x 0.200 = 0.0080 1 x RPM: ??????

| Machine         | Pos     | Dir       | Units | Date              |
|-----------------|---------|-----------|-------|-------------------|
| BOS 1 YORK 53HZ | 1       | H         | IN/S  | Feb 21 1989 10:23 |
| Frequency       | 1 x RPM | Amplitude |       |                   |
| 5227.5          | -       | 0.0098    |       |                   |
| 5276.2          | -       | 0.0509    |       |                   |
| 6363.7          | -       | 0.0166    |       |                   |
| 6371.2          | -       | 0.2292    |       |                   |
| 6423.7          | -       | 0.0089    |       |                   |

**FIGURE 6.12U**  
**SIDE BANDS AT 1/3 LINE FREQUENCY DUE TO**  
**ELECTRICAL PHASING PROBLEMS**



001% x 2.0000 = 0.02000 1 x RPM: 3600.00

| Machine         | Pos     | Dir       | Units | Date              |
|-----------------|---------|-----------|-------|-------------------|
| #1 VS YORK CHIL | 1       | H         | IN/S  | Mar 20 1989 08:39 |
| Frequency       | 1 x RPM | Amplitude |       |                   |
| 3600            | 1.00    | 0.06068   |       |                   |
| 6000            | 1.67    | 0.02669   |       |                   |
| 10800           | 3.00    | 0.02192   |       |                   |
| 28500           | 7.92    | 0.02383   |       |                   |
| 50100           | 13.92   | 0.02288   |       |                   |

004% x 2.0000 = 0.08000 1 x RPM: 3600.00

| Machine         | Pos     | Dir       | Units | Date              |
|-----------------|---------|-----------|-------|-------------------|
| #1 VS YORK CHIL | 1       | H         | IN/S  | Feb 21 1989 08:14 |
| Frequency       | 1 x RPM | Amplitude |       |                   |
| 300             | 0.08    | 0.10008   |       |                   |
| 7200            | 2.00    | 1.03002   |       |                   |

001% x 2.0000 = 0.02000 1 x RPM: 3600.00

| Machine         | Pos     | Dir       | Units | Date              |
|-----------------|---------|-----------|-------|-------------------|
| #1 VS YORK CHIL | 1       | H         | IN/S  | Nov 15 1988 13:09 |
| Frequency       | 1 x RPM | Amplitude |       |                   |
| 3600            | 1.00    | 0.05274   |       |                   |
| 6000            | 1.67    | 0.03272   |       |                   |
| 21300           | 5.92    | 0.02097   |       |                   |
| 49800           | 13.83   | 0.03781   |       |                   |

000% x 2.0000 = 0.00000 1 x RPM: 3600.00

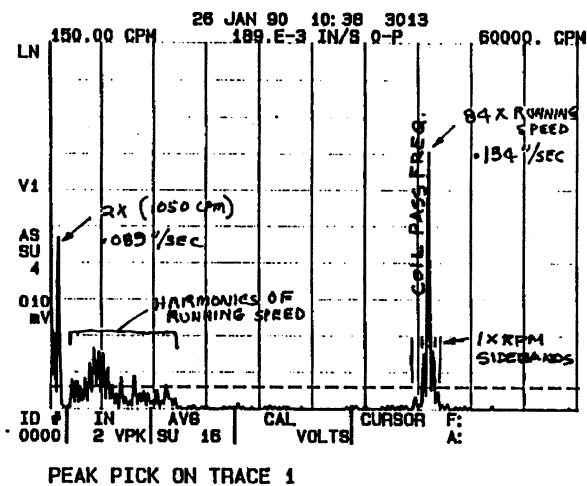
| Machine         | Pos     | Dir       | Units | Date              |
|-----------------|---------|-----------|-------|-------------------|
| #1 VS YORK CHIL | 1       | H         | IN/S  | Sep 20 1988 09:35 |
| Frequency       | 1 x RPM | Amplitude |       |                   |
| 300             | 0.08    | 0.00731   |       |                   |
| 900             | 0.25    | 0.00697   |       |                   |
| 3600            | 1.00    | 0.04988   |       |                   |
| 6000            | 1.67    | 0.03590   |       |                   |
| 7200            | 2.00    | 0.01875   |       |                   |
| 10800           | 3.00    | 0.01334   |       |                   |
| 21300           | 5.92    | 0.01843   |       |                   |
| 35700           | 9.92    | 0.01176   |       |                   |

FIGURE 6.12V

### 6.126 Synchronous Motors (Loose Stator Coils):

Figure 6.12W shows a spectrum for a synchronous motor having a problem with loose stator coils. This motor was driving a reciprocating air compressor and had experienced several bearing failures at intervals of only about 6 months. As Figure 6.12W shows, this spectrum resulted in a high vibration of .134 in/sec at about 43,200 CPM. A zoom spectrum showed that this corresponded to exactly 84X RPM. Further investigation revealed that this synchronous motor had a total of 14 poles with 6 coils per pole. Therefore, the coil pass frequency (CPF) equaled 14 poles X 6 coils/pole = 84X RPM. Note that the zoom spectrum showed not only high amplitude at coil pass frequency, but also sidebands spaced at 1X RPM of the motor.

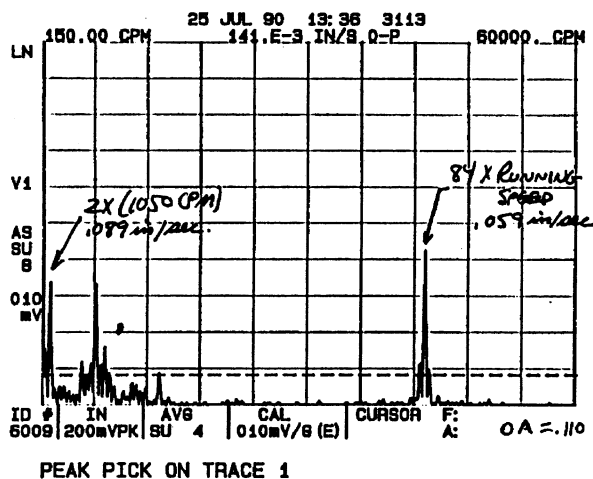
Here, the dominant problem was diagnosed to be loose stator coils. The "After" spectrum on Figure 6.12X shows the significant improvement made by repairing the poles and filling in all accessible voids between each of the coils dropping CPF from .134 in/sec to .059 in/sec.



| IN/S 0-P. | CPM    |
|-----------|--------|
| 40.6E-3   | 450.00 |
| 88.9E-3   | 1050.0 |
| 14.7E-3   | 2550.0 |
| 13.8E-3   | 3000.0 |
| 15.5E-3   | 4049.9 |
| 23.3E-3   | 4650.0 |
| 31.1E-3   | 5100.0 |
| 29.3E-3   | 5700.0 |
| 28.5E-3   | 6149.7 |
| 20.7E-3   | 6749.7 |
| 12.1E-3   | 7200.0 |
| 15.5E-3   | 8250.0 |
| 16.4E-3   | 9750.0 |
| 12.1E-3   | 13349. |
| 28.5E-3   | 42600. |
| 134.E-3   | 43198. |
| 30.2E-3   | 43650. |

29-JAN-90 FULL LOAD - 450 AMPS  
COMPRESSOR NO. 3  
POS 1 VERT  
14: 11: 57 JAN 26, 1990

FIGURE 6.12W  
LOOSE COILS IN SYNCHRONOUS MOTOR (BEFORE SPECTRUM)  
(14 POLES X 6 COILS/POLE = 84 X RPM)



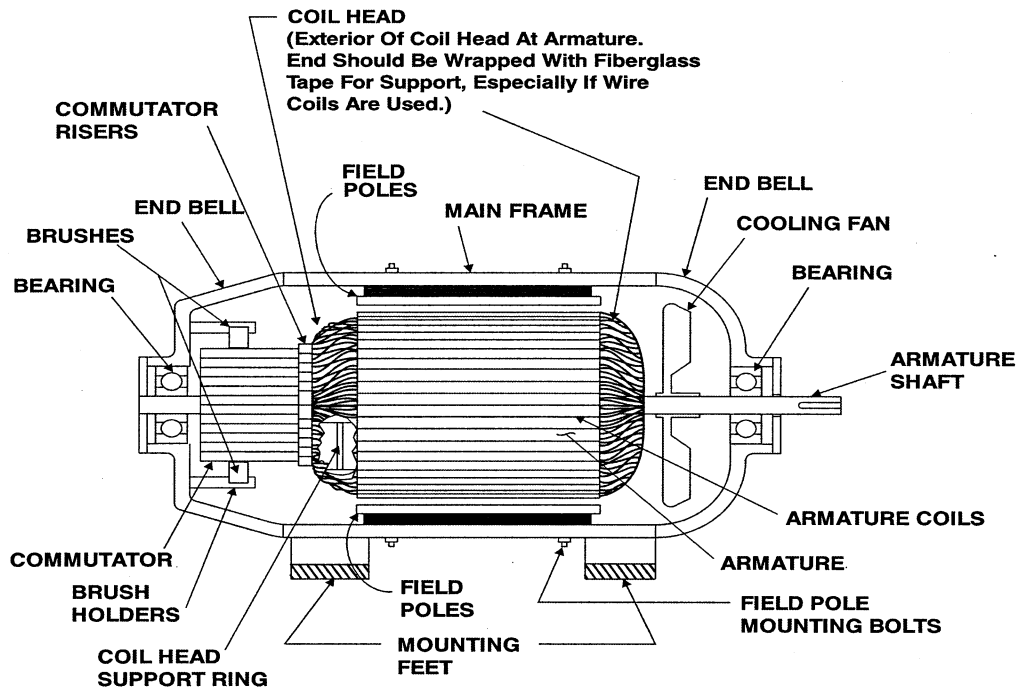
| IN/S 0-P. | CPM    |
|-----------|--------|
| 21.9E-3   | 450.00 |
| 47.6E-3   | 1050.0 |
| 16.7E-3   | 4650.0 |
| 11.6E-3   | 5100.0 |
| 16.1E-3   | 5700.0 |
| 47.0E-3   | 6149.7 |
| 15.5E-3   | 6749.7 |
| 22.5E-3   | 7200.0 |
| 12.2E-3   | 7650.0 |
| 12.2E-3   | 13349. |
| 16.1E-3   | 42748. |
| 59.9E-3   | 43198. |
| 13.5E-3   | 43650. |

26-JUL-90 FULL LOAD - 420 AMPS  
COMPRESSOR NO. 3  
POS 1 VERTICAL  
14: 27: 31 JAN 25, 1990

FIGURE 6.12X  
RESOLUTION OF LOOSE COIL PROBLEM (AFTER REPAIR)

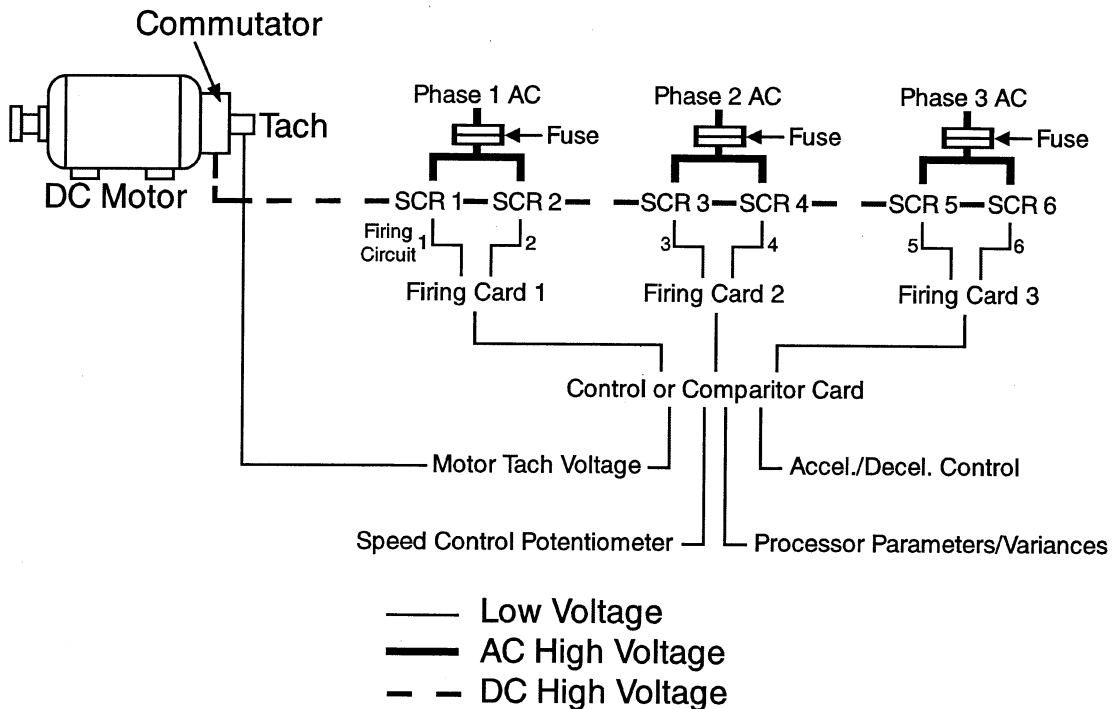
### 6.127 DC Motor Problems:

Figure 6.12Y shows the standard internal components of a common DC motor. Many DC motor and control problems can be detected through vibration analysis. A detailed discussion on DC motor construction and this functioning of each of its controls is beyond the scope of this section but is taught in a higher level seminar (Level III). However, the purpose here is to introduce the analyst to the fact that he can use vibration analysis to detect a series of problems, not only within the DC motor itself, but also with the controls serving it. Interestingly, when a DC motor is powered by a silicon controlled rectifier (SCR) circuit, many of the vibration problems found are actually SCR firing card or other control problems. SCR circuits have come into wide use throughout industry as a method to produce DC electricity from AC electricity. SCR circuits are either half-wave rectified with 3 SCR's or full-wave rectified with 6 SCR's, with each SCR creating a pulse at a rate of 3600 CPM, or 60 Hz (see Figure 6.12Z). A 3 SCR circuit electronically processes the 3 time waveforms of 3-phase 60 Hz (in U.S.) industrial electricity into a single waveform with only the positive AC peaks remaining. The result for a 3 SCR circuit is DC electricity with a 180 Hz (10,800 CPM) pulsing rate (i.e., 3 SCR's X 3600 = 10,800 CPM). A 6 SCR circuit processes both the positive and negative peaks of the alternating 3 phase waveform into a single DC waveform with 360 Hz (21,600 CPM) pulsing rate (i.e., 6 SCR's X 3600 CPM = 21,600 CPM). These pulsing rates of 10,800 CPM (3 SCR's) and 21,600 CPM (6 SCR's) affect the rotation of the motor armature and are detectable through vibration analysis. These pulsing rates are referred to as the SCR firing frequencies. Other characteristic vibration from SCR circuit problems can result in the appearance of line frequency harmonics of  $F_L$  and harmonics of the SCR firing frequency ( $F_L = 60$  Hz in U.S.). Amplitudes exceeding about .100 in/sec pk at the SCR firing frequency are usually considered excessive (although there are exceptions depending on size, operating speed, mounting method, etc.).



**FIGURE 6.12Y**  
**GENERAL DC MOTOR CONSTRUCTION**

Figures 6.12Y and 6.12Z picture the general construction of a DC motor as well as a 6 SCR (full-wave rectified) control circuit. The Diagnostic Chart (Figure 6.0) shows the many problems detectable through motor vibration signatures including broken armature windings, loose connectors, loose or blown fuses, shorted control cards and electrical fluting damage due to passage of electrical current through rolling element bearings. Each of these conditions is discussed in detail in higher level seminar texts.



**FIGURE 6.12Z**  
**TYPICAL FULL-WAVE RECTIFIED, 6 SCR DC MOTOR CIRCUIT WITH**  
**2 SCR'S PER FIRING CARD**

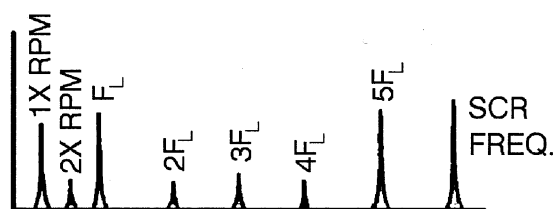
Figure 6.12AA illustrates a typical spectrum from a DC motor having a control problem due to a faulty SCR, shorted control card, loose connections and/or blown fuse. The problem is characterized by the pronounced appearance of multiple harmonics of the electrical line frequency ( $F_L$ ) (none of which should be present in a good running DC motor with properly functioning controls). Only the inherent pulsing of the SCR's at 10,800 CPM (for a 3 SCR circuit) or 21,600 CPM (for a 6 SCR circuit) should be present, but at amplitudes typically below about .100 in/sec pk. Figure 6.12BB is a spectrum from a 6 SCR powered DC motor with a blown fuse and a shorted control card (universal summing amplifier card). Note the presence of  $F_L$ ,  $2F_L$ ,  $3F_L$ ,  $4F_L$ , and  $5F_L$ . These  $F_L$  harmonics should not be present in a properly operating 6 SCR circuit. Figure 6.12CC was collected after repairs were made and shows no  $F_L$  harmonics were present after corrective actions were taken.

Again, a much more complete paper has been written concentrating not only on how each of the problems listed in the Diagnostic Chart (Figure 6.0) can be detected using vibration analysis, but also much information is provided on the construction of DC motors and controls and how they operate. This comprehensive DC motor paper (including real-world examples) is included within a higher level (Level III) seminar text, since it will require the analyst to have higher expertise in several related disciplines.

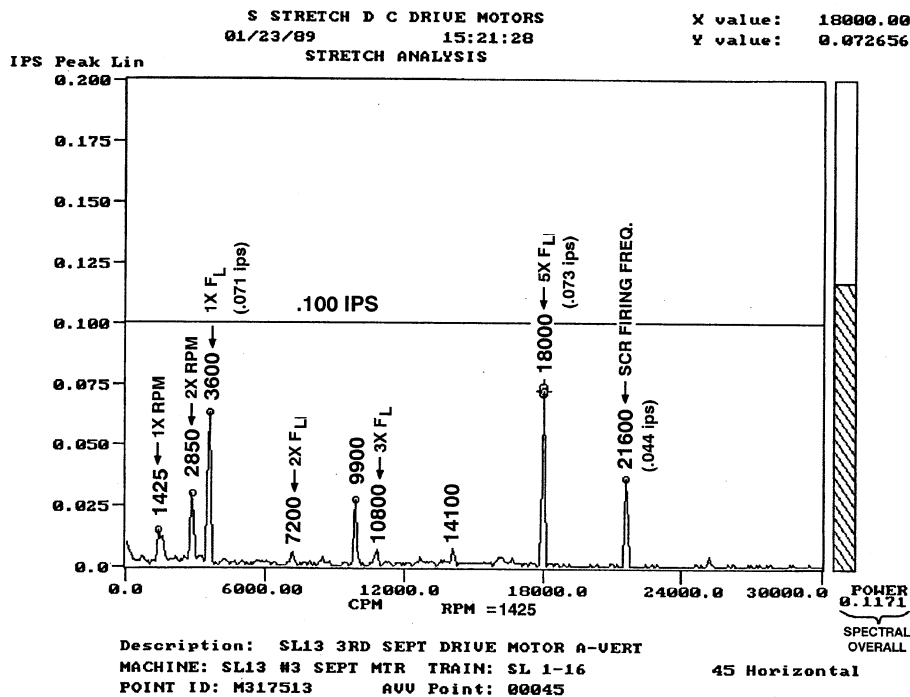
#### 6.128 Torque Pulse Problems:

Electric motors inherently have vibration due to "torque pulses" created as the rotating magnetic field energizes the stator poles. Normally, this vibration is quite low and not a problem. The torque pulses themselves will occur at 2X line frequency (7200 CPM for a 60 Hz line frequency). This occurs since each motor pole is essentially energized 2X for each cycle of AC current.

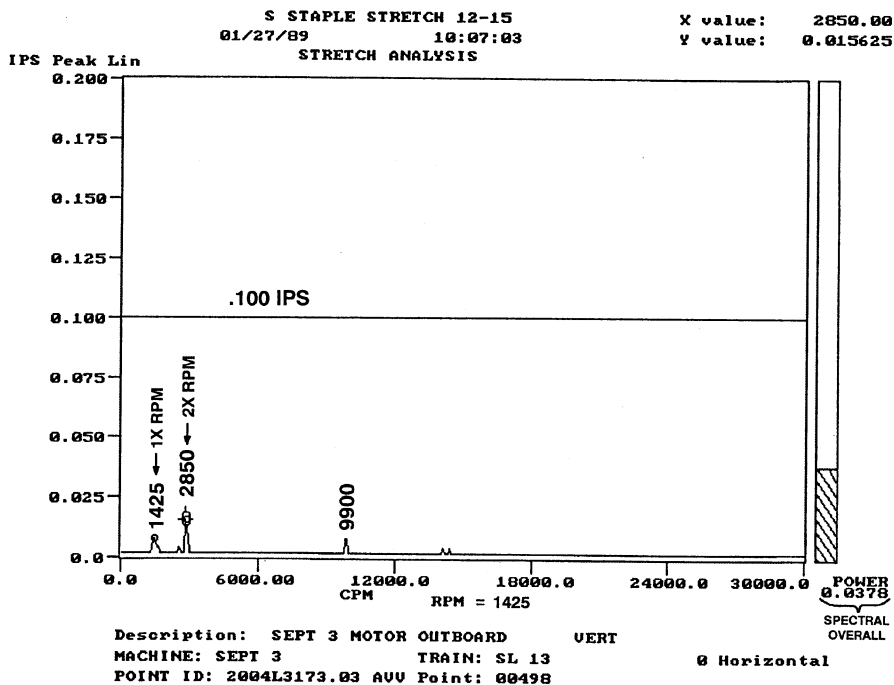
This vibration problem is extremely rare except in those cases where exceptionally low vibration levels are mandated as in the case of machine tools, or if these torque pulses should happen to excite a natural frequency in the machine or structure located at or near twice line frequency. Torque pulses also can excite loose rotor bars and loose stator windings at frequencies of 2X, 3X and even 4X torque pulse frequency (or 14,400; 21,600; and 28,800 CPM).



**FIGURE 6.12AA**  
**FAULTY SCR, SHORTED CONTROL CARD, LOOSE**  
**CONNECTIONS AND/OR BLOWN FUSE**



**FIGURE 6.12BB**  
**DC MOTOR SPECTRUM WITH A BLOWN FUSE AND SHORTED**  
**UNIVERSAL SUMMING CARD IN A 6 SCR CONTROL CIRCUIT**



**FIGURE 6.12CC**  
**DC MOTOR SPECTRUM AFTER REPAIRS COMPLETED TO THE**  
**6 SCR CONTROL CIRCUIT**

### 6.13 BELT DRIVE PROBLEMS

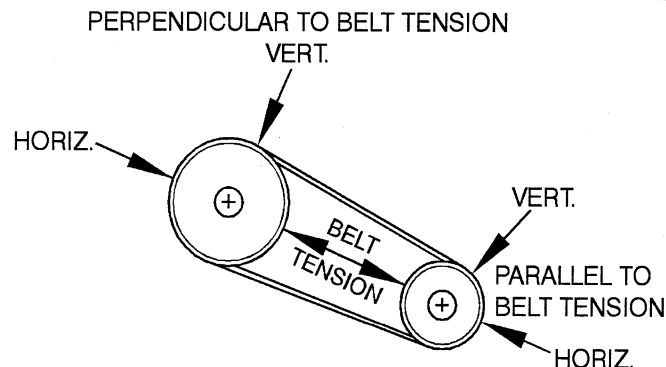
There are a whole variety of belt-driven machines ranging from those with inherently high vibration such as reciprocating air compressors to those requiring very low levels such as machine tool spindles. However, if the proper precautions are taken, the great majority of such machines should have low vibration. Probably 3 of the greatest factors affecting how much vibration a belt-driven machine will experience are:

- Alignment of Sheaves
- Sheave Concentricity
- Sheave Construction and Attachment Method

If each of these factors are carefully considered, there is no reason to have inherently high vibration in belt-driven machines.

Before discussing belt drive problems detectable by vibration analysis in particular, general statements concerning belt drives should be made:

1. It is best to take radial measurements in line with belt direction as shown in Figure 6.13A.



**FIGURE 6.13A**  
**PROPER MEASUREMENTS ON A BELT DRIVE**

2. Adjustable V-belt sheaves create undue vibration and premature belt and sheave deterioration. These devices have inherent vibration problems since it is not possible to keep sheave faces parallel with one another which allows belts to ride up and down in the grooves with each revolution. As a result, this creates belt tension variation which generates high vibration and accelerates belt and sheave wear.
3. Another critical factor in belt drives is the amount of sheave eccentricity (i.e., runout). Unfortunately, sheaves which are purchased for general utility machinery almost always have inherently high eccentricity much greater than that of other components which are used in general rotating machinery. As a result, as soon as these are assembled to the machine, they themselves can generate high vibration and even cause noticeable variations in belt length and tension with each revolution. It is up to industry to demand tighter concentricity tolerance sheaves and to enforce these specifications right at machine acceptance.

- 
4. V-belt drives are often blamed as the source of high vibration when, in fact, they are simply reacting to other problems such as unbalance, misalignment, mechanical looseness, etc. When these other problems are present, they can cause high vibration in the belts which themselves are not the source of the problem.

Following below are discussions on many of the belt drive problems which can be detected by vibration analysis:

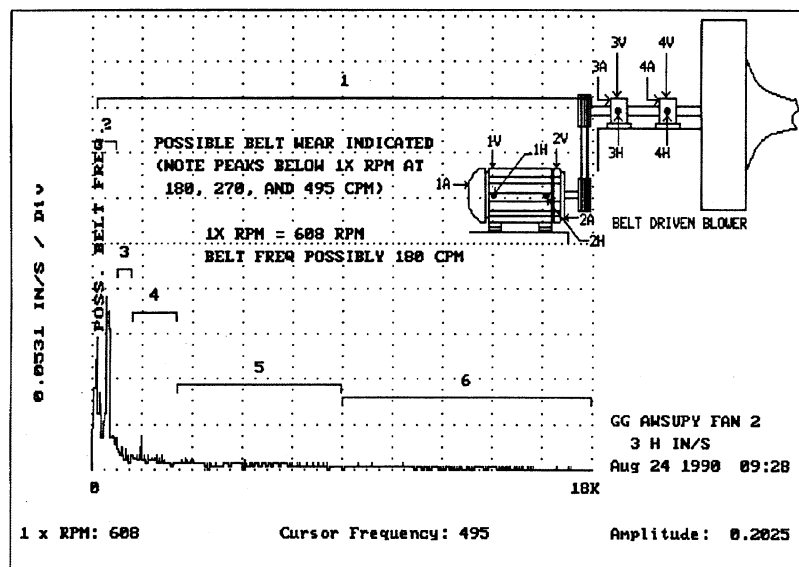
#### **6.131 Worn, Loose or Mismatched Belts:**

A typical spectrum indicating worn belts is given on page 4 of the **Illustrated Vibration Diagnostic Chart**. Note the belt frequency harmonics, all of which happen to be below both the driver and driven RPM in this example. The belt frequency (or belt RPM) is calculated as follows:

$$\text{Belt Freq.} = \frac{(3.142)(\text{Pulley RPM})(\text{Pulley Pitch Dia.})}{\text{Belt Length}}$$

When using the above equation, it is important to enter both the RPM and pitch diameter of the same pulley. It does not matter which pulley is used as long as the variables both come from the same pulley. Note that in all cases the belt RPM will be less than either the driver or driven RPM. However, belt frequency harmonics often will be higher than one or both of these. Worn, loose or mismatched belts display the following characteristics:

1. Worn belt defects detectable by vibration analysis include cracks, broken-off pieces of belt, hard and soft spots, lumps on belt faces, and also a crooked belt which has taken a set deformed shape during packing and storage.
2. When the problem is a worn belt(s), they will normally generate 3 to 4 multiples of belt frequency. Often, the 2X belt frequency peak may be dominant; in other cases, the fundamental belt frequency peak itself may not even show. In addition, worn belts sometimes lift the baseline of the spectrum throughout the subsynchronous frequency region, and just beyond driver and driven speeds as demonstrated in Figure 6.13B. In each case, worn belts normally cause unsteady amplitudes that sometimes pulsate with either the driver or driven RPM if any of these harmonics are close to either the driver or driven speed.
3. Belt defects usually show higher amplitudes in the direction parallel to belt tension. To get an idea how much vibration is caused by the belt defects themselves, compare amplitudes for the belt RPM frequencies themselves in a direction parallel with belt tension versus that in a direction perpendicular to belt tension.
4. Other belt specific problems which show up at belt RPM harmonics include belt width variations which cause the belts to ride up and down pulley grooves, creating vibration due to belt tension variations.
5. A loose cog belt is indicated by high vibration at the #cogs X RPM, and/or high vibration at the cog belt frequency itself which is calculated using the above equation.
6. Multiple V-belt drives can generate high vibration in the axial direction if they are unequally tensioned. This can result in excessive thrust bearing wear. These problems are sometimes solved by replacing several individual belts with one multi-belt which has been molded into one piece. If these are used, they place even greater importance on carefully aligning the sheaves.
7. Worn, loose or mismatched belts normally generate highest vibration in the radial direction, particularly in line with belt tension.



| Machine         | Pos     | Dir             | Units     | Date              |           |
|-----------------|---------|-----------------|-----------|-------------------|-----------|
| GG AWSUPY FAN 2 | 3       | H               | IN/S      | Aug 24 1990 09:28 |           |
| Frequency       | 1 x RPM | Amplitude       | Frequency | 1 x RPM           | Amplitude |
| 180             | 0.296   | 0.1550          | 2520      | 4.145             | 0.0175    |
| 270             | 0.444   | 0.0900          | 2610      | 4.293             | 0.0100    |
| 495             | 0.814   | 0.2025          | 3240      | 5.329             | 0.0100    |
| 585             | 0.962   | 0.1850 = 1x RPM | 3375      | 5.551             | 0.0125    |
| 765             | 1.258   | 0.0400          | 3600      | 5.921             | 0.0100    |
| 900             | 1.480   | 0.0375          | 5490      | 9.030             | 0.0100    |
| 1080            | 1.776   | 0.0250          |           |                   |           |
| 1215            | 1.998   | 0.0225          |           |                   |           |
| 1305            | 2.146   | 0.0150          |           |                   |           |
| 1440            | 2.368   | 0.0175          |           |                   |           |
| 1620            | 2.664   | 0.0175          |           |                   |           |
| 1800            | 2.961   | 0.0400          |           |                   |           |
| 1890            | 3.109   | 0.0175          |           |                   |           |
| 2025            | 3.331   | 0.0150          |           |                   |           |
| 2160            | 3.553   | 0.0125          |           |                   |           |
| 2340            | 3.849   | 0.0175          |           |                   |           |
| 2430            | 3.997   | 0.0125          |           |                   |           |

FIGURE 6.13B

---

### 6.132 Belt/Sheave Misalignment:

Probably one of the greatest sources of belt drive vibration is misalignment of driver and driven sheaves. It is often amazing to see how much vibration can be reduced simply by employing a chalk line between sheaves. Misaligned sheaves display the following characteristics:

1. Axial vibration is most always generated by sheave misalignment which can highly accelerate the rate of wear of thrust bearings.
2. Misaligned sheaves produce high vibration at 1X RPM, predominately in the axial direction. Dominant vibration is quite often at driver RPM, but occasionally at driven RPM. The ratio of amplitudes of driver to driven RPM depends exactly on where the data is taken as well as on relative mass and frame stiffness.
3. Often with sheave misalignment, the highest axial vibration on the motor will be at fan RPM, while the highest axial when measuring on the fan will be at motor RPM. However, this is not always the case.

### 6.133 Eccentric Sheaves:

Eccentric sheaves are one of the greatest contributors to high vibration in belt-driven machines today, often due to a lack of emphasis on specifying good concentricity in purchase specs. Sheave eccentricity displays the following characteristics:

1. Eccentric and/or unbalanced sheaves cause high vibration at 1X RPM of the eccentric sheave.
2. Highest amplitude is normally in line with the belts and should show up on both the driver and driven sheaves.
3. Unlike unbalance, reaction forces caused by an eccentric pulley are not equally applied throughout the entire 360° rotation of the pulley. The force instead is concentrated in the direction of belt tension along a line passing through the centerline of the 2 shafts. As a result, this highly directional vibration will show comparative horizontal and vertical phase readings which either are identical or 180° opposite one another, depending on which side of the bearing the transducer is located for the measurement. In any case, both phase readings show that the bearing is moving in one line.
4. Since the forces are so directional in nature, the resultant vibration cannot be totally corrected by balancing via attaching washers to taperlock bolts. Even if balanced, the sheave eccentricity will still induce vibration in the belt due to belt length and tension variations, and will result in premature accelerated wear of belts and/or sheaves along with the driver and driven bearings.

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#### 6.134 Belt Resonance:

Just like everything else in nature, belts themselves have a natural frequency which corresponds to the stiffness of the belt, the belt mass, and how much deflection is induced in the belt during operation. The natural frequency of a belt can be determined simply by pulling on the belt, releasing it and measuring the response. Belt resonance will display the following characteristics:

1. A spectrum indicating a belt resonance problem is shown on page 5 of the Table 6.0 **Illustrated Vibration Diagnostic Chart**. In this example, the belt natural frequency lies very close to 1X RPM of either the driver or driven machine. If this occurs, this will cause great flapping of the belt, particularly on the tension side at a frequency corresponding to the belt natural frequency.
2. Not only can a pulley speed excite a belt resonance, but also this can occur if a belt RPM harmonic should line up with its natural frequency.
3. Both the amplitude and phase of vibration at the belt resonant frequency will be unsteady.
4. The belt natural frequency itself can be changed either by altering the belt tension, belt length, sheave center distance, adding an idler pulley, etc..

#### 6.135 Excessive Motor Vibration At Fan Speed Due to Motor Frame/Foundation Resonance:

Often, in the case of belt-driven machines, there will be excessive vibration on the motor. However, when a spectrum analysis is taken, it will show low vibration on the motor at motor speed, but high vibration on it at fan RPM. This can occur in either radial or axial directions. This will be evident by an excessive vibration at 1X RPM of the driven unit, particularly in one direction on the motor. Often, when this occurs, the problem source is excitation of a motor frame or foundation natural frequency by the incoming fan speed vibration. This can be confirmed by simple impulse natural frequency tests explained in Section 6.05. In this case, the solution will normally be stiffening of the frame or foundation by addition of bracing or addition of concrete on the base if it is resonant.

#### 6.136 Loose Pulley or Fan Hub:

Excessive vibration sometimes occurs in belt-driven machines due to looseness either of a pulley or a fan hub itself. This is evidenced by each of the following characteristics:

1. There will be excessive vibration particularly at 1X RPM, but also at several running speeds harmonics.
2. One of the best indicators will be unstable phase. If operating properly, the phase in horizontal, vertical and axial directions should be steady. If a fan hub held on by set screws is loose on the shaft, it may show a difference in both phase and amplitude each time the unit is started up. In these cases, balancing would be only a temporary solution. It may help for a "couple of hours", but later when the fan hub or pulley rotates slightly on the shaft, it will upset the whole balance possibly dramatically changing the phase and amplitude. This same thing can occur if either the pulley or fan hub has a taper fit, and is not properly pulled up and fastened to the shaft.
3. The solution to each of these cases is ensuring that all rotors attached to the shaft are securely fastened, and properly oriented in the case of a taper fit.

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## 6.14 BEAT VIBRATION

Beat vibration can occur either when 2 frequencies lie close to one another, or a single frequency is continually changing in amplitude and frequency. More often than not "beat vibration" occurs when 2 or more steady-state vibration sources generate frequencies in close proximity to one another.

For example, if 2 fans are mounted on the same foundation and happen to operate at nearly identical speeds, they will generate a beat vibration which can both be felt and heard. For example, if one fan were operating at 900 RPM and the other at 950 RPM, they would generate a beat frequency of 50 RPM. When viewed in an analyzer with insufficient resolution to separate the 2 nearly identical frequencies, the analyst will see one peak pulsating up and down. For example, if his maximum frequency were set at 60,000 CPM (150 CPM resolution), the analyzer would display one peak at 900 CPM. If he could see this in "real-time", the peak would pulsate. If the analyst has "zoom capability", he can zoom in on this so called 900 CPM frequency and resolve it into the 2 individual peaks. Also, if the analyst had a low frequency seismic transducer, he could likewise detect the beat frequency itself down at 50 CPM.

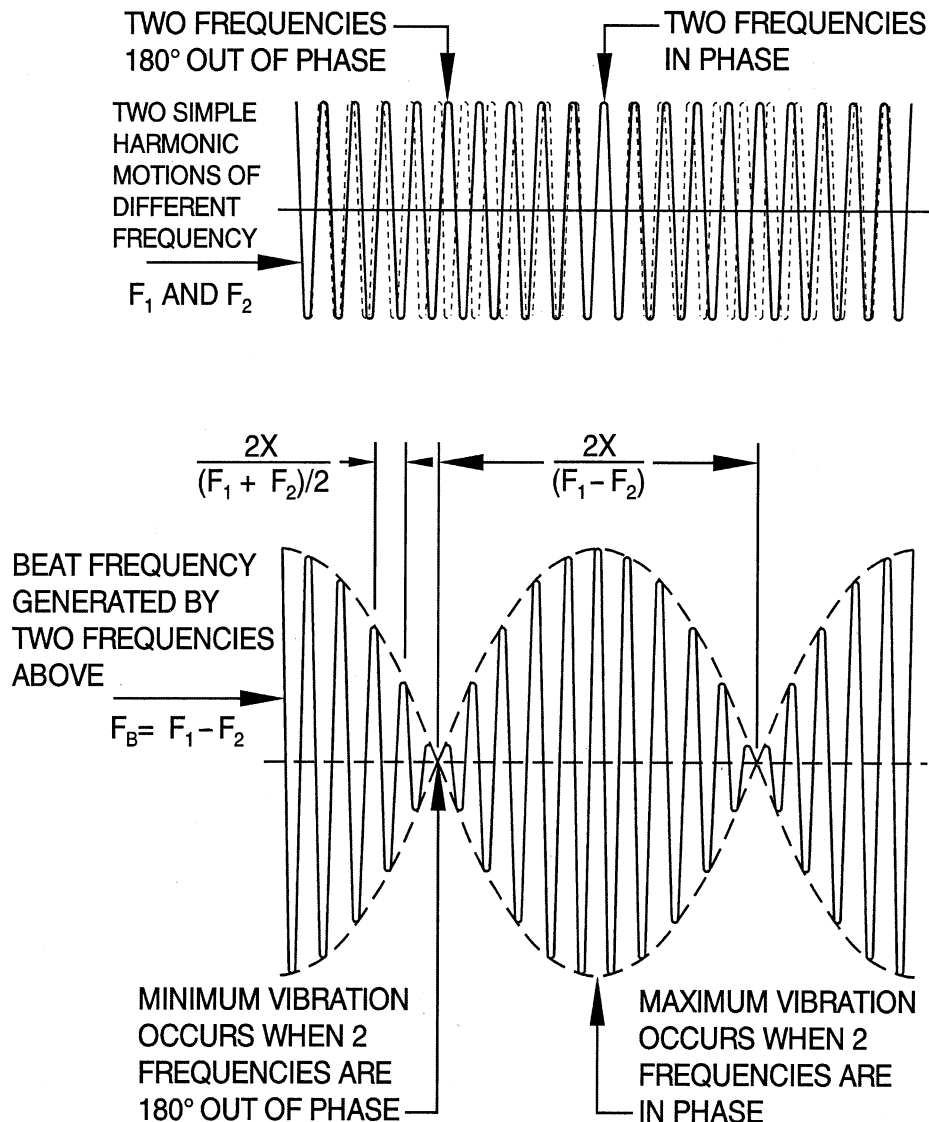
Audible beat frequencies are common in the case of induction electric motors which have stator or rotor problems. In these cases, a running speed harmonic will often occur close to twice line frequency, again causing a beat. For example, if a 3580 CPM motor developed an eccentric rotor problem causing high levels at 7200 CPM, its 2nd running speed harmonic at 7160 CPM would beat with 2X line frequency (7200 CPM), and would likewise generate a beat frequency itself at 40 CPM.

Figure 6.14A demonstrates what happens in a beat frequency. Here each of 2 time waveforms that correspond to 2 individual frequencies are shown in the upper part of the figure. Note that the time waveforms are slightly different meaning that the frequencies are likewise slightly different. However, periodically both waveforms peak at the same time as they come into phase with one another. At this point, they effectively add to one another causing the highest amplitudes as seen in the beat frequency time waveform itself in the lower diagram in this figure. Referring back to the upper figure, note that at other times, the time waveforms of the 2 frequencies are 180° out of phase, effectively cancelling one another. Looking directly below this point on the lower time waveform, note that the amplitude here goes almost to 0. Therefore, this explains why one feels and hears pulsating noise and vibration. Note that the lower time waveform is in fact the beat frequency generated by the 2 frequencies. As the figure shows, this beat frequency ( $F_b$ ) is in fact the difference frequency ( $F_1 - F_2$ ).

Figure 6.14B illustrates beat vibration on a real machine. This spectrum was taken on a large gearbox whose output was to a long jackshaft. Note that the input to the gearbox from the drive motor was at 1477 RPM and that one of its outputs to the jackshaft was at 1395 RPM (a difference of only 82 CPM). When looking at a much wider frequency scale, it combined these 2 individual peaks into one and showed them to be pulsating greatly. Figure 6.14C shows the beat frequency itself which was the difference between the higher drive motor and lower jackshaft operating speeds.

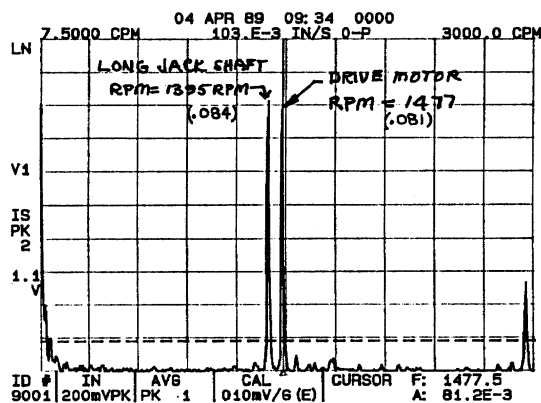
Beat frequencies themselves are not necessarily a problem. However, if one desires to balance one of the machines, for example, he will find it very difficult, particularly if attempting to do so with a strobe light instrument. If a beat frequency is involved, often the strobe light image will continually rotate at the beat frequency itself. It becomes impossible to proceed with any type of balance since phase cannot be measured.

In another scenario, 2 machines running alone may be within vibration severity criteria. However, when both are running, the resultant is that both machines are thrown above alarm limits. In this case, one solution is to change the speed of one or both units so that they will no longer beat. Another solution is to place vibration isolation materials under one or both machines to keep the vibration within each from affecting the other. In general, it is a good idea to keep significant forcing frequencies a minimum of approximately 10% away from one another in order to prevent the pulsating beat frequency vibration. In any case, beat frequencies are not normally a problem when the differences exceed 150 to 200 CPM.



**FIGURE 6.14A**  
**ILLUSTRATION OF A BEAT FREQUENCY**



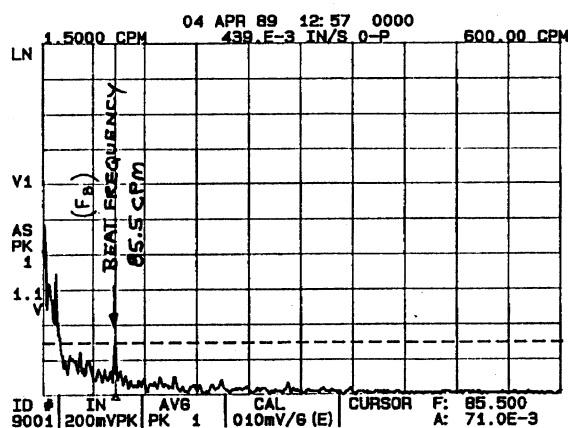


PEAK PICK ON TRACE 1

| IN/S 0-P                   | CPM    |            |
|----------------------------|--------|------------|
| 20.7E-3                    | 37.500 |            |
| 9.88E-3                    | 60.000 |            |
| 84.2E-3 = .084             | 1394.9 | JACK SHAFT |
| 81.4E-3 = .081             | 1477.4 | MOTOR      |
| 27.7E-3                    | 2955.0 |            |
| $\Delta \text{RPM} = 82.5$ |        |            |
| DRAW FRAME 1 - HIGH SPEED  |        |            |

DRAW MACHINE NO. 1  
VERTICAL ON FLOOR  
OVERALL VIB. = .128"/sec

**FIGURE 6.14C**  
**PULSATING AMPLITUDES OF JACKSHAFT & MOTOR SPEED**  
**FREQUENCIES DUE TO BEAT VIBRATION**  
(AMPLITUDES PULSATED APPROX.  $\pm 50\%$ , OR .04 UP TO .12 IN/SEC EACH)



PEAK PICK ON TRACE 1

| IN/S 0-P | CPM    |   |
|----------|--------|---|
| 214.E-3  | 3.0000 |   |
| 140.E-3  | 7.5000 |   |
| 152.E-3  | 16.500 |   |
| 70.2E-3  | 85.500 | ← |

DRAW MACHINE NO. 1  
VERTICAL ON FLOOR

**FIGURE 6.14D**  
**GENERATION OF BEAT FREQUENCY ITSELF**

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## REFERENCES

1. Parker, Sybil P., Editor and Chief; McGraw-Hill Dictionary of Mechanical and Design Engineering; McGraw-Hill Book Company; New York, NY; 1984.
2. Mitchell, John S.; An Introduction To Machinery Analysis and Monitoring; Pennwell Publishing Company; Tulsa, Oklahoma; 1981; Pages 141 - 151, 172 - 204.
3. Piotrowski, John D.; "Alignment Condition and Its Effect on The Vibration Response of Rotating Machinery"; Vibrations, Volume 1, No. 4; March, 1986; Pages 11 - 17.
4. Buscarello, Ralph T.; Practical Solutions to Machinery and Maintenance Vibration Problems; Published by Update International, Inc.; Denver, CO; Revised April, 1987.
5. Buehler, Mark W. and Bertin, C. D.; Proceedings Machinery Vibration Monitoring and Analysis Meeting - The Vibration Institute; Houston, TX; April 19 - 21, 1983; Pages 191 - 206.
6. Bently, Donald E.; President and Chairman, Bently Nevada, Inc.; "Rubs Research - Studies Reveal Physical Phenomena of Rotor Rubs"; Orbit, Volume 4, No. 3; October, 1983; Pages 3 - 5.
7. Maxwell, J. Howard; "Induction Motor Magnetic Vibration"; Proceedings Machinery Vibration Monitoring and Analysis Meeting - The Vibration Institute; April, 1983, Pages 39 - 51.
8. Advanced Audio-Visual Customer Training Instruction Manual - "Vibration Measurement and Analysis"; IRD Mechanalysis; Columbus, OH; 1985.
9. Makay, Elemer and Barrett, J. A.; "Field Experience Brings Help to Embattled Pump Users"; Power Magazine; July, 1987; Pages 27 - 30.
10. Salamone, Dana J.; "Introduction to Hydrodynamic Journal Bearings"; Mini-Course Notes Machinery Vibration Monitoring and Analysis Meeting - The Vibration Institute; New Orleans, LA; May 22 - 24, 1985; Pages 41 - 56.
11. Middleton, Ben; "Rolling Element Bearing Failure Detection Methods"; Presented at the Acoustical Society of America, Raleigh, NC, October 8 - 9, 1987; Pages 1 - 14.
12. Taylor, James I.; "Determination of Antifriction Bearing Condition by Spectral Analysis"; Sixth in a Series of Technology Interchange on Machinery Vibration Monitoring and Analysis; The Vibration Institute; Clarendon Hills, IL; Pages 1 - 26; 1978.
13. Szrom, David B.; "Low Speed Bearing Analysis"; Proceedings Tenth Annual Meeting - The Vibration Institute; Las Vegas, NE; June 24 - 26, 1986; Pages 183 - 188.
14. Springer, C. W.; "Spectral Analysis of Double-Row Antifriction Bearings"; Vibrations; Volume 4, No. 1; March, 1988; Pages 16 - 17.
15. Springer, C. W.; "The Role of the Time Domain in Analyzing Bearing Defects"; Vibrations; Volume 4, No. 3; September, 1988; Pages 14 - 15.
16. Berggren, J. Charles; "Diagnosing Faults in Rolling Element Bearings - Part I. Assessing Bearing Condition"; Vibrations; Volume 4, No. 1; March, 1988; Pages 5 - 14.  
And "Diagnosing Faults in Rolling Element Bearings - Part II. Alternative Analytical Methods"; Vibrations; Volume 4, No. 2; June, 1988; Pages 12 - 23.

- 
17. Bently, Donald E.; Zimmer, Siegfried; Palmatier, George E.; and Muszynska, Agnes; "Interpreting Vibration Information from Rotating Machinery"; Sound and Vibration Magazine; Volume 20, No. 2; February, 1986; Pages 14 - 23.
  18. Schlitz, Richard L.; "Forcing Frequency Identification of Rolling Element Bearings"; Sound and Vibration Magazine; Volume 24, No. 5; May, 1990.
  19. Berggren, J. Charles; "Diagnosing Faults in Rolling Element Bearings, Part III. Electronic Data Collector Applications"; Vibrations; Volume 5, No. 2; June, 1989; Pages 8 - 19.
  20. Fundamentals of Balancing; Published by Schenck Trebel Corp; Deer Park, L.I., New York; Second Edition; March, 1983, Pages 46 - 47.
  21. "Vibration and Noise Analysis, Dynamic Balancing and Preventive Maintenance"; Audio-Visual Customer Training Instruction Manual; Published by IRD Mechanalysis, Inc.; Columbus, OH; 1975; Page 108.
  22. Fasig, Paul; Gahagan, Richard; and Abernathy, Joe; "DC Motor and SCR Firing Vibrations" (Article Submitted to Vibration Institute); 1989; Pages 1 - 12.
  23. Szrom, David B.; "Determining Gear Condition With FFT Spectrum Analysis"; Proceedings 11th Annual Meeting - The Vibration Institute; June, 1987; Pages 1 - 5.
  24. Winterton, John G.; "Component Identification of Gear Generated Spectra"; Proceedings 11th Annual Meeting - The Vibration Institute; June, 1987; Pages 11 - 17.
  25. Vibration Technology - 1; Published by IRD Mechanalysis; Columbus, OH; 1989; Pages 2-2 thru 2-10; 3-2 thru 3-15; and 5-2 thru 5-31.
  26. Berry, James E.; "Required Vibration Analysis Techniques and Instruments on Low Speed Machines"; Advanced vibration Diagnostics and Reduction Techniques Seminar Text; Technical Associates of Charlotte, P.C.; Charlotte, NC; 1993.
  27. Hoskins, Robert; "Electrically Induced High Frequencies: Their Resulting Effect on Bearings and What to Do About It"; as given in "Case History No. 106 " by Update International, Inc.; Denver, CO.
  28. Berry, James E.; "What is High Frequency Enveloping and Demodulation Spectral Analysis and How Can it Be Used to Detect Problems with Rolling Element Bearings, Gears, & Lubrication?"; Vibration Analysis Level II Seminar Text - "Special Vibration Analysis Techniques and How to Best Employ Narrowband Spectral Envelope Alarms"; Technical Associates of Charlotte, P.C.; Charlotte, NC; 1993.
  29. Bowers, Dr. Stuart; "Amplitude Demodulation - A Powerful Tool for Predictive Maintenance"; CSI Technical Paper AN-P-027; Computational Systems, Incorporated.
  30. Ryckalts, Eddy; "Effective Enhanced Techniques for Early Detection, Monitoring, Trending and Diagnosis of Bearing Damage"; SKF Condition Monitoring; Utrecht, The Netherlands.
  31. McLain, Douglas A. and Hartman, David L.; "New Instrumentation, Techniques Accurately Predict Bearing Life"; Pulp & Paper Magazine; February, 1981.

## APPENDIX

### AFBMA NOMENCLATURE — BALL BEARINGS

| Bearing Bore<br>D D D                       | Bearing Type<br>L L L                 | Dimension Series<br>D D | Modifications<br>L LL* L*                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Fitup & Tolerance<br>D** D**                                                                                             | Lubricant<br>L**                                                                                                            | L designates a letter<br>D designates a digit |
|---------------------------------------------|---------------------------------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| Bearing Bore in Millimeters (2 or 3 Digits) | Type of Ball bearing (2 or 3 Letters) | Dimension Series        | Type of Cage or Ball Retainer<br>Column 2 or 3 Used for R, T, or U                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Internal Clearance                                                                                                       |                                                                                                                             |                                               |
|                                             |                                       |                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Tolerances                                                                                                               |                                                                                                                             |                                               |
|                                             |                                       |                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                          | R - Low temperature silicone grease 100°F<br>H - High temperature grease 275°F<br>S - High temperature silicon grease 300°F |                                               |
|                                             |                                       |                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0 - Standard fitup - AFBMA 0 or C/0<br>3 - Loose internal fitup - AFBMA 3 or C/3<br>4 - Looser than C/3 - AFBMA 4 or C/4 |                                                                                                                             |                                               |
|                                             |                                       |                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | G - Snap ring and groove in outer race<br>D - Single bearing modified for duplex mounting (DB, DF, or DT)                |                                                                                                                             |                                               |
|                                             |                                       |                         | P - Single shielded - permanently fastened<br>PP - Double shielded - permanently fastened<br>E - Single sealed - permanently fastened<br>EE - Double sealed - permanently fastened<br>KK - Double labyrinth seals for cartridge bearings<br>R - Pair of bearings modified for duplex mounting - back to back (DB)<br>T - Pair of bearings modified for duplex mounting - in tandem (DT)<br>U - Pair of bearings modified for duplex mounting - face to face (DF)<br>X - None of the above - see footnote *<br>XX - None of the above - see footnote * |                                                                                                                          |                                                                                                                             |                                               |
|                                             |                                       |                         | J - Standard - steel, sheet or strip form, centered by the balls<br>K - Bronze or brass, not sheet or strip, centered by one race<br>M - Bronze or brass, not sheet or strip, centered by the balls<br>D - Non-metallic (phenolic), centered by one race<br>X - Any type of cage<br>Y - Non-ferrous metal, sheet or strip form, centered by the balls                                                                                                                                                                                                 |                                                                                                                          |                                                                                                                             |                                               |
|                                             |                                       |                         | 10 - Extra light, e.g., 100-KS (MRC) or 6000 (SKF)<br>02 - Light, e.g., 200-S (MRC) or 6200 (SKF)<br>03 - Medium, e.g., 300-S (MRC) or 6300 (SKF)<br>32 - Light series wide cartridge type sealed bearings, e.g., 200-SFFC (MRC)<br>33 - Medium series wide cartridge type sealed bearings, e.g., 300-SFFC (MRC)                                                                                                                                                                                                                                      |                                                                                                                          |                                                                                                                             |                                               |
|                                             |                                       |                         | BC - Standard deep groove ball bearing, e.g., 206-S (MRC) 6308 (SKF)<br>BL - Maximum capacity ball bearing with filling slot, e.g., 222-M (MRC), 222-W (FAFNIR)<br>BH - Maximum capacity ball bearing double assembly, e.g., 222-R (MRC)<br>BT - Angular contact thrust bearing, e.g., 7313P (MRC)                                                                                                                                                                                                                                                    |                                                                                                                          |                                                                                                                             |                                               |
|                                             |                                       |                         | e.g., 203 = 17, 204 = 20, 307 = 35, 222 = 110, i.e., the last two digits are multiplied by 5 to obtain the bore, excepting the 203 or 303 size bearing which is 17.                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                          |                                                                                                                             |                                               |
|                                             |                                       |                         | Example: 35BC02JPP0 - Standard deep groove, light series, standard steel cage, double shielded, standard fitup and standard tolerances MRC 207-SFF or equivalent.                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                          |                                                                                                                             |                                               |
|                                             |                                       |                         | Example: 110BT03MT03 - This is a pair of 7322DT angular contact bearings mounted in tandem, standard fitup and ABEC3 tolerances.                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                          |                                                                                                                             |                                               |
|                                             |                                       |                         | * The letters for columns 2, 3 and 4 of modifications are omitted if none are applicable. If column 4 is applicable but not 3, or 2 and 3, an X or XX is used in column 3, or 2 and 3, e.g., 35 BC02J03 or 35BT03MXXD03.                                                                                                                                                                                                                                                                                                                              |                                                                                                                          |                                                                                                                             |                                               |
|                                             |                                       |                         | ** If these three columns are omitted, standard fitup, tolerances, and greases are implied.                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                          |                                                                                                                             |                                               |

Table 2-10

## APPENDIX

### AFBMA NOMENCLATURE — ROLLER BEARINGS

| Bearing Bore<br>D D D                                                                           | Bearing Type<br>L L L                                                                                                                                                                                                                                                                                                                                                             | Dimension Series<br>D D                                                                                                                                                                                                                                                                                       | Modifications<br>L L                                                                                                                                                                                                                 | Fitup & Tolerance<br>D D             | L designates a letter<br>D designates a digit                                                                                                                                                                                        |
|-------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bearing Bore in Millimeters (2 or 3 Digits)                                                     | Type of Roller Bearing (2 or 3 Letters)                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                               | Type of Cage or Roller Retainer                                                                                                                                                                                                      | Internal Clearance                   | 0 - Standard tolerances RBEC 1<br>3 - More precision than 0 RBEC 3<br>5 - More precision than 3 RBEC 5<br><br>0 - Normal internal clearance<br>2 - Internal clearance less than normal<br>3 - Internal clearance greater than normal |
|                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                      | NOT APPLICABLE - THIS COLUMN OMITTED |                                                                                                                                                                                                                                      |
|                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                               | X - Any type of cage<br>K - Brass or bronze, not in sheet or strip form, ring centered<br>M - Brass or bronze, not in sheet or strip form, centered by rolling element<br>H - Light metal, not in sheet or strip form, ring centered |                                      |                                                                                                                                                                                                                                      |
|                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                   | 02 Light series, e.g., N208 (SKF)<br>03 Medium series, e.g., N308 (SKF)<br>22 Series used for self aligning spherical roller bearings, e.g., 22222 (SKF)<br>23 Series used for self aligning spherical roller bearings, e.g., 22322 (SKF)<br>94 Series used for spherical roller thrust bearings, e.g., 29430 |                                                                                                                                                                                                                                      |                                      |                                                                                                                                                                                                                                      |
|                                                                                                 | RN - Cylindrical roller bearing, shoulders on inner ring<br>RU - Cylindrical roller bearing, shoulders on outer ring<br>RJ - Cylindrical roller bearing, shoulders on each side of outer ring and one side of inner ring<br>SC - Double row self-aligning roller bearing<br>SD - Double row self-aligning roller bearing<br>TS - Spherical roller thrust bearing, e.g., SKF 29430 |                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                      |                                      |                                                                                                                                                                                                                                      |
| e.g., N208 = 40, 22222 = 110, i.e., the last two digits are multiplied by 5 to obtain the bore. |                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                      |                                      |                                                                                                                                                                                                                                      |

Table 2-11

### RADIAL BALL BEARINGS INTERCHANGEABILITY CHART

|                                                     | SKF    | MRC    | FAFNIR | NORMA<br>HOFFMAN | HOOVER | NEW<br>DEPARTURE |
|-----------------------------------------------------|--------|--------|--------|------------------|--------|------------------|
| Deep groove, radial, light series, open type        | 6200   | 200S   | 200K   | 200              | 200    | 3200             |
| Deep groove, radial, medium series, open type       | 6300   | 300S   | 300K   | 300              | 300    | 3300             |
| Deep groove, radial, light series, single shielded  | 6200Z  | 200SF  | 200KD  | 200P             | 7200   | 7500             |
| Deep groove, radial, medium series, single shielded | 6300Z  | 300SF  | 300KD  | 300P             | 7300   | 7600             |
| Deep groove, radial, light series, double shielded  | 62002Z | 200SFF | 200KDD | 200PP            | 77200  | 77500            |
| Deep groove, radial, medium series, double shielded | 63002Z | 300SFF | 300KDD | 300PP            | 77300  | 77600            |

Table 2-12

